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题目：3/2 波动率模型下的期权定价：多方法比较与极端场景下的新框架

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Precision Option Pricing under the 3/2 Volatility Model: A Multi-Method Comparison and a Novel Framework for Extreme Scenarios

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ABSTRACT

The 3/2 stochastic volatility model, by virtue of its nonlinear diffusion structure, exhibits significant theoretical advantages in capturing extreme asset price volatility and fitting the volatility smile, making it a core tool for derivatives pricing in highly volatile markets. However, existing performance evaluations of pricing methods under this model are largely confined to idealized academic benchmark cases in stable markets, lacking empirical validation in actual extreme market environments. Consequently, the engineering applicability boundaries of these methods remain systematically unclarified.

To address this gap, this paper systematically constructs a comprehensive theoretical framework for nine European option pricing methods under the 3/2 model within a pure diffusion framework. These methods are categorized into three major paradigms: time-stepping conditional Monte Carlo methods, exact and near-exact terminal simulation methods, and Fourier frequency-domain pricing methods. The implementation logic, error characteristics, and convergence properties of each paradigm are rigorously derived. Building upon this, through seven sets of numerical experiments covering both conventional markets and extreme parameter environments, this study quantitatively compares the pricing accuracy, computational efficiency, and full-scenario numerical stability of these methods, thereby establishing a clear hierarchical performance framework.

Furthermore, taking the extreme Bitcoin flash crash of October 2025 as a case study, this paper develops a complete empirical framework consisting of “long-term fixed-parameter time-series estimation —time-varying parameter cross-sectional calibration —pricing method re-evaluation in extreme scenarios.” This validates the 3/2 model’s adaptability to extreme cryptocurrency market dynamics, identifies the event-driven time-varying patterns of core model parameters throughout the crash cycle, and executes a performance re-evaluation of the pricing methods under genuine extreme market conditions.

The findings indicate that the Fourier-Cosine (Fourier-Cos) method is the globally optimal scheme for European option pricing, achieving exceptional pricing precision, numerical stability, and computational efficiency across all scenarios. The Poisson-Gamma and Quasi-Exponential (QE) near-exact stepping methods are the sole robust solutions across all scenarios within the Monte Carlo framework, offering a reliable implementation pathway for pricing path-dependent derivatives and dynamic hedging. Conversely, the nominally "exact" terminal simulation methods can only be theoretically validated under idealized benchmark parameters; they prevalently experience numerical overflows and systematic pricing deviations in extreme market environments, thus lacking suitability for practical engineering applications.

This research bridges the existing disconnect between theoretical simulations and real-world markets in 3/2 model pricing literature, establishes method selection guidelines for diverse application scenarios, and provides systematic theoretical support and empirical foundations for derivatives pricing, model calibration, and risk management in highly volatile markets.

KEY WORDS: Option Pricing, 3/2 Volatility Model, Exact Simulation, Almost Exact Simulation, Numerical Methods

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Chapter 1 Introduction

1.1 Research Background

1.1.1 Current Development of Global and Domestic Option Markets

Since the Chicago Board Options Exchange (CBOE) officially listed standardized options for trading in 1973, the global financial derivatives market has undergone a profound evolution from a single-product market to a multi-layered, highly complex system. In mature international markets, options are no longer merely basic risk-hedging tools, but have evolved into core instruments for asset price discovery, tail risk management, and strategic gaming. In recent years, with the intensification of global macroeconomic uncertainty and frequent geopolitical conflicts, the demand among institutional investors to guard against extreme tail risks has become increasingly urgent, pushing global option trading volumes to continuous record highs. In particular, the massive popularity of ultra-short-term products represented by Zero-Days-To-Expiration (0DTE) options in the U.S. stock market has not only profoundly altered the microstructure of market liquidity but also posed more stringent theoretical challenges to the instantaneous response capabilities of pricing models at extremely short time scales, as well as their ability to accurately capture extreme volatility skew. Consequently, the limitations of traditional constant volatility models and linear stochastic volatility models have become increasingly prominent.

At the same time, the Chinese options market, as an important emerging force in the global derivatives landscape, is in a critical window of leapfrog development. In terms of exchange-traded markets, since the successful pilot of SSE 50 ETF options in 2015, China has rapidly built an equity option matrix covering core indices such as the CSI 300, CSI 500, and CSI 1000, as well as characteristic indices like the STAR 50 and ChiNext Index. The commodity options sector has achieved full-category coverage, ranging from non-ferrous metals and the ferrous industry chain to agricultural products and new energy raw materials. This provides a rich set of risk-hedging tools for real-economy enterprises and serves as an important financial pillar for their robust operations. In the over-the-counter (OTC) derivatives sector, the OTC options market—with the China Securities inter-institution quotation system as its core link—has developed steadily. The increasing abundance of personalized and customized risk solutions has further perfected the risk management functions of the multi-layered capital market.

1.1.2 Evolution of Option Pricing Models and Computational Methods

The evolution of option pricing theory constitutes the core logical thread in the development of modern financial engineering. Since Black, Scholes, and Merton proposed the classic BSM pricing model, the global financial derivatives market has undergone a profound evolution from a single-product market to a multi-layered, highly complex system. In mature international markets, options are no longer merely basic risk-hedging tools, but have evolved into core instruments for asset price discovery, tail risk management, and strategic gaming. In recent years, with the intensification of global macroeconomic uncertainty and frequent geopolitical conflicts, the demand among institutional investors to guard against extreme tail risks has become increasingly urgent, pushing global option trading volumes to continuous record highs. In particular, the massive popularity of ultra-short-term products represented by Zero-Days-To-Expiration (0DTE) options in the U.S. stock market has not only profoundly altered the microstructure of market liquidity but also posed more stringent theoretical challenges to the instantaneous response capabilities of pricing models at extremely short time scales, as well as their ability to accurately capture extreme volatility skew. Consequently, the limitations of traditional constant volatility models and linear stochastic volatility models have become increasingly prominent.

In the early explorations of stochastic volatility models, the Heston model (also known as the 1/2 model) became the standard benchmark for global derivatives pricing due to its elegant characterization of volatility's mean-reverting property and the closed-form expression of the underlying log-price characteristic function. Subsequently, to address the complex volatility dynamics in interest rate and foreign exchange markets, the SABR model introduced a stochastic volatility term and a correlation structure between volatility and the underlying price, further strengthening its ability to capture the shape of the implied volatility curve, thereby becoming the mainstream model for OTC derivatives pricing. However, when dealing with the steep implied volatility curves of ultra-short-term options and the explosive jumps in volatility under extreme market conditions, the linear diffusion structure of the traditional 1/2 model reveals rigid theoretical limitations—its linear drift term struggles to explain the nonlinear, accelerated mean-reverting characteristics of volatility during extreme crises, nor can it adequately capture the heavy-tailed distribution of volatility in highly volatile markets.

Against this backdrop, the 3/2 stochastic volatility model has gradually entered the core research focus of both academia and industry. Unlike the Heston model, the 3/2 model assumes that the instantaneous variance process follows a nonlinear power-law diffusion process, where the diffusion term is proportional to the instantaneous variance raised to the power of 3/2, and the drift term exhibits a nonlinear mean-reverting structure. This nonlinear specification endows the model with tremendous flexibility, enabling it to more sensitively reflect volatility clustering effects under extreme market stress and more accurately fit the volatility smile across different maturities and strike prices. Consequently, it demonstrates fitting precision significantly superior to traditional affine models in the pricing practices of cross-asset classes such as equities, foreign exchange, and cryptocurrencies.

Accompanied by the increasing dimensionality and structural complexity of pricing models, the evolution of numerical computational methods has simultaneously driven the expansion of financial engineering practices. Early Finite Difference Methods (FDM) provided robust pricing pathways for path-dependent derivatives like American options by discretizing partial differential equations (PDEs); however, under multi-factor stochastic volatility models, they often face the challenge of the "curse of dimensionality," with computational complexity growing exponentially as model dimensions increase. Monte Carlo Simulation, with its powerful versatility, has become the mainstream tool for handling high-dimensional models and path-dependent exotic options, but it suffers from slow convergence rates, and computational costs rise significantly as precision requirements increase. To balance pricing efficiency and precision, Conditional Monte Carlo introduces variance reduction techniques to decouple the underlying price process from the volatility process, greatly optimizing the simulation effectiveness and convergence speed of stochastic volatility models. In recent years, Fourier Transform Methods based on characteristic functions, combined with numerical algorithms like the Fast Fourier Transform (FFT) and Fourier-Cosine (COS) methods, have achieved high-speed analytical solutions for complex log-prices. This has laid a solid technical foundation for the real-time pricing and full-surface calibration of non-affine stochastic volatility models such as the 3/2 model.

1.1.3 Research Questions

Although the 3/2 model possesses significant theoretical advantages in characterizing extreme market volatility and fitting volatility surfaces, its non-affine mathematical structure also brings severe numerical implementation challenges to option pricing. Existing research has proposed various option pricing methods for the 3/2 model, encompassing three major paradigms: time-stepping Monte Carlo simulations, exact/near-exact terminal simulations, and Fourier time-domain pricing. These cumulatively form nine mainstream implementation schemes; however, three core unresolved issues remain in current research and practice:

First, existing performance evaluations of various pricing methods under the 3/2 model are mostly limited to idealized academic benchmark cases in stable market environments, lacking systematic testing of the methods' full-scenario numerical stability. The majority of nominally "exact" pricing methods can only achieve theoretical validation under the benchmark parameters given in literature. Under extreme parameter combinations—such as high volatility-of-volatility (vol-of-vol), strong price-volatility correlation, and extremely short time-to-maturity—problems like numerical overflow of special functions, distribution fitting failures, and non-numerical calculation results (NaN/Inf) are prevalent. The engineering applicability boundaries of these methods have yet to be clearly defined.

Second, there is a lack of empirical re-validation for pricing methods under genuinely extreme market conditions; theoretical precision in idealized examples does not equate to practical performance in real markets. The actual parameters of extremely volatile markets, such as cryptocurrencies, often exceed the parameter ranges of academic benchmark cases. Structural mutations of model parameters during extreme market conditions further amplify the performance divergence among different pricing methods. Existing research has not yet clarified the pricing accuracy, computational efficiency, and robustness of different methods under real market extreme shocks, failing to provide a reliable basis for method selection in derivatives pricing and risk management during extreme conditions.

Third, empirical adaptation research of the 3/2 model in high-volatility markets remains insufficient, lacking a dynamic evolution analysis of model parameters throughout the full cycle of extreme events. Current research focuses on static parameter calibration and the derivation of pricing formulas for the 3/2 model. It has neither constructed a comprehensive analytical framework of "long-term fixed-parameter estimation - time-varying parameter cross-sectional calibration" for extreme crash events, nor validated the model's ability to capture the evolution of the volatility smile under extreme conditions within a pure diffusion framework. The practical value of the model in highly volatile markets has yet to receive adequate empirical support.

Based on this, the core research questions of this paper focus on three levels: Firstly, to systematically clarify the theoretical logic, error sources, and convergence characteristics of the nine mainstream option pricing methods under the 3/2 model, and to quantitatively evaluate their pricing accuracy, computational efficiency, and full-scenario numerical stability through multi-scenario numerical experiments, thereby forming a clear hierarchical method performance system. Secondly, to execute an extreme-scenario re-evaluation of pricing methods based on model parameters calibrated from real extreme market events, comparing the performance differences between idealized cases and real markets to define the engineering applicability boundaries of different methods. Thirdly, to validate the empirical adaptability of the 3/2 model in extreme cryptocurrency market conditions, identifying the time-varying patterns of core model parameters throughout the full cycle of extreme events, thus providing theoretical support and empirical foundations for derivatives pricing and risk management in high-volatility environments.

1.1.4 Research Objectives and Main Contents

Addressing the core research questions mentioned above, this paper takes European option pricing under the 3/2 stochastic volatility model as its central research object. Through systematic research encompassing theoretical construction, numerical experiments, and empirical analysis, this study comprehensively evaluates the full-scenario performance of different pricing methods, clarifies their applicability boundaries, and simultaneously validates the empirical adaptability of the 3/2 model in extreme market environments. The core research objectives and main contents of this paper are divided into the following three aspects:

First, to construct a comprehensive methodological framework for European option pricing under the 3/2 model, accomplishing the theoretical derivation and numerical implementation of various methods. This paper categorizes the nine mainstream pricing methods into three major paradigms: time-stepping conditional Monte Carlo methods, exact and near-exact terminal simulation methods, and Fourier frequency-domain pricing methods. It systematically derives the core theoretical logic, numerical implementation processes, error characteristics, and convergence properties of each method paradigm. This addresses the problem of scattered methodological frameworks and unclear theoretical boundaries in existing research, building a unified theoretical structure and reproducible implementation schemes for subsequent numerical comparisons and empirical testing.

Second, to execute a systematic performance evaluation of different pricing methods through multi-scenario numerical experiments. This paper designs seven sets of test cases covering both conventional markets and extreme parameter environments. Using numerical stability, pricing accuracy, and computational efficiency as core evaluation metrics, it quantitatively compares the performance of the nine methods under different parameter combinations. This identifies the advantageous scenarios and core defects of each method, forming a clear hierarchical performance system. It resolves the issues of unilateral method performance evaluation and insufficient extreme-scenario robustness testing found in existing literature, providing a quantitative basis for method selection across different application scenarios.

Third, to perform empirical calibration of the 3/2 model and extreme-scenario re-evaluation of pricing methods based on real extreme market events. Taking the extreme Bitcoin market flash crash of October 2025 as the research sample, this paper constructs a two-step empirical framework of "long-term fixed-parameter time-series estimation - time-varying parameter cross-sectional calibration." It identifies the time-varying patterns of the core 3/2 model parameters throughout the crash cycle and validates the model's ability to fit the evolution of the volatility smile during extreme conditions. Simultaneously, based on the real market parameters obtained from calibration, it constructs extreme scenario test cases to complete a performance re-evaluation of the pricing methods. By conducting a horizontal comparison with the academic benchmark results, it clarifies the core differences in method performance between idealized environments and real markets, ultimately forming method selection guidelines for extreme market environments, thus bridging the gap between theoretical examples and real-world markets in current research.

1.1.5 Structure of the Thesis

This thesis consists of seven chapters, organized along a logical thread of "Theoretical Foundation - Method Construction - Numerical Experiments - Empirical Testing - Conclusions and Outlook." The core contents and structural arrangements of each chapter are as follows:

Chapter One is the Introduction. This chapter systematically expounds on the research background and significance of this study, reviews the developmental trajectory of global and domestic option pricing theories, defines the core research questions, objectives, and main contents concerning the 3/2 stochastic volatility model, and outlines the overall structural arrangement of the thesis.

Chapter Two is the Literature Review. This chapter systematically combs through the theoretical evolution of stochastic volatility models and the current research status of exact simulation pricing methods. It summarizes applied research of the 3/2 model in financial derivatives markets and relevant domestic research, ultimately clarifying the deficiencies in existing studies through literature review to establish the academic positioning and marginal contributions of this paper.

Chapter Three covers the Specification of the 3/2 Stochastic Volatility Model. This chapter details the core theoretical setup of the 3/2 model, deriving the stochastic differential equations for the underlying asset price and the instantaneous variance process under the risk-neutral measure. It derives the conditional distribution properties of the integrated variance and the moment generating function representation of the underlying price, clarifying the economic implications of the model's core parameters to lay the theoretical foundation for the subsequent construction and empirical calibration of pricing methods.

Chapter Four discusses Option Pricing Methods and Algorithmic Implementation. Within the theoretical framework of the 3/2 model, this chapter systematically constructs a complete implementation system for the nine option pricing methods across three paradigms. It derives the core theoretical formulas for time-stepping conditional Monte Carlo simulations, exact and near-exact terminal simulation methods, and Fourier time-domain pricing methods. It also provides the numerical implementation flow for each method and conducts theoretical analyses on their error characteristics, convergence properties, and applicable scenarios, establishing a unified methodological foundation for subsequent numerical experiments.

Chapter Five focuses on Numerical Experiment Setup and Comparative Analysis of Results. This chapter establishes a unified numerical experiment environment and evaluation metric system, designing seven sets of test cases covering both conventional and extreme parameter environments. It systematically completes numerical testing for the nine pricing methods, quantitatively comparing their numerical stability, pricing accuracy, and computational efficiency, ultimately forming a hierarchical system of method performance and scenario-based selection recommendations.

Chapter Six presents the Empirical Calibration and Pricing Testing of the 3/2 Stochastic Volatility Model in Extreme Cryptocurrency Markets. Taking the extreme Bitcoin market flash crash of October 2025 as the research sample, this chapter constructs a complete empirical framework of "long-term fixed-parameter time-series estimation - time-varying parameter cross-sectional calibration - pricing method extreme-scenario re-evaluation." It identifies the time-varying patterns of the model's core parameters throughout the crash event and the model's fitting effectiveness, validates the model's ability to fit the evolution of the volatility smile during extreme conditions, and compares these findings horizontally with the academic benchmark results from Chapter Five, defining the applicability boundaries of different pricing methods in extreme market environments.

Chapter Seven covers Conclusions and Outlook. This chapter systematically summarizes the core research conclusions of the entire thesis, objectively analyzes the limitations of this study, and provides an outlook on future research directions from the dimensions of model extension, method optimization, empirical scope expansion, and risk management applications.

Motivated by the operational failures and severe latencies of standard pricing algorithms under boundary conditions, this thesis seeks to overhaul the valuation framework for the 3/2 stochastic volatility model.

Our research trajectory is anchored in two primary phases. First, we reconstruct the five prevailing numerical schemes, benchmarking their performance against vanilla options across a diverse grid of market regimes. By dissecting the resulting simulation data, the study maps out the exact compromise each algorithm makes between computational throughput and pricing fidelity, providing a transparent view of their operational boundaries.

The core contribution of this work, however, is a methodological breakthrough designed specifically to bypass the severe limitations these traditional models face with at-the-money (ATM) and short-tenor options. To resolve these bottlenecks, we introduce a highly robust pricing mechanism. This advanced approach consistently delivers rapid, high-precision valuations, retaining its structural integrity regardless of whether the market environment is tranquil or under extreme stress.

Chapter 2 Literature Review

In the developmental history of stochastic volatility models, the 3/2 model has gradually become a research focal point in the fields of option pricing and risk management within financial engineering, owing to its unique nonlinear diffusion structure and superior ability to capture extreme market volatility. This chapter follows a logical thread of “model paradigm evolution —theoretical framework development —pricing method iteration —market application expansion —domestic research status.” It systematically reviews the domestic and international research achievements related to the 3/2 model, ultimately clarifying the core gaps in existing research through a literature summary to establish the academic positioning and marginal contributions of this study.

2.1 Evolution of Stochastic Volatility Models

The evolution of option pricing theory is essentially a process of market participants continuously deepening their understanding of the “non-normalit” of financial asset returns and the “time-varying risk” of volatility. Since the Black-Scholes-Merton (BSM) model Black et al. (1973), Merton (1973) established the basic paradigm of arbitrage-free pricing, its core assumptions—constant volatility and the underlying price following geometric Brownian motion—have provided a concise and efficient analytical framework for derivatives pricing, laying the foundation of modern financial engineering. However, as the complexity of global financial markets continues to escalate, the core assumptions of the BSM model have systematically diverged from true market characteristics. A large number of empirical studies have shown that the return distributions of various global assets universally exhibit significant “heavy-tailed” characteristics. Meanwhile, implied volatility in the options market presents a “volatility smile” varying with strike prices and a “volatility skew” varying with time to maturity Rubinstein (1985). These phenomena are particularly prominent during market crises and extreme conditions, revealing the nonlinear pricing characteristics of the market’s tail risk premium and driving a paradigm shift in volatility modeling from determinism to stochasticity.

To correct the theoretical deficiencies of the BSM model, academia first developed local volatility models. Derman et al. (1994) and Dupire (1994) constructed deterministic local volatility frameworks dependent on the underlying price and time from the perspectives of discrete trees and continuous partial differential equations, respectively, which can perfectly fit the implied volatility surface at a specific moment. However, local volatility models fundamentally remain within the realm of deterministic volatility; they cannot characterize the stochastic fluctuation features of volatility itself and possess weak predictive power for future volatility surfaces, making it difficult to meet the practical demands of dynamic hedging and long-term derivatives pricing.

Against this backdrop, Stochastic Volatility (SV) models emerged. Their core concept is to define volatility itself as a stochastic diffusion process, thereby simultaneously characterizing the stochastic fluctuations of the underlying price and the time-varying features of volatility itself. Affine SV Models, represented by Heston (1997), introduced a square-root diffusion process (also known as the 1/2 model) to characterize the mean-reverting property of volatility, deriving an affine closed-form solution for the characteristic function of the underlying log-price. This greatly reduced the computational complexity of model calibration and option pricing, becoming a standard pricing cornerstone in the global derivatives industry. Subsequently, targeting the complex volatility dynamics in interest rate and foreign exchange markets, the SABR model proposed by Hagan et al. (2002) further enhanced the ability to capture the shape of implied volatility curves by introducing a stochastic volatility term and a correlation structure between volatility and the underlying price, becoming the mainstream model for OTC interest rate derivatives pricing.

However, with the evolution of market microstructure and the frequent occurrence of extreme market conditions, the endogenous limitations of the traditional 1/2 model have gradually become apparent. Scholars have found that the Heston model possesses theoretical shortcomings in fitting the steep implied volatility of ultra-short-term options, and its linear drift term struggles to explain the explosive growth and nonlinear mean-reverting characteristics of volatility during extreme crises (Jones, 2003). This tension between "modeling precision" and "real-world dynamics" drove academia to transition from affine frameworks to non-affine frameworks. As a typical representative of non-affine SV models, the 3/2 model features a diffusion term proportional to the instantaneous variance raised to the power of 3/2, and a drift term exhibiting a nonlinear mean-reverting structure. This stronger nonlinearity endows the model with a more flexible kurtosis characterization and tail risk capture capability. Consequently, when dealing with scenarios involving severe underlying price jumps and highly clustered volatility, it demonstrates a deeper theoretical foundation and superior empirical performance compared to the traditional 1/2 model.

2.2 Development of the 3/2 Model and Its Extended Frameworks

Since Lewis (2000) systematically derived the fundamental analytical properties of the 3/2 stochastic volatility model, academic research has extensively explored the theoretical framework optimization, analytical property expansion, and numerical implementation schemes of the 3/2 model, forming a relatively complete system of theoretical achievements.

In terms of constructing the model's fundamental analytical framework, Carr et al. (2007) developed a novel pricing framework by modeling the variance swap rate directly. This bypassed the complex process of modeling the market price of instantaneous volatility risk, providing a completely new implementation path for the risk-neutral pricing of the 3/2 model. Addressing the option pricing and calibration challenges brought by the 3/2 model's non-affine nature, Medvedev et al. (2007) proposed a comprehensive analytical framework based on short-maturity asymptotic expansions. They derived a closed-form expansion for implied volatility under non-affine local volatility models, providing a core theoretical tool for the rapid cross-sectional calibration of non-affine models, including the 3/2 model, and serving as a crucial foundation for subsequent empirical research on the 3/2 model.

Regarding the extension and optimization of the model structure, to achieve consistent modeling of spot equities and VIX derivatives, Baldeaux et al. (2012b) introduced a simultaneous jump term into the pure diffusion 3/2 model, constructing the "3/2 plus jumps model" to further enhance the model's ability to characterize extreme volatility in severe markets. Grasselli (2017), in their pioneering research, proposed the "4/2 stochastic volatility model," which defines the diffusion term of the instantaneous variance as a linear combination of a 1/2 term and a 3/2 term. This organically integrated the Heston model and the 3/2 model within the same framework, forming a generalized SV model framework that effectively combines the advantages of both models, greatly expanding the theoretical application boundaries of the 3/2 model. To tackle the simulation difficulties caused by the 3/2 model's non-affine characteristics, Kouarfate et al. (2021) developed a novel simulation framework based on an explicit solution for the 3/2 model's variance process. This framework ingeniously leverages the core mathematical property that the variance process of the 3/2 model is the reciprocal of a CIR process, providing a new implementation approach for Monte Carlo simulations of the 3/2 model.

2.3 Evolution of Exact Simulation Methods

In the pricing practice of stochastic volatility models, Monte Carlo simulation is the core tool for handling path-dependent derivatives and high-dimensional model pricing. The central developmental direction of simulation methods is to eliminate time-discretization errors while balancing computational efficiency and full-scenario numerical stability. Since exact simulation methods were introduced, academia has conducted continuous optimization research on exact simulation schemes for both affine models and the non-affine 3/2 model.

In foundational research on the exact simulation of affine SV models, Broadie et al. (2006) first proposed an unbiased exact simulation method for the underlying stock price and variance process under the Heston model and other affine jump-diffusion processes, known as the Exact MC scheme. By sampling the conditional distribution of the integrated variance, this method completely eliminated the systematic errors caused by time discretization, laying the theoretical foundation for the exact simulation of SV models. Subsequently, Glasserman et al. (2011) significantly optimized the Broadie-Kaya algorithm. By utilizing Pitman et al. (1982)'s conclusions on the decomposition of integral series of diffusion processes, they decomposed the conditional integrated variance into an infinite series of Gamma random variables. This avoided the computationally expensive numerical inverse Laplace transform in the original algorithm, markedly improving the computational efficiency of exact simulations.

Focusing on the exact simulation of the non-affine 3/2 model, Baldeaux (2012a) conducted pioneering research, exploring the exact simulation scheme for the stock price process under the 3/2 model for the first time. Utilizing conclusions regarding the transition density of diffusion processes derived via Lie symmetry analysis by Craddock et al. (2009), this study successfully adapted and applied Broadie et al. (2006)'s exact simulation algorithm for affine processes to the 3/2 model. They derived a closed-form solution for the conditional Laplace transform of the integrated variance under the 3/2 model, providing the first theoretically viable implementation scheme for its unbiased simulation.

In recent years, exact simulation algorithms for the 3/2 model have achieved further breakthroughs in numerical stability and computational efficiency. Choi et al. (2024) proposed a novel exact simulation scheme that does not require the use of highly oscillatory modified Bessel functions. Their core finding was that when conditioning on the Poisson variable used to simulate the terminal variance, the conditional integrated variance of the 3/2 model can be significantly simplified, fundamentally resolving the core pain point of modified Bessel functions easily causing numerical overflow under extreme parameters in the original Baldeaux algorithm. Additionally, Brignone et al. (2026) proposed a new exact simulation framework utilizing the conditional Fourier-Cosine (COS) method. By constructing the conditional cumulative distribution function of the integrated variance through cosine series expansion, they achieved highly efficient and accurate sampling of the integrated variance. This resolved the long-standing implementation difficulties of error control and high computational overhead in exact simulations since Broadie et al. (2006), further expanding the engineering application boundaries of exact simulation methods.

2.4 Research on Fourier Transform Methods

The Fourier transform pricing method is the core technical pathway for the rapid pricing and comprehensive calibration of European options under SV models. Its central idea is to convert the risk-neutral expectation of the option price into a complex-domain integral of the underlying log-price's characteristic function, achieving rapid solutions through numerical integration algorithms and completely circumventing the statistical and time-discretization errors of Monte Carlo simulations.

In foundational research on Fourier transform pricing methods, Carr et al. (1999) proposed an option pricing method based on the Fast Fourier Transform (FFT). By introducing a damping factor to resolve the non-integrability issue of the option payoff function's Fourier transform, they derived a closed-form inverse Fourier transform expression for European option prices. Combined with the FFT algorithm, this enabled the simultaneous calculation of option prices across a full range of strike prices within an $O(N \log N)$ time complexity, becoming the mainstream engineering method for option pricing under SV models. Addressing the limitations of the FFT method, such as the picket-fence effect and the restriction to equally spaced strike prices, Fang et al. (2009) proposed the COS pricing method based on Fourier cosine series expansions. By expanding the probability density function of underlying returns into a cosine series over an effective support interval, they derived a closed-form series summation for option prices. This achieved exponential convergence rates while allowing flexible calculations for arbitrary strike prices, demonstrating extremely strong applicability and computational efficiency advantages in non-affine SV models.

For the 3/2 model, due to its non-affine structural nature, the underlying log-price lacks the simple exponential-affine characteristic function found in affine models. The analytical expression of its characteristic function involves complex special functions, posing certain challenges to the application of Fourier transform methods. Lewis (2000)'s early research on complex-domain integral pricing provided the core theoretical framework for exploring the analytical properties of the 3/2 model; Dufresne (2001)'s research further pointed out that the integral distribution of the 3/2 process is closely related to Whittaker functions and modified Bessel functions. He fully derived the distributional properties of the integrated variance under the 3/2 model, laying the mathematical foundation for the closed-form solution of the characteristic function. In practical applications, the FFT pricing method is widely used for parameter calibration and option pricing of the 3/2 model, but it still faces challenges of insufficient numerical stability and excessive computation time under extremely short maturities and extreme volatility parameters Drimus (2011). In contrast, the COS method, with its exponential convergence rate and strong numerical stability, has demonstrated significant performance advantages in the pricing and calibration of the 3/2 model, becoming a primary focus of research in this field in recent years.

2.5 Research on Stochastic Volatility Models in the Cryptocurrency Options Market

As crypto assets represented by Bitcoin (BTC) and Ethereum (ETH) enter the era of institutional trading, the scale of the cryptocurrency options market has grown exponentially, and its unique market dynamics have become a frontier research direction in financial engineering. Unlike traditional equity and foreign exchange markets, the cryptocurrency market is characterized by 7×24 continuous trading, no price limits, extremely high endogenous volatility, and frequent extreme market events. Its return distribution exhibits significant heavy-tailed features and an extremely high volatility of volatility (Vol-of-Vol, VoV), posing severe challenges to traditional option pricing models.

Hou et al. (2020)'s research points out that the volatility of crypto assets exhibits stronger persistence and a higher Vol-of-Vol than traditional stock assets, with extreme tail risks significantly higher than those in traditional financial markets. This frequently causes severe valuation deviations in crypto option pricing when using the BSM model or even the basic Heston model; especially under extreme conditions, the models' ability to fit the volatility smile sharply declines. Addressing this phenomenon, academia primarily explores model frameworks suitable for the crypto market from two directions: one stream of research attempts to capture the instantaneous extreme jumps in cryptocurrency prices by introducing jump-diffusion processes into SV models Scaillet et al. (2020); the other focuses on improving the nonlinear structure of the volatility process, exploring the adaptability of non-affine SV models in the crypto market.

In recent years, due to the 3/2 model's natural advantages in characterizing implied volatility skew and capturing nonlinear mean-reverting features, research on its adaptability to the cryptocurrency market has significantly increased. Studies show that when crypto assets face regulatory policy shifts or macro liquidity shocks, the mean reversion of their volatility exhibits obvious nonlinear acceleration, which aligns highly with the mathematical properties of the 3/2 model (Fassas et al., 2022). Furthermore, given the crypto market's unique 7×24 continuous trading mechanism, utilizing the 3/2 model combined with high-frequency data for real-time volatility forecasting and dynamic option hedging has become a current research hotspot in the field. However, overall, existing research on the 3/2 model in the crypto market mostly focuses on static parameter calibration and model fit comparisons, lacking dynamic evolutionary analysis of model parameters throughout the full cycle of extreme crash events. Moreover, there has been no systematic empirical testing of the performance of different pricing methods under extreme market conditions. This research gap provides critical space for the empirical analysis conducted in this thesis.

2.6 Current Status of Domestic Research

Compared to the rich achievements in international academia, domestic research on the 3/2 model is still in its infancy. Currently, mainstream domestic literature primarily focuses on the Heston model and its application in the A-share market, or the performance of the SABR model in interest rate derivatives. Although some scholars have begun to pay attention to nonlinear SV models, studies focusing on the 3/2 model regarding high-dimensional calibration, computational robustness under extreme parameters, and applicability in high-volatility environments like the crypto market remain relatively rare. This research status indicates that constructing a 3/2 pricing system that can simultaneously balance nonlinear characterization capabilities and computational efficiency holds significant theoretical meaning and application value.

Overall, existing domestic research still suffers from three core deficiencies: First, there is a lack of systematic theoretical research on the 3/2 model, and analyses of numerical stability under extreme market parameters are quite scarce. Second, empirical research targeting high-volatility markets such as cryptocurrencies is lacking, with no systematic testing of different pricing methods' performance in extreme market environments. Third, existing studies have not clarified the engineering applicability boundaries of different pricing methods under the 3/2 model, failing to provide actionable method guidelines for practitioners. This research landscape suggests that building a pricing system for the 3/2 model that ensures both non-linear volatility characterization and global computational efficiency and stability possesses significant theoretical supplementary meaning and practical application value domestically.

2.7 Literature Review Summary and Research Positioning of this Paper

In summary, international academia has conducted extensive research around the theoretical construction, pricing method iteration, and market application expansion of the 3/2 stochastic volatility model. Significant achievements have been made in exploring analytical properties, constructing extended frameworks, optimizing exact simulation methods, and improving Fourier pricing methods, laying a solid theoretical foundation for this study. Concurrently, domestic research largely focuses on traditional SV models like the Heston and SABR models, while systematic research on the 3/2 model remains relatively scarce. There are still prominent research gaps concerning performance evaluations of pricing methods in extreme market environments and empirical adaptability tests in high-volatility crypto asset markets.

Despite the relative abundance of existing research, three core shortcomings remain. First, performance evaluations of existing 3/2 model pricing methods are largely confined to idealized cases using benchmark parameters from literature. There is a lack of full-scenario numerical stability testing covering both conventional markets and extreme parameter combinations. Some methods experience issues like numerical overflow and computational failure under real-world extreme parameters, such as extreme Vol-of-Vol, strong correlation coefficients, and extremely short maturities, leaving their engineering applicability boundaries vaguely defined. Second, existing research has not yet conducted empirical re-evaluations of pricing methods using parameters calibrated from genuinely extreme markets. Performance in theoretical cases cannot fully equate to practical efficacy in real markets, and empirical evidence is still lacking regarding the differences in accuracy, efficiency, and robustness among various pricing methods during extreme conditions. Third, empirical research on the 3/2 model in highly volatile cryptocurrency markets remains imperfect, lacking dynamic evolutionary analysis of model parameters and volatility smile fitting tests over the full cycle of extreme crash events; the pure diffusion 3/2 model's adaptability to extreme markets has not yet been fully validated.

Based on the deficiencies in existing research, this paper clarifies its core research positioning: taking European option pricing under the 3/2 stochastic volatility model as the research subject, this study systematically constructs a unified implementation framework for nine mainstream pricing methods across three paradigms. Through multi-scenario numerical experiments and empirical testing in extreme markets, it comprehensively evaluates the numerical stability, pricing accuracy, and computational efficiency of each method to clarify their engineering applicability boundaries. Concurrently, using a severe Bitcoin crash event as a sample, it validates the 3/2 model's adaptability to extreme markets and identifies the time-varying patterns of its model parameters. This research aims to bridge the gaps in existing studies regarding performance testing in extreme scenarios and empirical analysis of real markets, providing reliable method selection and empirical support for 3/2 model pricing practices in high-volatility markets.

Chapter 3 Specification of the 3/2 Volatility Model

3.1 Introduction to the 3/2 Model

As an advanced stochastic volatility framework, the 3/2 model is a complex evolutionary form of the Square-Root Process (popularized by Cox-Ingersoll-Ross and later further developed by Steven Heston).

In the current financial industry, the Heston model (i.e., the 1/2 model) has long been regarded as the industry standard. However, when facing extreme market stress, the Heston model often fails to accurately capture the "volatility of volatilit" observed in the market and struggles to perfectly fit the steepness of the short-term volatility skew.

To overcome this limitation, researchers proposed a completely new approach in the late 1990s and early 2000s: modifying the exponent of the diffusion term from 1/2 to 3/2. This specification change allows the model to generate more explosive volatility dynamics and a nonlinear mean reversion speed. Consequently, the 3/2 model allows volatility to spike and presents a steeper skew under market stress, which highly aligns with actual empirical data. However, it should be noted that under certain normal market conditions, the 3/2 model may lack the stability inherent in the Heston model.

3.2 Stochastic Differential Equation (SDE) Specification

Under the risk-neutral measure, the asset price process and the instantaneous variance/volatility process of the 3/2 model can be characterized by the following system of stochastic differential equations:

The dynamics of the underlying asset price process S_t satisfy:

$$dS_t = rS_t dt + S_t\sqrt{V_t}(\rho dW_t^1 + \sqrt{1-\rho^2}dW_t^2) \tag{3.1}$$

Where r represents the risk-free interest rate, W_t^1 and W_t^2 are two independent standard Brownian motions, and ρ is the correlation coefficient between the two Brownian motions. Meanwhile, the dynamic evolution of the variance process V_t satisfies:

$$dV_t = \kappa V_t(\theta - V_t)dt + \epsilon V_t^{3/2} dW_t^1 \tag{3.2}$$

In the above equations:

- κ controls the speed of mean reversion.
- θ represents the long-term average variance level.
- ϵ is the volatility of volatility parameter, controlling the degree of fluctuation of the variance process.
- W_t^1 and W_t^2 are two independent standard Brownian motions driving the randomness in the variance process and the price process, respectively.

It is worth noting that this variance process can also be equivalently expressed in terms of dV_t/V_t, thereby more intuitively reflecting the drift and diffusion characteristics of its relative changes.

3.3 Integrated Variance and Conditional Distribution

In the option pricing framework of the 3/2 model, the integrated variance $\mathcal{V}_T$ over the time interval $[0, T]$ plays a crucial role. It is defined as follows:

$$\mathcal{V}_T = \int_0^T V_t dt \tag{3.3}$$

Since the randomness of the asset price is partially driven by volatility, when we condition on the integrated variance $\mathcal{V}_T$, the terminal price of the underlying asset S_T follows a lognormal distribution. This important property means that, given $\mathcal{V}_T$, we can borrow the pricing framework of the Black-Scholes (BS) model.

According to Itô's isometry principle:

$$\int_0^T \rho^* \sigma_t dX_t \sim \rho^* \sqrt{\mathcal{V}_T} X_1 \sim N(0, (\rho^*)^2 \mathcal{V}_T)$$

Accordingly, the effective volatility σ between time 0 and T can be expressed as:

$$\sigma = \rho^* \sqrt{\mathcal{V}_T / T}$$

Through further stochastic calculus derivation, the log-return formula of the terminal price relative to the initial price can be expanded as:

$$\log\left(\frac{S_T}{S_0}\right) = \frac{\rho}{\epsilon} \log\left(\frac{V_T}{V_0}\right) + \left(\kappa + \frac{\epsilon^2}{2}\right) \mathcal{V}_T - \kappa\theta T - \frac{1}{2}\mathcal{V}_T + \rho^* \sqrt{\mathcal{V}_T} X_1 \tag{3.4}$$

This formula is the core foundation for pricing via subsequent exact simulation methods.

3.4 Moment Generating Function of the 3/2 Model

Under the risk-neutral measure $\mathbb{Q}$, the stochastic differential equations (SDEs) of the 3/2 model are defined by Equation (3.1) and Equation (3.2).

To utilize Fourier transform methods for option pricing, the core lies in solving the moment generating function (MGF) of the log-asset price $x_T = \ln S_T$.

According to the derivations by Lewis (2000) and Carr and Sun (2007), let $\phi(u, \tau) = \mathbb{E}[e^{ux_T} | \mathcal{F}_t]$ be defined as the moment generating function of the log-price. Within the 3/2 model framework, because the reciprocal of its variance process V_t follows a CIR process (square-root process), this MGF does not possess the common exponential-affine form found in affine models. Instead, it manifests as an analytical form involving Kummer's Confluent Hypergeometric Function (${}_1F_1$).

Specifically, given the initial state $V_0 = \sigma_0^2$, the auxiliary variables are defined as follows: Parameter transformation terms:

$$\mu = \frac{1}{2} + \frac{\kappa - u\rho\epsilon}{\epsilon^2}, \quad \tilde{c} = \frac{u(1-u)}{\epsilon^2}$$

Characteristic parameters:

$$\delta = \sqrt{\mu^2 + \tilde{c}}, \quad \alpha = -\mu + \delta, \quad \beta = 1 + 2\delta$$

Time-dependent term: define

$$X = \frac{2\kappa\theta}{\epsilon^2\sigma^2(e^{\kappa\theta\tau} - 1)}$$

This term characterizes the nonlinear impact of the mean reversion strength and the maturity τ on the volatility structure. In summary, the moment generating function of the log-price under the 3/2 model can be expressed as:

$$M(u, \tau) = \frac{\Gamma(\beta - \alpha)}{\Gamma(\beta)} X^{\alpha} \, {}_1F_1(\alpha, \beta, -X)$$

In the process of numerical computation, the calculation of ${}_1F_1(\alpha, \beta, -X)$ often involves the processing of special functions in the complex domain. In code implementation, to prevent numerical overflow in large-parameter scenarios (such as extreme volatility or long maturities), the log-gamma function (Log-Gamma) is used for reconstruction. For the MGF pre-factor:

$$M(u, \tau) = \frac{\Gamma(\beta - \alpha)}{\Gamma(\beta)} X^{\alpha} \, {}_1F_1(\alpha, \beta, -X)$$

Use

$$\ln P = \ln \Gamma(\beta - \alpha) - \ln \Gamma(\beta) + \alpha \ln X, \qquad P = \exp(\ln P),$$

then

$$M = P \cdot {}_1F_1(\alpha, \beta, -X)$$

to replace direct multiplication/exponentiation. This approach guarantees the numerical stability of the operations.

Chapter 4 Option Pricing Methods and Algorithmic Implementation

Under the unified framework of the 3/2 stochastic volatility model, this chapter systematically constructs a methodological system of nine classes for European option pricing. All methods utilize the arbitrage-free risk-neutral pricing principle as their theoretical benchmark. Complete theoretical derivations, convergence analyses, and error characteristic characterizations are developed across three major paradigms: time-discretization approximation, exact/near-exact terminal simulation, and Fourier frequency-domain integration. This establishes a rigorous theoretical foundation for subsequent numerical comparisons and empirical testing.

Under the risk-neutral measure $\mathbb{Q}$ in a complete market, the underlying asset price and variance processes of the 3/2 stochastic volatility model satisfy the equations given by (3.1) and (3.2).

For a European call option with time to maturity T and strike price K, its arbitrage-free price satisfies the risk-neutral pricing formula:

$$C(S_0, v_0; K, T) = e^{-rT}\mathbb{E}^{\mathbb{Q}}\Big[\max(S_T - K, 0) \,\Big|\, S_0, v_0\Big], \tag{4.1}$$

where S_0 and v_0 are the initial price and initial instantaneous variance of the underlying asset, respectively. Unlike the affine structure of the Heston model, the 3/2 model belongs to the class of non-affine stochastic volatility models; thus, its European option price lacks a closed-form analytical solution. Consequently, numerical methods must be employed to approximate the aforementioned conditional expectation. The nine classes of pricing methods constructed in this chapter cover the complete technical spectrum from time-domain path simulation to frequency-domain integral transformation, effectively approximating option prices from various dimensions.

4.1 Time-Stepping Monte Carlo Pricing Methods

The core idea of time-stepping methods is to discretize the continuous-time diffusion process, generating sample paths of the variance process through step-by-step iteration. The log-price of the underlying asset is then generated using the closed-form analytical solution (3.4) given the variance path, thereby circumventing the discretization error of the price process and reducing the statistical variance of the Monte Carlo simulation. This section sequentially derives the theoretical construction of the Euler discretization scheme, the Milstein discretization scheme, the Exact Stepping method, the Poisson-Gamma Near-Exact Stepping method, and the Quasi-Exponential (QE) Near-Exact Stepping method, and analyzes the convergence characteristics and error sources of each method.

4.1.1 Euler and Milstein Discretization Schemes

4.1.1.1 Euler Discretization Scheme for the Variance Process

The Euler discretization scheme is the most fundamental discretization method for numerically solving stochastic differential equations. Its core lies in performing a first-order Taylor expansion approximation on the increments of the diffusion process. For a general Itô diffusion process

$$dX_t = \mu(X_t)dt + \sigma(X_t)dW_t,$$

the discrete form of the Euler scheme is

$$X_{t+\Delta t} \approx X_t + \mu(X_t)\Delta t + \sigma(X_t)\Delta W_t,$$

where Δt is the discrete time step, and ΔW_t is the Brownian motion increment following a normal distribution $N(0, \Delta t)$ with mean 0 and variance Δt. Applying this scheme to the variance process of the 3/2 model, the drift term is $\mu(v_t) = \kappa(\theta - v_t)v_t$, and the diffusion term is $\sigma(v_t) = \epsilon v_t^{3/2}$. From this, the Euler discrete iterative formula for the variance process can be obtained. The specific discretization scheme is as follows:

Divide the maturity interval $[0, T]$ equally into M time steps, with a time step length of $\Delta t = T/M$, and a time grid of $\{t_0 = 0, t_1, \dots, t_M = T\}$. For the variance process of the 3/2 model, the Euler iterative formula at each time step is:

$$v_{t_{n+1}} = v_{t_n} + \kappa(\theta - v_{t_n})v_{t_n}\Delta t + \epsilon v_{t_n}^{3/2}\Delta W_n^1, \tag{4.2}$$

where $\Delta W_n^1 \sim N(0, \Delta t)$ is the Brownian motion increment at the n-th time step, and the increments at each time step are mutually independent.

After completing the iteration for all time steps, the integrated variance V_T is calculated using the trapezoidal numerical integration method to improve integration accuracy:

$$I_T = \int_0^T v_t dt \approx \sum_{n=0}^{M-1} \frac{v_{t_n} + v_{t_{n+1}}}{2} \cdot \Delta t. \tag{4.3}$$

By substituting the iterated terminal variance $v_T = v_{t_M}$ and the numerically integrated V_T into Equation (3.4), the logarithmic terminal price corresponding to that path can be generated, subsequently allowing for the calculation of the option's payoff at maturity.

4.1.1.2 Milstein Discretization Scheme for the Variance Process

The Milstein discretization scheme is a second-order correction to the Euler scheme. By introducing the first-order derivative term of the diffusion term, it eliminates the first-order discrete bias caused by the state dependence of the diffusion term, significantly improving the simulation accuracy of the variance process.

For the variance process of the 3/2 model, its diffusion term is $\sigma_v(v_t) = \epsilon v_t^{3/2}$. Taking the first derivative with respect to v_t yields:

$$\sigma_v'(v_t) = \frac{\partial \sigma_v(v_t)}{\partial v_t} = \frac{3}{2}\epsilon v_t^{1/2}. \tag{4.4}$$

Substituting this into the general form of the Milstein scheme yields the Milstein discrete iterative formula for the variance process:

$$v_{t_{n+1}} = v_{t_n} + \kappa(\theta - v_{t_n})v_{t_n}\Delta t + \epsilon v_{t_n}^{3/2}\Delta W_n^1 + \frac{1}{2}\sigma_v(v_{t_n})\sigma_v'(v_{t_n})\left((\Delta W_n^1)^2 - \Delta t\right). \tag{4.5}$$

After substituting the diffusion term and its derivative, the final Milstein iterative formula is:

$$v_{t_{n+1}} = v_{t_n} + \kappa(\theta - v_{t_n})v_{t_n}\Delta t + \epsilon v_{t_n}^{3/2}\Delta W_n^1 + \frac{3}{4}\epsilon^2 v_{t_n}^2\left((\Delta W_n^1)^2 - \Delta t\right). \tag{4.6}$$

Consistent with the Euler scheme, after completing the iteration of the variance path, the integrated variance V_t is calculated via Equation (4.3), which is then substituted into Equation (3.4) to generate the logarithmic terminal price.

4.1.1.3 Error Characteristics and Convergence Analysis

Based on the Euler/Milstein schemes under the conditional Monte Carlo framework, the error sources can be decomposed into three independent parts. The convergence characteristics of each part are as follows:

1. Time discretization error of the variance process: The Euler scheme has a strong order of convergence of 1/2 and a weak order of convergence of 1 for the diffusion process; the Milstein scheme elevates the strong order of convergence to 1 through a second-order correction, while the weak order remains 1. The discrete biases of both schemes monotonically decay as the time step Δt decreases. Given the same step size, the discrete bias of the Milstein scheme is significantly lower than that of the Euler scheme.
2. Numerical integration error of the integrated variance: Calculating the integrated vari-

ance using the trapezoidal method has a convergence order of 2, which is significantly higher than the discrete convergence order of the variance process. Therefore, in practical calculations, the integration error is negligible and will not become the dominant term in the overall error.

3. Statistical error of the Monte Carlo simulation: Because the conditional Monte Carlo method is employed, the discrete noise of the price process is eliminated through an analytical mapping. The statistical variance of the simulation is significantly lower than that of fully discretized standard Monte Carlo methods. The statistical error decays at a rate of $O(N^{-1/2})$ as the number of simulated paths N increases.

4.1.2 Exact Stepping Method

The Exact Stepping method provides the highest precision among time-stepping conditional Monte Carlo methods. Its core relies on utilizing the specific mathematical structure of the variance process in the 3/2 model. Through a variable transformation, it maps the process to an affine process with a known transition distribution, thereby achieving exact sampling of the variance process without discrete bias and completely eliminating systematic errors caused by time discretization.

4.1.2.1 Reciprocal Transformation and Exact Distribution of the Variance Process

Applying a reciprocal transformation to the variance process of the 3/2 model, let $x_t = 1/v_t$. The dynamic process of x_t can be derived via Itô's Lemma. First, compute the first and second derivatives of x_t with respect to v_t:

$$\frac{\partial x_t}{\partial v_t} = -\frac{1}{v_t^2}, \quad \frac{\partial^2 x_t}{\partial v_t^2} = \frac{2}{v_t^3}. \tag{4.7}$$

Substituting this into the Itô Lemma formula $dx_t = \frac{\partial x_t}{\partial v_t} dv_t + \frac{1}{2}\frac{\partial^2 x_t}{\partial v_t^2}(dv_t)^2$, and inserting the SDE of the 3/2 model's variance $dv_t = \kappa v_t(\theta - v_t)dt + \epsilon v_t^{3/2} dW_t^1$, where the quadratic variation term is $(dv_t)^2 = \epsilon^2 v_t^3 dt$. Rearranging yields:

$$dx_t = \left(\kappa - \kappa\theta x_t\right) dt - \epsilon\sqrt{x_t} dW_t^1. \tag{4.8}$$

Equation (4.8) demonstrates that the transformed process x_t is a classical CIR (Cox-Ingersoll-Ross) square-root diffusion process, characterized by an affine drift structure and a square-root diffusion structure. A core property of the CIR process is that its transition probability distribution has an explicit analytical form, following a non-central chi-squared distribution.

Specifically, given y_{t_n} at the initial time t_n, after a time step Δt, the conditional distribution of $x_{t_{n+1}}$ at time t_{n+1} is:

$$x_{t_{n+1}} \mid x_{t_n} \sim \frac{1}{c} \cdot \chi'^2 (d, \lambda), \tag{4.9}$$

where $\chi'^2(d, \lambda)$ represents a non-central chi-squared random variable with degrees of freedom d and a non-centrality parameter λ. The distribution parameters are determined by the 3/2 model parameters and the time step:

$$d = \frac{4\kappa\theta}{\epsilon^2}, \quad c = \frac{\epsilon^2 \left(1 - e^{-\kappa\Delta t}\right)}{4\kappa}, \quad \lambda = c \cdot y_{t_n} e^{-\kappa\Delta t}. \tag{4.10}$$

Through the inverse transformation $v_{t_{n+1}} = 1/x_{t_{n+1}}$, exact samples of the variance process for the 3/2 model can be obtained. This sampling procedure introduces no time discretization bias.

4.1.2.2 Implementation Under the Conditional Monte Carlo Framework

Under the conditional Monte Carlo framework, the implementation procedure of the Exact Stepping method is as follows:

Divide the maturity interval $[0, T]$ equally into M time steps, with a time step length of $\Delta t = T/M$, and a time grid of $\{t_0 = 0, t_1, \dots, t_M = T\}$. At each time step:

1. Given the current variance v_{t_n}, calculate $y_{t_n} = 1/v_{t_n}$;
2. Generate a non-central chi-squared random variable $\chi'^2(d, \lambda)$ according to the distribution parameters in Equation (4.9);
3. Calculate $x_{t_{n+1}} = \chi'^2(d, \lambda)/c$;
4. Apply the inverse transformation to obtain $v_{t_{n+1}} = 1/x_{t_{n+1}}$.

After completing the iterations for all time steps, calculate the integrated variance V_T using the trapezoidal numerical integration method:

$$V_T = \int_0^T v_t dt \approx \sum_{n=0}^{M-1} \frac{v_{t_n} + v_{t_{n+1}}}{2} \cdot \Delta t. \tag{4.11}$$

Finally, substitute the iterated terminal variance $v_T = v_{t_M}$ and the numerically integrated V_T into the core closed-form mapping (3.4).

4.1.2.3 Error Characteristics and Convergence Analysis

The sources of error in the Exact Stepping method consist of only two components, completely eliminating the time discretization error of the variance process:

1. Numerical integration error of the integrated variance: Using the trapezoidal rule to

calculate the integrated variance results in a convergence order of 2. In practical applications, even with a small number of time steps, this error is negligible and will not dominate the overall error. If further elimination of the integration error is desired, one could combine the integral moment generating function of the CIR process to analytically compute the conditional distribution of V_T.

2. Statistical error of the Monte Carlo simulation: Due to the use of the conditional Monte Carlo method and exact sampling of the variance process, the statistical variance of the simulation is minimized. The statistical error decays at a rate of $O(N^{-1/2})$ as the number of simulated paths N increases.

Among all time-stepping methods, the Exact Stepping method exhibits the smallest bias.

4.1.3 Poisson-Gamma Near-Exact Stepping Method

The core of the Poisson-Gamma Near-Exact Stepping method lies in leveraging the classical statistical properties of the CIR process derived from the reciprocal transformation of the 3/2 model's variance process. Its transition distribution can be precisely represented as an infinite mixture series of Gamma distributions weighted by Poisson probabilities. By truncating this series at a finite number of terms, an efficient, near-exact sampling of the variance process is achieved. This maintains extremely low bias while avoiding the high numerical computation costs associated with non-central chi-squared distribution sampling. This method is developed entirely based on the reciprocal-transformed CIR process and is an equivalent numerical implementation of the CIR transition distribution.

4.1.3.1 Reciprocal Transformation and Poisson-Gamma Mixture Representation of the CIR Process

Consistent with the Exact Stepping method, we first apply a reciprocal transformation to the variance process of the 3/2 model, let $x_t = 1/v_t$, which via Itô's lemma yields Equation (4.8). According to the classic mixture property of the non-central chi-squared distribution, a non-central chi-squared random variable $\chi'^2(d, \lambda)$ with d degrees of freedom and non-centrality parameter λ can be exactly decomposed into an infinite mixture of Poisson and Gamma distributions:

$$\chi'^2(d, \lambda) \mid N \sim \text{Gamma}\left(\frac{d}{2} + N, \frac{1}{2}\right), \quad N \sim \text{Poisson}\left(\frac{\lambda}{2}\right), \tag{4.12}$$

where N is a latent variable following a Poisson distribution, and $\text{Gamma}(\alpha, \beta)$ is a Gamma distribution with shape parameter α and rate parameter β, whose probability density function is:

$$f_{\text{Gamma}}(x;\alpha,\beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0. \tag{4.13}$$

Combining this with the conclusion regarding the transition distribution of the CIR process in the Exact Stepping method, given an initial value x_{t_n}, the conditional distribution of $x_{t_{n+1}}$ after a time step Δt is:

$$y_{t_{n+1}} \mid y_{t_n} \sim \frac{1}{c} \cdot \chi'^2(d, \lambda), \tag{4.14}$$

where the distribution parameters are:

$$d = \frac{4\kappa\theta}{\epsilon^2}, \quad c = \frac{\epsilon^2\left(1 - e^{-\kappa\Delta t}\right)}{4\kappa}, \quad \lambda = c \cdot y_{t_n} e^{-\kappa\Delta t}. \tag{4.15}$$

Substituting Equation (4.12) into the above yields the direct Poisson-Gamma mixture representation of $y_{t_{n+1}}$:

$$y_{t_{n+1}} \mid N \sim \frac{1}{c} \cdot \text{Gamma}\left(\frac{d}{2} + N, \frac{1}{2}\right), \quad N \sim \text{Poisson}\left(\frac{\lambda}{2}\right). \tag{4.16}$$

Equation (4.16) represents the core of this method: the transition distribution of the CIR process is inherently an infinite mixture of Gamma distributions weighted by Poisson probabilities. There is zero distribution fitting error; one only needs to apply a finite truncation to the Poisson distribution values to achieve near-exact sampling.

4.1.3.2 Implementation Under the Conditional Monte Carlo Framework

Under the conditional Monte Carlo framework, the implementation procedure of the Poisson-Gamma Near-Exact Stepping method aligns closely with the Exact Stepping method. It merely replaces the non-central chi-squared sampling with the more efficient Poisson-Gamma mixture sampling. The specific procedure is as follows:

Divide the maturity interval $[0, T]$ equally into M time steps, with a time step length of $\Delta t = T/M$, and a time grid of $\{t_0 = 0, t_1, \dots, t_M = T\}$. At each time step:

1. Given the current variance v_{t_n}, calculate $x_{t_n} = 1/v_{t_n}$;
2. Pre-calculate the fixed parameters d, c of the CIR transition distribution, and the non-centrality parameter $\lambda = c \cdot x_{t_n} e^{-\kappa\Delta t}$;
3. Generate a Poisson latent variable $N \sim \text{Poisson}\left(\frac{\lambda}{2}\right)$. In practical calculations, an upper bound truncation is applied to N (setting $N_{\max} = \lfloor \lambda/2 + 5\sqrt{\lambda/2} \rfloor$, ensuring the truncation probability is less than 10^{-6});
4. Generate a Gamma random variable $G \sim \text{Gamma}\left(\frac{d}{2} + N, \frac{1}{2}\right)$;

5. Calculate $x_{t_{n+1}} = G/c$, then apply the inverse transformation to obtain the variance at the next time step $v_{t_{n+1}} = 1/x_{t_{n+1}}$.

After completing the iterations for all time steps, compute the integrated variance V_T using the trapezoidal numerical integration method. Then, substitute v_T and V_T into Equation (3.4) to generate the logarithmic terminal price.

4.1.3.3 Error Characteristics and Convergence Analysis

The error sources for the Poisson-Gamma Near-Exact Stepping method contain only three parts, and its bias level is substantially lower than that of the Euler/Milstein discrete schemes:

1. Truncation error of the Poisson distribution: This is the sole inherent bias of the method. Because the probability mass of the Poisson distribution decays exponentially as N increases, the truncation error decreases exponentially with the increase of the truncation upper bound $N_{\max}$. Under normal parameters, setting $N_{\max}$ to the Poisson mean plus five standard deviations can control the truncation error to below 10^{-6}, making it negligible.
2. Numerical integration error of the integrated variance: Using the trapezoidal method yields a convergence order of 2. In practical applications, this error is far smaller than the truncation error and will not dominate the overall error.
3. Statistical error of the Monte Carlo simulation: Since the sampling bias of the variance process is minuscule, and the conditional Monte Carlo framework eliminates the discrete noise of the price process, the statistical error decays at a rate of $O(N^{-1/2})$ as the number of paths N increases, placing it on the same level as the Exact Stepping method.

The core advantage of this method is the balance between computational efficiency and precision. Compared to generating non-central chi-squared variables in the Exact Stepping method, generating Poisson and Gamma random numbers incurs extremely low computational costs, offering a significant speed boost. Compared to the Euler/Milstein schemes, this method entirely eliminates the discrete bias of the diffusion process, leaving only a negligible truncation error, improving precision by several orders of magnitude.

4.1.4 Quasi-Exponential (QE) Near-Exact Stepping Method

The Quasi-Exponential (QE) Near-Exact Stepping method was initially designed by Andersen (2007) for the Heston model. Here, we extend it to the 3/2 model. Its core logic lies in matching the conditional first and second moments of the variance process to construct a simple exponential-type approximate distribution. This avoids sampling from complex distributions and maintains excellent numerical stability even in extreme scenarios featuring high volatility and thick tails.

4.1.4.1 Moment Matching of the Variance Process and QE Distribution Construction

The central idea of the QE method is "moment matching". No matter how complex the true transition distribution is, one only needs to match its first two moments to construct a sufficiently precise approximate distribution. For the variance process of the 3/2 model, given the initial variance v_t, let its reciprocal process be $x_t = 1/v_t$. After a time step Δt, its conditional first moment $\mathbb{E}[x_{t+\Delta t} \mid x_t]$ and conditional second moment $\mathbb{E}[x^2_{t+\Delta t} \mid x_t]$ can be derived analytically by solving the moment differential equations of the variance process. Let $m_1 = \mathbb{E}[v_{t+\Delta t} \mid v_t]$ and $m_2 = \mathbb{E}[v^2_{t+\Delta t} \mid v_t]$ denote the conditional first and second moments, respectively. Compute the conditional variance $\sigma^2 = m_2 - m_1^2$. The QE method selects two different approximate distributions based on the magnitude of the variance:

1. Low variance scenario: When the conditional variance is relatively small, a Shifted Log-Normal Distribution is used as an approximation;
2. High variance scenario: When the conditional variance is large, an Exponential-Two-Point Mixture distribution is used as an approximation.

By satisfying the moment matching conditions $m_1 = \mathbb{E}[X]$ and $\sigma^2 = \mathrm{Var}(X)$, all parameters of the QE distribution can be solved analytically, enabling efficient sampling.

4.1.4.2 Implementation Under the Conditional Monte Carlo Framework

Under the conditional Monte Carlo framework, the implementation procedure of the QE Near-Exact Stepping method is as follows: Divide the maturity interval $[0, T]$ equally into M time steps, with a time step length of $\Delta t = T/M$, and a time grid of $\{t_0 = 0, t_1, \dots, t_M = T\}$. At each time step:

1. Given the current reciprocal variance x_{t_n}, analytically compute the conditional mean m_1 and conditional variance σ^2 via the moment equations;
2. Evaluate the current scenario against a predefined threshold to determine the appropriate form of the QE distribution;
3. Based on whether the conditional variance is high or low, select the corresponding type of approximate distribution, and calculate the distribution parameters via moment matching;

4. Sample from the constructed QE distribution to obtain the approximate sample $x_{t_{n+1}}$, and take its reciprocal to obtain $v_{t_{n+1}}$.

After completing the iterations for all time steps, compute the integrated variance V_T using the trapezoidal numerical integration method. Then, substitute v_T and V_T into Equation (3.4) to generate the logarithmic terminal price.

4.1.4.3 Error Characteristics and Convergence Analysis

The sources of error in the QE Near-Exact Stepping method consist of three parts:

1. Approximation bias of moment matching: This error is the inherent bias of the method, featuring a weak convergence order of 1. Because only the first two moments are matched, omitting information from higher-order moments, slight deviations may occur in extreme heavy-tailed scenarios. Nonetheless, it remains significantly superior to the Euler/Milstein schemes.
2. Threshold error in distribution selection: This error can be controlled to a minimal level by setting an appropriate threshold (typically based on the variance coefficient σ/m_1).
3. Numerical integration error of the integrated variance and Monte Carlo statistical error: These are consistent with the previously discussed methods.

The greatest advantage of the QE method is its exceptionally high computational efficiency and robust numerical stability, though the reciprocal operation here may mildly impact stability.

4.2 Exact and Near-Exact Simulation Pricing Methods

The terminal simulation class of methods discussed in this section completely circumvents the step-by-step iterative discretization process seen in time-stepping methods. Instead, it directly samples the joint distribution of the terminal value v_T of the variance process and the integrated variance $V_T = \int_0^T v_s ds$ over the maturity interval $[0, T]$. It then directly generates the terminal price of the underlying asset using the closed-form logarithmic price mapping derived in Section 4.1, thoroughly eradicating the discretization error caused by time steps. All methods strictly follow the conditional Monte Carlo framework: core statistics of the variance process are sampled first, followed by the generation of the underlying price through a conditional normal distribution, maximizing the reduction of statistical variance in the simulation.

4.2.1 Exact Simulation Method

The exact simulation method by Baldeaux (2012a) is the first unbiased terminal exact simulation method under the 3/2 model. Its core exploits the reciprocal transformation property of the 3/2 model's integrated variance process, combined with modified Bessel functions and inverse Laplace transforms, to achieve the joint exact sampling of v_T and V_T, completely eliminating all systematic biases.

4.2.1.1 Theoretical Derivation

Consistent with time-stepping methods, a reciprocal transformation is first applied to the variance process of the 3/2 model, let $x_t = 1/v_t$, leading to the CIR square-root diffusion process (4.8). The conditional distribution of the terminal value $x_T = 1/v_T$ of this process is a non-central chi-squared distribution, which can be exactly sampled, with the specific form aligning with Section 4.2.2.

The core contribution of Baldeaux (2012a) lies in deriving the closed-form expression of the conditional Laplace transform of the integrated variance $V_T = \int_0^T v_s ds$, given the initial value y_0 and the terminal value y_T:

$$\mathscr{L}(\beta) = \mathbb{E}\left[e^{-\beta V_T} \mid y_0, y_T\right] = \frac{I_\nu\left(\sqrt{\frac{4\kappa\theta}{\epsilon^2} + \frac{8\beta}{\epsilon^2}} \cdot \frac{2\sqrt{y_0 y_T} e^{-\kappa T/2}}{1-e^{-\kappa T}}\right)}{I_\nu\left(\sqrt{\frac{4\kappa\theta}{\epsilon^2}} \cdot \frac{2\sqrt{y_0 y_T} e^{-\kappa T/2}}{1-e^{-\kappa T}}\right)}, \tag{4.17}$$

where $\nu = \frac{2\kappa\theta}{\epsilon^2} - 1$ is the order of the modified Bessel function, and $I_\nu(\cdot)$ is the modified Bessel function of the first kind of order ν.

Based on this closed-form Laplace transform, the conditional probability density function of V_T can be obtained via the inverse Fourier transform, which then allows for the exact sampling of V_T through inverse transform sampling. Ultimately, the unbiased simulation of the underlying asset price is achieved by the joint sampling of V_T and v_T in the core mapping that generates the underlying asset's terminal price, as per Equation (3.4).

4.2.1.2 Implementation Under the Conditional Monte Carlo Framework

Under the conditional Monte Carlo framework, the implementation flow of Baldeaux's Exact Simulation method completely avoids time discretization, proceeding as follows:

1. Exact sampling of terminal variance v_T: Based on the non-central chi-squared transition distribution of the CIR process, directly sample to obtain y_T at maturity, then take the reciprocal to get $v_T = 1/x_T$. This sampling incurs zero systematic bias.

2. Exact sampling of integrated variance V_T: Given $y_0 = 1/v_0$ and $y_T = 1/v_T$, and based on the Laplace transform in Equation (4.17), derive the conditional cumulative distribution function of V_T via a numerical inverse Fourier transform, then draw an exact sample of V_T using uniform inverse transform sampling.
3. Finally, substitute $v_T = 1/x_T$ and V_T into Equation (3.4) to generate the underlying asset's terminal price samples.

4.2.1.3 Error Characteristics and Convergence Analysis

Baldeaux's Exact Simulation method is the only unbiased terminal simulation method under the 3/2 model. Its error sources are limited to two parts:

1. Numerical truncation error of the inverse Laplace transform: This error decays exponentially as the number of truncated terms in the Fourier cosine series increases. Under typical parameters, truncating at 256 terms can restrain the error to below 10^{-8}, making it negligible.
2. Statistical error of the Monte Carlo simulation: Since both the variance process and integrated variance are sampled exactly, and the conditional Monte Carlo framework eliminates discrete noise from the price process, the statistical variance of the simulation is at the theoretical minimum. The statistical error decays at a rate of $O(N^{-1/2})$ as the number of simulated paths N increases.

4.2.2 Conditional Fourier-Cosine Simulation Method

The Conditional Fourier-Cosine simulation method was proposed by Brignone et al. (2026). Its core approach involves constructing the conditional cumulative distribution function of the integrated variance V_T via a Fourier cosine series expansion based on the conditional characteristic function of the underlying logarithmic price. This enables highly efficient and accurate sampling, massively boosting computational efficiency compared to Baldeaux's exact simulation method while maintaining comparable accuracy.

4.2.2.1 Theoretical Derivation

Consistent with the Baldeaux method, the terminal variance v_T is first exactly sampled via the non-central chi-squared distribution of the CIR process. The core problem then shifts to how to efficiently sample the integrated variance V_T given v_0 and v_T.

Let the conditional characteristic function of I_T be denoted as $\varphi(u) = \mathbb{E}\left[e^{iuI_T} \mid v_0, v_T\right]$. This characteristic function can be obtained through the analytical continuation of the Laplace transform in Equation (4.17), possessing a definitive closed-form expression. According to Fourier cosine expansion theory, the conditional probability density function of I_T can be expanded into a cosine series over the effective support interval $[a, b]$:

$$f_{I_T|v_0,v_T}(x) = \frac{2}{b-a} \sum_{n=0}^{\infty} \mathrm{Re}\left[\varphi\left(\frac{n\pi}{b-a}\right) e^{-i\frac{n\pi a}{b-a}}\right] \cos\left(\frac{n\pi(x-a)}{b-a}\right), \tag{4.18}$$

where $[a, b]$ is the effective support interval of I_T, usually set to $a = 0$ and $b = \mathbb{E}[I_T] + 10\sqrt{\mathrm{Var}(I_T)}$ to cover over 99.99% of the probability mass.

Integrating the density function yields the closed-form series expansion of the conditional cumulative distribution function of V_T:

$$F_{V_T|v_0,v_T}(x) = \int_a^x f(z)dz = \frac{2}{b-a} \sum_{n=0}^{\infty} \mathrm{Re}\left[\varphi\left(\frac{n\pi}{b-a}\right) e^{-i\frac{n\pi a}{b-a}}\right] \cdot \Psi_n(x), \tag{4.19}$$

where $\Psi_n(x)$ is the closed-form integral of the cosine term, which can be computed analytically without numerical integration:

$$\Psi_n(x) = \begin{cases} x-a, & n = 0, \\ \dfrac{b-a}{n\pi} \sin\left(\dfrac{n\pi(x-a)}{b-a}\right), & n \geq 1. \end{cases} \tag{4.20}$$

4.2.2.2 Implementation Under the Conditional Monte Carlo Framework

Under the conditional Monte Carlo framework, the implementation flow of the Conditional Fourier-Cosine simulation method is as follows:

1. Exact sampling of terminal variance v_T: As in the Baldeaux method, v_T is exactly sampled through the non-central chi-squared distribution of the CIR process;
2. Efficient sampling of integrated variance V_T: Given v_0 and v_T, calculate the conditional characteristic function of V_T, construct the conditional cumulative distribution function using the truncated cosine series, and draw a sample of V_T via uniform inverse transform sampling;
3. Generation of underlying asset terminal price: Finally, substitute $v_T = 1/x_T$ and V_T into Equation (3.4) to generate the underlying asset terminal price samples.

4.2.2.3 Error Characteristics and Convergence Analysis

The error sources for this method primarily consist of three components:

1. Truncation error of the cosine series: This error decays exponentially with an increasing number of truncated terms. Under typical parameters, truncating at merely 128 terms

reaches an accuracy of 10^{-6}, converging much faster than ordinary polynomials;

2. Truncation error of the support interval: By appropriately setting the $[a, b]$ interval, this error can be kept below 10^{-8}, rendering it negligible;
3. Statistical error of the Monte Carlo simulation: Because the variance process sampling is unbiased, and the conditional Monte Carlo framework eliminates the discrete noise of the price process, the statistical error decays at a rate of $O(N^{-1/2})$ as the number of simulated paths N increases, performing at the same level as the Baldeaux method.

The core advantage of this method lies in its computational efficiency. Compared to the path-by-path inverse Laplace transform of the Baldeaux method, the coefficients of the cosine series can be pre-computed in batches, lifting speed by more than an order of magnitude.

4.2.3 Conditional Moment Matching Simulation Method

The core of the conditional moment matching simulation method involves matching the first few moments of the integrated variance V_T to construct an analytical approximate distribution for sampling. This avoids complex characteristic function calculations and series expansions, achieving extreme computational efficiency while ensuring a baseline of precision.

4.2.3.1 Theoretical Derivation

The first four moments of the integrated variance V_T in the 3/2 model can be analytically derived via the recursive moment differential equations of the CIR process:

1. First moment (mean): $\mathbb{E}[V_T] = \int_0^T \mathbb{E}[v_s]ds$, where $\mathbb{E}[v_s]$ is the unconditional mean of the 3/2 model variance process, which possesses a closed-form expression;
2. Second moment: $\mathbb{E}[V_T^2] = 2\int_0^T \int_0^s \mathbb{E}[v_s v_t]dtds$, where the covariance $\mathbb{E}[v_s v_t]$ can be analytically computed through the transition moments of the CIR process;
3. Third and fourth moments: These can be derived analytically through the recursive relations of the moment equations, subsequently enabling the calculation of the skewness and excess kurtosis of V_T.

Based on these first four moments, an approximate probability density function of V_T can be constructed via the Gram-Charlier Type A expansion:

$$f(x) = \phi_{\mu,\sigma}(x)\left(1 + \frac{\kappa_3}{6}H_3\left(\frac{x-\mu}{\sigma}\right) + \frac{\kappa_4}{24}H_4\left(\frac{x-\mu}{\sigma}\right)\right), \tag{4.21}$$

where $\phi_{\mu,\sigma}(x)$ is the normal density function with mean μ and variance σ^2; κ_3 and κ_4 are the standardized third and fourth moments of V_T; and $H_k(\cdot)$ represents the k-th order Hermite polynomial, whose first four orders take the forms:

$$H_0(x) = 1, \quad H_1(x) = x, \quad H_2(x) = x^2 - 1, \quad H_3(x) = x^3 - 3x, \quad H_4(x) = x^4 - 6x^2 + 3. \tag{4.22}$$

Integrating this density function provides an approximate cumulative distribution function, yielding a sample of V_T through inverse transform sampling.

4.2.3.2 Implementation Under the Conditional Monte Carlo Framework

Under the conditional Monte Carlo framework, the implementation procedure for the Conditional Moment Matching simulation method is as follows:

1. Exact sampling of terminal variance v_T: Exact sampling of v_T is accomplished through the non-central chi-squared distribution of the CIR process;
2. Moment-matching sampling of integrated variance V_T: Given v_0 and v_T, analytically calculate the first four moments of V_T, construct the Gram-Charlier approximate distribution, and obtain a sample of V_T via inverse transform sampling;
3. Generation of terminal price: Substitute v_T and V_T into the core closed-form mapping to generate the logarithmic terminal price.

4.2.3.3 Error Characteristics and Convergence Analysis

1. Approximation error of moment matching: This is the method's inherent bias. Matching only the first four moments disregards higher-order moment information. The bias is minimal under standard parameters but increases in extreme heavy-tailed scenarios, displaying a weak convergence order of $O(N^{-2})$;
2. Statistical error of Monte Carlo simulation: The convergence speed is $O(N^{-1/2})$.

The central advantage of this method is extreme computational efficiency. All moments can be pre-computed, circumventing path-by-path complex numerical calculations, making it well-suited for massive-scale path simulations and parameter calibration scenarios.

4.2.4 Inverse Gaussian (IG) Near-Exact Simulation Method

Proposed by Choi et al. (2024), the Inverse Gaussian Near-Exact Simulation method capitalizes on the Inverse Gaussian distribution's exceptional ability to fit positively skewed, heavy-tailed non-negative random variables. By matching the first two moments of the integrated variance V_T to construct an approximate distribution, it establishes a prime balance between accuracy and efficiency.

4.2.4.1 Theoretical Derivation

The Inverse Gaussian distribution is a class of two-parameter continuous distributions defined on the domain of positive real numbers, with a probability density function of:

$$f_{\mathrm{IG}}(x;\mu,\lambda)=\sqrt{\frac{\lambda}{2\pi x^3}}\exp\left(-\frac{\lambda(x-\mu)^2}{2\mu^2 x}\right),\quad x>0, \tag{4.23}$$

where $\mu > 0$ is the mean of the distribution, and $\lambda > 0$ is the shape parameter. Its variance satisfies $\mathrm{Var}(x) = \mu^3/\lambda$.

The integrated variance I_T of the 3/2 model is a non-negative random variable that exhibits significant positive skewness and thick tails, which is highly consistent with the statistical properties of the Inverse Gaussian distribution. Through moment matching conditions, the parameters of the Inverse Gaussian distribution can be solved analytically:

1. Given v_0 and v_T, analytically calculate the conditional mean $\mu = \mathbb{E}[I_T \mid v_0, v_T]$ and conditional variance $\sigma^2 = \mathrm{Var}(I_T \mid v_0, v_T)$ of I_T;
2. Force the mean and variance of the Inverse Gaussian distribution to match the target values, solving for:

$$\mu_{\mathrm{IG}}=\mu,\quad \lambda_{\mathrm{IG}}=\frac{\mu^3}{\sigma^2}. \tag{4.24}$$

4.2.4.2 Implementation Under the Conditional Monte Carlo Framework

Under the conditional Monte Carlo framework, the execution steps for the Inverse Gaussian Near-Exact Simulation method are:

1. Exact sampling of terminal variance v_T: Exact sampling of v_T is accomplished through the non-central chi-squared distribution of the CIR process;
2. IG distribution sampling of integrated variance V_T: Given v_0 and v_T, analytically evaluate the conditional mean and variance of V_T, calibrate the parameters of the IG distribution, and generate random numbers from the IG distribution to obtain the V_T sample;
3. Generation of terminal price: Substitute v_T and V_T into the core closed-form mapping to yield the logarithmic terminal price.

4.2.4.3 Error Characteristics and Convergence Analysis

The primary error sources of this approach entail two components:

1. Distribution fitting error: This represents the method's inherent bias, as the IG distribution fits solely the first two moments, neglecting the influence of higher-order moments;
2. Statistical error of Monte Carlo simulation: The convergence speed is $O(N^{-1/2})$.

The principal asset of this method lies in bridging accuracy with efficiency. The generation of IG distribution random numbers is exceptionally swift, and its fitting accuracy verges on that of exact simulation methods, cementing it as the most widely used simulation scheme for the 3/2 model in practical applications.

4.2.5 Gamma Near-Exact Simulation Method

The Gamma distribution is a two-parameter continuous distribution defined on the positive real numbers, whose probability density function is:

$$f_{\text{Gamma}}(x; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0, \tag{4.25}$$

where $\alpha > 0$ is the shape parameter, and $\beta > 0$ is the rate parameter. Its mean and variance satisfy $\mathbb{E}[x] = \alpha/\beta$ and $\mathrm{Var}(x) = \alpha/\beta^2$.

Via moment matching conditions, the parameters of the Gamma distribution can be solved analytically:

1. Given v_0 and v_T, analytically calculate the conditional mean μ and conditional variance σ^2 of I_T;
2. Force the mean and variance of the Gamma distribution to align with the target values, solving for:

$$\alpha = \frac{\mu^2}{\sigma^2}, \quad \beta = \frac{\mu}{\sigma^2}. \tag{4.26}$$

4.2.5.1 Implementation Under the Conditional Monte Carlo Framework

Under the conditional Monte Carlo framework, the implementation of the Gamma Near-Exact Simulation method mimics the IG method exactly, simply substituting the IG distribution with the Gamma distribution for sampling. Ultimately, v_T and I_T are plugged into the core closed-form mapping to generate the terminal underlying price.

4.2.5.2 Error Characteristics and Convergence Analysis

The error characteristics of this method parallel the IG method. The essential difference is that the Gamma distribution's fit accuracy for the thick tails of I_T is slightly inferior to the IG distribution, but its random number generation exhibits higher computational efficiency, offering a notable advantage in ultra-large-scale path simulation scenarios. Its error sources remain the distribution fitting error and the Monte Carlo statistical error, scaling at a convergence speed of $O(N^{-1/2})$.

4.3 Fourier Frequency-Domain Pricing Methods

Fourier frequency-domain pricing methods entirely bypass the path generation process characteristic of Monte Carlo simulations. The core technique translates the risk-neutral expectation of the option price into a Fourier integral of the characteristic function of the underlying logarithmic price, securing swift solutions through numerical integration algorithms. This approach commands core advantages—high computational efficiency, zero statistical error, and aptitude for batch pricing across strike prices—establishing it as the dominant method for 3/2 model option pricing and parameter calibration.

4.3.1 Fast Fourier Transform (FFT) Pricing Method

Proposed by Carr et al. (1999), the FFT pricing method centers on expressing the European option price as a Fourier transform of the characteristic function. Employing the Fast Fourier Transform algorithm, it simultaneously computes option prices for all equidistant strike prices within a time complexity of $O(N \log N)$.

4.3.1.1 Theoretical Derivation

For a European call option with a time to maturity T and strike price K, its risk-neutral price can be represented as:

$$C(K) = e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[\max(S_T - K, 0)\right] = e^{-rT}\int_{-\infty}^{\infty} \max(S_0 e^x - K, 0) f(x)\, dx, \tag{4.27}$$

where $x = \log(S_T/S_0)$ is the logarithmic return of the underlying asset, and $f(x)$ is the probability density function of x.

Let $k = \log(K/S_0)$ denote the logarithmic strike price. Rewriting the option price as a function $C(k)$ of k, one can derive the closed form of $C(k)$'s Fourier transform through the convolution property of Fourier transforms:

$$\hat{C}(u) = \int_{-\infty}^{\infty} e^{iuk} C(k)\, dk = e^{-rT}\frac{\phi(u - i)}{iu - u^2}, \tag{4.28}$$

where $\phi(u) = \mathbb{E}^{\mathbb{Q}}\left[e^{iux}\right]$ is the characteristic function of the underlying logarithmic return, serving as the nucleus of frequency-domain pricing methods.

For the 3/2 model, the characteristic function of the logarithmic return can be expressed in closed form via the confluent hypergeometric function ${}_1F_1$:

$$\phi(u) = \exp(iurT)\cdot\frac{\Gamma(\beta-\alpha)}{\Gamma(\beta)}\cdot\left(\frac{2\kappa\theta}{\epsilon^2 v_0}\left(e^{\kappa\theta T} - 1\right)\right)^{\alpha}\cdot {}_1F_1\left(\alpha, \beta, -\frac{2\kappa\theta}{\epsilon^2 v_0}\left(e^{\kappa\theta T} - 1\right)\right), \tag{4.29}$$

where parameters α, β are analytically determined by the model parameters and the Fourier variable u:

$$\alpha = -\frac{1}{2} + \frac{\kappa - \rho\epsilon u}{\epsilon^2} + \sqrt{\left(\frac{1}{2} - \frac{\kappa - \rho\epsilon u}{\epsilon^2}\right)^2 + \frac{u(iu+1)}{\epsilon^2}}, \quad \beta = 1 + 2\alpha. \tag{4.30}$$

Through the inverse Fourier transform, the integral expression for the option price emerges:

$$C(k) = \frac{e^{-rT}}{\pi} \int_0^\infty \mathrm{Re}\left[e^{-iuk} \frac{\phi(u-i)}{iu - u^2}\right] du. \tag{4.31}$$

Discretizing this continuous integral into a Riemann sum over an equidistant grid yields a summation structure fitting the Discrete Fourier Transform perfectly, enabling rapid calculation via the FFT algorithm.

4.3.1.2 Numerical Implementation Flow

The complete implementation workflow of the FFT pricing method is as follows:

1. Parameter initialization: Specify the 3/2 model parameters, option contract parameters, FFT grid parameter N (typically 2^{12}), and frequency step size Δu;
2. Characteristic function calculation: Compute the values of the 3/2 model's characteristic function over the discrete frequency grid;
3. FFT execution: Perform the FFT operation on the weighted characteristic function values to secure the discrete Fourier transform results;
4. Price recovery: Scale and perform inverse operations on the FFT results to obtain the option prices corresponding to varying logarithmic strike prices;
5. Interpolation handling: Acquire the option price for any arbitrary strike price via interpolation methods.

4.3.1.3 Error Characteristics and Convergence Analysis

The error footprint of the FFT pricing method primarily contains two parts:

1. Integral truncation error in the frequency domain: This error exponentially diminishes as the truncation frequency escalates. Judicious calibration of grid parameters curtails this error beneath 10^{-6};
2. Discretization error of the Riemann sum: This error decays linearly relative to the reduction in the frequency step size, and can be further minimized by compounding grid points.

Absent Monte Carlo statistical error, this method flaunts superb computational efficiency, making it highly suited for batch pricing across the full volatility smile and parameter calibration arenas.

4.3.2 Fourier-Cosine (COS) Pricing Method

Developed by (Fang et al., 2009), the COS pricing method marks a substantial upgrade over the FFT method. Its fundamental innovation lies in utilizing a cosine series expansion of the probability density function of the underlying logarithmic return, attaining an exponential convergence rate for option pricing, and retaining commanding computational efficiency and precision even within the non-affine structure of the 3/2 model.

4.3.2.1 Theoretical Derivation

The core logic of the COS method entails expanding the probability density function of the underlying logarithmic return x into a cosine series over an effective support interval $[a, b]$:

$$f(x) = \frac{2}{b-a} \sum_{n=0}^{\infty} \mathrm{Re}\left[\phi\left(\frac{n\pi}{b-a}\right) e^{-i\frac{n\pi a}{b-a}} \cos\left(\frac{n\pi(x-a)}{b-a}\right)\right], \tag{4.32}$$

where $\phi(u)$ is the characteristic function of x, and $[a, b]$ is the effective support interval of x, characteristically assigned as $a = \mathbb{E}[x] - 10\sqrt{\mathrm{Var}(x)}$ and $b = \mathbb{E}[x] + 10\sqrt{\mathrm{Var}(x)}$.

Fusing this series expansion into the integral expression of the option price manifests a closed-form series summation representing the option price:

$$C(K) = e^{-rT} \sum_{n=0}^{N-1} \mathrm{Re}\left[\phi\left(\frac{n\pi}{b-a}\right) e^{-i\frac{n\pi a}{b-a}}\right] \cdot \frac{2}{b-a} V_n(k), \tag{4.33}$$

where $k = \log(K/S_0)$, and $V_n(k)$ defines the cosine coefficients of the European call option payoff function. These coefficients hold fully analytical closed-form representations, dodging any numerical integration:

$$V_n(k) = \begin{cases} S_0 e^{rT}\left(\frac{b-k}{b-a}\right) - K\left(\frac{b-k}{b-a}\right), & n = 0, \\ \frac{2S_0e^{rT}}{b-a} \cdot \frac{\cos\left(\frac{n\pi(b-a)}{b-a}\right) - \cos\left(\frac{n\pi(k-a)}{b-a}\right)}{1+\left(\frac{n\pi}{b-a}\right)^2} & \\ \quad - \frac{2K}{b-a} \cdot \frac{\sin\left(\frac{n\pi(b-a)}{b-a}\right) - \sin\left(\frac{n\pi(k-a)}{b-a}\right)}{\frac{n\pi}{b-a}}, & n \geq 1. \end{cases} \tag{4.34}$$

4.3.2.2 Numerical Implementation Flow

The complete operational steps of the COS pricing method are configured as follows:

1. Parameter initialization: Specify the 3/2 model parameters, option contract parameters, the truncation term limit N for the COS series (typically 128-256), and the effective support interval $[a, b]$;
2. Characteristic function calculation: Compute the feature function values of the 3/2 model across discrete frequency points;
3. Payoff coefficient calculation: Analytically compute the cosine coefficients $V_n(k)$ associated with the option payoff;
4. Series summation: Compute the resultant option price via the truncated series summation.

4.3.2.3 Error Characteristics and Convergence Analysis

The paramount edge of the COS pricing method is its exponential convergence velocity. Its truncation error diminishes exponentially aligned with the incremental number of series terms N. Merely 128 truncated terms accomplish the exactness equivalent to an FFT scheme relying on 4096 grid points, heightening computational efficiency by over an order of magnitude. Concurrently, this method bypasses the picket-fence effect intrinsic to the FFT procedure, deftly evaluating option prices tied to any strike price, displaying formidable applicability within non-affine architectures like the 3/2 model.

4.4 Chapter Summary

Within the integrated framework of the 3/2 stochastic volatility model, this chapter has systematically constructed a theoretical structure comprising nine classes of methods for European option pricing. This corpus embraces three principal paradigms: time-stepping conditional Monte Carlo methods, exact and near-exact terminal simulation methods, and Fourier frequency-domain pricing methods. All configurations stringently conform to academic conventions bound by the conditional Monte Carlo framework or frequency-domain integration. The theoretical anatomy, execution workflow, and error signatures of each method class were comprehensively deduced and analyzed. Time-stepping methods champion ease of implementation and flexibility, adapting well to short-maturity options and dynamic hedging contexts; here, the Exact Stepping method and Poisson-Gamma method capably mitigate discrete bias. Terminal simulation methods utterly avoid time-discretization errors, with Baldeaux's exact simulation acting as the theoretical benchmark, whilst the Inverse Gaussian and Gamma approximation methods unlock an outstanding synergy between accuracy and computational speed. Fourier frequency-domain methods, unshackled from statistical errors and boasting stratospheric efficiency, excel in batch pricing across volatility smiles and executing model parameter calibrations. The comprehensive system of nine distinct methods articulated in this chapter institutes a rigorous theoretical backbone and consolidated procedural framework for subsequent comparative accuracy investigations, extreme-scenario performance stress tests, and empirical pricing evaluations.

Chapter 5 Numerical Experiment Setup and Comparative Analysis of Results

Based on the nine classes of option pricing methods under the 3/2 stochastic volatility model constructed in Chapter 4, this chapter conducts systematic numerical experiments. The core objective is to quantitatively compare the performance differences of various methods across two core dimensions: pricing accuracy and computational efficiency. Furthermore, it aims to verify the applicability of different methods under both conventional financial scenarios and extreme parameters, thereby providing a quantitative basis for the practical engineering application of the 3/2 model. All experiments are executed in a unified hardware environment to ensure that the results are reproducible and comparable.

5.1 Numerical Experiment Environment and Setup

5.1.1 Hardware and Software Environment

The hardware utilized for these numerical experiments is a consumer-grade laptop, the Lenovo ThinkBook 14, with the following specific configurations: the processor is an AMD Ryzen 7 6800H with Radeon Graphics, featuring a base clock speed of 3.20 GHz, 8 cores, 16 threads, and a standard TDP of 45W; the memory is 16.0 GB DDR5 dual-channel RAM, with 13.7 GB available during the experiments, which is sufficient to support large-scale Monte Carlo path simulations and frequency-domain grid calculations; the operating system is Windows 11 Enterprise (Version 26200.7840), 64-bit architecture. The development and computational environment is implemented in Python 3.9. Core numerical computations rely on the vectorized operations library `numpy`, and the special functions and probability distributions library `scipy`. All methods are implemented using vectorization to maximize the utilization of hardware computational power. GPU acceleration is not employed; thus, the results reflect the actual performance of the methods in a general-purpose CPU environment.

5.1.2 Test Case Design

A total of 7 sets of test cases were designed for this experiment, covering the most common conventional scenarios and extreme scenarios in financial derivatives pricing. These include at-the-money (ATM), in-the-money (ITM), and out-of-the-money (OTM) options, as well as typical scenarios such as short time-to-maturity, high leverage effects (strong negative correlation between price and volatility), and high volatility of volatility (vol-of-vol). The benchmark prices for all cases are derived from exact analytical results or high-precision numerical solutions in corresponding literature, serving as the baseline for accuracy evaluation. The specific parameters and design logic for the 7 sets of cases are outlined in the table below:

Table 5.1 Test Cases and Parameter Configurations

Case	σ	vov	ρ	κ	θ	r	T	K	S_0	Exact Price (p_{exact})	Scenario Description
Case I	1	0.2	-0.5	2	1.5	0.05	1	1	1	0.443059	Baldeaux (2012)
Case II	0.06	8.56	-0.99	22.84	0.218	0	0.5	95	100	10.364	Kouarfate et al. (2021), ATM
								100	100	7.386	
								105	100	4.938	
Case III	0.06	3.2	-0.99	20.48	0.218	0	0.5	95	100	11.724	Kouarfate et al. (2021), ATM
								100	100	8.999	
								105	100	6.710	
Case IV	1	0.3	0.5	0.7	0.6	0	0.1	47	49	7.040	Near expiration and ATM
								48	49	6.500	
								49	49	6.121	
								50	49	5.7006	
								51	49	5.305	
Case V	1	0.3	0.5	0.7	0.6	0	0.1	10	50	39.9985	Near exp only
								20	50	30.002	
								40	50	11.9266	
Case VI	1	3.3	0.5	0.7	0.6	0	1	47	49	12.7468	ATM only
								48	49	12.3995	
								49	49	12.0658	
								50	49	11.7452	
								51	49	11.4371	
Case VII	0.06	3.2	-0.99	20.48	0.218	0	0.5	95	100	11.7235	Lewis (2000), ATM
								100	100	8.9978	
								105	100	6.7091	

Case I is a classic benchmark example from Baldeaux (2012a), utilized to verify the fundamental theoretical pricing accuracy of the methods. Case II represents an extreme scenario with high vol-of-vol and strong negative price-volatility correlation, designed to simulate a highly leveraged pricing environment during a market crisis. Case III is a scenario with moderately high vol-of-vol and strong negative correlation, serving as a control for Case II to test the impact of the vol-of-vol parameter on method performance. Case IV involves short time-to-maturity and near-ATM parameters, corresponding to the short-term option pricing demands in high-frequency trading. Case V features short time-to-maturity and deep ITM options, used to examine the pricing stability of the methods for deep ITM options. Case VI presents a high vol-of-vol and ATM scenario to evaluate the long-term pricing performance of the methods in medium-term, high-volatility environments. Case VII is the classic numerical example proposed by Lewis (2000), serving as a benchmark validation scenario for frequency-domain pricing methods. These 7 sets of cases comprehensively cover the vast majority of practical application scenarios for the 3/2 model, encompassing both benchmark examples for baseline accuracy validation and extreme parameter scenarios for robustness testing, thereby allowing for a comprehensive comparison of the performance boundaries of different methods.

5.1.3 Definition of Evaluation Metrics

This experiment establishes a three-tier evaluation metric system to quantitatively assess the comprehensive performance of the methods. The metrics, in descending order of priority, are numerical stability, pricing accuracy, and computational efficiency. Numerical stability serves as the fundamental criterion for determining the engineering utility of a method; if a method outputs Not-a-Number (NaN) or infinity (Inf), it is deemed to have failed numerically in that scenario and lost its pricing capability. Pricing accuracy is measured by both absolute bias and relative bias. Absolute bias is the difference between the method's calculated price and the benchmark price, while relative bias is the percentage of the absolute bias relative to the benchmark price, which facilitates horizontal accuracy comparisons across different strike prices and price levels. Computational efficiency is quantified by the total runtime required for a single set of cases (including all corresponding strike prices) measured in seconds, reflecting the method's computational demands and its suitability for engineering applications.

5.2 Result Analysis of Time-Stepping Conditional Monte Carlo Methods

The time-stepping methods comprise 4 schemes: Euler/Milstein discretization scheme, Exact Stepping, Poisson-Gamma Near-Exact Stepping, and QE Near-Exact Stepping. The experiment configures 3 sets of time step sizes, $dt = 1/50, 1/500,$ and $1/5000$, representing a progression from coarse to fine time discretization precision to examine the impact of step size on method performance. The full results are presented in the following table:

The experimental results indicate a significant divergence in numerical stability and pricing performance among the four methods. The core limitation of the Euler/Milstein discretization scheme is its high sensitivity to the time step size and its numerical failure in extreme scenarios. With a coarse time step of 1/50, the relative bias of this method in the Case II OTM scenario reached 714.93%, and it produced infinite results in all Case VI high vol-of-vol scenarios, completely losing its pricing capability. When the step size was reduced to 1/500, the method still failed numerically across all Case VI scenarios, and the relative bias for Case II ATM options remained at 3.89%, failing to meet engineering accuracy requirements. Only

Table 5.2 Time-Stepping Conditional Monte Carlo Results

Case	dt	Strike	Euler/Milstein scheme		Exact Stepping		Poisson-Gamma Almost Exact Stepping		QE Almost Exact Stepping	
			Time (s)	Option Bias	Time (s)	Option Bias	Time (s)	Option Bias	Time (s)	Option Bias
Case I	1/50	1	0.405223	−0.01165 (−2.63%)	0.428566	−0.0009242 (−0.21%)	0.6209759	−0.0005159 (−0.12%)	0.6758925	−0.0002905 (−0.07%)
	1/500	1	2.8924741	−0.001347 (−0.30%)	3.848103	−0.000335 (−0.07%)	5.5486124	−0.0004362 (−0.10%)	6.66017	−0.0001411 (−0.03%)
	1/5000	1	29.065831	0.0002249 (0.06%)	40.3844027	−0.0001585 (−0.04%)	88.8646271	−0.000115 (−0.02%)	101.9920107	−0.000367 (−0.04%)
Case II		95		230.2249 (−9.06%)		−0.0114 (−0.11%)		−0.004389 (−0.05%)		−0.000977 (−0.01%)
	1/50	100	0.2677657	2.302 (31.17%)	0.2455173	−0.0079 (−0.23%)	0.4654895	−0.007585 (−0.10%)	0.390341	−0.01131 (−0.15%)
		105		7.401 (714.93%)		0.001158 (0.02%)		0.002399 (0.05%)		0.01003 (0.20%)
		105		7.427 (150.41%)		0.0002409 (0.00%)		−0.004103 (−0.08%)		0.01213 (0.25%)
	1/500	100	1.504743	0.15086 (3.89%)	1.9639806	0.003497 (0.03%)	2.9796648	−0.01347 (−0.13%)	3.397378	0.01355 (0.18%)
		95		0.0909 (1.23%)		−0.001302 (−0.02%)		−0.01213 (−0.16%)		−0.01645 (−0.22%)
		105		0.0909 (1.89%)		−0.002073 (−0.06%)		−0.01034 (−0.21%)		−0.01676 (−0.34%)
	1/5000	100	14.525341	0.36318 (1.23%)	20.2330929	−0.00273 (−0.07%)	29.5629414	−0.007066 (−0.10%)	48.5850875	−0.0309 (−0.46%)
		95		0.093112 (0.89%)		−0.00142 (−0.01%)		−0.009375 (−0.34%)		−0.02017 (−0.17%)
Case III		95		364.9 (9054.87%)		−0.0126 (−0.14%)		−0.05426 (−0.66%)		−0.03312 (−0.37%)
	1/50	100	0.238946	36.4 (5438.78%)	0.2417619	−0.005401 (−0.08%)	0.3870248	−0.0472 (−0.70%)	0.4423578	−0.0326 (−0.50%)
		105		0.05495 (−0.57%)		−0.0107 (−0.12%)		−0.0188 (−0.21%)		0.01358 (0.17%)
		105		−0.05096 (−0.72%)		−0.009423 (−0.14%)		−0.01699 (−0.25%)		0.0144 (0.20%)
	1/500	100	1.4755198	−0.02709 (−0.23%)	1.9966579	−0.00122 (−0.01%)	2.8634368	−0.00975 (−0.11%)	3.3736463	−0.0155 (−0.17%)
		95		−0.02338 (−0.33%)		−0.006346 (−0.07%)		−0.0375 (−0.34%)		−0.02017 (−0.17%)
		105		−0.02866 (−0.40%)		−0.008676 (−0.13%)		−0.00941 (−0.14%)		−0.02275 (−0.31%)
	1/5000	100	14.4603692	−0.01184 (−0.17%)	19.967838	−0.01091 (−0.16%)	28.5484009	−0.03466 (−0.52%)	48.5172484	−0.0207 (−0.25%)
		95		0.018024 (0.24%)		−0.005735 (−0.06%)		−0.00736 (0.11%)		−0.00109 (−0.02%)
Case IV		48		0.01468 (0.26%)		−0.008412 (−0.15%)		0.004264 (0.07%)		−0.0002527 (−0.00%)
		49		0.01468 (0.23%)		−0.007911 (−0.13%)		0.004212 (0.07%)		−0.0002652 (−0.00%)
	1/50	50	0.0891715	0.01468 (0.26%)	0.0836116	0.006557 (0.10%)	0.1061546	0.003103 (0.04%)	0.1498051	−0.004188 (−0.06%)
		51		0.00985 (−0.04%)		0.007421 (0.14%)		0.00358 (0.09%)		−0.00351 (−0.08%)
		47		−0.00295 (−0.04%)		−0.01091 (−0.16%)		0.007103 (0.10%)		−0.003395 (−0.05%)
		49		0.006214 (0.10%)		0.00662 (0.11%)		0.003458 (0.06%)		0.00635 (0.09%)
		48		−0.001867 (−0.03%)		0.006706 (0.12%)		0.003327 (0.06%)		−0.003306 (−0.06%)
	1/500	50	0.323018	−0.001804 (−0.03%)	0.4360611	0.006374 (0.12%)	0.6052143	0.00312 (0.06%)	0.7028849	−0.003293 (−0.06%)
		51		−0.001804 (−0.03%)		0.006539 (0.12%)		0.002685 (0.04%)		−0.003207 (−0.06%)
		47		0.006706 (0.10%)		0.006259 (0.09%)		0.00703 (0.10%)		0.00693 (0.10%)
		49		0.00152 (0.03%)		−0.004082 (−0.06%)		0.00318 (0.05%)		0.001356 (0.00%)
		48		0.00152 (0.03%)		−0.00989 (−0.96%)		0.003308 (0.05%)		0.000693 (0.01%)
	1/5000	50	3.0013151	0.0002269 (0.00%)	4.0081004	−0.003774 (−0.07%)	5.7636176	0.003322 (0.06%)	10.02923375	8.264×10^{-5} (0.00%)
		51		−0.0002143 (−0.00%)		−0.00356 (−0.07%)		0.003245 (0.06%)		6.51×10^{-5} (0.00%)
		47		−0.001015 (−0.01%)		−0.01999 (−0.30%)		0.005225 (0.07%)		0.0005575 (0.01%)
Case V		20		−0.00994 (−0.03%)		−0.01497 (−0.07%)		0.005325 (0.02%)		0.000554 (0.00%)
	1/50	40	0.0873017	0.005247 (0.04%)	0.0845473	0.01425 (0.12%)	0.1103219	0.00362 (0.02%)	0.129903	0.000675 (0.01%)
		10		−0.002526 (−0.01%)		0.01425 (0.04%)		0.008478 (0.02%)		−0.00144 (−0.00%)
		40		−0.001768 (−0.01%)		0.01129 (0.09%)		0.00529 (0.03%)		−0.001399 (−0.02%)
	1/500	20	0.3298876	0.002483 (0.01%)	0.4290495	0.008224 (0.05%)	0.5894391	0.006829 (0.06%)	0.7007175	−0.00217 (−0.02%)
		10		0.00053 (0.00%)		−0.008224 (−0.02%)		0.007159 (0.02%)		0.000674 (0.00%)
		40		0.0005752 (0.00%)		−0.008224 (−0.02%)		0.007159 (0.02%)		0.0008315 (0.01%)
	1/5000	20	2.9262788	0.0005752 (0.00%)	4.0733267	−0.008224 (−0.03%)	5.7201683	0.007159 (0.02%)	9.9833601	0.000687 (0.00%)
		10		0.000949 (0.00%)		−0.005735 (−0.01%)		0.00539 (0.01%)		0.000933 (0.00%)
Case VI		48		inf (inf%)		225.7 (171.53%)		36.17 (283.9%)		inf (inf%)
		49		inf (inf%)		225.6 (1871.65%)		36.17 (300.01%)		inf (inf%)
	1/50	50	0.3831229	inf (inf%)	0.4302237	225.6 (1922.8%)	0.7873058	36.17 (308.2%)	0.7045123	inf (inf%)
		51		inf (inf%)		225.6 (1974.65%)		36.17 (316.52%)		inf (inf%)
		47		inf (inf%)		225.7 (1771.21%)		36.17 (291.99%)		inf (inf%)
		49		inf (inf%)		−0.0573 (−0.46%)		0.01475 (0.12%)		4.106×10^{68} (nan%)
		48		inf (inf%)		−0.0575 (−0.47%)		0.01395 (0.11%)		4.106×10^{68} (nan%)
	1/500	50	2.8947977	inf (inf%)	3.8898247	−0.05727 (−0.49%)	5.8676512	0.01336 (0.11%)	6.6754493	4.106×10^{68} (nan%)
		51		inf (inf%)		−0.05716 (−0.50%)		0.01297 (0.11%)		4.106×10^{68} (nan%)
		47		inf (inf%)		−0.05716 (−0.47%)		0.0136 (0.11%)		4.106×10^{68} (nan%)
		49		468.6 (3678.98%)		0.00935 (0.08%)		0.02743 (0.22%)		3.057×10^{21} (nan%)
		48		468.6 (3782.16%)		0.00958 (0.08%)		0.02739 (0.22%)		3.057×10^{21} (nan%)
	1/5000	50	29.229539	468.6 (3893.10%)	55.4278611	0.00962 (0.08%)	57.7701995	0.02713 (0.23%)	99.3793894	3.057×10^{21} (nan%)
		51		468.6 (4100.83%)		0.00943 (0.08%)		0.02707 (0.24%)		3.057×10^{21} (nan%)
		47		468.6 (3678.98%)		0.00958 (0.08%)		0.02743 (0.22%)		3.057×10^{21} (nan%)
Case VII		95		364.9 (4055.43%)		−0.0114 (−0.13%)		−0.05822 (−0.66%)		−0.03192 (−0.36%)
	1/50	100	0.2103698	36.4 (5439.54%)	0.2497061	−0.004501 (−0.07%)	0.3766218	−0.0436 (−0.65%)	0.3742216	−0.03236 (−0.48%)
		105		0.05495 (−0.46%)		−0.001013 (−0.01%)		−0.0198 (−0.16%)		0.015 (0.13%)
		105		−0.04372 (−0.71%)		−0.009498 (−0.11%)		−0.0176 (−0.20%)		0.01678 (0.21%)
	1/500	100	1.4952733	−0.01256 (−0.18%)	1.9388048	−0.01523 (−0.13%)	2.9082565	−0.01609 (−0.24%)	3.36681	0.01431 (0.19%)
		95		−0.02709 (−0.23%)		−0.00742 (−0.01%)		−0.03935 (−0.42%)		−0.01967 (−0.17%)
		105		−0.02589 (−0.31%)		−0.005146 (−0.06%)		−0.0321 (−0.50%)		−0.02155 (−0.24%)
	1/5000	100	14.7206092	−0.0218 (−0.39%)	34.2497818	−0.00076 (−0.01%)	48.1214847	−0.03376 (−0.40%)	52.2054604	−0.01987 (−0.22%)
		95		−0.02596 (−0.23%)		−0.009742 (−0.12%)		−0.03621 (−0.34%)		−0.01931 (−0.17%)

when the step size was further refined to 1/5000 did the relative bias drop below 1% for conventional scenarios; however, the corresponding computation time surged from the 0.2 seconds range to over 30 seconds—an increase of two orders of magnitude in time cost, rendering the computational efficiency extremely low. The primary cause of this phenomenon is the strong nonlinear nature of the variance process in the 3/2 model. Under coarse step sizes, the first-order discretization error of the Euler scheme is significantly amplified, making it prone to generating non-physical negative variances, which ultimately leads to pricing failures.

In stark contrast to the Euler/Milstein scheme, the Exact Stepping, Poisson-Gamma Near-Exact Stepping, and QE Near-Exact Stepping methods did not exhibit any numerical failures across all 7 test cases, all strike prices, and all time step sizes. These methods consistently output valid pricing results and are the only Monte Carlo methods demonstrating full-scenario stability in this experiment. Even under the coarse step size of 1/50, the relative bias of the Exact Stepping method for Case II ATM options was merely -0.03%, while the Poisson-Gamma and QE methods contained their bias within 0.2%. This accuracy significantly surpasses that of the Euler scheme with the fine step size of 1/5000, while computation times hovered around 0.2 seconds—less than 1% of the time required by the fine-step Euler scheme. In the extreme high vol-of-vol scenario of Case VI, with a 1/500 step size, the relative bias was -0.46% for the Exact Stepping method and 0.12% for the Poisson-Gamma method, whereas the Euler scheme failed entirely. The performance differences among these three methods can also be quantified: the Exact Stepping method yields the highest pricing accuracy, keeping relative bias under 0.5% across all scenarios, with no time discretization bias, setting the accuracy benchmark for time-stepping methods. The Poisson-Gamma and QE methods offer slightly lower accuracy than the Exact Stepping method but share the same order of magnitude in computation time as the Euler scheme, maintaining numerical stability in extreme scenarios and achieving an optimal balance between pricing accuracy and computational efficiency.

5.3 Result Analysis of Exact and Near-Exact Terminal Simulation Methods

Terminal simulation methods entirely bypass the time-discretization process by directly sampling the joint distribution of the terminal variance and the integrated variance. This category includes 5 schemes: Baldeaux Exact Simulation, Fourier-Cosine Exact Simulation, Moment Matching Exact Simulation, Inverse Gaussian (IG) Near-Exact Simulation, and Gamma Near-Exact Simulation. The complete results are as follows:

As the table shows, the three "exact simulation" terminal methods were only able to output

Table 5.3 Terminal Exact and Almost Exact Simulation Results

Case	Strike	Exact Simulation		Exact Simulation (Use Fourier-Cosine)		Exact Simulation (Use Moment Matching)		Almost Exact Simulation (Inverse Gaussian)		Almost Exact Simulation (Gamma)	
		Time(s)	Option Bias	Time(s)	Option Bias	Time(s)	Option Bias	Time(s)	Option Bias	Time(s)	Option Bias
Case I	1	59.5239847	-1.571×10^{-5} (−0.00%)	25.4983391	−0.04977 (−11.23%)	0.025079	−0.05216 (−11.77%)	2.6390925	0.001326 (0.30%)	2.164342	0.0008747 (0.20%)
Case I	95		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−8.215 (−79.26%)		−7.956 (−76.77%)
	100		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−6.213 (−84.12%)		−5.994 (−81.16%)
Case II	105	69.0326374	NaN (NaN%)	8.3304516	NaN (NaN%)	0.034264	NaN (NaN%)	0.2152987	−4.361 (−88.31%)	0.204456	−4.195 (−84.95%)
	95		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−11.64 (−89.30%)		−11.97 (−99.56%)
	100		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−8.96 (−99.57%)		−8.975 (−99.73%)
Case III	105	73.7386331	NaN (NaN%)	6.6512523	NaN (NaN%)	0.032648	NaN (NaN%)	0.256394	−6.693 (−99.74%)	0.241701	−6.699 (−99.84%)
	95		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−11.64 (−99.30%)		−11.975 (−99.56%)
	47		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−5.559 (−78.96%)		38.31 (544.13%)
	48		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−5.287 (−81.34%)		37.91 (583.29%)
Case IV	49	79.0467981	NaN (NaN%)	6.5286679	NaN (NaN%)	0.026266	NaN (NaN%)	20.5061735	−5.135 (−83.88%)	13.86428	37.37 (610.49%)
	50		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−4.903 (−86.01%)		36.87 (646.82%)
	51		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−4.664 (−87.91%)		36.36 (685.41%)
	10		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−5.255 (−13.14%)		43.83 (109.58%)
Case V	20	75.2066147	NaN (NaN%)	6.0646468	NaN (NaN%)	0.025757	NaN (NaN%)	22.739208	−5.259 (−17.53%)	13.84294	43.83 (146.08%)
	40		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−6.356 (−53.29%)		41.99 (352.08%)
	47		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−0.06061 (−0.48%)		−0.1563 (−1.23%)
	48		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−0.05685 (−0.46%)		−0.1489 (−1.20%)
Case VI	49	69.5002801	NaN (NaN%)	21.3088199	NaN (NaN%)	0.033759	NaN (NaN%)	0.1932632	−0.05316 (−0.44%)	0.186914	−0.1419 (−1.18%)
	50		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−0.04984 (−0.42%)		−0.1354 (−1.15%)
	51		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−0.04649 (−0.41%)		−0.1291 (−1.13%)
	95		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−11.64 (−99.30%)		−11.67 (−99.56%)
Case VII	100	70.4046528	NaN (NaN%)	6.2345658	NaN (NaN%)	0.032753	NaN (NaN%)	0.2169977	−8.959 (−99.57%)	0.235242	−8.974 (−99.73%)
	105		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−6.692 (−99.74%)		−6.698 (−99.84%)

valid results in the single strike scenario of the Case I benchmark example. In all other test scenarios, they suffered complete numerical failure and do not possess full-scenario engineering practicality.

Specifically, the Baldeaux (2012) Exact Simulation method yielded valid pricing results only in the single strike scenario of Case I, where the relative bias approached 0, confirming its theoretical unbiasedness. However, across the remaining 6 sets of test cases (Case II to Case VII) involving 22 strike price scenarios, it experienced numerical overflow and computational failure, outputting Not-a-Number (NaN) results. The core cause lies in the fact that the method requires the path-by-path calculation of high-order modified Bessel functions and inverse Laplace transforms. Under non-benchmark parameters, this easily triggers numerical overflow in the Bessel functions and integration convergence failures. It can only compute successfully under the literature's benchmark parameters, failing to adapt to the complex parameter environments of actual markets. Both the Fourier-Cosine Exact Simulation method and the Moment Matching Exact Simulation method behaved nearly identically to the Baldeaux method: they only succeeded in Case I's single strike scenario, and even there exhibited terrible pricing accuracy with relative biases of −11.23% and −11.77%, respectively. They failed entirely in all other scenarios. The root causes of failure for these two methods are: for the Fourier-Cosine method, the cosine series truncation for the integrated variance distribution amplifies truncation error in the distribution tail under non-benchmark parameters, leading to failure in evaluating the characteristic function; for the moment matching method, matching only the first four moments via Gram-Charlier expansion cannot capture the non-normal heavy-tailed feature of the 3/2 model's variance distribution. This leads to non-physical negative densities, ultimately causing the inverse transform sampling of the cumulative distribution function to fail.

Regarding the two near-exact simulation methods, their pricing biases were only controllable under the medium-term, positive-correlation ATM scenario of Case VI. In the vast majority of other scenarios, they exhibited systematic and massive pricing errors. In the 5 near-ATM scenarios of Case VI, the relative bias of the IG method ranged from -0.41% to -0.48%, and the Gamma method ranged from -1.13% to -1.23%. This bias is manageable, and the computation time was roughly 0.19 seconds, indicating high efficiency. However, in all other scenarios, both methods demonstrated significant pricing deviations: in the high vol-of-vol, strongly negative correlation scenarios (Cases II, III, VII), their relative biases reached -80% to -99.8% across all strike prices, completely deviating from the benchmark. In short-term, deep ITM scenarios (Cases IV, V), the Gamma method exhibited positive systematic biases exceeding 500%, and the IG method showed negative systematic biases of -78% to -87%, losing all reference value. This happens because, under strong negative correlation, high vol-of-vol, and short-term scenarios, the distribution of the integrated variance becomes extremely heavy-tailed and highly right-skewed. The two-parameter IG and Gamma distributions, which only match the first two moments, cannot fit the higher-order features and tail behaviors of the distribution, resulting in systematic pricing errors.

5.4 Result Analysis of Fourier Frequency-Domain Pricing Methods

Frequency-domain pricing methods entirely bypass the path generation process of Monte Carlo simulations, solving for option prices directly through the Fourier integral of the characteristic function. This category encompasses the Fast Fourier Transform (FFT) and Fourier-Cosine (COS) methods. The complete results are detailed in the following table:

As the table shows, these two methods are the only ones in this experiment that experienced zero numerical failures across all 7 test cases and all strike prices, while simultaneously maintaining exceptionally high pricing accuracy. Their comprehensive performance is significantly superior to all Monte Carlo-based methods.

In terms of numerical stability, the FFT and COS methods never produced non-numerical or infinite results in any scenario. Whether evaluating benchmark ATM scenarios, extreme high vol-of-vol and strong negative correlation cases, short maturity scenarios, or deep ITM/OTM scenarios, they stably output pricing results, entirely circumventing the sampling failures and distribution fitting failures associated with Monte Carlo methods. In terms of pricing accuracy,

Table 5.4 Fourier Frequency Domain Pricing Results

Case	Strike	Fast Fourier Transformation		Fourier-Cosine Method	
		Time (s)	Option Bias	Time (s)	Option Bias
Case I	1	0.3843623	-2.434×10^{-7} (−0.00%)	0.1115441	-2.437×10^{-7} (−0.00%)
Case II	95	0.1986538	−0.0004891 (−0.00%)	0.0517513	−0.0004891 (−0.00%)
	100		−0.0001461 (−0.00%)		−0.0001461 (−0.00%)
	105		−0.0009548 (−0.02%)		−0.0009548 (−0.02%)
Case III	95	0.2813563	−0.0005218 (−0.00%)	0.0735898	−0.0005218 (−0.00%)
	100		−0.001246 (−0.01%)		−0.001246 (−0.01%)
	105		−0.0008511 (−0.01%)		−0.0008511 (−0.01%)
Case IV	47	15.8604601	0.0003048 (0.00%)	3.7037431	0.0003048 (0.00%)
	48		0.068 (1.05%)		0.068 (1.05%)
	49		0.001088 (0.02%)		0.001088 (0.02%)
	50		0.001157 (0.02%)		0.001157 (0.02%)
	51		0.001138 (0.02%)		0.001138 (0.02%)
Case V	10	16.0524815	0.0015 (0.00%)	3.788367	0.0015 (0.00%)
	20		0.001557 (0.01%)		0.001557 (0.01%)
	40		0.002021 (0.02%)		0.002021 (0.02%)
Case VI	47	0.2070812	0.01801 (0.14%)	0.0686066	0.01785 (0.14%)
	48		0.01801 (0.15%)		0.01791 (0.15%)
	49		0.01807 (0.15%)		0.01791 (0.15%)
	50		0.01794 (0.15%)		0.01778 (0.15%)
	51		0.01794 (0.16%)		0.01775 (0.16%)
Case VII	95	0.3207817	-2.18×10^{-5} (−0.00%)	0.0993632	-2.179×10^{-5} (−0.00%)
	100		-4.598×10^{-5} (−0.00%)		-4.597×10^{-5} (−0.00%)
	105		4.895×10^{-5} (0.00%)		4.895×10^{-5} (0.00%)

the relative biases of both methods were contained within 1.05% across all scenarios, with the vast majority falling below 0.02%. In the extreme high vol-of-vol environment of Case II, their maximum relative bias was merely −0.02%. In the short maturity Case IV, the maximum bias was 1.05%, while for all other strike prices it remained under 0.02%. In the high vol-of-vol Case VI, the relative bias stood around 0.15%. This precision overwhelmingly surpasses all Monte Carlo methods. The core reason is that frequency-domain methods execute pricing directly via numerical integration relying on the closed-form solution of the 3/2 model's characteristic function. They are immune to Monte Carlo sampling errors, distribution fitting errors, and time-discretization truncation errors, suffering only from negligible numerical integration truncation errors.

Regarding computational efficiency, the COS method comprehensively outperformed the FFT method. Across all test scenarios, the execution time for the COS method was only 1/3 to 1/4 of that required by FFT. In the multi-strike Case IV, FFT took 15.86 seconds, whereas COS took a mere 3.70 seconds; in single-strike scenarios, COS execution times ranged from 0.05 to 0.11 seconds, boasting efficiency far superior to any Monte Carlo scheme. The theoretical driver behind this efficiency gap is that the COS method, based on cosine series expansion, converges exponentially. It requires only 128 truncation terms to match the precision of an FFT grid with 4096 points. Moreover, the COS method is free from the FFT's picket-fence effect, allowing flexible pricing at arbitrary strike prices without extra interpolation steps, thereby further reducing computational overhead.

5.5 Comprehensive Performance Comparison and Research Conclusions

Based on the full-scenario experimental results from the 7 test cases, the 9 classes of pricing methods can be systematically tiered across three core dimensions: numerical stability, pricing accuracy, and computational efficiency.

The first tier consists of the COS and FFT frequency-domain methods, which are the only approaches that ensure full-scenario numerical stability while retaining extreme accuracy and efficiency. Specifically, the COS method comprehensively outperforms the FFT method, cementing its status as the optimal engineering solution for pricing European options under the 3/2 model.

The second tier includes the Exact Stepping, Poisson-Gamma Near-Exact Stepping, and QE Near-Exact Stepping methods from the time-stepping category. They represent the only stable solutions across all scenarios within the Monte Carlo framework. They can stably generate complete variance paths and offer a favorable balance between pricing accuracy and efficiency, making them the sole reliable options for pricing path-dependent options or for dynamic hedging scenarios where complete path generation is mandatory.

The third tier is the Euler/Milstein discretization scheme, which is only valid under fine step sizes and conventional parameters. It fails numerically in extreme scenarios and exhibits extremely poor efficiency; thus, it serves merely as a baseline for theoretical accuracy comparisons.

The fourth tier comprises the IG and Gamma Near-Exact Terminal Simulation methods, which provide controllable bias only in specific ATM scenarios. In the overwhelming majority of cases, they suffer massive systematic pricing errors. They are limited to specific academic comparisons and possess no engineering practicality.

The fifth tier features the Baldeaux Exact Simulation, Fourier-Cosine Exact Terminal Simulation, and Moment Matching Exact Terminal Simulation methods. These function solely in a singular benchmark scenario, utterly failing in all others. They are strictly limited to validating theoretical unbiasedness.

Based on these full-scenario experimental validations, three conclusions can be drawn. First, the theoretical unbiasedness of a 3/2 model pricing method does not equate to full-scenario stability in engineering applications. Nominally "exact" terminal simulation methods are highly susceptible to special function overflows and integration convergence failures under non-benchmark parameters. They only pass theoretical validation under literature-provided benchmarks and fail to adapt to the complex parameters of actual markets. Second, time-stepping sampling methods built upon the distributional properties of the variance process constitute the optimal implementation schemes within the Monte Carlo framework. They inherently guarantee the non-negativity of the variance process, ensuring numerical stability in all extreme environments, and consistently outperform traditional Euler discretization in accuracy, providing a robust path-generation basis for derivatives pricing. Third, the characteristic function-based COS frequency-domain method is the absolute optimal choice for pricing European options under the 3/2 model. It preserves supreme precision and stability across all test scenarios with zero numerical failures, and its computational efficiency far exceeds all Monte Carlo techniques, granting it overwhelming performance superiority in batch option pricing, model parameter calibration, and extreme market conditions.

Deriving from this holistic validation, we provide corresponding method recommendations for distinct application scenarios. For batch pricing of European options and model parameter calibration, the COS frequency-domain method is the premier choice; its stability, peak precision, and supreme efficiency seamlessly satisfy these demands. For scenarios necessitating complete variance paths, such as path-dependent option pricing or dynamic hedging, the Poisson-Gamma or QE Near-Exact Stepping methods are prioritized, given their comprehensive stability and well-balanced precision and efficiency. For baseline theoretical precision comparisons, the Baldeaux Exact Simulation method may be utilized, but strictly under benchmark parameter settings. The remaining terminal simulation methods are not recommended for practical pricing environments and should be reserved solely for specific academic comparative analyses.

Chapter 6 Empirical Calibration and Pricing Evaluation of the 3/2 Stochastic Volatility Model in Extreme Cryptocurrency Crash Events

6.1 Chapter Introduction

Chapter 5 systematically evaluated the accuracy, efficiency, and numerical stability of nine European option pricing methods under the 3/2 stochastic volatility model based on benchmark parameter cases from classical financial literature, clarifying the theoretical performance boundaries and applicable scenarios of different methods in idealized parameter environments. However, it must be noted that the benchmark test cases in existing academic research mostly utilize idealized parameter settings in stable market environments. The parameter values involve no extreme volatility or sudden market shocks. Although they can validate the theoretical effectiveness of the methods, they cannot reflect the time-varying characteristics of model parameters and the actual engineering adaptability of pricing methods in real financial markets, especially in highly volatile markets concentrated with tail risks.

The cryptocurrency market, as one of the major asset classes with the highest global volatility levels, possesses core characteristics such as violent underlying price fluctuations, a high proportion of leveraged trading, and the rapid release of tail risks during extreme conditions. The volatility smile in its options market can undergo rapid and severe deformations during extreme events. This provides a natural empirical testing ground to examine the market explanatory power of stochastic volatility models, their dynamic parameter characteristics, and the robustness of pricing methods in extreme scenarios. This chapter takes the BTC flash crash event on October 10, 2025 (UTC) as the research subject, expanding the pricing analysis of the 3/2 model from academic benchmark cases to real extreme market environments, executing a full-cycle parameter calibration of the model and an empirical re-evaluation of the pricing methods.

The core research objectives of this chapter are divided into three aspects: First, to identify the time-varying patterns of the 3/2 model's core parameters across three stages—pre-shock, crash, and recovery—clarifying the impact characteristics of extreme conditions on model parameters and validating the 3/2 model's ability to fit the volatility smile in the cryptocurrency options market. Second, based on parameters calibrated from the real market, to construct brand-new test cases and replicate the performance comparison of pricing methods from

Chapter 5. This tests the numerical stability, pricing accuracy, and computational efficiency of different methods in extreme market scenarios, bridging the disconnect between academic benchmark cases and real market environments. Third, to define the practical adaptability of the 3/2 model in cryptocurrency option pricing, providing actionable empirical evidence and method selection guidelines for derivatives pricing and risk management under extreme market conditions.

The research logic of this chapter strictly follows the complete chain of "long-term fixed-parameter estimation —cross-sectional time-varying parameter calibration —extreme scenario re-evaluation of pricing methods." All empirical analyses are completed based on real tick-by-tick transaction data from the Deribit exchange. The subsequent structure of this chapter is as follows: Section 6.2 defines the time window division of the research event, data sources, and preprocessing rules. Section 6.3 constructs the two-step calibration framework of the 3/2 model, explaining the theoretical foundations and implementation logic of long-term parameter estimation and cross-sectional parameter calibration. Section 6.4 conducts a descriptive statistical analysis of option transaction data throughout the full cycle of the crash event, clarifying the criteria for selecting calibration samples. Section 6.5 presents the full-cycle parameter calibration results, analyzing the time-varying characteristics of the model's core parameters and the model's fitting efficacy under extreme conditions. Section 6.6 executes a performance re-evaluation of the pricing methods based on the calibrated real market parameters, conducting a comparative analysis with the academic cases from Chapter 5. Section 6.7 summarizes the core empirical conclusions of this chapter.

6.2 Research Event, Time Window Division, and Data Preprocessing

The empirical analysis in this chapter focuses on the global cryptocurrency market's BTC flash crash event from October 10 to 11, 2025. During this event, the spot price of BTC plummeted drastically within a short period, accompanied by a violent jump in implied volatility and a significant magnification of trading volume in the options market. This provides a natural experimental scenario to test the dynamic parameter characteristics and pricing method robustness of the 3/2 stochastic volatility model under extreme market conditions. To systematically capture the market characteristics throughout the full cycle of the crash event, this chapter divides the research interval into three consecutive 10-minute time windows, corresponding respectively to the stable pre-shock market stage, the extreme shock crash stage, and the post-crash market recovery stage. The specific time window divisions are as follows: The pre-shock stage is selected from 2025-10-10 20:50:00 to 2025-10-10 21:00:00 (UTC time, same below), used to capture the model's benchmark parameters before the extreme condition occurs; the crash stage is selected from 2025-10-10 21:15:00 to 2025-10-10 21:20:00, which is the core shock period where BTC prices fell rapidly and market volatility expanded violently, used to identify the abrupt parameter changes under extreme conditions; the recovery stage is selected from 2025-10-10 23:50:00 to 2025-10-11 00:00:00, representing the gradual market repair period after the crash event, used to observe the reversion characteristics of model parameters following extreme conditions. The length of all three time windows is set to 10 minutes, which not only ensures a sufficient sample of option transactions within each window but also accurately captures the instantaneous market characteristics at different stages of the event, preventing the time-varying information of parameters from being smoothed out by overly long time windows.

The data used for the empirical research in this chapter entirely comes from Deribit, a leading global cryptocurrency options exchange. The BTC option products offered by this exchange possess the highest global liquidity and standardized market data, and its public historical data API provides an authoritative and traceable data source for this research. Specifically, this chapter utilizes two types of core data: The first type is the daily historical data of the Deribit BTC DVOL index. The DVOL index is a Bitcoin implied volatility index published by Deribit, reflecting the market's consensus expectation of BTC volatility over the next 30 days. This chapter selects daily DVOL data from April 11, 2025, to October 10, 2025 (half a year before the crash), to be used for the subsequent estimation of the 3/2 model's long-term fixed parameters. The second type is the tick-by-tick BTCUSDT option transaction data within the aforementioned three 10-minute time windows. The data fields include the option contract code, transaction time, transaction price, transaction volume, trade direction, underlying spot index price, implied volatility, etc., used for the cross-sectional calibration of the 3/2 model's time-varying parameters. Both types of data are obtained via Deribit's public historical data API. The data acquisition process employs multi-threading concurrency and exponential back-off retry mechanisms to balance data scraping efficiency with API rate limits, while utilizing strict deduplication rules to ensure data uniqueness.

Regarding the acquired raw option transaction data, this chapter executes a systematic preprocessing flow to ensure that subsequent parameter calibration is based on highly liquid, highly representative valid samples. First, the option contract parameters are decomposed and time-standardized. The expiration date, strike price, and option type (call/put) are extracted from the option contract code. All transaction timestamps are uniformly converted to the UTC time zone to avoid calculation errors caused by cross-timezone differences, and the annualized remaining time to maturity for each transaction is calculated based on the transaction time and the option's expiration time. Second, outliers and low-liquidity samples are eliminated. Contracts with a time to maturity of less than 1 day or greater than 365 days are removed; such contracts have extremely poor liquidity and significant price noise, lacking market representativeness. Simultaneously, abnormal transaction records with negative implied volatility or zero transaction volume are removed; such data mostly results from trading system failures or extreme inactivity, which would interfere with subsequent calibration. Next, out-of-the-money (OTM) options are filtered. OTM contracts—where call option strike prices are greater than or equal to the underlying spot price, and put option strike prices are less than or equal to the underlying spot price—are retained. OTM options are the most liquid contract type in the options market, and their prices are more sensitive to changes in market tail risk, making them the core sample for volatility smile analysis. Finally, target maturities are selected and samples are aggregated. The maturities with the best liquidity—6 days and 20 days—are selected as the core analysis targets. For each event stage, expiration dates with sufficient sample sizes are sought near these two target maturities, ensuring that the valid sample size for each maturity is no less than 3, to guarantee the reliability of subsequent volatility smile fitting. Multiple transactions of the same contract within the same time window are aggregated at the contract level, taking the average of the underlying spot index price and implied volatility to prevent repetitive data from interfering with the analysis results. The preprocessed data is saved separately according to the event stage and target maturity, laying the foundation for subsequent parameter calibration and pricing analysis.

6.3 Two-Step Calibration Framework for the 3/2 Stochastic Volatility Model

The 3/2 stochastic volatility model adopted in this paper is a pure diffusion specification, excluding price jump processes. The complete dynamics of the model are jointly determined by four core parameters: the long-term mean reversion rate κ, the long-term variance level θ, the volatility of volatility (VoV) ϵ, and the correlation coefficient ρ between the price and volatility processes. To balance the stability of the model's long-term parameters with the ability to capture instantaneous market structure changes during extreme events, this chapter constructs a "two-step calibration method" to complete full parameter estimation. The first step estimates the model's long-term fixed parameters κ and θ based on the DVOL index time series data from the half-year preceding the crash event. These parameters reflect the long-term volatility characteristics of the cryptocurrency market and can be considered stable and invariant within short time windows. The second step calibrates the time-varying parameters ϵ and ρ using the asymptotic expansion method based on cross-sectional option data from the three stages of the crash event. These parameters are more sensitive to instantaneous market shocks and can accurately capture the structural changes of the volatility smile throughout the full cycle of the crash event. The two-step calibration method ensures the reliability of long-term parameter estimation through long-cycle time series while realizing flexible capture of time-varying market structure characteristics via cross-sectional data. This is a mainstream analytical framework in the empirical calibration of stochastic volatility models Andersen et al. (2003).

6.3.1 Time Series Estimation of Long-Term Fixed Parameters

Under the pure diffusion framework, the variance process of the 3/2 stochastic volatility model satisfies the following dynamic specification Baldeaux (2012a):

$$dv_t = \kappa v_t(\theta - v_t)dt + \epsilon v_t^{3/2} dW_t^v$$

where v_t is the instantaneous variance of the underlying asset; κ is the mean reversion rate, reflecting the speed at which the variance process reverts to its long-term level; θ is the long-term variance level; ϵ is the volatility of volatility, characterizing the degree of fluctuation of the variance process itself; and W_t^v is the standard Brownian motion corresponding to the variance process. The DVOL index published by the Deribit exchange is a standardized implied volatility index for the Bitcoin options market, reflecting the market's consensus expectation of the underlying's volatility over the next 30 days. Its squared term can serve as an unbiased proxy variable for the underlying's instantaneous variance v_t, providing a reliable data foundation for estimating the model's long-term parameters based on time series.

To convert the continuous-time model into an estimable discrete form, referring to the stochastic volatility model discretization estimation method proposed by Andersen et al. (2002), the aforementioned variance dynamics are Euler-discretized, yielding the following linear regression equation:

$$\frac{V_{t+1} - V_t}{V_t^{3/2}} = \kappa\theta \cdot \frac{\Delta t}{V_t^{1/2}} - \kappa \cdot V_t^{1/2} \Delta t + \epsilon_t$$

where V_t is the squared term of the DVOL index on day t, i.e., $V_t = (\text{DVOL}_t/100)^2$; $\Delta t = 1/365.25$ is the annualized time step corresponding to daily data; and ϵ_t is the random disturbance term generated by the discretization process. This regression equation transforms the parameter identification problem of the continuous-time model into a standard linear regression problem. Consistent parameter estimation can be achieved through Ordinary Least Squares (OLS).

The implementation logic for long-term fixed parameters is as follows: First, read the daily DVOL data for the half-year preceding the crash event, convert the DVOL index series into a variance series V_t, and clean the series, replacing non-positive variance values with extremely small values to avoid division-by-zero calculation issues. Second, construct the dependent and independent variables required for regression: the dependent variable is $\frac{V_{t+1}-V_t}{V_t^{3/2}}$, and the two independent variables are $\frac{\Delta t}{V_t^{1/2}}$ and $V_t^{1/2}\Delta t$. Subsequently, perform OLS regression on the constructed variable series to obtain the regression coefficients corresponding to the two independent variables. Finally, solve backwards for the model's long-term parameters based on the regression coefficients, where the long-term mean reversion rate κ is the negative of the coefficient for the second independent variable, and the long-term variance level θ is the ratio of the coefficient for the first independent variable to κ. Concurrently, the long-term implied volatility level can be obtained via the square root of θ, achieving an intuitive economic interpretation of the parameter.

6.3.2 Cross-Sectional Calibration of Time-Varying Parameters

After obtaining the long-term fixed parameters κ and θ, this chapter utilizes the implied volatility asymptotic expansion method for pure diffusion stochastic volatility models proposed by Medvedev et al. (2007) to complete the calibration of the time-varying parameters ϵ and ρ, based on the cross-sectional option data from the three stages of the crash event. Targeting short-maturity options, this method leverages the short-maturity asymptotic properties of stochastic volatility models to express market implied volatility as a quadratic function of log-moneyness. It can directly map the skew and curvature features of the volatility smile to the model's core time-varying parameters without requiring complex numerical integration calculations, making it an efficient mainstream method in the field of short-term option cross-sectional calibration.

Under the pure diffusion framework, the corresponding implied volatility asymptotic expansion formula for the 3/2 model is Medvedev et al. (2007):

$$\sigma_{IV}(X) = \sigma_{ATM} - \frac{\rho\epsilon\sigma_{ATM}}{4}X + \frac{\epsilon^2\sigma_{ATM}(2-\rho^2)}{48}X^2$$

where $\sigma_{IV}(X)$ is the option implied volatility corresponding to log-moneyness X; σ_{ATM} is the implied volatility of at-the-money options; $X = -\ln(K/S) + rT$ is log-moneyness, with the risk-free interest rate r set to 0, K representing the option strike price, S representing the underlying spot price, and T representing the option's remaining time to maturity. This expansion formula clearly characterizes the correspondence between model parameters and the shape of the volatility smile: the skew feature of the volatility smile is jointly determined by the correlation coefficient ρ and the volatility of volatility ϵ, while the curvature feature is primarily determined by the volatility of volatility ϵ.

The implementation logic for the cross-sectional calibration of time-varying parameters is as follows: First, for each stage of the crash event and target maturity (6 days and 20 days), filter the preprocessed option samples, extracting the core trading interval with log-moneyness $X \in [-0.15, 0.15]$. This interval is the most liquid region in the options market and aligns with the applicable range of the asymptotic expansion method, avoiding expansion approximation errors under extreme moneyness. Second, extract the ATM implied volatility σ_{ATM} from the sample, where the ATM option is defined as the contract with the smallest absolute log-moneyness. Subsequently, construct a nonlinear least squares optimization objective function aiming to minimize the mean squared error between the model's implied volatility and the market's implied volatility. The parameters to be estimated are ϵ and ρ, with physical bounds set to guarantee the economic rationality of the optimization results, where the bound for ϵ is $[0.1, 3.0]$, and the bound for ρ is $[-0.99, 0.99]$. Finally, employ the L-BFGS-B algorithm to complete the constrained nonlinear optimization solution. This algorithm is suitable for nonlinear optimization problems with boundary constraints, featuring fast convergence and good numerical stability. Independent calibrations are performed on the three event stage datasets for the 6-day and 20-day maturities, yielding the corresponding time-varying parameters ϵ and ρ for each stage.

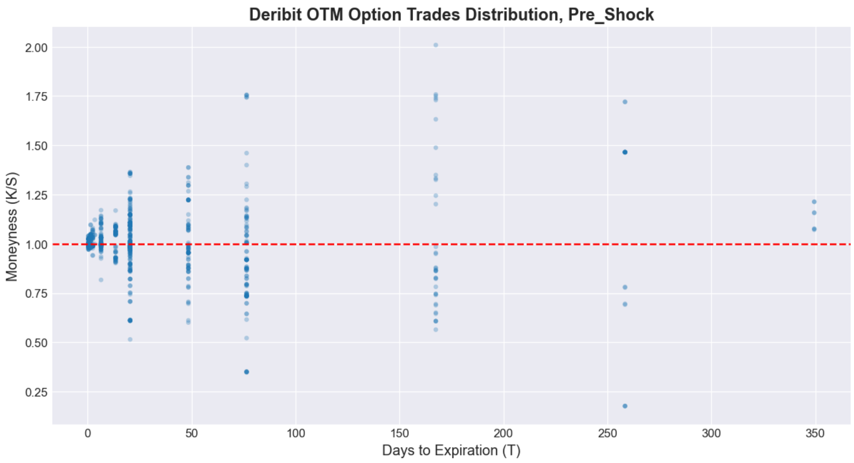
6.4 Descriptive Statistics of Full-Cycle Option Transaction Data and Calibration Sample Selection

Based on the preprocessed tick-by-tick BTC option transaction data from the Deribit exchange, this section conducts a systematic descriptive statistical analysis of the transaction distribution, term structure, and liquidity characteristics of out-of-the-money (OTM) options across the pre-shock, crash, and recovery stages of the crash event. This clarifies the trading structure changes in the crypto options market under extreme events, providing a rigorous empirical basis for the subsequent sample selection for 3/2 model parameter calibration.

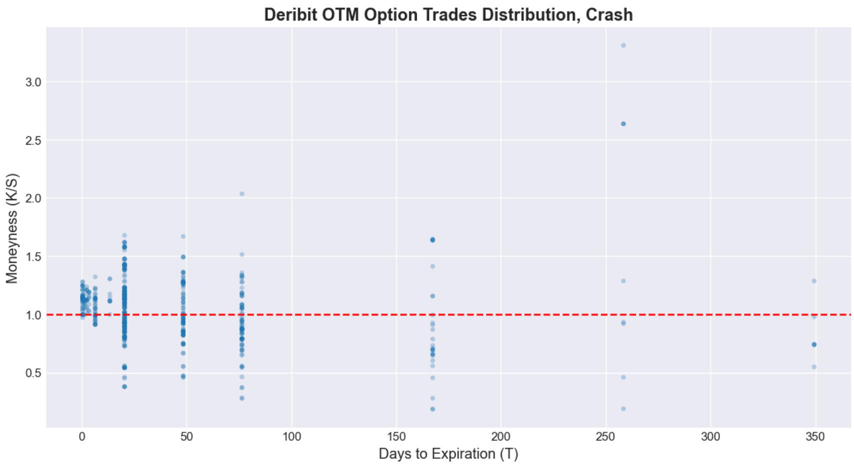
6.4.1 OTM Option Transaction Distribution Characteristics Across the Crash Event Cycle

OTM options are the contract type with the most concentrated liquidity and highest sensitivity to tail risk changes in the cryptocurrency options market. Their transaction distribution characteristics directly reflect market trading demands and risk expectations across different stages. Figure 6-1 displays the OTM option transaction distribution during the three stages of the crash event. The horizontal axis represents the days to maturity of the options, the vertical axis represents the moneyness K/S (the ratio of strike price to underlying spot price), and the red dashed line represents the at-the-money (ATM, $M = 1.0$) baseline.

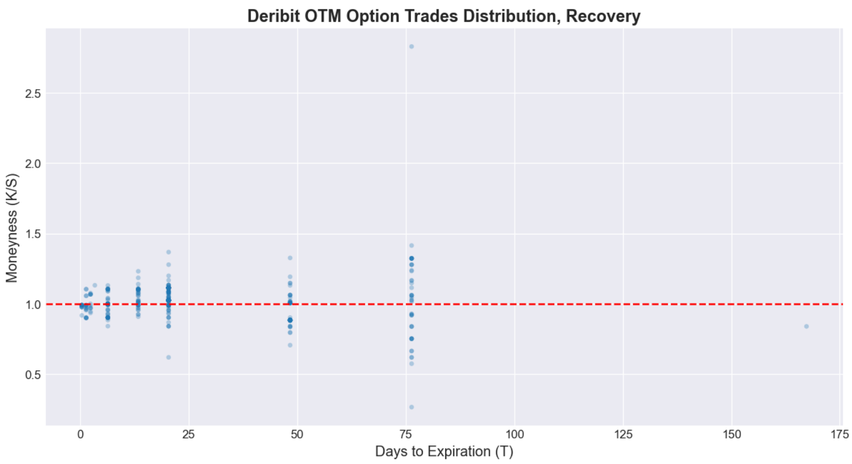
Observing the transaction distribution in the pre-shock stage, option trading exhibits a significant concentration. Transactions are primarily clustered within 100 days to maturity and a moneyness range of 0.5 to 2.0. The transaction density near ATM is significantly higher than that of deep ITM and deep OTM contracts, conforming to the conventional liquidity distribution patterns of global option markets. This reflects that pre-crash market trading was dominated by conventional hedging and speculative demands near ATM, with low demand for extreme tail risk trading. Upon entering the crash stage, the option transaction distribution underwent a significant structural change. The transaction density near ATM increased further, while the distribution range of moneyness expanded substantially compared to the pre-shock stage. The transaction upper bound for deep OTM call options rose to around 1.8, and the lower bound for deep OTM put options fell to around 0.2. This reflects that after the crash event occurred, market demand for tail risk hedging rose rapidly, with investors utilizing OTM contracts across a wider moneyness range to hedge against massive price fluctuation risks. Simultaneously, the distribution of option maturities became more dispersed. The transaction proportion of medium-term contracts between 50 and 150 days increased, marking a shift in market trading behavior from short-term speculation to medium-term risk hedging. In the recovery stage, the option transaction distribution gradually returned to pre-crash normality. The distribution range of moneyness contracted notably, and transactions at extreme moneyness decreased significantly. Trading re-clustered within 75 days to maturity and a moneyness range of 0.5 to 1.5, with transaction density near ATM gradually subsiding. This indicates that the shock of the extreme event was gradually fading, and the market trading structure and risk expectations began to repair.



(a) OTM Option Transaction Distribution in the Pre-Shock Stage



(b) OTM Option Transaction Distribution in the Crash Stage



(c) OTM Option Transaction Distribution in the Recovery Stage

Figure 6.1 Characteristics of OTM Option Transaction Distribution Throughout the Crash Event Cycle

6.4.2 Liquidity Statistics and Term Structure Analysis of Option Transactions

To quantify the liquidity levels of contracts with different maturities, this section statistically sorts the number of valid OTM option transactions across the three stages based on days to maturity. The complete statistical results are shown in Table 6.1.

Table 6.1 Statistical Number of OTM Option Transactions by Maturity Across the Crash Event Cycle

Days to Maturity	Pre-Shock	Crash	Recovery
0.0 Days	100	52	–
6.0 Days	128	43	66
13.0 Days	–	–	41
20.0 Days	244	238	214
48.0 Days	87	103	57
76.0 Days	103	93	45

Note: "–" indicates no valid transaction records for that maturity in the corresponding stage.

From the statistical results in Table 6.1, it is clear that the liquidity of contracts with different maturities exhibits significant heterogeneity throughout the crash event cycle. In the pre-shock stage, OTM option transactions showed a clear short-term characteristic. Contracts with 20 days to maturity had the highest transaction volume, reaching 244 trades, followed by contracts with 6 days to maturity at 128 trades. Together, these two accounted for over 50% of valid transactions across all maturities, representing the two most liquid contract types in the market. Medium-term contracts of 76 days and 48 days followed, with 103 and 87 trades respectively. Ultra-short-term contracts expiring on the same day had 100 trades. Overall, it presented a conventional term structure characteristic where shorter maturities correspond to higher liquidity. After the crash event occurred, the market's liquidity term structure underwent a significant shift. Contracts with 20 days to maturity still maintained the highest transaction volume at 238 trades, merely a slight decrease of 2.46% compared to the pre-shock stage. Their liquidity level remained basically flat compared to pre-crash, demonstrating extremely strong stability. However, the transaction volume for short-term contracts with 6 days to maturity dropped drastically to 43 trades, a 66.41% decrease compared to pre-crash, slipping from second to fifth place in the all-maturity transaction volume ranking. Concurrently, the transaction volumes for medium-term contracts of 48 days and 76 days remained stable at 103 and 93 trades, moving up to second and third place, respectively, in the all-maturity ranking. This reflects that under extreme shocks, investors' willingness to trade short-term contracts declines significantly; they prefer using medium-term contracts for risk hedging to avoid the Gamma risks and liquidity exhaustion associated with short-term contracts. Entering the recovery stage, the market liquidity structure gradually repaired. Contracts with 20 days to maturity still led with 214 trades, maintaining the highest liquidity level throughout the entire cycle. The transaction volume for short-term contracts with 6 days to maturity rebounded to 66 trades, an increase of 53.49% compared to the crash period, returning to second place in the all-maturity ranking, indicating a gradual recovery in short-term contract trading activity. Transaction volumes for medium-term contracts of 48 days and 76 days fell back to 57 and 45 trades, respectively, returning the market's term structure to pre-crash normality.

Overall, option contracts with 20 days to maturity maintained the highest trading activity across all three stages of the crash event, making them the most stable target in terms of liquidity throughout the full cycle. Short-term contracts with 6 days to maturity possessed ample liquidity in a stable market environment but were highly sensitive to extreme market shocks, with activity dropping significantly during the crash and recovering gradually afterward. Meanwhile, ultra-short-term (same-day expiration) and long-term contracts over 76 days consistently maintained low transaction volumes throughout the cycle, lacking liquidity and containing high noise in price formation, making them unsuitable as valid samples for model calibration.

6.4.3 Criteria for Selecting Calibration Samples

Based on the full-cycle transaction distribution characteristics and liquidity statistical results, combined with the applicable conditions of the 3/2 model calibration method, this section ultimately selects option contracts with 6 days and 20 days to maturity as the core samples for parameter calibration. The selection criteria involve rigorous considerations on three levels.

First, sample liquidity and stability requirements. The reliability of parameter calibration results highly depends on the liquidity level of the samples. Prices of illiquid contracts cannot effectively reflect true market expectations, leading to systematic biases in calibration results. Contracts with 20 days to maturity maintained the highest number of transactions in the market across the pre-shock, crash, and recovery stages, boasting stable liquidity levels and suffering the least impact from extreme events. They ensure horizontal comparability of calibration results across the three stages and are the optimal sample for capturing long-term changes in model parameters. Contracts with 6 days to maturity possess ample liquidity before the crash, and the significant change in their transaction volume during the crash can accurately reflect the instantaneous shock of extreme events on short-term market expectations, providing a valid sample for capturing abrupt parameter changes under extreme conditions.

Second, the applicable range requirements of the calibration method. The Medvedev-Scaillet asymptotic expansion calibration method adopted in this chapter performs best in approximating short-maturity, near-ATM contracts. Contracts with extreme moneyness or long maturities will produce significant approximation errors. Both the 6-day and 20-day maturity contracts fall within the short-to-medium maturity range, and their transactions are primarily concentrated in the core near-ATM interval, aligning with the applicable conditions of the asymptotic expansion method. This effectively minimizes the negative impact of approximation errors on calibration results, ensuring parameter estimation accuracy.

Third, sample validity requirements. For both maturities across all three stages of the crash event, the number of valid OTM transactions meets the minimum sample requirement for calibration, allowing for the construction of complete volatility smile curves and providing ample sample points for cross-sectional calibration. Conversely, ultra-short-term, long-term, and deep OTM/ITM contracts either lack sufficient transaction volume to construct valid volatility smiles or possess excessively high price noise; thus, they are excluded from the calibration samples.

6.4.4 Parameter Calibration Results and Analysis

Based on the two-step calibration framework constructed in Section 6.3, this section sequentially presents the time series estimation results for the 3/2 model's long-term fixed parameters, as well as the cross-sectional calibration results for time-varying parameters throughout the full cycle of the crash event. It systematically analyzes the time-varying patterns of the model's core parameters and the volatility smile fitting efficacy under extreme conditions.

6.4.5 Time Series Estimation Results of Long-Term Fixed Parameters

Based on the Deribit DVOL daily data from the 180 days prior to the crash event, the estimation of the 3/2 model's long-term mean reversion rate κ and long-term variance level θ was completed using the linear regression model established in Section 6.3.1. The regression results are presented in Table 6.2.

Judging by the statistical significance of the regression results, the coefficients for both core explanatory variables reject the null hypothesis at the 1% significance level. The P-value corresponding to the F-statistic is 0.0215, meaning the overall regression equation is significant at the 5% level. This indicates that the model specification possesses statistical explanatory power over the dependent variable. Regarding model diagnostics, the Durbin-Watson statistic

Table 6.2 OLS Regression Estimation Results for 3/2 Model Long-Term Parameters

Variable	Coefficient	Std. Error	t-statistic	p-value	95% Confidence Interval
X1 ($\kappa\theta$)	20.6809	7.404	2.793	0.006	[6.070, 35.292]
X2 (κ)	-126.1498	45.477	-2.774	0.006	[-215.894, -36.406]

Dependent Variable: y *No. Observations*: 180
Model: OLS without intercept R^2 *(uncentered)*: 0.042
Adjusted R^2 *(uncentered)*: 0.031 *F-statistic*: 3.923
AIC: -179.7 *BIC*: -173.3 *Durbin-Watson*: 2.170

Diagnostic Statistics
Omnibus: 14.016 *Prob (Omnibus)*: 0.001
Jarque-Bera (JB): 21.634 *Prob (JB)*: 2.01e-05
Skewness: 0.441 *Kurtosis*: 4.451
Cond. No.: 29.4 *Covariance Type*: nonrobust

Notes: [1] R^2 is computed without centering (uncentered); [2] Standard Errors assume the covariance matrix of errors is correctly specified.

is 2.170, a value close to 2, indicating no significant serial autocorrelation issues in the regression residuals, satisfying classical OLS assumptions. The condition number is 29.4, below the critical value of 30, indicating no severe multicollinearity among the explanatory variables, making the estimated regression coefficients stable. Both the Omnibus and JB statistics for residual distribution reject the null hypothesis of normal distribution at the 1% level. The residuals exhibit right-skewed and leptokurtic distribution characteristics, consistent with the non-normal nature of cryptocurrency volatility data, and this does not affect the consistent estimation of the regression coefficients.

We solve backwards for the 3/2 model's long-term core parameters based on the regression coefficients: according to the theoretical setup of the regression model, the coefficient for X1 is the product of the long-term mean reversion rate κ and long-term variance level θ ($\kappa\theta$), and the coefficient for X2 is $-\kappa$. Calculating from this, the long-term mean reversion rate $\kappa = 126.1498$. This value reflects a rapid speed at which the BTC market variance process reverts to its long-term level, fitting the market characteristics of cryptocurrencies, which exhibit significant mean-reverting volatility and swift returns to normality following extreme fluctuations. The long-term variance level $\theta = \frac{20.6809}{126.1498} \approx 0.1639$, corresponding to an annualized long-term implied volatility of $\sqrt{\theta} \times 100 \approx 40.48\%$. This result aligns highly with the central level of long-term volatility in the BTC market, bearing clear economic meaning.

6.4.6 Cross-Sectional Calibration Results and Full-Cycle Characteristic Analysis of Time-Varying Parameters

After obtaining the long-term fixed parameters, the Medvedev-Scaillet asymptotic expansion method from Section 6.3.2 was applied to option data for 6-day and 20-day maturities.

This completed the calibration of time-varying parameters (volatility of volatility ϵ, price-volatility correlation coefficient ρ) across the pre-shock, crash, and recovery stages, while also yielding the ATM implied volatility (ATM IV) for each stage. The visual fitting results of the calibration are shown in Figure 6.2 and Figure 6.3.

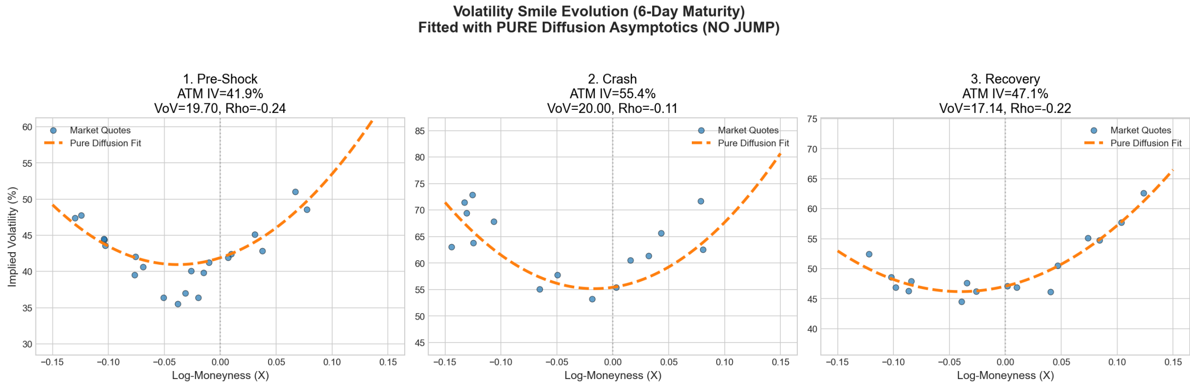


Figure 6.2 Full-Cycle Evolution of Volatility Smile for 6-Day Maturity and 3/2 Model Fitting Results

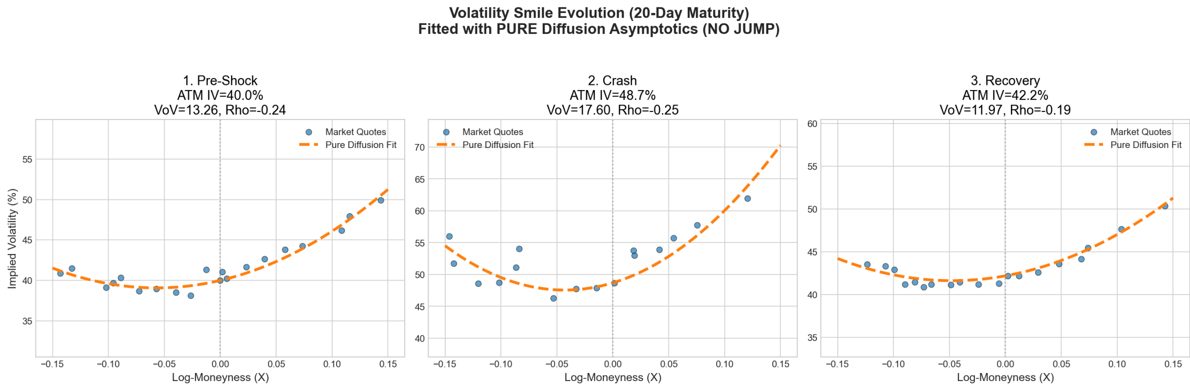


Figure 6.3 Full-Cycle Evolution of Volatility Smile for 20-Day Maturity and 3/2 Model Fitting Results

To clearly present the time-varying patterns of the parameters, the full-cycle calibration results for both maturities are consolidated in Table 6.3.

Judging by the model fitting efficacy, the asymptotic expansion fitting curves of the 3/2 model under the pure diffusion framework exhibit a high degree of conformance with real market quote scatter plots across all three stages. For both the short-term 6-day maturity and the medium-term 20-day maturity, the model accurately captures the morphological changes of the volatility smile throughout the full cycle of the crash event. This indicates that the 3/2 model effectively characterizes option pricing features under extreme cryptocurrency market conditions, displaying excellent market adaptability.

Table 6.3 Calibration Results for Time-Varying Parameters of the 3/2 Model Across the Crash Event Cycle

Maturity	Event Stage	ATM Implied Volatility	Volatility of Volatility (VoV) ϵ	Correlation Coefficient ρ
6 Days	Pre-Shock	41.9%	19.70	-0.24
	Crash	55.4%	20.00	-0.11
	Recovery	47.1%	17.14	-0.22
20 Days	Pre-Shock	40.0%	13.26	-0.24
	Crash	48.7%	17.60	-0.24
	Recovery	42.2%	11.97	-0.19

Analyzing the full-cycle time-varying characteristics of core parameters yields three core conclusions:

First, ATM implied volatility exhibits a significant event-driven pattern of "pre-shock stability —crash spike —recovery pullback," aligning with the volatility clustering effect under extreme conditions. The ATM IV for the 6-day maturity spiked from 41.9% pre-shock to 55.4% during the crash, an increase of 32.22%; the ATM IV for the 20-day maturity rose from 40.0% to 48.7%, an increase of 21.75%. The volatility surge in short-term maturities is significantly higher than in medium-term maturities, reflecting the market's heightened sensitivity to short-term volatility expectations regarding extreme events, conforming to classical laws of the options market volatility term structure. Entering the recovery stage, ATM IV for both maturities pulled back significantly, gradually returning to pre-crash central levels, reflecting the progressive dissipation of market panic.

Second, the volatility of volatility (VoV) ϵ rose significantly during the crash and fell below pre-shock levels in the recovery stage. This indicates that extreme events not only push up the overall market volatility level but also amplify the uncertainty of volatility itself, significantly elevating tail risks during the crash. The VoV for the 6-day maturity slightly increased from 19.70 pre-shock to 20.00 during the crash; the VoV for the 20-day maturity surged from 13.26 to 17.60, an increase of 32.73%. Medium-term VoV reacted more strongly to extreme events, suggesting a massive recalibration of market pricing for medium-term volatility uncertainty during the crash. Moving into the recovery stage, VoV for both maturities fell below pre-shock levels—down to 17.14 for 6 days and 11.97 for 20 days—reflecting a gradual normalization of the market's pricing of volatility uncertainty after the extreme shock, and even a degree of risk premium compression.

Third, the price-volatility correlation coefficient ρ exhibited significant structural changes throughout the crash cycle, indicating a staged distortion of the market's leverage effect under extreme conditions. For the 6-day maturity, the correlation coefficient was -0.24 pre-shock, showing a weak negative leverage effect (price drops accompanied by volatility increases), consistent with conventional features of global equity and crypto markets. However, during the crash, the absolute value of the correlation coefficient shrank massively to -0.11, denoting a significant weakening of the leverage effect. This implies that panic trading during the crash broke the conventional correlation between price and volatility; market trading behavior was dominated more by liquidity shocks and tail risk hedging rather than unconventional leverage effects. In the recovery stage, the correlation coefficient rebounded to -0.22, reverting toward pre-crash normality.

In contrast, the correlation coefficient for the 20-day medium-term maturity remained relatively stable throughout the cycle: -0.24 pre-shock, slightly down to -0.25 during the crash, and up to -0.19 in the recovery stage, with no significant distortion. This indicates that the impact of extreme events on the leverage effect is concentrated mainly in short-term option contracts; the price-volatility correlation for medium-term contracts is more stable and less sensitive to instantaneous extreme shocks.

Looking at parameter differences across maturities, the VoV level for the short-term 6-day maturity is significantly higher than that of the medium-term 20-day maturity throughout the entire cycle. This reflects higher volatility uncertainty and a greater curvature in the volatility smile for short-term options, matching perfectly the volatility smile shapes shown in Figures 6-2 and 6-3. Differences in correlation coefficients across maturities manifest primarily during the crash, where the weakening of the leverage effect is much more pronounced for short-term contracts, further validating that extreme events hit short-term option pricing structures more violently.

6.5 Pricing Method Performance Re-Evaluation and Comparative Analysis in Extreme Market Scenarios

Chapter 5 evaluated the performance of nine European option pricing methods under the 3/2 model based on benchmark parameter cases from classical academic literature, defining their theoretical performance boundaries within idealized parameter environments. However, these academic benchmark parameters are largely rooted in stable markets and cannot reflect the impact of real market parameters on pricing methods during extreme conditions. Based on the real market parameters across the full crash cycle calibrated in Section 6.5, this section constructs 6 new sets of test cases, duplicating the performance testing flow from Chapter 5. It quantitatively analyzes the numerical stability, pricing accuracy, and computational efficiency of different methods in extreme crypto crash scenarios, and cross-compares these with the academic benchmark results from Chapter 5. This clarifies the fundamental performance differences between idealized environments and real extreme markets, offering empirical backing for selecting option pricing methods under extreme conditions.

6.5.1 Test Setup and Case Construction

The cases constructed for this test are based entirely on real market calibration results from the crash event, featuring 6 sets of test cases. They correspond respectively to 3 market scenarios (pre-shock, crash, recovery) for the 6-day maturity, and 3 market scenarios for the 20-day maturity. For all cases, the long-term mean reversion parameter κ and long-term variance level θ utilize the time series estimation results from Section 6.5.1; the time-varying VoV ϵ and price-volatility correlation coefficient ρ use the cross-sectional calibration results for the corresponding scenario. The risk-free rate r is set to 0, and the underlying spot price S_0 is set to the average BTC spot price during that specific time window. Option strike prices cover five tiers: 85, 98, 100, 102, 115, comprehensively covering deep ITM, ATM, and deep OTM scenarios, maintaining structural consistency with the test cases in Chapter 5.

The establishment of benchmark prices aligns with the accuracy baseline in Chapter 5, employing the Euler-Milstein discretization scheme with an extremely fine time step $dt = \frac{1}{50000}$ via a Monte Carlo simulation of 1 million paths to obtain the benchmark option price. Chapter 5's test results verified that the Euler-Milstein method at this step size yields the highest precision and strongest stability, serving as a reliable benchmark for calculating pricing deviations across methods.

The evaluation metric system for this test is identical to Chapter 5 to guarantee horizontal comparability. The metrics, in descending priority, are numerical stability, pricing accuracy, and computational efficiency. Numerical stability is judged primarily by whether a method outputs Not-a-Number (nan) or infinity (inf); such outputs constitute a numerical failure for that scenario. Pricing accuracy is gauged via dual metrics: absolute bias and relative bias, where relative bias is the percentage of absolute bias to the benchmark price, annotated in parentheses trailing the absolute bias. Computational efficiency is measured by the complete runtime for a single set of cases, in seconds.

6.5.2 Performance Results and Analysis of Time-Stepping Conditional Monte Carlo Methods

Time-stepping methods include the Euler/Milstein discrete scheme, Exact Stepping method, Poisson-Gamma Near-Exact Stepping method, and QE Near-Exact Stepping method. Tests were configured with 3 time steps: 1/50, 1/500, and 1/5000. Complete results are shown in Table 6.4.

From the results in Table 6.4, we can see that the performance of the four time-stepping methods in real extreme market scenarios is highly consistent with the conclusions from the academic benchmark cases in Chapter 5. However, extreme market conditions impose higher requirements on the accuracy and stability of the methods. The Euler/Milstein discrete scheme remains highly sensitive to time step size. Under a coarse step size of $dt = 1/50$, this method exhibited massive pricing deviations across all 6 test cases. The relative bias for ATM options during the 6D crash reached 1612.23%, and for deep OTM options before the 20D crash, it even reached 762.56%, completely losing its pricing capability. Even when the step size was reduced to $dt = 1/500$, the relative bias in most scenarios still exceeded 50%. Only when the step size was further reduced to $dt = 1/5000$ did the bias fall below 1%, but the corresponding computation time increased by an order of magnitude compared to the coarse step size, rendering its efficiency extremely low.

In stark contrast, the Exact Stepping, Poisson-Gamma Near-Exact Stepping, and QE Near-Exact Stepping methods did not suffer any numerical failures across all 6 test cases and all time step sizes, consistently maintaining exceptionally high pricing stability. Even under the coarse step size of $dt = 1/50$, the relative bias of these three methods was generally kept within 2%, demonstrating accuracy far superior to the Euler/Milstein scheme under the fine step size of $dt = 1/5000$, while computation times remained on the same order of magnitude as the Euler/Milstein scheme at the same step size. Among them, the Exact Stepping method achieved the highest accuracy, with relative biases controlled within 1% across all scenarios, serving as the accuracy benchmark for time-stepping methods. The accuracy of the Poisson-Gamma and QE methods was slightly lower than that of the Exact Stepping method, but they offered better computational efficiency, maintaining stable pricing performance even in extreme scenarios like the crash period, thereby achieving an optimal balance between accuracy and efficiency.

Table 6.4 Time-Stepping Monte Carlo Option Pricing Results

Situation	dt	Strike	Euler/Milstein scheme		Exact Stepping		Poisson-Gamma Almost Exact Stepping		QE Almost Exact Stepping	
			Time (s)	Option Bias	Time (s)	Option Bias	Time (s)	Option Bias	Time (s)	Option Bias
		85		52.78 (343.90%)		-0.3025 (-1.97%)		-0.303 (-1.97%)		-0.4921 (-3.21%)
		98		50.99 (1008.18%)		-0.06433 (-1.27%)		-0.0648 (-1.28%)		-0.116 (-2.29%)
	1/50	100	0.097975	50.25 (1264.36%)	0.073905	-0.02048 (-0.52%)	0.08308	-0.02098 (-0.53%)	0.090593	-0.05895 (-1.48%)
		102		49.36 (1614.22%)		0.01505 (0.49%)		0.01452 (0.47%)		-0.01246 (-0.41%)
		115		40.6 (11478.07%)		0.03456 (9.77%)		0.03419 (9.67%)		0.03758 (10.63%)
		85		8.006 (52.17%)		-0.007846 (-0.05%)		-0.008731 (-0.06%)		-0.03622 (-0.24%)
		98		7.206 (142.48%)		-0.004967 (-0.10%)		-0.005825 (-0.12%)		-0.01238 (-0.24%)
20D during crash	1/500	100	0.260825	7.024 (176.73%)	0.376691	-0.004271 (-0.11%)	0.541429	-0.004983 (-0.13%)	0.500772	-0.009548 (-0.24%)
		102		6.834 (223.50%)		-0.003733 (-0.12%)		-0.0043 (-0.14%)		-0.007244 (-0.24%)
		115		5.497 (1554.06%)		-0.002588 (-0.73%)		-0.002764 (-0.78%)		-0.00155 (-0.44%)
		85		0.1869 (1.22%)		-0.004063 (-0.03%)		-0.01244 (-0.08%)		-0.006032 (-0.04%)
		98		0.1493 (2.95%)		-0.002526 (-0.05%)		-0.004968 (-0.10%)		-0.003897 (-0.08%)
	1/5000	100	2.147697	0.1432 (3.60%)	4.123019	-0.002434 (-0.06%)	3.745557	-0.004031 (-0.10%)	4.445529	-0.00364 (-0.09%)
		102		0.1373 (4.49%)		-0.002466 (-0.08%)		-0.003371 (-0.11%)		-0.003511 (-0.11%)
		115		0.1051 (29.71%)		-0.002438 (-0.69%)		-0.002065 (-0.58%)		-0.002741 (-0.77%)
		85		70.57 (458.80%)		-0.1383 (-0.90%)		-0.1359 (-0.88%)		-0.2368 (-1.54%)
		98		68.33 (1302.82%)		-0.03066 (-0.58%)		-0.03023 (-0.58%)		-0.05632 (-1.07%)
	1/50	100	0.10964	67.57 (1618.57%)	0.072515	-0.01057 (-0.25%)	0.087479	-0.01042 (-0.25%)	0.087417	-0.02948 (-0.71%)
		102		66.67 (2044.48%)		0.006084 (0.19%)		0.005995 (0.18%)		-0.007262 (-0.22%)
		115		58.32 (13521.37%)		0.01885 (4.37%)		0.01855 (4.30%)		0.02139 (4.96%)
		85		2.87 (18.66%)		-0.006586 (-0.04%)		-0.008506 (-0.06%)		-0.007351 (-0.05%)
		98		2.522 (48.08%)		-0.002124 (-0.04%)		-0.003617 (-0.07%)		-0.002112 (-0.04%)
20D pre-shock	1/500	100	0.280515	2.457 (58.86%)	0.36221	-0.001463 (-0.04%)	0.539243	-0.002796 (-0.07%)	0.574392	-0.001406 (-0.03%)
		102		2.394 (73.40%)		-0.0008839 (-0.03%)		-0.002039 (-0.06%)		-0.0007904 (-0.02%)
		115		2.019 (468.13%)		0.0003308 (0.08%)		0.000147 (0.03%)		0.0005106 (0.12%)
		85		0.03389 (0.22%)		-0.001573 (-0.01%)		-0.005295 (-0.03%)		-0.0009402 (-0.01%)
		98		0.01982 (0.38%)		0.0004488 (0.01%)		-0.001427 (-0.03%)		4.042e-05 (0.00%)
	1/5000	100	1.995124	0.01777 (0.43%)	4.798202	0.0004454 (0.01%)	3.812062	-0.0009235 (-0.02%)	4.348587	7.606e-05 (0.00%)
		102		0.01599 (0.49%)		0.0003961 (0.01%)		-0.0004937 (-0.02%)		9.859e-05 (0.00%)
		115		0.01084 (2.51%)		0.000117 (0.03%)		0.0003757 (0.09%)		0.0001203 (0.03%)
		85		59.05 (383.33%)		-0.1175 (-0.76%)		-0.1147 (-0.74%)		-0.1813 (-1.18%)
		98		56.68 (1054.03%)		-0.009567 (-0.18%)		-0.009586 (-0.18%)		-0.02326 (-0.43%)
	1/50	100	0.098142	55.91 (1294.30%)	0.088713	0.007429 (0.17%)	0.079911	0.007149 (0.17%)	0.095661	-0.001574 (-0.04%)
		102		55.02 (1611.90%)		0.02082 (0.61%)		0.02034 (0.60%)		0.01568 (0.46%)
		115		46.85 (9134.21%)		0.02293 (4.47%)		0.02233 (4.35%)		0.02815 (5.49%)
		85		1.687 (10.96%)		-0.005204 (-0.03%)		-0.00616 (-0.04%)		-0.004593 (-0.03%)
		98		1.419 (26.38%)		-0.001239 (-0.02%)		-0.002311 (-0.04%)		-0.0008583 (-0.02%)
20D recovery	1/500	100	0.269408	1.372 (31.75%)	0.370337	-0.0007568 (-0.02%)	0.554489	-0.001756 (-0.04%)	0.474414	-0.0004048 (-0.01%)
		102		1.327 (38.87%)		-0.0003489 (-0.01%)		-0.001255 (-0.04%)		-1.756e-05 (-0.00%)
		115		1.081 (210.68%)		0.000451 (0.09%)		0.0001603 (0.03%)		0.0006369 (0.12%)
		85		0.01856 (0.12%)		-0.001722 (-0.01%)		-0.002718 (-0.02%)		-0.0009435 (-0.01%)
		98		0.006872 (0.13%)		-0.0001248 (-0.00%)		-6.51e-05 (-0.00%)		0.0001071 (0.00%)
	1/5000	100	2.857551	0.005187 (0.12%)	2.678423	2.452e-05 (0.00%)	3.641867	0.0001319 (0.00%)	4.265534	0.0001094 (0.00%)
		102		0.003788 (0.11%)		0.000152 (0.00%)		0.0002834 (0.01%)		0.0001107 (0.00%)
		115		0.001108 (0.22%)		0.000323 (0.06%)		0.0004545 (0.09%)		0.00019 (0.04%)
		85		13.3 (88.27%)		-0.1617 (-1.07%)		-0.1608 (-1.07%)		-0.2239 (-1.49%)
		98		13.12 (336.33%)		0.1601 (4.10%)		0.1599 (4.10%)		0.1684 (4.32%)
	1/50	100	0.089805	12.64 (452.63%)	0.063753	0.2069 (7.41%)	0.088638	0.2067 (7.40%)	0.068912	0.2211 (7.92%)
		102		11.96 (622.57%)		0.2226 (11.59%)		0.2223 (11.58%)		0.2407 (12.53%)
		115		5.002 (5437.06%)		0.01928 (20.95%)		0.01918 (20.85%)		0.03486 (37.89%)
		85		3.487 (23.13%)		-0.004953 (-0.03%)		-0.003007 (-0.02%)		-0.02175 (-0.14%)
		98		2.883 (73.87%)		-0.0007794 (-0.02%)		-0.0008907 (-0.02%)		0.006112 (0.16%)
6D during crash	1/500	100	0.115985	2.699 (96.68%)	0.188744	0.0002888 (0.01%)	0.213808	-0.0001072 (-0.00%)	0.241169	0.008969 (0.32%)
		102		2.506 (130.52%)		0.0007156 (0.04%)		0.0001468 (0.01%)		0.01058 (0.55%)
		115		1.133 (1231.99%)		-0.0006745 (-0.73%)		-0.0007963 (-0.87%)		0.006959 (7.56%)
		85		0.1193 (0.79%)		-0.004369 (-0.03%)		-0.005645 (-0.04%)		-0.004032 (-0.03%)
		98		0.08104 (2.08%)		0.0004828 (0.01%)		-0.0009292 (-0.02%)		-0.001528 (-0.04%)
	1/5000	100	0.584799	0.07398 (2.65%)	1.375289	0.0008826 (0.03%)	1.153838	-0.0002692 (-0.01%)	1.281406	-0.001203 (-0.04%)
		102		0.0673 (3.50%)		0.001027 (0.05%)		9.225e-05 (0.00%)		-0.001018 (-0.05%)
		115		0.03297 (35.84%)		0.0001429 (0.16%)		-0.0002349 (-0.26%)		-0.0004011 (-0.44%)
		85		31.94 (211.97%)		-0.1701 (-1.13%)		-0.1694 (-1.12%)		-0.2781 (-1.85%)
		98		32.14 (858.70%)		0.00318 (0.08%)		0.003333 (0.09%)		-0.01203 (-0.32%)
	1/50	100	0.096497	31.76 (1220.41%)	0.070505	0.05443 (2.09%)	0.089836	0.05451 (2.09%)	0.073266	0.0469 (1.80%)
		102		31.18 (1815.89%)		0.08528 (4.97%)		0.0853 (4.97%)		0.08313 (4.84%)
		115		24.17 (50996.85%)		0.0007269 (1.53%)		0.000708 (1.49%)		0.003945 (8.32%)
		85		5.31 (35.24%)		-0.00964 (-0.06%)		-0.0065 (-0.04%)		-0.03089 (-0.21%)
		98		4.813 (128.59%)		-0.005615 (-0.15%)		-0.004322 (-0.12%)		-0.007194 (-0.19%)
6D pre-shock	1/500	100	0.114386	4.662 (179.14%)	0.163379	-0.004361 (-0.17%)	0.24423	-0.003463 (-0.13%)	0.230178	-0.004502 (-0.17%)
		102		4.503 (262.28%)		-0.003424 (-0.20%)		-0.002864 (-0.17%)		-0.002548 (-0.15%)
		115		3.4 (7173.56%)		-0.002647 (-5.59%)		-0.00267 (-5.63%)		-0.001087 (-2.29%)
		85		0.1641 (1.09%)		-0.009055 (-0.06%)		-0.009074 (-0.06%)		-0.007244 (-0.05%)
		98		0.1332 (3.56%)		-0.002905 (-0.08%)		-0.005887 (-0.16%)		-0.00452 (-0.12%)
	1/5000	100	0.595634	0.1273 (4.89%)	1.268211	-0.0024 (-0.09%)	1.184932	-0.004927 (-0.19%)	1.384249	-0.004142 (-0.16%)
		102		0.1217 (7.09%)		-0.002158 (-0.13%)		-0.00412 (-0.24%)		-0.003842 (-0.22%)
		115		0.08972 (189.29%)		-0.002234 (-4.71%)		-0.002566 (-5.41%)		-0.002595 (-5.48%)
		85		34.92 (231.58%)		-0.1713 (-1.14%)		-0.1696 (-1.12%)		-0.2575 (-1.71%)
		98		34.87 (893.86%)		0.0171 (0.44%)		0.01731 (0.44%)		0.005745 (0.15%)
	1/50	100	0.101825	34.46 (1241.16%)	0.065492	0.06363 (2.29%)	0.0785	0.06369 (2.29%)	0.068987	0.05845 (2.11%)
		102		33.85 (1792.31%)		0.09178 (4.86%)		0.09174 (4.86%)		0.09097 (4.82%)
		115		26.86 (40036.99%)		0.007004 (10.44%)		0.006926 (10.32%)		0.01063 (15.85%)
		85		4.375 (29.01%)		-0.00604 (-0.04%)		-0.001276 (-0.01%)		-0.01968 (-0.13%)
		98		3.92 (100.50%)		-0.002504 (-0.07%)		-0.001241 (-0.03%)		-0.003214 (-0.08%)
6D recovery	1/500	100	0.131068	3.796 (136.75%)	0.157947	-0.001505 (-0.05%)	0.21773	-0.0007573 (-0.03%)	0.256783	-0.001224 (-0.04%)
		102		3.669 (194.25%)		-0.0007158 (-0.04%)		-0.000418 (-0.02%)		0.0002462 (0.01%)
		115		2.834 (4223.79%)		-0.001056 (-1.57%)		-0.001193 (-1.78%)		0.0004148 (0.62%)
		85		0.09437 (0.63%)		-0.004436 (-0.03%)		-0.005493 (-0.04%)		-0.004303 (-0.03%)
		98		0.07127 (1.83%)		-0.0006069 (-0.02%)		-0.002377 (-0.06%)		-0.002236 (-0.06%)
	1/5000	100	0.613319	0.06712 (2.42%)	1.29491	-0.000296 (-0.01%)	1.184146	-0.001994 (-0.07%)	1.296308	-0.001931 (-0.07%)
		102		0.06345 (3.36%)		-0.0001041 (-0.01%)		-0.001623 (-0.09%)		-0.001645 (-0.09%)
		115		0.04743 (70.68%)		-0.0006742 (-1.00%)		-0.0008689 (-1.30%)		-0.001032 (-1.54%)

6.5.3 Performance Results and Analysis of Terminal Simulation Methods

Terminal simulation methods encompass five schemes: Baldeaux Exact Simulation, Fourier-Cosine Exact Terminal Simulation, Moment Matching Exact Terminal Simulation, Inverse Gaussian (IG) Near-Exact Simulation, and Gamma Near-Exact Simulation. The complete test results are shown in Table 6.5.

Table 6.5 Pricing Performance Results of Terminal Simulation Methods in Extreme Market Scenarios

Scenario	Strike	Exact Simulation		Exact Simulation (Fourier-Cosine)		Exact Simulation (Moment Matching)		Almost Exact Simulation (IG)		Almost Exact Simulation (Gamma)	
		Time (s)	Option Bias	Time (s)	Option Bias	Time (s)	Option Bias	Time (s)	Option Bias	Time (s)	Option Bias
	85		1.534 (10.18%)		nan (nan%)		nan (nan%)		1.665 (11.05%)		2.965 (19.68%)
	98		-0.03761 (-1.00%)		nan (nan%)		nan (nan%)		0.00733 (0.02%)		1.292 (34.53%)
6D Pre-Shock	100	55.2718	-0.6423 (-24.68%)	21.9335	nan (nan%)	0.0377	nan (nan%)	0.3837	-0.7651 (-29.40%)	2.965	0.447 (17.18%)
	102		-1.031 (-60.04%)		nan (nan%)		nan (nan%)		-1.275 (-74.28%)		-0.5019 (-29.23%)
	115		-0.0474 (-99.99%)		nan (nan%)		nan (nan%)		-0.0474 (-100.00%)		-0.0474 (-100.00%)
	85		0.7671 (5.09%)		nan (nan%)		nan (nan%)		0.8457 (5.61%)		1.455 (9.66%)
	98		-0.8963 (-22.97%)		nan (nan%)		nan (nan%)		-0.9757 (-25.00%)		-0.3736 (-9.57%)
6D Crash	100	63.5117	-1.384 (-49.57%)	19.1247	nan (nan%)	0.0360	nan (nan%)	0.3375	-1.71 (-61.26%)	0.3797	-1.221 (-43.74%)
	102		-1.471 (-76.62%)		nan (nan%)		nan (nan%)		-1.805 (-93.99%)		-1.72 (-89.59%)
	115		-0.09199 (-99.99%)		nan (nan%)		nan (nan%)		-0.092 (-100.00%)		-0.092 (-100.00%)
	85		1.567 (10.39%)		nan (nan%)		nan (nan%)		1.686 (11.18%)		3.032 (20.11%)
	98		-0.1522 (-3.90%)		nan (nan%)		nan (nan%)		-0.126 (-3.23%)		1.211 (31.05%)
6D Recovery	100	46.3179	-0.7739 (-27.88%)	19.1559	nan (nan%)	0.0358	nan (nan%)	0.2984	-0.9046 (-32.59%)	0.3350	0.3499 (12.61%)
	102		-1.17 (-61.93%)		nan (nan%)		nan (nan%)		-1.419 (-75.13%)		-0.6054 (-32.05%)
	115		-0.0671 (-99.99%)		nan (nan%)		nan (nan%)		-0.0671 (-100.00%)		-0.0671 (-100.00%)
	85		2.732 (17.76%)		nan (nan%)		nan (nan%)		0.1622 (3.88%)		0.6949 (16.64%)
	98		0.6052 (11.54%)		nan (nan%)		nan (nan%)		0.9414 (17.95%)		1.522 (29.01%)
20D Pre-Shock	100	57.0627	0.1364 (3.27%)	10.6008	nan (nan%)	0.0356	nan (nan%)	0.3099	0.1267 (3.04%)	0.2807	0.6949 (16.64%)
	102		-0.2787 (-8.55%)		nan (nan%)		nan (nan%)		-0.5563 (-17.06%)		-0.1177 (-3.61%)
	115		-0.3418 (-79.25%)		nan (nan%)		nan (nan%)		-0.4291 (-99.49%)		-0.4287 (-99.40%)
	85		2.087 (13.60%)		nan (nan%)		nan (nan%)		3.585 (23.36%)		1.43 (28.29%)
	98		0.4211 (8.33%)		nan (nan%)		nan (nan%)		0.9223 (18.24%)		1.43 (28.29%)
20D Crash	100	57.4672	0.0838 (2.11%)	17.3398	nan (nan%)	0.0434	nan (nan%)	0.3143	0.1267 (3.19%)	0.2824	0.5963 (15.00%)
	102		-0.2042 (-6.68%)		nan (nan%)		nan (nan%)		-0.6188 (-20.24%)		-0.2366 (-7.74%)
	115		-0.237 (-67.01%)		nan (nan%)		nan (nan%)		-0.353 (-99.79%)		-0.3529 (-99.78%)
	85		2.468 (16.02%)		nan (nan%)		nan (nan%)		3.14 (20.39%)		3.662 (23.77%)
	98		0.1784 (3.32%)		nan (nan%)		nan (nan%)		0.2658 (4.94%)		0.7524 (13.99%)
20D Recovery	100	59.4667	-0.308 (-7.13%)	7.6416	nan (nan%)	0.0387	nan (nan%)	0.3289	-0.4767 (-11.03%)	0.2639	-0.04502 (-1.04%)
	102		-0.7179 (-21.03%)		nan (nan%)		nan (nan%)		-1.106 (-32.44%)		-0.7729 (-22.64%)
	115		-0.4415 (-86.08%)		nan (nan%)		nan (nan%)		-0.5112 (-99.66%)		-0.5111 (-99.64%)

The results in Table 6.5 clearly show that terminal simulation methods suffer from severe numerical failure and pricing bias issues in real extreme market scenarios. Compared to the results from the academic benchmark cases in Chapter 5, the stability and accuracy of the methods have deteriorated significantly. The Fourier-Cosine Exact Terminal Simulation method and the Moment Matching Exact Terminal Simulation method completely failed across all 6 test cases, outputting nan results and losing their pricing capability entirely; they failed to perform valid calculations even in the stable pre-shock market scenarios. The core reason is that the high volatility of volatility parameter calibrated from the real market caused severe numerical errors in truncating the cosine series and executing the moment-matching expansion for the distribution, eventually leading to computational failure.

Although the Baldeaux Exact Simulation method did not experience complete numerical failure, its pricing bias was massive. The relative bias for ATM options during the 6D crash reached -49.57%, and for deep OTM options before the 20D crash, it plunged to -79.25%, completely failing to meet pricing accuracy requirements. Furthermore, the computational efficiency of this method was extremely low, with a single case runtime exceeding 45 seconds —a further increase compared to the runtime in the academic benchmark cases. The core reason is that under extreme market parameters, the computational complexity of the high-order modified Bessel function and the inverse Laplace transform increased drastically, while the risk of numerical overflow also rose significantly, ultimately leading to degraded accuracy and worsened computational efficiency.

While the IG and Gamma Near-Exact Simulation methods avoided complete numerical failure, they exhibited massive systematic pricing biases in the vast majority of scenarios. The relative bias for deep OTM options generally exceeded -90%, and for ATM options, it was frequently above 10%. Only for ITM options in certain stable market scenarios did they maintain relatively controllable bias. The core reason is that in extreme market scenarios, the distribution of the integrated variance exhibits extreme thick tails and rightward skewness. The IG and Gamma distributions, which only match the first two moments, cannot adequately fit the true shape of the distribution, resulting in systematic pricing biases. This phenomenon is particularly pronounced in the extreme scenarios of the crash period.

6.5.4 Performance Results and Analysis of Fourier Frequency-Domain Pricing Methods

Fourier frequency-domain pricing methods comprise two schemes: the Fast Fourier Transform (FFT) method and the Fourier-Cosine (COS) method. The complete test results are presented in Table 6.6.

The results from Table 6.6 indicate that Fourier frequency-domain methods maintain ultimate stability and pricing accuracy in real extreme market scenarios, proving to be the optimal category among all evaluated methods. Neither the FFT nor the COS method encountered any numerical failures across all 6 test cases, reliably outputting pricing results even under the extreme parameter environments of the crash period. Regarding pricing accuracy, the relative biases for both methods were generally maintained within 0.2%, with the relative biases for the vast majority of ATM and ITM options approaching 0. They only exhibited relative biases up to 6% in deep OTM options, showcasing an accuracy that significantly surpasses all Monte Carlo-based methods.

In terms of computational efficiency, the COS method's performance comprehensively outshines that of the FFT method. Its runtime for a single case stabilizes around 0.05 seconds, which is only 1/4 of the FFT method's time and far quicker than any time-stepping or terminal

Table 6.6 Pricing Performance Results of Fourier Frequency-Domain Methods in Extreme Market Scenarios

Scenario	Strike	Fast Fourier Transformation		Fourier-Cosine Method	
		Time (s)	Option Bias	Time (s)	Option Bias
6D Pre-Shock	85	0.2520	-0.006751 (-0.04%)	0.0645	-0.006749 (-0.04%)
	98		-0.003863 (-0.10%)		-0.003861 (-0.10%)
	100		-0.003562 (-0.14%)		-0.00356 (-0.14%)
	102		-0.003347 (-0.19%)		-0.003346 (-0.19%)
	115		-0.002481 (-5.23%)		-0.002479 (-5.23%)
6D Crash	85	0.2182	-0.003449 (-0.02%)	0.0530	-0.00345 (-0.02%)
	98		-0.0009857 (-0.03%)		-0.0009859 (-0.03%)
	100		-0.0007312 (-0.03%)		-0.0007315 (-0.03%)
	102		-0.0006061 (-0.03%)		-0.0006102 (-0.03%)
	115		-0.0003863 (-0.42%)		-0.0003865 (-0.42%)
6D Recovery	85	0.2378	-0.004141 (-0.03%)	0.0551	-0.004141 (-0.03%)
	98		-0.001658 (-0.04%)		-0.001658 (-0.04%)
	100		-0.001403 (-0.05%)		-0.001403 (-0.05%)
	102		-0.001176 (-0.06%)		-0.001176 (-0.06%)
	115		-0.0008781 (-1.31%)		-0.0008779 (-1.31%)
20D Pre-Shock	85	0.2050	-0.001781 (-0.01%)	0.0503	-0.001781 (-0.01%)
	98		-0.0002175 (-0.00%)		-0.0002176 (-0.00%)
	100		-9.566e-05 (-0.00%)		-9.566e-05 (-0.00%)
	102		3.404e-06 (0.00%)		3.399e-06 (0.00%)
	115		0.002053 (0.05%)		0.002053 (0.05%)
20D Crash	85	0.1893	-0.006654 (-0.01%)	0.0447	-0.006654 (-0.01%)
	98		-0.004071 (-0.08%)		-0.004071 (-0.08%)
	100		-0.003738 (-0.09%)		-0.003738 (-0.09%)
	102		-0.003539 (-0.12%)		-0.003539 (-0.12%)
	115		-0.002622 (-0.74%)		-0.002622 (-0.74%)
20D Recovery	85	0.2092	-0.001571 (-0.01%)	0.0496	-0.001571 (-0.01%)
	98		-4.587e-05 (-0.00%)		-4.587e-05 (-0.00%)
	100		2.363e-05 (0.00%)		2.363e-05 (0.00%)
	102		8.295e-05 (0.00%)		8.295e-05 (0.00%)
	115		0.0002768 (0.05%)		0.0002768 (0.05%)

simulation methods. The core reason lies in the exponential convergence property of the COS method, which allows it to reach extremely high computational precision using only a small number of series terms. Furthermore, it completely sidesteps the picket-fence effect present in the FFT method, eliminating the need for extra interpolation calculations and thereby further reducing computational overhead. Even in the extreme parameter environment of the crash stage, the computational time for both methods did not increase significantly, displaying immense robustness.

6.5.4.1 Comparative Analysis with Academic Benchmark Case Results

Combining the test results from the academic benchmark cases in Chapter 5, this re-evaluation in real extreme market scenarios yielded three core supplementary conclusions, clarifying the essential differences in pricing method performance between idealized environments and real markets.

First, extreme market scenarios significantly amplify the divergence in numerical stability among different methods. In Chapter 5's academic benchmark cases, the Fourier-Cosine and moment-matching exact terminal simulation methods only failed numerically in select high-volatility scenarios. However, in this real extreme market test, both methods failed completely across all scenarios, and the Baldeaux exact simulation method also exhibited severe accuracy degradation. This result demonstrates that terminal simulation methods deemed "exact" in academic literature can only be theoretically validated under idealized parameters in stable markets. In real extreme market environments, they are highly prone to numerical overflow and computational failure, lacking engineering-level adaptability. Conversely, time-stepping distribution sampling methods and Fourier frequency-domain methods maintained 100% numerical stability in both academic cases and real extreme scenarios, demonstrating robust resilience.

Second, extreme market scenarios impose higher demands on the accuracy of pricing methods, and the hierarchy of accuracy differences between methods is further magnified. In Chapter 5's academic benchmark cases, the relative biases of the Euler/Milstein scheme at coarse step sizes were mostly within 100%. However, in this real extreme scenario, relative biases under coarse step sizes universally exceeded 1000%, and acceptable accuracy was only achieved at extremely fine step sizes. The pricing biases of terminal simulation methods also magnified several times over compared to the academic cases. Only the frequency-domain methods and the time-stepping distribution sampling methods maintained stable accuracy. This indicates that only exact sampling methods built on model distributional properties and frequency-domain methods based on characteristic functions can maintain stable pricing accuracy in extreme market environments. Traditional discretization schemes and moment-matching approximation methods will suffer from severe accuracy failures under extreme conditions.

Third, the efficiency ranking of the methods remained highly consistent between the academic cases and the real extreme scenarios, but the extreme scenarios further highlighted the efficiency advantage of frequency-domain methods. Across both sets of tests, the COS method was consistently the most efficient high-accuracy method, and the Baldeaux exact simulation method was consistently the least efficient. However, in real extreme scenarios, the computation time for terminal simulation methods increased further, while the computation time for frequency-domain methods remained stable, thereby widening the efficiency gap between the two. This demonstrates that the Fourier frequency-domain COS method not only enjoys an efficiency advantage in stable markets but sees that advantage become even more prominent under extreme conditions, establishing it as the optimal solution for batch pricing and parameter calibration in real market environments.

6.5.4.2 Summary of Method Applicability in Extreme Market Scenarios

Based on the test results from these real extreme market scenarios, combined with the conclusions from the academic cases in Chapter 5, the applicability of different methods in extreme conditions within the cryptocurrency options market can be clearly defined: The Fourier frequency-domain COS method is the absolute optimal scheme for extreme market scenarios. It combines full-scenario numerical stability, ultimate pricing precision, and the highest computational efficiency, making it suitable for batch option pricing, real-time valuation, and model parameter calibration under extreme conditions. The time-stepping Poisson-Gamma and QE Near-Exact Stepping methods are the sole reliable solutions in scenarios requiring complete variance path generation, such as path-dependent option pricing and dynamic hedging; they maintain stable accuracy and efficiency even under extreme conditions. The Euler/Milstein discretization scheme only achieves acceptable accuracy at extremely fine step sizes, resulting in abysmal efficiency; it is only suitable as a tool for generating benchmark prices and is not recommended for practical engineering applications. All terminal simulation methods exhibited severe numerical failures or massive pricing biases in extreme market scenarios; they should only be used for academic theoretical validation under idealized parameters in stable markets and are not recommended for use in real market environments.

6.6 Chapter Summary

Focusing on the extreme BTC flash crash event from October 10 to 11, 2025, this chapter expanded the analysis of the 3/2 stochastic volatility model from the idealized academic benchmark cases in Chapter 5 to the extreme market scenarios of real financial markets. By constructing a complete empirical framework comprising "long-term parameter time series estimation —cross-sectional time-varying parameter calibration —extreme scenario re-evaluation of pricing methods," it systematically identified the time-varying patterns of the 3/2 model's core parameters throughout the crash event cycle. Concurrently, it completed performance validations for nine option pricing methods in a genuine extreme market environment, yielding three core empirical conclusions.

First, the 3/2 stochastic volatility model can effectively capture option pricing characteristics under extreme cryptocurrency market conditions, and its core parameters exhibit significant event-driven time-varying patterns throughout the crash cycle. The long-term mean reversion parameter estimated from daily DVOL data over the half-year preceding the crash aligns with the core characteristic of rapid volatility mean reversion in cryptocurrency markets. The central level of long-term volatility closely matches the historical volatility features of the BTC market, providing a stable long-term parameter foundation for model calibration. Cross-sectional calibration results based on the Medvedev-Scaillet asymptotic expansion method show that ATM implied volatility spiked massively during the crash. The increase for the short-term 6-day maturity was significantly higher than that for the medium-term 20-day maturity, reflecting the market's heightened sensitivity to short-term volatility expectations surrounding extreme events, which conforms to classical laws of option market volatility term structures. The volatility of volatility rose synchronously during the crash, indicating that the extreme event not only drove up the overall market volatility level but also amplified the uncertainty of volatility itself, significantly boosting market tail risk pricing. The negative correlation coefficient between price and volatility weakened structurally during the crash, with the leverage effect of short-term option contracts dropping notably. This reflects that panic trading and liquidity shocks under extreme conditions severed the conventional price-volatility correlation in the market, whereas the correlation in medium-term contracts remained relatively stable and less sensitive to instantaneous extreme shocks. Overall, the 3/2 model under a pure diffusion framework precisely fits the evolution of the volatility smile across the crash event cycle, demonstrating excellent market adaptability for option pricing under extreme cryptocurrency market conditions.

Second, real extreme market scenarios impose requirements on the numerical stability and robustness of option pricing methods that far exceed those of academic benchmark cases; theoretical precision in an idealized environment does not equal engineering practicality in a real market. In the academic benchmark cases from Chapter 5, some nominally "exact" terminal simulation methods only failed numerically in a few high-volatility scenarios. However, in this real extreme market test, the exact terminal simulation methods based on Fourier-Cosine and moment matching failed completely across all 6 test cases, unable to output valid pricing results. While the Baldeaux exact simulation method did not fail entirely numerically, it exhibited systematic and massive pricing biases in the vast majority of scenarios, and the computation time for a single case exceeded 45 seconds, making it extremely inefficient. This result indicates that such nominally "exact" terminal simulation methods can only be validated for unbiasedness theoretically under idealized parameters in stable markets. In real extreme conditions, they are highly prone to special function numerical overflows and distribution fitting failures, lacking adaptability to extreme scenarios at the engineering application level. In stark contrast, time-stepping Exact Stepping, Poisson-Gamma, and QE Near-Exact Stepping methods, along with the Fourier frequency-domain FFT and COS methods, maintained 100% numerical stability across all extreme market scenarios, suffering zero computational failures. This aligns highly with their performance in the academic benchmark cases of Chapter 5, displaying formidable robustness against extreme market conditions.

Third, performance re-evaluations based on real market parameters further clarified the scenario-based applicability boundaries of different pricing methods in the cryptocurrency options market. The Fourier frequency-domain COS method is the absolute optimal scheme for pricing European options in extreme market scenarios. It maintains a pricing bias approaching 0 across all scenarios, with relative biases controlled within 0.2% in the vast majority of cases. Simultaneously, its computation time per case stabilizes around 0.05 seconds, delivering computational efficiency far superior to all other methods. It possesses an absolute performance edge in real-time option valuation, batch pricing under extreme conditions, and model cross-sectional parameter calibration. The time-stepping Poisson-Gamma and QE Near-Exact Stepping methods are the sole reliable solutions in scenarios demanding complete variance path generation, such as path-dependent option pricing and dynamic hedging. Under extreme conditions, they balance pricing accuracy, computational efficiency, and numerical stability, maintaining controllable pricing bias even with coarse time steps, thoroughly overcoming the core flaws of traditional Euler/Milstein discrete schemes which are step-size sensitive and highly in-accurate in extreme scenarios. The Euler/Milstein discrete scheme achieves acceptable pricing accuracy only at extremely fine time steps, incurring massive computational costs; it is only suitable as a tool for generating benchmark prices and is not recommended for practical engineering applications. All terminal simulation methods lack application value in real extreme market environments and should only be utilized for academic theoretical comparisons under stable market conditions.

The empirical research in this chapter bridges the gap between idealized simulations and real market environments present in existing 3/2 model pricing literature. It not only validates the market explanatory power of the 3/2 model under extreme cryptocurrency conditions but also provides rigorous empirical evidence and clear practical guidelines for selecting option pricing methods in extreme market environments, establishing an actionable analytical framework for pricing and risk management in the cryptocurrency derivatives market.

Chapter 7 Conclusions and Outlook

This thesis takes the European option pricing problem under the 3/2 stochastic volatility model as its core research subject. Addressing the shortcomings of existing research—where performance evaluations of pricing methods are largely confined to idealized academic benchmark cases and lack empirical validation in real extreme market environments—this study systematically constructed a comprehensive theoretical framework for nine European option pricing methods under the 3/2 model. Through multi-scenario numerical experiments, it quantitatively evaluated the pricing accuracy, computational efficiency, and numerical stability of different methods. Furthermore, based on the extreme BTC market flash crash event of October 2025, it completed a full-cycle parameter calibration of the model and an extreme-scenario re-evaluation of the pricing methods. This clarified the performance boundaries of different methods in both idealized environments and real markets, providing a systematic analytical framework and empirical evidence for the theoretical application and engineering practice of the 3/2 model in derivatives pricing. This chapter will systematically summarize the core research conclusions of the entire thesis, objectively analyze the limitations of the study, and provide an outlook on future related research directions.

7.1 Main Research Conclusions

Through systematic research across three dimensions—theoretical construction, numerical experiments, and empirical analysis—this thesis has formed four core research conclusions that comprehensively cover the pricing method performance, market adaptability, and practical application boundaries of the 3/2 model.

First, this thesis systematically clarified the theoretical logic and performance boundaries of nine option pricing methods for the 3/2 model within a pure diffusion framework, forming a clear hierarchical system of method performance. This thesis categorized the nine pricing methods into three major paradigms: time-stepping conditional Monte Carlo methods, terminal exact and near-exact simulation methods, and Fourier frequency-domain pricing methods, clarifying the implementation pathways and theoretical assumptions of each method. Based on 7 sets of academic benchmark cases covering conventional markets and extreme parameter environments, a systematic performance evaluation of these methods was completed. The research results indicate that the nine methods can be classified into five tiers based on engineering practicality: The first tier consists of the Fourier-domain COS and FFT methods, which are the only methods that maintain numerical stability across all scenarios while possessing extremely high pricing accuracy and computational efficiency, with the COS method demonstrating comprehensive superiority. The second tier includes the time-stepping Exact Stepping, Poisson-Gamma, and QE Near-Exact Stepping methods; these are the only solutions within the Monte Carlo framework experiencing zero numerical failures across all scenarios, balancing pricing accuracy and computational efficiency, and serving as the sole reliable choices for scenarios requiring complete path generation. The third tier is the Euler/Milstein discretization scheme, which achieves acceptable pricing accuracy only at extremely fine time steps and has extremely low computational efficiency, serving merely as a benchmark tool for theoretical accuracy comparisons. The fourth tier comprises the IG and Gamma near-exact simulation methods, whose biases are controllable only in specific ATM scenarios but exhibit systematic and massive pricing deviations in the vast majority of scenarios, making them suitable only for academic comparisons in specific settings. The complete failure tier consists of the Baldeaux exact simulation, Fourier-Cosine exact terminal simulation, and moment matching exact terminal simulation methods, which can only output valid results in a single benchmark scenario and completely fail numerically in all other scenarios. The core finding is that the theoretical unbiasedness and "exact" label of a pricing method do not equate to full-scenario stability in engineering practice; most nominally "exact" terminal simulation methods can only complete theoretical validation under idealized benchmark parameters and fail to adapt to non-benchmark extreme parameter environments.

Second, the 3/2 stochastic volatility model can effectively capture option pricing characteristics in highly volatile cryptocurrency markets, demonstrating good market adaptability under extreme conditions. This thesis constructed a two-step calibration framework of "long-term parameter time series estimation - time-varying parameter cross-sectional calibration." It estimated the long-term mean reversion parameters based on daily DVOL data from the half-year preceding the crash event and completed the cross-sectional calibration of time-varying parameters throughout the crash event cycle using the Medvedev-Scaillet asymptotic expansion method. Empirical results show that the 3/2 model under a pure diffusion framework can accurately fit the morphological evolution of the volatility smile across the crash event cycle. The model's core parameters exhibit significant event-driven time-varying patterns: ATM implied volatility and the volatility of volatility (VoV) spiked synchronously during the crash, with the increase for the 6-day short-term maturity being significantly higher than that for the

20-day medium-term maturity. This reflects a significant elevation in the market's short-term volatility expectations and tail risk pricing regarding extreme events. The negative correlation coefficient between price and volatility weakened in stages during the crash, with the distortion of the leverage effect being more pronounced for short-term option contracts, while the correlation for medium-term contracts remained relatively stable. This embodies the significant impact of panic trading and liquidity shocks on the short-term option pricing structure under extreme conditions. This result validates the good adaptability of the 3/2 model in highly volatile cryptocurrency markets, providing a reliable model foundation for pricing and risk measurement in crypto derivatives markets.

Third, real extreme market scenarios impose requirements on the numerical stability and robustness of pricing methods that are far higher than those of academic benchmark cases, further amplifying the performance divergence among different methods. Based on 6 sets of real market parameters obtained from calibration, this thesis constructed brand-new extreme scenario test cases, completed a performance re-evaluation of the pricing methods, and conducted a horizontal comparison with the academic benchmark results. The study found that terminal exact simulation methods, which only failed in a few scenarios in the academic benchmark cases, experienced full-scenario numerical failures or systematic and massive pricing biases in real extreme scenarios, completely losing their engineering practicality. Conversely, time-stepping distribution sampling methods and Fourier frequency-domain methods maintained 100% numerical stability and stable pricing accuracy in both academic cases and real extreme scenarios, demonstrating extremely strong robustness against extreme conditions. Simultaneously, extreme scenarios further highlighted the performance advantages of the COS method, which maintained a pricing bias approaching 0 across all scenarios with a single-case computation time stabilizing around 0.05 seconds, proving to be the optimal scheme for batch pricing of European options, real-time valuation, and model calibration in extreme market environments. The Poisson-Gamma and QE near-exact stepping methods are the optimal choices for path-dependent option pricing and dynamic hedging scenarios, completely compensating for the core defects of the traditional Euler/Milstein schemes, which are sensitive to step size and exhibit extremely poor accuracy in extreme scenarios.

Fourth, based on theoretical and empirical results, this thesis has formed scenario-based application guidelines for option pricing methods under the 3/2 model. For scenarios such as the batch pricing of European options and model parameter calibration, the COS frequency-domain method should be prioritized; its full-scenario stability, supreme accuracy, and optimal efficiency fully satisfy the demands of these scenarios. For scenarios requiring complete variance path generation, such as path-dependent option pricing and dynamic hedging, the Poisson-Gamma or QE near-exact stepping methods should be prioritized; their balance of full-scenario stability, accuracy, and efficiency meets the engineering requirements of these scenarios. For theoretical accuracy benchmark comparison scenarios, the Baldeaux exact simulation method may be used, but strictly under benchmark parameter environments. The remaining terminal simulation methods are not recommended for use in practical pricing scenarios and should only be utilized for specific academic comparative analyses.

7.2 Research Limitations

This thesis has conducted systematic theoretical and empirical research centered on the option pricing problem under the 3/2 stochastic volatility model, forming a relatively complete analytical framework. However, constrained by the research scope, data conditions, and model specifications, certain limitations remain, primarily reflected in the following four aspects:

First, there is room for expansion in the dimensions of model specification. This thesis focuses on the 3/2 stochastic volatility model under a pure diffusion framework and has not introduced market friction factors such as underlying price jump terms, stochastic interest rates, or stochastic liquidity into the model. Although the pure diffusion model can already fit the volatility smile morphology throughout the full cycle of crash events quite well, the jump characteristics of underlying prices and instantaneous changes in market liquidity during extreme conditions may still exert a systematic impact on option pricing. There is still further room to enrich the model specification to portray real market structures.

Second, the coverage scope of empirical research needs to be broadened. The empirical analysis in this thesis is based solely on BTC option data from the Deribit exchange, without covering other mainstream crypto assets like ETH, nor extending to underlying assets in traditional financial markets such as equities, commodities, or foreign exchange. Significant differences exist in the volatility characteristics, market microstructures, and investor structures across different underlying assets. The adaptability and parameter characteristics of the 3/2 model across various markets still require cross-validation using multi-dimensional empirical data.

Third, there are limitations in the evaluation scope of pricing methods. The performance evaluation of pricing methods in this thesis focuses solely on European options and does not conduct analyses on mainstream path-dependent derivatives such as American options, barrier options, and Asian options. Method construction, accuracy optimization, and performance evaluation for pricing path-dependent options under the 3/2 model remain inadequately covered by existing research and represent a core limitation of the research scope of this thesis.

Fourth, there is room for improvement in the richness of the parameter calibration framework. The Medvedev-Scaillet asymptotic expansion method adopted in this thesis is more suitable for short-maturity options, and there is still room to optimize calibration accuracy for ultra-long-maturity options. Concurrently, this thesis only employs a single calibration method for parameter estimation and has not introduced methods like characteristic function-based global calibration or Bayesian calibration to conduct robust comparisons of parameter estimation results. The diversity and robustness of the calibration framework can still be further enhanced.

7.3 Future Research Outlook

Based on the research conclusions and existing limitations of this thesis, further in-depth research can be conducted in the following five directions in the future to perfect the pricing theory and practical application system of the 3/2 stochastic volatility model:

First, expand the theoretical specifications of the 3/2 stochastic volatility model. Building upon the pure diffusion model, factors such as synchronous jumps, independent jumps, stochastic interest rates, and stochastic liquidity can be introduced to construct an extended 3/2 model that better aligns with real market structures. Derive the closed-form solutions of characteristic functions and option pricing formulas for the corresponding extended models, quantify the marginal impact of different expansion factors on option pricing, and further enhance the model's adaptability to extreme market environments. Concurrently, conduct horizontal comparisons of pricing accuracy between the extended models and the benchmark model to clearly define the applicable scenarios of different model setups.

Second, expand the applicable scope of pricing methods and optimize computational performance. Targeting mainstream path-dependent derivatives such as American options, barrier options, and Asian options, construct highly efficient pricing methods adapted to the variance process characteristics of the 3/2 model. Complete systematic evaluations of the pricing accuracy, computational efficiency, and numerical stability of different methods to form a 3/2 model pricing system covering a full range of derivative products. Meanwhile, by integrating technologies such as GPU parallel computing and vectorization optimization, further elevate the computational efficiency of large-scale Monte Carlo simulations and frequency-domain methods to meet the high-frequency real-time pricing demands in actual business operations.

Third, expand the market coverage dimensions of empirical research. Extend the empirical analysis from BTC options to other mainstream crypto assets like ETH, as well as to traditional financial sectors such as global equity markets, commodity markets, and foreign exchange markets. Validate the adaptability of the 3/2 model across different volatility characteristics and market structures to clearly delineate the model's universal application boundaries. Concurrently, based on longer-cycle historical data, conduct large-sample horizontal comparisons between the 3/2 model and mainstream stochastic volatility models like the Heston model and SABR model. Quantify the differences in pricing accuracy and market explanatory power among various models to provide empirical evidence for model selection across different markets.

Fourth, optimize the parameter calibration framework of the 3/2 model. Addressing the shortcomings of the asymptotic expansion method in calibrating long-term options, introduce methods such as characteristic function-based global calibration, Bayesian calibration, and machine learning-assisted calibration to elevate parameter calibration accuracy and stability in extreme market scenarios. Simultaneously, construct a real-time dynamic calibration system for model parameters to achieve high-frequency updates of model parameters under extreme conditions, adapting to the real-time risk management demands of derivatives markets.

Fifth, construct a full-process risk management system for derivatives based on the 3/2 model. Building on pricing research, derive closed-form solutions and efficient numerical computation methods for option Greeks under the 3/2 model. Establish model-based tail risk measurement systems such as VaR and CVaR to quantify the risk exposure of option portfolios during extreme conditions. Concurrently, based on the time-varying characteristics of model parameters, build measurement models for volatility risk premiums, providing theoretical support and empirical evidence for hedging strategy design and dynamic asset allocation in the crypto derivatives market.

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摘要

3/2 随机波动率模型凭借其非线性扩散结构，在刻画资产价格极端波动、拟合波动率微笑形态上具备显著的理论优势，是高波动市场衍生品定价的核心工具之一。然而现有研究对该模型下各类定价方法的性能评价，多局限于平稳市场的理想化学术基准算例，缺乏真实极端市场环境下的实证检验，不同方法的工程适用性边界尚未得到系统性厘清。针对这一问题，本文在纯扩散框架下系统构建了 3/2 模型九类欧式期权定价方法的完整理论体系，将其划分为时间步进类条件蒙特卡洛方法、终端精确与近似精确模拟方法、傅里叶频域定价方法三大范式，完整推导了各类方法的实现逻辑、误差特性与收敛性质。在此基础上，本文通过 7 组覆盖常规市场与极端参数环境的数值实验，量化对比了不同方法的定价精度、计算效率与全场景数值稳定性，形成了清晰的方法性能分层体系。进一步地，本文以 2025 年 10 月比特币市场闪崩极端事件为研究对象，构建“长期固定参数时间序列估计 - 时变参数横截面校准 - 定价方法极端场景再检验”的完整实证框架，验证了 3/2 模型对加密货币极端行情的市场适配性，识别了崩盘事件全周期模型核心参数的事件驱动型时变规律，并完成了定价方法在真实极端市场环境下的性能再检验。研究结果表明：傅里叶余弦（Fourier-Cos）方法是欧式期权定价的全局最优方案，在全场景兼具极致的定价精度、数值稳定性与计算效率；Poisson-Gamma 与 Quasi-Exponential（QE）近似精确步进方法，是蒙特卡洛框架下唯一全场景无失效的方案，为路径依赖型衍生品定价与动态对冲提供了可靠实现路径；标称“精确”的终端模拟方法仅能在理想化基准参数下完成理论验证，在极端市场环境中普遍出现数值溢出与系统性定价偏差，不具备工程实践层面的适配性。本文的研究弥补了现有 3/2 模型定价研究中理论算例与真实市场脱节的不足，形成了不同应用场景下的方法选择指引，为高波动市场环境下的衍生品定价、模型校准与风险管理提供了系统的理论支撑与实证依据。

关键词：期权定价, 3/2 波动率模型, 精确模拟, 近似精确模拟, 数值方法

目录

第一章　引论

1.1　研究背景

1.1.1　全球及国内期权市场的发展现状

自 1973 年芝加哥期权交易所（CBOE）正式挂牌交易标准化期权以来，全球金融衍生品市场经历了从单一品种向多层次、高复杂度体系的深刻演变。在国际成熟市场中，期权已不仅是基础的风险对冲工具，更演化为资产价格发现、尾部风险管理与策略博弈的核心载体。近年来，随着全球宏观经济不确定性加剧、地缘政治冲突频发，机构投资者对极端尾部风险的防范需求日益迫切，推动全球期权成交量持续创下历史新高。特别是以零日期权（0DTE）为代表的超短期品种在美股市场的大规模流行，不仅深刻改变了市场流动性的微观结构，更对定价模型在极短时间尺度下的瞬时响应能力、对极端波动率偏斜（Skew）的精准捕捉能力提出了更为严苛的理论挑战，传统常数波动率模型与线性随机波动率模型的局限性愈发凸显。

与此同时，中国期权市场作为全球衍生品版图中的重要新兴力量，正处于跨越式发展的关键窗口期。在场内市场层面，自上证 50ETF 期权 2015 年成功试点以来，国内已快速构建起覆盖沪深 300、中证 500、中证 1000 等核心指数，以及科创 50、创业板指等特色指数的权益期权矩阵；商品期权领域则实现了从有色金属、黑色系产业链到农产品、新能源原材料的全品类覆盖，为实体企业提供了丰富的风险对冲工具，成为实体经济稳健经营的重要金融支撑。在场外（OTC）衍生品领域，以中证报价系统为核心纽带的场外期权市场稳步发展，个性化、定制化的风险解决方案日益丰富，进一步完善了多层次资本市场的风险管理功能。

1.1.2　期权定价模型与计算方法的演进

期权定价理论的演进，构成了现代金融工程学发展的核心逻辑主线。自 Black、Scholes 与 Merton 提出经典的 BSM 定价模型以来，全球金融衍生品市场经历了从单一品种向多层次、高复杂度体系的深刻演变。在国际成熟市场中，期权已不仅是基础的风险对冲工具，更演化为资产价格发现、尾部风险管理与策略博弈的核心载体。近年来，随着全球宏观经济不确定性加剧、地缘政治冲突频发，机构投资者对极端尾部风险的防范需求日益迫切，推动全球期权成交量持续创下历史新高。特别是以零日期权（0DTE）为代表的超短期品种在美股市场的大规模流行，不仅深刻改变了市场流动性的微观结构，更对定价模型在极短时间尺度下的瞬时响应能力、对极端波动率偏斜（Skew）的

精准捕捉能力提出了更为严苛的理论挑战，传统常数波动率模型与线性随机波动率模型的局限性愈发凸显。

在随机波动率模型的早期探索中，Heston 模型（亦称 1/2 模型）凭借其对波动率均值回归特性的优雅刻画，以及标的对数价格特征函数的封闭式表达，成为全球衍生品定价的标准基准。随后，针对利率与外汇市场的复杂波动动态，SABR 模型通过引入随机波动率项与波动率-标的价格的相关结构，进一步强化了对隐含波动率曲线形态的捕捉能力，成为场外衍生品定价的主流模型。然而，传统的 1/2 模型在处理极短期期权的隐含波动率曲线陡峭、以及极端行情下波动率的爆发式跳变时，其线性的扩散项结构显露出生硬的理论局限性——线性漂移项难以解释波动率在极端危机时刻的非线性加速均值回归特征，也无法充分刻画高波动市场中波动率的厚尾分布特性。

在此背景下，3/2 随机波动率模型逐渐进入学术与业界的核心研究视野。与 Heston 模型不同，3/2 模型假设瞬时方差过程遵循非线性的幂律扩散过程，其扩散项与瞬时方差的 3/2 次方成正比，漂移项则呈现出非线性的均值回归结构。这种非线性设定赋予了模型极强的灵活性，使其能够更灵敏地反映市场在极端压力下的波动聚集效应，更精准地拟合不同期限、不同行权价下的波动率微笑形态，从而在股票、外汇、加密货币等跨资产类别的定价实践中展现出显著优于传统仿射模型的拟合精度。

伴随着定价模型维度的增加与结构的复杂化，数值计算方法的演进同步推动着金融工程实践的边界拓展。早期的有限差分法（Finite Difference Method, FDM）通过离散化偏微分方程（PDE），为美式期权等路径依赖型衍生品提供了稳健的定价路径，但在多因素随机波动率模型下往往面临“维数灾难”的挑战，计算复杂度随模型维度呈指数级上升。蒙特卡洛模拟（Monte Carlo Simulation）方法以其强大的通用性，成为处理高维模型、路径依赖型奇异期权的主流工具，但其收敛速度较慢，计算成本随精度要求的提升而显著上升。为兼顾定价效率与精度，条件蒙特卡洛（Conditional Monte Carlo）通过引入方差缩减技术，将标的价格过程与波动率过程解耦，极大地优化了随机波动率模型的模拟效果与收敛速度。近年来，基于特征函数的傅里叶变换定价方法（Fourier Transform Method）结合快速傅里叶变换（FFT）、傅里叶余弦（COS）等数值算法，实现了复对数价格的高速解析求解，为 3/2 模型等非仿射随机波动率模型的实时定价、全曲面校准奠定了坚实的技术基础。

1.1.3 研究问题

尽管 3/2 模型在刻画市场极端波动、拟合波动率曲面方面具备显著的理论优势，但其非仿射的数学结构也为期权定价带来了严峻的数值实现挑战。现有研究针对 3/2 模型提出了多类期权定价方法，涵盖时间步进类蒙特卡洛模拟、终端精确/近似精确模拟、

傅里叶时域定价三大范式，累计形成九类主流实施方案，但现有研究与实践中仍存在三大核心待解决问题：

第一，现有研究对 3/2 模型下各类定价方法的性能评价，多局限于平稳市场环境下的理想化学术基准算例，缺乏对方法全场景数值稳定性的系统性检验。多数标称“精确”的定价方法，仅能在文献给定的基准参数下完成理论验证，在高波动率之波动率、强价格-波动率相关、极短到期期限等极端参数组合下，普遍存在特殊函数数值溢出、分布拟合失效、计算结果非数值化（nan/inf）等问题，方法的工程适用边界尚未得到清晰界定。

第二，真实极端市场环境下的定价方法实证再检验不足，理想化算例中的理论精度不等于真实市场中的实践性能。加密货币等极端波动市场的真实参数，往往超出学术基准算例的参数范围，极端行情下模型参数的结构性突变，会进一步放大不同定价方法的性能分化。现有研究尚未厘清不同方法在真实市场极端冲击下的定价精度、计算效率与鲁棒性，无法为极端行情下的衍生品定价与风险管理提供可靠的方法选择依据。

第三，3/2 模型在高波动市场中的实证适配性研究仍存在不足，缺乏极端事件全周期的模型参数动态演化分析。现有研究聚焦于 3/2 模型的静态参数校准与定价公式推导，尚未针对极端崩盘事件构建“长期固定参数估计-时变参数横截面校准”的完整分析框架，也未验证纯扩散框架下 3/2 模型对极端行情下波动率微笑形态演化的捕捉能力，模型在高波动市场中的实践价值尚未得到充分的实证支撑。

基于此，本文的核心研究问题聚焦于三个层面：其一，系统厘清 3/2 模型下九类主流期权定价方法的理论逻辑、误差来源与收敛特性，通过多场景数值实验量化不同方法的定价精度、计算效率与全场景数值稳定性，形成清晰的方法性能分层体系；其二，基于真实极端市场事件校准的模型参数，完成定价方法的极端场景再检验，对比理想化算例与真实市场中方法性能的差异，明确不同方法的工程适用边界；其三，验证 3/2 模型在加密货币极端行情下的市场适配性，识别极端事件全周期模型核心参数的时变规律，为高波动市场环境下的衍生品定价与风险管理提供理论支撑与实证依据。

1.1.4　研究目的与主要内容

针对上述核心研究问题，本文以 3/2 随机波动率模型下的欧式期权定价为核心研究对象，通过理论构建、数值实验与实证分析三个层面的系统研究，全面评估不同定价方法的全场景性能，厘清其适用边界，同时验证 3/2 模型在极端市场环境下的实证适配性。本文的核心研究目的与主要研究内容分为以下三个方面：

第一，构建 3/2 模型下欧式期权定价的完整方法体系，完成各类方法的理论推导与

数值实现。本文将九类主流定价方法划分为时间步进类条件蒙特卡洛方法、终端精确与近似精确模拟方法、傅里叶频域定价方法三大范式，系统推导每类方法的核心理论逻辑、数值实现流程、误差特性与收敛性质，解决现有研究中方法体系分散、理论边界不清晰的问题，为后续的数值对比与实证检验构建统一的理论框架与可复现的实现方案。

第二，通过多场景数值实验，完成不同定价方法的系统性性能评价。本文设计 7 组覆盖常规市场与极端参数环境的测试案例，以数值稳定性、定价精度、计算效率为核心评价指标，量化对比九类方法在不同参数组合下的性能表现，明确各类方法的优势场景与核心缺陷，形成清晰的方法性能分层体系，解决现有研究中方法性能评价片面、极端场景鲁棒性检验不足的问题，为不同应用场景下的方法选择提供量化依据。

第三，基于真实极端市场事件，完成 3/2 模型的实证校准与定价方法的极端场景再检验。本文以 2025 年 10 月比特币市场闪崩极端事件为研究样本，构建“长期固定参数时间序列估计 - 时变参数横截面校准”的两步实证框架，识别崩盘事件全周期 3/2 模型核心参数的时变规律，验证模型对极端行情下波动率微笑演化的拟合能力；同时基于校准得到的真实市场参数，构建极端场景测试案例，完成定价方法的性能再检验，并与学术基准算例结果进行横向对比，厘清理想化环境与真实市场中方法性能的核心差异，最终形成极端市场环境下的方法选择指引，弥补现有研究中理论算例与真实市场脱节的不足。

1.1.5 论文结构安排

本文共设置七个章节，遵循“理论基础-方法构建-数值实验-实证检验-结论展望”的逻辑主线展开，各章节的核心内容与结构安排如下：

第一章为引论。本章系统阐述本文的研究背景与研究意义，梳理全球及国内期权定价理论的发展脉络，明确本文针对 3/2 随机波动率模型的核心研究问题、研究目的与主要研究内容，并介绍全文的章节结构安排。

第二章为文献综述。本章系统梳理随机波动率模型的理论演进脉络，精确模拟定价方法的研究现状，总结 3/2 模型在金融衍生品市场中的应用研究与国内相关研究现状，最终通过文献述评明确现有研究的不足，确立本文的学术定位与研究边际贡献。

第三章为 3/2 随机波动率模型设定。本章详细介绍 3/2 随机波动率模型的核心理论设定，推导风险中性测度下标的资产价格与瞬时方差过程的随机微分方程，推导模型积分方差的条件分布性质与标的价格的矩母函数表达，厘清模型核心参数的经济含义，为后续定价方法的构建与实证校准奠定理论基础。

第四章为期权定价方法与算法实现。本章在 3/2 模型的理论框架下，系统构建三大

范式九类期权定价方法的完整实现体系，分别推导时间步进类条件蒙特卡洛模拟、终端精确与近似精确模拟方法、傅里叶时域定价方法的核心理论公式，给出每类方法的数值实现流程，并对其误差特性、收敛性质与适用场景进行理论分析，为后续的数值实验提供统一的方法基础。

第五章为数值实验设定与结果对比分析。本章设定统一的数值实验环境与评价指标体系，设计 7 组覆盖常规市场与极端参数环境的测试案例，系统完成九类定价方法的数值测试，量化对比不同方法的数值稳定性、定价精度与计算效率，最终形成方法性能的分层体系与场景化选择建议。

第六章为 3/2 随机波动率模型在加密货币极端行情中的实证校准与定价检验。本章以 2025 年 10 月比特币市场闪崩等极端事件为研究样本，构建“长期固定参数时间序列估计-时变参数横截面校准-定价方法极端场景再检验”的完整实证框架，识别崩盘事件全周期模型核心参数的时变规律与模型拟合效果，验证模型对极端行情下波动率微笑演化的拟合能力，并与第五章的学术基准算例结果进行横向对比，明确极端市场环境下不同定价方法的适用边界。

第七章为结论与展望。本章系统总结全文的核心研究结论，客观分析本研究存在的局限性，并从模型扩展、方法优化、实证范围拓展、风险管理应用等维度，对未来相关研究方向进行展望。

第二章　文献综述

在随机波动率模型的发展历程中，3/2 模型凭借其独特的非线性扩散结构与对极端市场波动的优异捕捉能力，逐渐成为金融工程领域期权定价与风险管理方向的研究热点。本章将遵循“模型范式演进 - 理论框架发展 - 定价方法迭代 - 市场应用拓展 - 国内研究现状”的逻辑主线，系统梳理 3/2 模型相关领域的国内外研究成果，最终通过文献述评明确现有研究的核心缺口，确立本研究的学术定位与边际贡献。

2.1　随机波动率模型的演进

期权定价理论的演进过程，本质上是市场参与者对金融资产收益率“非正态性”与波动率“时变风险”认知不断深化的过程。自 Black-Scholes-Merton（BSM）模型Black et al. (1973), Merton (1973) 确立了无套利定价的基本范式以来，常数波动率与标的价格服从几何布朗运动的核心假设，为衍生品定价提供了简洁且高效的解析框架，奠定了现代金融工程的基石。然而，随着全球金融市场复杂性的持续提升，BSM 模型的核心假设与真实市场特征产生了系统性背离。大量实证研究表明，全球各类资产的收益率分布普遍存在显著的“尖峰厚尾”特征，期权市场隐含波动率层面则呈现出随行权价格变化的“波动率微笑”与随到期期限变化的“波动率偏斜”现象Rubinstein (1985)，这一现象在市场危机与极端行情中表现得尤为突出，揭示了市场对尾部风险溢价的非线性定价特征，也推动了波动率模型从确定性向随机性的范式转换。

为修正 BSM 模型的理论缺陷，学术界首先发展出局部波动率模型。Derman et al. (1994) 与Dupire (1994) 分别从离散图与连续偏微分方程视角，构建了依赖于标的价格与时间的确定性局部波动率框架，能够完美拟合某一时刻的隐含波动率曲面。但局部波动率模型本质上仍属于确定性波动率范畴，无法刻画波动率自身的随机波动特征，对未来波动率曲面的预测能力较弱，难以适配动态对冲与长期衍生品定价的实践需求。

在此背景下，随机波动率（Stochastic Volatility, SV）模型应运而生，其核心思想是将波动率本身设定为一个随机扩散过程，从而同时刻画标的价格的随机波动与波动率自身的时变特征。以Heston (1997) 为代表的仿射随机波动率模型（Affine SV Models），通过引入平方根扩散过程（也称 1/2 模型）刻画波动率的均值回归属性，推导标的对数价格特征函数的仿射闭式解，极大地降低了模型校准与期权定价的计算复杂度，成为全球衍生品行业的定价标准基石。随后，针对利率与外汇市场的复杂波动动态，Hagan et al. (2002) 提出的 SABR 模型通过引入随机波动率项与波动率 - 标的价格的相关结构，进一步强化了对隐含波动率曲线形态的捕捉能力，成为场外利率衍生品定价的主流模

型。

然而，随着市场微观结构的演变与极端行情的频繁出现，传统 1/2 模型的内生局限性逐渐凸显。学者们发现，Heston 模型在拟合极超短期期限权的隐含波动率陡峭时存在理论不足，其线性漂移项难以解释波动率在极端危机时刻的爆发式增长与非线性均值回归特征(Jones, 2003)。这种“建模精度”与“现实动态”之间的张力，驱动了学术界从仿射框架向非仿射框架的跨越。3/2 模型作为非仿射随机波动率模型的典型代表，其扩散项与瞬时方差的 3/2 次方成正比，漂移项呈现出非线性的均值回归结构，这种更强的非线性特征赋予了模型更灵活的峰度刻画能力与尾部风险捕捉能力，使其在处理资产价格剧烈跳变与波动率高度聚集的场景时，展现出比传统 1/2 模型更深厚的理论底蕴与更优的实证表现。

2.2 3/2 模型及其扩展框架的发展

自 Lewis (2000) 对 3/2 随机波动率模型的基础解析性质进行系统性推导以来，学术研究围绕 3/2 模型的理论框架优化、解析性质拓展与数值实现方案开展了大量研究，形成了较为完备的理论成果体系。

在模型基础解析框架的构建方面，Carr et al. (2007) 开发了新颖的定价框架，通过直接对方差互换率进行建模，绕过了对瞬时波动率风险价格建模的复杂过程，为 3/2 模型的风险中性定价提供了全新的实现路径。针对 3/2 模型作为非仿射模型带来的期权定价与校准难题，Medvedev et al. (2007) 提出了基于小期限渐近展开的综合分析框架，推导出非仿射局部波动率模型下隐含波动率的闭式展开式，为包括 3/2 模型在内的非仿射模型的快速截面校准提供了核心理论工具，也成为后续 3/2 模型实证研究的重要校准方法基础。

在模型结构的扩展与优化方面，为实现股票现货与 VIX 衍生品的一致性建模，Baldeaux et al. (2012b) 在纯扩散 3/2 模型的基础上引入同步跳跃项，构建了“带跳跃的 3/2 模型”（3/2 plus jumps model），进一步提升了模型对极端市场极端波动的刻画能力。Grasselli (2017) 在其开创性研究中提出了“4/2 随机波动率模型”，将瞬时方差的扩散项设定为 1/2 项与 3/2 项的线性组合，从而将 Heston 模型与 3/2 模型有机地整合在同一框架内，形成了通用的随机波动率模型框架，有效结合了两类模型的优势，极大地拓展了 3/2 模型的理论应用边界。针对 3/2 模型的非仿射特性带来的模拟难题，Kouarfate et al. (2021) 基于 3/2 模型方差过程的显式解式，开发了一套全新的模拟框架，该框架巧妙利用了 3/2 模型的方差过程是 CIR 过程倒数的核心数学性质，为 3/2 模型的蒙特卡洛模拟提供了新的实现思路。

2.3　精确模拟方法的演进

在随机波动率模型的定价实践中，蒙特卡洛模拟是处理路径依赖型衍生品、高维模型定价的核心工具，而模拟方法的核心发展方向，是在消除时间离散化误差的同时，兼顾计算效率与全场景数值稳定性。自精确模拟方法被提出以来，学术界围绕仿射模型与非仿射 3/2 模型的精确模拟方案开展了持续的优化研究。

在仿射随机波动率模型的精确模拟奠基性研究中，Broadie et al. (2006) 首次提出了 Heston 随机波动率模型及其他仿射跳跃扩散过程下，标的股票价格与方差过程的无偏精确模拟方法，被称为精确蒙特卡洛（Exact MC）格式。该方法通过对积分方差的条件分布进行抽样，彻底消除了时间离散化带来的系统误差，为随机波动率模型的精确模拟奠定了理论基础。随后，Glasserman et al. (2011) 对 Broadie-Kaya 算法进行了关键优化，通过利用 Pitman et al. (1982) 关于扩散过程积分级数的分解结论，将条件积分方差分解为无限个伽马随机变量的级数形式，避免了原算法中计算成本极高的数值拉普拉斯逆变换，显著提升了精确模拟的计算效率。

针对非仿射 3/2 模型的精确模拟，Baldeaux (2012a) 开展了开创性研究，首次探讨了 3/2 模型下股票价格过程的精确模拟方案。该研究利用 Craddock et al. (2009) 基于李对称分析得出的扩散过程转移密度结论，成功将 Broadie et al. (2006) 针对仿射过程的精确模拟算法改编并应用于 3/2 模型，推导出 3/2 模型下积分方差的条件拉普拉斯变换闭式解，为 3/2 模型的无偏模拟提供了首个理论可行的实现方案。

近年来，3/2 模型的精确模拟算法在数值稳定性与计算效率上取得了进一步突破。Choi et al. (2024) 提出了一种无需使用高维修正贝塞尔函数的新型精确模拟方案，其核心发现是当用于模拟终端方差的泊松变量为条件时，3/2 模型的条件积分方差可以得到显著简化，彻底解决了原 Baldeaux 算法中修正贝塞尔函数在极端参数下易出现数值溢出的核心痛点。此外，Brignone et al. (2026) 提出了一种利用条件傅里叶余弦（COS）方法的新型精确模拟框架，通过余弦级数展开构造积分方差的条件累积分布函数，实现了积分方差的高效高精度抽样，解决了自 Broadie et al. (2006) 以来精确模拟中长期存在的误差控制难、计算开销大的实施难题，进一步拓展了精确模拟方法的工程应用边界。

2.4　傅里叶变换方法的研究

傅里叶变换定价方法是随机波动率模型下欧式期权快速定价与全面校准的核心技术路径，其核心思想是将期权价格的风险中性期望转化为标的对数价格特征函数的复数域积分，通过数值积分算法实现快速求解，完全规避了蒙特卡洛模拟的统计误差与时间离散化误差。

在傅里叶变换定价方法的奠基性研究中，Carr et al. (1999) 提出了基于快速傅里叶变换（Fast Fourier Transform, FFT）的期权定价方法，通过引入衰减因子解决了期权支付函数的傅里叶变换不可积问题，推导出欧式期权价格的傅里叶逆变换闭式表达式，结合 FFT 算法实现了在 $O(N \log N)$ 时间复杂度内一次性计算全行权价区间的期权价格，成为随机波动率模型下期权定价的主流工程方法。针对 FFT 方法存在的栅栏效应、仅能计算等距行权价格的局限，Fang et al. (2009) 提出了基于傅里叶余弦级数展开的 COS 定价方法，通过将标的收益率的概率密度函数在有效支撑区间上展开为余弦级数，推导出期权价格的闭式级数求和形式，实现了指数级的收敛速度，同时可灵活计算任意行权价对应的期权价格，在非仿射随机波动率模型中展现出极强的适用性与计算效率优势。

对于 3/2 模型而言，由于其非仿射的模型结构，标的对数价格并不具备仿射模型中简单的指数仿射特征函数，其特征函数的解析表达涉及复杂的特殊函数，为傅里叶变换方法的应用带来了一定挑战。Lewis (2000) 关于复数域积分定价的早期研究，为 3/2 模型的解析性质探索提供了核心理论框架；Dufresne (2001) 的研究进一步指出，3/2 过程的积分分布与 Whittaker 函数及修正贝塞尔函数密切相关，完整推导了 3/2 模型下积分方差的分布性质，为特征函数的闭式求解奠定了数学基础。在实务应用中，FFT 定价法被广泛应用于 3/2 模型的参数校准与期权定价，但其在极短到期时间、极端波动率参数下仍面临数值稳定性不足、计算耗时过长的挑战Drimus (2011)。而 COS 方法凭借其指数级数收敛速度与强数值稳定性，在 3/2 模型的定价与校准中展现出显著的性能优势，成为近年来该领域的研究重点。

2.5 加密货币期权市场中随机波动率模型的研究

随着以比特币（BTC）和以太坊（ETH）为代表的加密资产进入机构化交易时代，加密货币期权市场规模呈现指数级增长，其独特的市场动力学特征也成为金融工程领域的前沿研究方向。不同于传统权益、外汇市场，加密货币市场具有 7×24 小时不间断交易、无涨跌幅限制、内生波动率极高、极端行情频发的核心特征，其收益率分布呈现出显著的尖峰厚尾特征与极高的波动率的波动率（Vol-of-Vol, VoV），对传统期权定价模型提出了严峻挑战。

Hou et al. (2020) 的研究指出，加密资产的波动率展现出比传统股票资产更强的持久性和更高的波动率的波动率，市场极端尾部风险显著高于传统金融市场，这使得 BSM 模型甚至基础的 Heston 模型在加密期权定价中经常出现严重的估值偏差，尤其是在极端行情下，模型对波动率微笑的拟合能力大幅下降。针对这一现象，学术界主要从两个方向探索适配加密市场的模型框架：一类研究通过在随机波动率模型中引入跳跃扩

散过程（Jump-Diffusion），捕捉加密货币价格的瞬时极端跳变Scaillet et al. (2020)；另一类研究则侧重于改进波动率过程的非线性结构，探索非仿射随机波动率模型在加密市场的适配性。

近年来，由于 3/2 模型在刻画隐含波动率偏斜形态、捕捉非线性均值回归特征上的天然优势，其在加密货币市场的适配性研究显著增多。研究表明，加密资产在面临监管政策变动、宏观流动性冲击时，其波动率的均值回归表现出明显的非线性加速特征，这与 3/2 模型的数学性质高度契合（Fassas 等，2022）。此外，针对加密市场特有的 7×24 不间断交易机制，利用 3/2 模型结合高频数据进行实时波动率预测与期权动态对冲，已成为当前该领域的研究热点。但总体而言，现有关于 3/2 模型在加密市场的研究，多集中于静态参数校准与模型拟合效果对比，缺乏针对极端崩盘事件全周期的模型参数动态演化分析，也未对极端市场环境下不同定价方法的性能进行系统性实证检验，这一研究缺口为本文的实证分析提供了重要的研究空间。

2.6　国内研究现状

相较于国际学术界的丰富成果，国内关于 3/2 模型的研究尚处于起步阶段。目前国内主流文献多集中于 Heston 模型及其在 A 股市场的应用，或是 SABR 模型在利率衍生品中的表现。虽然部分学者开始关注非线性随机波动率模型，但针对 3/2 模型在高维校准、极端参数下的计算鲁棒性以及在 Crypto 市场等高波动环境下的适用性研究仍较为鲜见。这种研究现状表明，构建一个既能兼顾非线性刻画能力又能保证计算效率的 3/2 定价体系，具有显著的理论意义与应用价值。总体而言，现有国内研究仍存在三个核心不足：其一，针对 3/2 模型的理论系统性研究缺乏，对极端市场参数下的数值稳定性分析较为鲜见；其二，缺乏针对加密货币等高波动市场的实证研究，未对极端市场环境下不同定价方法的性能进行系统性检验；其三，现有研究未厘清 3/2 模型下不同定价方法的工程适用边界，无法为实务界提供可落地的方法指引。这种研究现状表明，构建一个既能兼顾非线性波动刻画能力，又能保证全局计算效率与数值稳定性的 3/2 模型定价体系，在国内具有显著的理论补充意义与实务应用价值。

2.7　文献述评与本文研究定位

综上所述，国际学术界围绕 3/2 随机波动率模型的理论构建、定价方法迭代与市场应用拓展已开展了丰富的研究，在模型解析性质、扩展框架构建、精确模拟方法优化、傅里叶定价方法改进等方面取得了一系列重要成果，为本文的研究奠定了坚实的理论基础。与此同时，国内研究多聚焦于 Heston 模型、SABR 模型等传统随机波动率

模型，针对 3/2 模型的系统性研究相对匮乏，在极端市场环境下的定价方法性能评价、高波动加密资产市场的实证适配性检验等方面仍存在显著的研究缺口。

尽管现有研究已较为丰富，但仍存在三方面核心不足。第一，现有 3/2 模型定价方法的性能评价多局限于文献基准参数下的理想算例，缺乏覆盖常规市场与极端参数组合的全场景数值稳定性检验，部分方法在极端波动率的波动率、强相关系数、极短到期期限等真实市场常见参数下会出现数值溢出、计算失效等问题，方法的工程适用边界尚未得到清晰界定。第二，现有研究尚未基于真实极端市场校准参数完成定价方法的实证再检验，理论算例中的性能表现无法完全等价于真实市场中的实践效果，不同定价方法在极端行情下的精度、效率与鲁棒性差异仍缺乏实证依据。第三，3/2 模型在高波动加密货币市场的实证研究仍不完善，缺乏极端崩盘事件全周期下的模型参数动态演化分析与波动率微笑拟合效果检验，纯扩散 3/2 模型对极端市场的适配能力尚未得到充分验证。

基于现有研究的不足，本文明确了核心研究定位：以 3/2 随机波动率模型下的欧式期权定价为研究对象，系统构建三大范式九类主流定价方法的统一实现框架，通过多场景数值实验与极端市场实证检验，全面评价各类方法的数值稳定性、定价精度与计算效率，厘清其工程适用边界；同时以比特币极端崩盘事件为样本，验证 3/2 模型在极端市场中的适配性，识别模型参数的时变规律。本文的研究旨在弥补现有研究在极端场景性能检验、真实市场实证分析等方面的不足，为 3/2 模型在高波动市场中的定价实践提供可靠的方法选择与实证支撑。

第三章 3/2 波动率模型设定

3.1 3/2 模型介绍

3/2 模型作为一种高级的随机波动率框架，是平方根过程 (Square-Root Process，由 Cox-Ingersoll-Ross 普及，后被 Steven Heston 进一步发展) 的复杂演进形式。

在当前的金融业界，Heston 模型（即 1/2 模型）长期以来被视为行业标准。然而，在面对极端的市场压力时，Heston 模型往往无法准确捕捉市场中观察到的 “波动率的波动率” (volatility of volatility)，并且难以完美拟合短期波动率偏斜 (short-term volatility skew) 的陡峭程度。

为了克服这一局限性，研究人员在 20 世纪 90 年代末和 21 世纪初提出了一种全新的思路：将扩散项的指数从 1/2 修改为 3/2 。这种设定的改变允许模型产生更具爆发力的波动率动态以及非线性的均值回归速度。因此，3/2 模型允许波动率出现尖峰，并在市场压力下呈现出更陡峭的偏斜，这与真实的实证数据高度吻合。然而，需要指出的是，在某些正常的市场条件下，3/2 模型可能会缺乏 Heston 模型所具备的稳定性。

3.2 随机微分方程 (SDE) 设定

在风险中性测度下，3/2 模型的资产价格过程和瞬时方差/波动率过程可以由以下随机微分方程组来刻画：

标的资产价格过程 S_t 的动态变化满足：

$$dS_t = rS_t dt + S_t\sqrt{V_t}(\rho dW_t^1 + \sqrt{1-\rho^2}dW_t^2) \tag{3.1}$$

其中，r 代表无风险利率，W_t^1 和 W_t^2 是两个相互独立的标准布朗运动，ρ 是两个布朗运动之间的相关系数。同时，方差过程 V_t 的动态演进满足：

$$dV_t = \kappa V_t(\theta - V_t)dt + \epsilon V_t^{3/2} dW_t^1 \tag{3.2}$$

在上述方程中：

- κ 控制着均值回归的速度。
- θ 代表长期的平均方差水平。
- ϵ 为波动率的波动率参数，控制着方差过程的波动程度。
- W_t^1 和 W_t^2 是两个独立的标准布朗运动，分别驱动着方差过程和价格过程中的随机性。

值得注意的是，该方差过程也可以等价地表示为一种关于 dV_t/V_t 的形式，从而更直观地体现其相对变化的漂移与扩散特征。

3.3 积分方差与条件分布

在 3/2 模型的期权定价框架中，时间区间 $[0,T]$ 内的积分方差 (Integrated variance) $\mathcal{V}_T$ 扮演着至关重要的角色。其定义如下：

$$\mathcal{V}_T = \int_0^T V_t dt \tag{3.3}$$

由于资产价格的随机性部分由波动率驱动，当我们以积分方差 $\mathcal{V}_T$ 为条件时，标的资产的终端价格 S_T 服从对数正态分布 (lognormal distribution) 。这一重要性质意味着，在给定 $\mathcal{V}_T$ 的条件下，我们可以借用 Black-Scholes (BS) 模型的定价框架。

根据伊藤等距同构 (Itô’ s isometry) 原理：

$$\int_0^T \rho^* \sigma_t dX_t \sim \rho^* \sqrt{\mathcal{V}_T} X_1 \sim N(0, (\rho^*)^2 \mathcal{V}_T)$$

据此，在时间 0 到 T 之间的有效波动率 (effective volatility) σ 可以表示为：

$$\sigma = \rho^* \sqrt{\mathcal{V}_T / T}$$

经过进一步的随机微积分推导，终端价格与初始价格的对数收益率公式可展开为：

$$\log\left(\frac{S_T}{S_0}\right) = \frac{\rho}{\epsilon}\log\left(\frac{V_T}{V_0}\right) + \left(\kappa + \frac{\epsilon^2}{2}\right)\mathcal{V}_T - \kappa\theta T - \frac{1}{2}\mathcal{V}_T + \rho^*\sqrt{\mathcal{V}_T}X_1 \tag{3.4}$$

该公式是后续精确模拟方法进行定价的核心基础。

3.4 3/2 模型矩母函数

在风险中性测度 $\mathbb{Q}$ 下，3/2 模型的随机微分方程（SDE）由公式(3.1)和公式(3.2)定义。

为了利用傅里叶变换方法进行期权定价，核心在于求解对数资产价格 $x_T = \ln S_T$ 的矩母函数（MGF）。

根据 Lewis (2000) 与 Carr 和 Sun (2007) 的推导，定义 $\phi(u,\tau) = \mathbb{E}[e^{ux_T} | \mathcal{F}_t]$ 为对数价格的矩母函数。在 3/2 模型框架下，由于其方差过程 V_t 的倒数遵循 CIR 过程（平方根过程），该 MGF 并不具备仿射模型中常见的指数仿射形式，而是表现为包含库默尔合流超几何函数（Kummer’s Confluent Hypergeometric Function, ${}_1F_1$）的解析形式。

具体而言，给定初始状态 $V_0 = \sigma_0^2$，定义辅助变量如下：参数变换项：

$$\mu = \frac{1}{2} + \frac{\kappa - u\rho\epsilon}{\epsilon^2}, \quad \tilde{c} = \frac{u(1-u)}{\epsilon^2}$$

特征参数：

$$\delta = \sqrt{\mu^2 + \tilde{c}}, \quad \alpha = -\mu + \delta, \quad \beta = 1 + 2\delta$$

时间相关项：定义

$$X = \frac{2\kappa\theta}{\epsilon^2\sigma^2(e^{\kappa\theta\tau} - 1)}$$

该项刻画了均值回归强度与期限 τ 对波动率结构的非线性影响。综上，3/2 模型下对数价格的矩母函数可表示为：

$$M(u,\tau) = \frac{\Gamma(\beta - \alpha)}{\Gamma(\beta)} X^{\alpha}\ {}_1F_1(\alpha, \beta, -X)$$

在数值计算过程中，${}_1F_1(\alpha, \beta, -X)$ 的计算往往涉及复数域的特殊函数处理。代码实现中，为了防止在大参数场景下（如极端波动率或长到期期限）出现数值溢出，采用了对数伽马函数（Log-Gamma）进行重构。对 MGF 前置因子：

$$M(u,\tau) = \frac{\Gamma(\beta - \alpha)}{\Gamma(\beta)}\, X^{\alpha}\ {}_1F_1(\alpha, \beta, -X)$$

使用

$$\ln P = \ln\Gamma(\beta - \alpha) - \ln\Gamma(\beta) + \alpha \ln X, \qquad P = \exp(\ln P),$$

然后

$$M = P \cdot {}_1F_1(\alpha, \beta, -X)$$

来替代直接的乘法/幂运算。这种处理方式可以保证运算的数值稳定性。

第四章　期权定价方法与算法实现

本章在 3/2 随机波动率模型的统一框架下，系统构建欧式期权定价的九类方法体系，所有方法均以无套利风险中性定价原理为理论基准，按照时间离散近似、终端精确 / 近似精确模拟、傅里叶频域积分三大范式展开完整的理论推导、收敛性分析与误差特性刻画，为后续的数值对比与实证检验奠定严格的理论基础。

在完备市场的风险中性测度 $\mathbb{Q}$ 下，3/2 随机波动率模型的标的资产价格与方差过程满足由式(3.1)和(3.2)给出。

对于到期时间为 T、执行价格为 K 的欧式看涨期权，其无套利价格满足风险中性定价公式：

$$C(S_0, v_0; K, T) = e^{-rT}\mathbb{E}^{\mathbb{Q}}\Big[\max(S_T - K, 0)\ \Big|\ S_0, v_0\Big], \tag{4.1}$$

其中 S_0 与 v_0 分别为标的资产初始价格与初始瞬时方差。不同于仿射结构的 Heston 模型，3/2 模型属于非仿射随机波动率模型，其欧式期权价格不存在闭形式解析解，因此必须通过数值方法逼近上述条件期望。本章所构建的九类定价方法，覆盖了从时域路径模拟到频域积分变换的完整技术谱系，分别从不同维度实现对期权价格的有效逼近。

4.1　时间步进类蒙特卡洛定价方法

时间步进类方法的核心思想是对连续时间的扩散过程进行离散化处理，通过逐期迭代生成方差过程的样本路径，通过给定方差路径的闭式解析解(3.4)生成标的资产的对数价格，从而规避标的价格过程的离散化误差，降低蒙特卡洛模拟的统计方差。本节依次推导 Euler 离散格式、Milstein 离散格式、精确步进方法、Poisson-Gamma 近似精确步进方法与 Quasi-Exponential 近似精确步进方法的理论构造，并对每类方法的收敛特性与误差来源进行分析。

4.1.1　Euler 与 Milstein 离散格式

4.1.1.1　方差过程的 Euler 离散格式

Euler 离散格式是随机微分方程数值求解中最基础的离散化方法，其核心是对扩散过程的增量进行一阶泰勒展开近似。对于一般的伊藤扩散过程

$$dX_t = \mu(X_t)dt + \sigma(X_t)dW_t,$$

Euler 格式的离散形式为

$$X_{t+\Delta t} \approx X_t + \mu(_t)\Delta t + \sigma(X_t)\Delta W_t,$$

其中 Δt 为离散时间步长，ΔW_t 为服从均值为 0、方差为 Δt 的正态分布 $N(0,\Delta t)$ 的布朗运动增量。将该格式应用于 3/2 模型的方差过程，其漂移项为 $\mu(v_t) = \kappa(\theta - v_t)v_t$，扩散项为 $\sigma(v_t) = \epsilon v_t^{3/2}$，由此可得到方差过程的 Euler 离散迭代式。具体的离散化方案如下：

将到期区间 $[0,T]$ 等分为 M 个时间步，时间步长 $\Delta t = T/M$，时间网格为 $\{t_0 = 0, t_1, \ldots, t_M = T\}$。对 3/2 模型的方差过程，在每个时间步上的 Euler 迭代式为：

$$v_{t_{n+1}} = v_{t_n} + \kappa(\theta - v_{t_n})v_{t_n}\Delta t + \epsilon v_{t_n}^{3/2}\Delta W_n^1, \tag{4.2}$$

其中 $\Delta W_n^1 \sim N(0,\Delta t)$ 为第 n 个时间步上的布朗运动增量，各时间步的增量相互独立。

在完成所有时间步的迭代后，通过梯形数值积分方法计算积分方差 V_T，以提升积分精度：

$$I_T = \int_0^T v_t dt \approx \sum_{n=0}^{M-1} \frac{v_{t_n} + v_{t_{n+1}}}{2} \cdot \Delta t. \tag{4.3}$$

将迭代得到的到期方差 $v_T = v_{t_M}$ 与数值积分得到的 V_T 代入式 (3.4)，即可生成该条路径对应的对数到期价格，进而计算期权到期收益。

4.1.1.2 方差过程的 Milstein 离散格式

Milstein 离散格式是对 Euler 格式的二阶修正，通过引入扩散项的一阶导数项，消除扩散项状态依赖带来的一阶离散偏差，显著提升方差过程的模拟精度。

对于 3/2 模型的方差过程，其扩散项为 $\sigma_v(v_t) = \epsilon v_t^{3/2}$，对 v_t 求一阶导数可得

$$\sigma_v'(v_t) = \frac{\partial \sigma_v(v_t)}{\partial v_t} = \frac{3}{2}\epsilon v_t^{1/2}. \tag{4.4}$$

代入 Milstein 格式的通用形式后，可得到方差过程的 Milstein 离散迭代式：

$$v_{t_{n+1}} = v_{t_n} + \kappa(\theta - v_{t_n})v_{t_n}\Delta t + \epsilon v_{t_n}^{3/2}\Delta W_n^1 + \frac{1}{2}\sigma_v(v_{t_n})\sigma_v'(v_{t_n})\left((\Delta W_n^1)^2 - \Delta t\right). \tag{4.5}$$

将扩散项及其导数代入后，最终的 Milstein 迭代式为：

$$v_{t_{n+1}} = v_{t_n} + \kappa(\theta - v_{t_n})v_{t_n}\Delta t + \epsilon v_{t_n}^{3/2}\Delta W_n^1 + \frac{3}{4}\epsilon^2 v_{t_n}^2\left((\Delta W_n^1)^2 - \Delta t\right). \tag{4.6}$$

与 Euler 格式一致，完成方差路径迭代后，通过式 (4.3) 计算积分方差 V_t，再代入

式 (3.4) 生成对数到期价格。

4.1.1.3　误差特性与收敛性分析

本节基于条件蒙特卡洛框架的 Euler/Milstein 格式，误差来源可分解为三个独立部分，各部分的收敛特性如下：

1. 方差过程的时间离散误差：Euler 格式对扩散过程的强收敛阶为 1/2，弱收敛阶为 1；Milstein 格式通过二阶修正将强收敛阶提升至 1，弱收敛阶仍为 1。两类格式的离散偏差均随时间步长 Δt 的减小而单调衰减，在相同步长下，Milstein 格式的离散偏差显著低于 Euler 格式。
2. 积分方差的数值积分误差：采用梯形法计算积分方差，其收敛阶为 2，远高于方差过程的离散收敛阶，因此在实际计算中，积分误差可忽略不计，不会成为整体误差的主导项。
3. 蒙特卡洛模拟的统计误差：由于采用了条件蒙特卡洛方法，通过解析映射消除了价格过程的离散噪声，模拟的统计方差显著低于全离散的标准蒙特卡洛方法，统计误差随模拟路径数 N 的增加以 $O(N^{-1/2})$ 的速度衰减。

4.1.2　精确步进（Exact Stepping）方法

精确步进方法是时间步进类条件蒙特卡洛方法中精度最高的方案，其核心是利用 3/2 模型方差过程的特殊数学结构，通过变量变换将其映射为具有已知转移分布的仿射过程，从而实现方差过程的无离散偏差抽样，完全消除时间离散带来的系统误差。

4.1.2.1　方差过程的倒数变换与精确分布

对 3/2 模型的方差过程做倒数变换，令 $x_t = 1/v_t$，通过伊藤引理可推导 x_t 的动态过程。首先计算 x_t 关于 v_t 的一阶与二阶导数:

$$\frac{\partial x_t}{\partial v_t} = -\frac{1}{v_t^2}, \quad \frac{\partial^2 x_t}{\partial v_t^2} = \frac{2}{v_t^3}. \tag{4.7}$$

代入伊藤引理公式 $dx_t = \frac{\partial x_t}{\partial v_t} dv_t + \frac{1}{2}\frac{\partial^2 x_t}{\partial v_t^2}(dv_t)^2$，并将 3/2 模型的方差 SDE $dv_t = \kappa v_t(\theta - v_t)dt + \epsilon v_t^{3/2} dW_t^1$ 代入，其中二次变差项 $(dv_t)^2 = \epsilon^2 v_t^3 dt$。整理后可得：

$$dx_t = \left(\kappa - \kappa\theta x_t\right) dt - \epsilon\sqrt{x_t} dW_t^1. \tag{4.8}$$

式 (4.8) 表明，变换后的过程 x_t 是一个经典的 CIR（Cox-Ingersoll-Ross）平方根扩散过程，其漂移项为仿射结构，扩散项为平方根结构。CIR 过程的一个核心性质是其转移概率分布具有明确的解析形式，服从非中心卡方分布（Non-central Chi-squared Distribution）。

具体而言，给定初始时刻 t_n 的 y_{t_n}，经过时间步长 Δt 后，t_{n+1} 时刻 $x_{t_{n+1}}$ 的条件分布为：

$$x_{t_{n+1}} \mid x_{t_n} \sim \frac{1}{c} \cdot \chi'^2(d, \lambda), \tag{4.9}$$

其中 $\chi'^2(d, \lambda)$ 表示自由度为 d、非中心参数为 λ 的非中心卡方分布随机变量，分布参数由 3/2 模型参数与时间步长确定：

$$d = \frac{4\kappa\theta}{\epsilon^2}, \quad c = \frac{\epsilon^2\left(1 - e^{-\kappa\Delta t}\right)}{4\kappa}, \quad \lambda = c \cdot y_{t_n} e^{-\kappa\Delta t}. \tag{4.10}$$

通过逆变换 $v_{t_{n+1}} = 1/x_{t_{n+1}}$，即可得到 3/2 模型方差过程的精确样本，该抽样过程不引入任何时间离散偏差。

4.1.2.2 条件蒙特卡洛框架下的实现

在条件蒙特卡洛框架下，精确步进方法的实现流程如下：

将到期区间 $[0, T]$ 等分为 M 个时间步，时间步长 $\Delta t = T/M$，时间网格为 $\{t_0 = 0, t_1, \ldots, t_M = T\}$。在每个时间步上：

1. 给定当前方差 v_{t_n}，计算 $y_{t_n} = 1/v_{t_n}$；
2. 根据式 (4.9) 的分布参数，生成非中心卡方随机变量 $\chi'^2(d, \lambda)$；
3. 计算 $x_{t_{n+1}} = \chi'^2(d, \lambda)/c$；
4. 通过逆变换得到 $v_{t_{n+1}} = 1/x_{t_{n+1}}$。

在完成所有时间步的迭代后，通过梯形数值积分方法计算积分方差 V_T：

$$V_T = \int_0^T v_t dt \approx \sum_{n=0}^{M-1} \frac{v_{t_n} + v_{t_{n+1}}}{2} \cdot \Delta t. \tag{4.11}$$

最后，将迭代得到的到期方差 $v_T = v_{t_M}$ 与数值积分得到的 V_T 代入核心闭式映射(3.4)

4.1.2.3 误差特性与收敛性分析

精确步进方法的误差来源仅包含两个部分，完全消除了方差过程的时间离散误差：

1. 积分方差的数值积分误差：采用梯形法计算积分方差，其收敛阶为 2，在实际应用中，即使使用较少的时间步数，该误差也可忽略不计，不会成为整体误差的主导项。若需进一步消除积分误差，可结合 CIR 过程的积分矩生成函数，通过解析方法计算 V_T 的条件分布。
2. 蒙特卡洛模拟的统计误差：由于采用了条件蒙特卡洛方法，且方差过程为精确抽样，模拟的统计方差处于最低水平，统计误差随模拟路径数 N 的增加以 $O(N^{-1/2})$ 的速度衰减。

在所有时间步进类方法中，精确步进方法具有最小的偏差。

4.1.3　Poisson-Gamma 近似精确步进方法

Poisson-Gamma 近似精确步进方法的核心，是利用 3/2 模型方差过程倒数变换后的 CIR 过程的经典统计性质——其转移分布可精确表示为 Poisson 分布加权的 Gamma 分布无穷混合级数，通过对该级数进行有限项截断，实现方差过程的高效近似精确抽样，在保持极低偏差的同时，避免了非中心卡方分布抽样的高数值计算成本。该方法完全基于倒数变换后的 CIR 过程展开，是 CIR 转移分布的等价数值实现形式。

4.1.3.1　倒数变换与 CIR 过程的 Poisson-Gamma 混合表示

与精确步进方法一致，我们首先对 3/2 模型的方差过程做倒数变换，令 $x_t = 1/v_t$，进而通过伊藤引理得到式(4.8)。根据非中心卡方分布的经典混合性质，自由度为 d、非中心参数为 λ 的非中心卡方随机变量 $\chi'^2(d,\lambda)$，可精确分解为 Poisson 分布与 Gamma 分布的无穷混合形式：

$$\chi'^2(d,\lambda) \mid N \sim \text{Gamma}\left(\frac{d}{2}+N, \frac{1}{2}\right), \quad N \sim \text{Poisson}\left(\frac{\lambda}{2}\right), \tag{4.12}$$

其中 N 为服从 Poisson 分布的潜变量，$Gamma(\alpha,\beta)$ 为形状参数 α、率参数 β 的 Gamma 分布，其概率密度函数为：

$$f_{\text{Gamma}}(x;\alpha,\beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0. \tag{4.13}$$

结合精确步进方法中 CIR 过程的转移分布结论，给定初始值 x_{t_n}，经过时间步长 Δt 后，$x_{t_{n+1}}$ 的条件分布为：

$$y_{t_{n+1}} \mid y_{t_n} \sim \frac{1}{c} \cdot \chi'^2(d,\lambda), \tag{4.14}$$

其中分布参数为：

$$d = \frac{4\kappa\theta}{\epsilon^2}, \quad c = \frac{\epsilon^2\left(1-e^{-\kappa\Delta t}\right)}{4\kappa}, \quad \lambda = c \cdot y_{t_n} e^{-\kappa\Delta t}. \tag{4.15}$$

将式 (4.12) 代入上式，可直接得到 $y_{t_{n+1}}$ 的 Poisson-Gamma 混合表示：

$$y_{t_{n+1}} \mid N \sim \frac{1}{c} \cdot \text{Gamma}\left(\frac{d}{2}+N, \frac{1}{2}\right), \quad N \sim \text{Poisson}\left(\frac{\lambda}{2}\right). \tag{4.16}$$

式 (4.16) 是该方法的核心：CIR 过程的转移分布本身就是 Poisson 加权的 Gamma 分布无穷混合，不存在任何分布拟合误差，仅需对 Poisson 分布的取值做有限项截断，即可实现近似精确抽样。

4.1.3.2 条件蒙特卡洛框架下的实现

在条件蒙特卡洛框架下，Poisson-Gamma 近似精确步进方法的实现流程与精确步进方法高度一致，仅将非中心卡方分布抽样替换为更高效的 Poisson-Gamma 混合抽样，具体流程如下：

将到期区间 $[0,T]$ 等分为 M 个时间步，时间步长 $\Delta t = T/M$，时间网格为 $\{t_0 = 0, t_1, \dots, t_M = T\}$。在每个时间步上：

1. 给定当前方差 v_{t_n}，计算 $x_{t_n} = 1/v_{t_n}$；
2. 预计算 CIR 转移分布的固定参数 d, c，以及非中心参数 $\lambda = c \cdot x_{t_n} e^{-\kappa\Delta t}$；
3. 生成 Poisson 潜变量 $N \sim \text{Poisson}\left(\frac{\lambda}{2}\right)$，实际计算中对 N 做上限截断（取 $N_{\max} = \lfloor \lambda/2 + 5\lambda/2 \rfloor$，保证截断概率小于 10^{-6}）；
4. 生成 Gamma 随机变量 $G \sim \text{Gamma}\left(\frac{d}{2} + N, \frac{1}{2}\right)$；
5. 计算 $x_{t_{n+1}} = G/c$，再通过逆变换得到下一时间步的方差 $v_{t_{n+1}} = 1/x_{t_{n+1}}$。

在完成所有时间步的迭代后，通过梯形数值积分方法计算积分方差 V_T，再将 v_T 与 V_T 代入式 (3.4)，即可生成对数到期价格。

4.1.3.3 误差特性与收敛性分析

Poisson-Gamma 近似精确步进方法的误差来源仅包含三个部分，且偏差水平远低于 Euler/Milstein 离散格式：

1. Poisson 分布的截断误差：该误差是方法唯一的固有偏差，由于 Poisson 分布的概率质量随 N 的增大呈指数衰减，截断误差随截断上限 $N_{\max}$ 的增大呈指数级下降。在常规参数下，取 $N_{\max}$ 为 Poisson 均值加 5 倍标准差，即可将截断误差控制在 10^{-6} 以下，可忽略不计。
2. 积分方差的数值积分误差：采用梯形法计算积分方差，收敛阶为 2，在实际应用中该误差远小于截断误差，不会成为整体误差的主导项。
3. 蒙特卡洛模拟的统计误差：由于方差过程的抽样偏差极小，且采用了条件蒙特卡洛框架消除了价格过程的离散噪声，统计误差随模拟路径数 N 的增加以 $O(N^{-1/2})$ 的速度衰减，与精确步进方法处于同一水平。

该方法的核心优势在于计算效率与精度的平衡：相较于精确步进方法的非中心卡方分布抽样，Poisson 与 Gamma 分布的随机数生成计算成本极低，速度提升显著；相较于 Euler/Milstein 格式，该方法完全消除了扩散过程的离散偏差，仅存在可忽略的截断误差，精度提升数个数量级。

4.1.4 Quasi-Exponential (QE) 近似精确步进方法

Quasi-Exponential（QE）近似精确步进方法最初由 Andersen (2007) 为 Heston 模型设计，现我们将其推广至 3/2 模型，其核心是通过匹配方差过程的条件一阶矩与二阶矩，构造一个指数型的简单近似分布，避免了复杂分布的抽样，在波动率较高、尾部较厚的极端场景下仍能保持良好的数值稳定性。

4.1.4.1 方差过程的矩匹配与 QE 分布构造

QE 方法的核心思想是"矩匹配"，无论真实转移分布多么复杂，仅需匹配其前两阶矩，即可构造一个足够精确的近似分布。对于 3/2 模型的方差过程，给定初始方差 v_t，取其倒数过程 $x_t = 1/v_t$，经过时间步长 Δt 后，其条件一阶矩 $\mathbb{E}[x_{t+\Delta t} \mid x_t]$ 与条件二阶矩 $\mathbb{E}[x_{t+\Delta t}^2 \mid x_t]$ 可通过求解方差过程的矩微分方程解析得到。记 $m_1 = \mathbb{E}[v_{t+\Delta t}|v_t]$，$m_2 = \mathbb{E}[v_{t+\Delta t}^2|v_t]$ 为条件二阶矩，计算条件方差 $\sigma^2 = m_2 - m_1^2$。QE 方法根据方差水平的高低，选择两种不同的近似分布：

1. 低方差场景：当条件方差相对较小时，采用移位对数正态分布（Shifted Log-Normal Distribution）作为近似；
2. 高方差场景：当条件方差较大时，采用指数分布与两点分布的混合（Exponential-Two-Point Mixture）作为近似。

通过矩匹配条件 $m_1 = \mathbb{E}[X]$ 与 $\sigma^2 = Var(X)$，可解析求解出 QE 分布的所有参数，进而实现高效抽样。

4.1.4.2 条件蒙特卡洛框架下的实现

在条件蒙特卡洛框架下，QE 近似精确步进方法的实现流程如下：将到期区间 $[0, T]$ 等分为 M 个时间步，时间步长 $\Delta t = T/M$，时间网格为 $\{t_0 = 0, t_1, \ldots, t_M = T\}$。在每个时间步上：

1. 给定当前方差倒数 x_{t_n}，通过矩方程解析计算条件均值 m_1 与条件方差 σ^2；
2. 根据预设的阈值判断当前场景，选择对应的 QE 分布形式；
3. 根据条件方差的高低，选择对应的近似分布类型，并通过矩匹配求解分布参数；
4. 从构造的 QE 分布中抽样，得到 $x_{t_{n+1}}$ 的近似样本，并取倒数得到 $v_{t_{n+1}}$。

在完成所有时间步的迭代后，通过梯形数值积分方法计算积分方差 V_T，再将 v_T 与 V_T 代入式 (3.4)，即可生成对数到期价格。

4.1.4.3 误差特性与收敛性分析

QE 近似精确步进方法的误差来源包含三个部分：

1. 矩匹配的近似偏差：该误差为方法的固有偏差，其弱收敛阶为 1，由于仅匹配前两阶矩，忽略了高阶矩的信息，在极端厚尾场景下可能存在微小偏差，但仍显著优于 Euler/Milstein 格式。
2. 分布选择的阈值误差：该误差可通过合理设置阈值（通常基于方差系数 σ/m_1）控制在极小水平。
3. 积分方差的数值积分误差与蒙特卡洛统计误差：与前述方法一致。

QE 方法的最大优势在于其极高的计算效率与极强的数值稳定性，但此处我们对其取倒数的操作，会导致数值稳定性受到一些影响。

4.2 精确与近似精确模拟定价方法

本节所述的终端模拟类方法，完全规避了时间步进类方法中逐期迭代的离散化过程，直接针对 [0,T] 到期区间内的方差过程终端值 v_T 与积分方差 $V_T = \int_0^T v_s ds$ 的联合分布进行抽样，再通过 4.1 节推导的对数价格闭式映射直接生成标的资产到期价格，彻底消除了时间步长带来的离散误差。所有方法仍严格遵循条件蒙特卡洛框架：先抽样方差过程的核心统计量，再通过条件正态分布生成标的价格，最大化降低模拟的统计方差。

4.2.1 精确模拟方法

Baldeaux (2012a) 精确模拟方法是 3/2 模型下首个无偏的终端精确模拟方法，其核心是利用 3/2 模型积分方差过程的倒数变换性质，结合修正贝塞尔函数与逆拉普拉斯变换，实现 v_T 与 V_T 的联合精确抽样，完全消除所有系统偏差。

4.2.1.1 理论推导

与时间步进类方法一致，首先对 3/2 模型的方差过程做倒数变换，令 $x_t = 1/v_t$，得到 CIR 平方根扩散过程(4.8)。该过程的终端值 $x_T = 1/v_T$ 的条件分布为非中心卡方分布，可实现精确抽样，具体形式与 4.2.2 节一致。

Baldeaux (2012a) 的核心贡献在于，推导了给定初始值 y_0 与终端值 y_T 的条件下，积分方差 $V_T = \int_0^T v_s ds$ 的条件拉普拉斯变换的闭式表达：

$$\mathscr{L}(\beta) = \mathbb{E}\left[e^{-\beta V_T} \mid y_0, y_T\right] = \frac{I_\nu\left(\sqrt{\frac{4\kappa\theta}{\epsilon^2} + \frac{8\beta}{\epsilon^2}} \cdot \frac{2\sqrt{y_0 y_T} e^{-\kappa T/2}}{1-e^{-\kappa T}}\right)}{I_\nu\left(\sqrt{\frac{4\kappa\theta}{\epsilon^2}} \cdot \frac{2\sqrt{y_0 y_T} e^{-\kappa T/2}}{1-e^{-\kappa T}}\right)}, \tag{4.17}$$

其中 $\nu = \frac{2\kappa\theta}{\epsilon^2} - 1$ 为修正贝塞尔函数的阶数，$I_\nu(\cdot)$ 为第一类 ν 阶修正贝塞尔函数。

基于该闭式拉普拉斯变换，可通过傅里叶逆变换得到 V_T 的条件概率密度函数，再通过逆变换抽样实现 V_T 的精确抽样。最终，通过式 (3.4) 生成标的资产到期价格的核心映射中，V_T 与 v_T 的联合抽样实现了对标的资产价格的无偏模拟。

4.2.1.2 条件蒙特卡洛框架下的实现

在条件蒙特卡洛框架下, Baldeaux 精确模拟方法的实现流程完全规避时间离散, 具体如下:

1. 终端方差 v_T 的精确抽样：根据 CIR 过程的非中心卡方转移分布，直接抽样得到到期时刻的 y_T，再通过求倒数得到 $v_T = 1/x_T$，该抽样无任何系统偏差；
2. 积分方差 V_T 的精确抽样：给定 $y_0 = 1/v_0$ 和 $y_T = 1/v_T$，基于式 (4.17) 的拉普拉斯变换，通过数值傅里叶逆变换得到 V_T 的条件累计分布函数，再通过均匀分布逆变换抽样得到 V_T 的精确样本；
3. 最后将 $v_T = 1/x_T$ 与 V_T 代入式 (3.4)，生成标的资产到期价格的样本。

4.2.1.3 误差特性与收敛性分析

Baldeaux 精确模拟方法是 3/2 模型下唯一的无偏终端模拟方法，其误差来源仅包含两个部分:

1. 拉普拉斯逆变换的数值截断误差：该误差随傅里叶余弦级数的截断项数增加呈指数衰减，在常规参数下，取 256 项截断即可将误差控制在 10^{-8} 以下，可忽略不计；
2. 蒙特卡洛模拟的统计误差：由于方差过程与积分方差均为精确抽样，且采用条件蒙特卡洛框架消除了价格过程的离散噪声，模拟的统计方差处于理论最低水平，统计误差随模拟路径数 N 的增加以 $O(N^{-1/2})$ 的速度衰减。

4.2.2 条件傅里叶余弦（Fourier-Cosine）模拟方法

条件傅里叶余弦模拟方法由 Brignone et al. (2026) 提出，其核心是基于标的资产对数价格的条件特征函数，通过傅里叶余弦级数展开构造积分方差 V_T 的条件累积分布函数，实现高效高精度的抽样，相较于相较于 Baldeaux 精确模拟方法，在保持相近精度的前提下大幅提升了计算效率。

4.2.2.1 理论推导

与 Baldeaux 方法一致，首先通过 CIR 过程的非中心卡方分布精确抽样得到终端方差 v_T，核心问题转化为给定 v_0 与 v_T 的条件下，如何高效抽样积分方差 V_T。

记 I_T 的条件特征函数为 $\varphi(u) = \mathbb{E}\left[e^{iuI_T} \mid v_0, v_T\right]$，该特征函数可通过式 (4.17) 的拉普拉斯变换解析延拓得到，具有明确的闭式表达。根据傅里叶余弦展开理论，I_T 的条件概率密度函数可在有效支撑区间 $[a, b]$ 上展开为余弦级数：

$$f_{I_T|v_0,v_T}(x) = \frac{2}{b-a}\sum_{n=0}^{\infty} \mathrm{Re}\left[\varphi\left(\frac{n\pi}{b-a}\right)e^{-i\frac{n\pi a}{b-a}}\right]\cos\left(\frac{n\pi(x-a)}{b-a}\right), \tag{4.18}$$

其中 $[a, b]$ 为 I_T 的有效支撑区间，通常取 $a = 0$，$b = \mathbb{E}[I_T]+10\sqrt{\mathrm{Var}(I_T)}$ 以覆盖 99.99% 以上的概率质量。

对密度函数积分，可得到 V_T 的条件累积分布函数的闭式级数展开：

$$F_{V_T|v_0,v_T}(x) = \int_a^x f(z)dz = \frac{2}{b-a}\sum_{n=0}^{\infty} \mathrm{Re}\left[\varphi\left(\frac{n\pi}{b-a}\right)e^{-i\frac{n\pi a}{b-a}}\right]\cdot \Psi_n(x), \tag{4.19}$$

其中 $\Psi_n(x)$ 为余弦项的积分闭式，可直接解析计算，无需数值积分：

$$\Psi_n(x) = \begin{cases} x-a, & n=0, \\ \dfrac{b-a}{n\pi}\sin\left(\dfrac{n\pi(x-a)}{b-a}\right), & n\geq 1. \end{cases} \tag{4.20}$$

4.2.2.2 条件蒙特卡洛框架下的实现

在条件蒙特卡洛框架下，条件傅里叶余弦模拟方法的实现流程如下：

1. 终端方差 v_T 的精确抽样：与 Baldeaux 方法一致，通过 CIR 过程的非中心卡方分布精确抽样得到 v_T；
2. 积分方差 V_T 的高效抽样：给定 v_0 与 v_T，计算 V_T 的条件特征函数，通过截断余弦级数构造条件累积分布函数，再通过均匀分布逆变换抽样得到 V_T 的样本；
3. 标的资产到期价格的生成：最后将 $v_T = 1/x_T$ 与 V_T 代入式 (3.4)，生成标的资产到期价格的样本。

4.2.2.3 误差特性与收敛性分析

该方法的误差来源主要包含三个部分：

1. 余弦级数的截断误差：该误差随截断项数的增加呈指数衰减，在常规参数下，仅需 128 项截断即可达到 10^{-6} 的精度，远快于普通多项式收敛；
2. 支撑区间的截断误差：通过合理设置 $[a, b]$ 区间，可将该误差控制在 10^{-8} 以下，可忽略不计；
3. 蒙特卡洛模拟的统计误差：由于方差过程的抽样无偏，且采用条件蒙特卡洛框架消除了价格过程的离散噪声，统计误差随模拟路径数 N 的增加以 $O(N^{-1/2})$

的速度衰减，与 Baldeaux 方法处于同一水平。

该方法的核心优势在于计算效率，相较于 Baldeaux 方法的逐路径逆拉普拉斯变换，余弦级数的系数可批量预计算，速度提升一个数量级以上。

4.2.3 条件矩匹配模拟方法

条件矩匹配模拟方法的核心是通过匹配积分方差 V_T 的前若干阶矩，构造解析近似分布实现抽样，避免了复杂的特征函数计算与级数展开，在保证一定精度的前提下实现了极致的计算效率。

4.2.3.1 理论推导

3/2 模型的积分方差 V_T 的前四阶矩，可通过 CIR 过程的矩微分方程解析递推得到：

1. 一阶矩（均值: $\mathbb{E}[V_T] = \int_0^T \mathbb{E}[v_s]ds$，其中 $\mathbb{E}[v_s]$ 为 3/2 模型方差过程的无条件均值，具有闭式表达；
2. 二阶矩: $\mathbb{E}[V_T^2] = 2\int_0^T \int_0^s \mathbb{E}[v_s v_t]dtds$，协方差 $\mathbb{E}[v_s v_t]$ 可通过 CIR 过程的转移矩解析计算；
3. 三阶矩、四阶矩: 可通过矩方程的递推关系解析得到，进而计算出 V_T 的偏度与超额峰度。

基于前四阶矩，可通过 Gram-Charlier A 类展开构造 V_T 的近似概率密度函数：

$$f(x) = \phi_{\mu,\sigma}(x)\left(1 + \frac{\kappa_3}{6}H_3\left(\frac{x-\mu}{\sigma}\right) + \frac{\kappa_4}{24}H_4\left(\frac{x-\mu}{\sigma}\right)\right), \tag{4.21}$$

其中 $\phi_{\mu,\sigma}(x)$ 为均值 μ、方差 σ^2 的正态密度函数，κ_3, κ_4 为 V_T 的标准化三阶、四阶矩，$H_k(\cdot)$ 为 k 阶埃尔米特多项式，其前四阶形式为：

$$H_0(x) = 1, \quad H_1(x) = x, \quad H_2(x) = x^2 - 1, \quad H_3(x) = x^3 - 3x, \quad H_4(x) = x^4 - 6x^2 + 3. \tag{4.22}$$

对该密度函数积分，可得到近似累积分布函数，通过逆变换抽样得到 V_T 的样本。

4.2.3.2 条件蒙特卡洛框架下的实现

在条件蒙特卡洛框架下，条件矩匹配模拟方法的实现流程如下：

1. 终端方差 v_T 的精确抽样: 通过 CIR 过程的非中心卡方分布精确抽样得到 v_T；
2. 积分方差 V_T 的矩匹配抽样: 给定 v_0 与 v_T，解析计算 V_T 的前四阶矩，构造 Gram-Charlier 近似分布，通过逆变变换抽样得到 V_T 的样本；
3. 标的到期价格的生成: 将 v_T 与 V_T 代入核心闭式映射，生成对数到期价格。

4.2.3.3 误差特性与收敛性分析

1. 矩匹配的近似误差: 该误差为方法的固有偏差，仅匹配前四阶矩忽略了高阶矩的信息，在常规参数下偏差较小，但在极端厚尾场景下偏差会有所增大，其弱收敛阶为 $O(N^{-2})$；
2. 蒙特卡洛模拟的统计误差: 收敛速度为 $O(N^{-1/2})$。

该方法的核心优势是极致的计算效率，所有矩均可预计算，无需逐路径的复杂数值运算，适合大规模路径模拟与参数校准场景。

4.2.4 逆高斯（Inverse Gaussian, IG）近似精确模拟方法

逆高斯近似精确模拟方法由 Choi et al. (2024) 提出，其核心是利用逆高斯分布对正偏、厚尾非负随机变量的优秀拟合能力，通过匹配积分方差 V_T 的前两阶矩构造近似分布，在精度与效率之间实现了极佳的平衡。

4.2.4.1 理论推导

逆高斯分布是一类定义在正实数域上的双参数连续分布，其概率密度函数为:

$$f_{\mathrm{IG}}(x;\mu,\lambda)=\sqrt{\frac{\lambda}{2\pi x^3}}\exp\left(-\frac{\lambda(x-\mu)^2}{2\mu^2 x}\right),\quad x>0, \tag{4.23}$$

其中 $\mu>0$ 为分布的均值，$\lambda>0$ 为形状参数，其方差满足 $\mathrm{Var}(x)=\mu^3/\lambda$。

3/2 模型的积分方差 I_T 为非负随机变量，具有显著的正偏与厚尾特征，与逆高斯分布的统计特性高度契合。通过矩匹配条件，可解析求解逆高斯分布的参数:

1. 给定 v_0 与 v_T，解析计算 I_T 的条件均值 $\mu=\mathbb{E}[I_T\mid v_0,v_T]$ 与条件方差 $\sigma^2=\mathrm{Var}(I_T\mid v_0,v_T)$；
2. 令逆高斯分布的均值与方差与目标值匹配，解得:

$$\mu_{\mathrm{IG}}=\mu,\quad \lambda_{\mathrm{IG}}=\frac{\mu^3}{\sigma^2}. \tag{4.24}$$

4.2.4.2 条件蒙特卡洛框架下的实现

在条件蒙特卡洛框架下，逆高斯近似精确模拟方法的实现流程如下:

1. 终端方差 v_T 的精确抽样: 通过 CIR 过程的非中心卡方分布精确抽样得到 v_T；
2. 积分方差 V_T 的 IG 分布抽样: 给定 v_0 与 v_T，解析计算 V_T 的条件均值与方差，校准 IG 分布参数，生成 IG 分布随机数得到 V_T 的样本；
3. 标的到期价格的生成: 将 v_T 与 V_T 代入核心闭式映射，生成对数到期价格。

4.2.4.3 误差特性与收敛性分析

该方法的误差来源主要包含两个部分：

1. 分布拟合误差：该误差为方法的固有偏差，由于 IG 分布仅匹配前两阶矩，忽略了高阶矩的影响；
2. 蒙特卡洛模拟的统计误差：收敛速度为 $O(N^{-1/2})$。

该方法的核心优势是兼顾精度与效率，IG 分布的随机数生成速度极快，且拟合精度接近精确模拟方法，是实际应用中最常用的 3/2 模型模拟方案。

4.2.5 Gamma 近似精确模拟方法

Gamma 分布是定义在正实数域上的双参数连续分布，其概率密度函数为：

$$f_{\text{Gamma}}(x;\alpha,\beta) = \frac{\beta^{\alpha}}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0, \tag{4.25}$$

其中 $\alpha > 0$ 为形状参数，$\beta > 0$ 为率参数，其均值与方差满足 $\mathbb{E}[x] = \alpha/\beta$，$\mathrm{Var}(x) = \alpha/\beta^2$。

通过矩匹配条件，可解析求解 Gamma 分布的参数：

1. 给定 v_0 与 v_T，解析计算 I_T 的条件均值 μ 与条件方差 σ^2；
2. 令 Gamma 分布的均值与方差与目标值匹配，解得：

$$\alpha = \frac{\mu^2}{\sigma^2}, \quad \beta = \frac{\mu}{\sigma^2}. \tag{4.26}$$

4.2.5.1 条件蒙特卡洛框架下的实现

在条件蒙特卡洛框架下，Gamma 近似精确模拟方法的实现流程与 IG 方法完全一致，仅将 IG 分布替换为 Gamma 分布进行抽样，最终将 v_T 与 I_T 代入核心闭式映射生成标的到期价格。

4.2.5.2 误差特性与收敛性分析

该方法的误差特性与 IG 方法类似，核心差异在于：Gamma 分布对 I_T 的厚尾拟合精度略低于 IG 分布，但其随机数生成的计算效率更高，在超大规模路径模拟场景下具有显著优势。其误差来源同样为分布拟合误差与蒙特卡洛统计误差，收敛速度为 $O(N^{-1/2})$。

4.3 傅里叶频域定价方法

傅里叶频域定价方法完全规避了蒙特卡洛模拟的路径生成过程，其核心是将期权价格的风险中性期望转化为标的对数价格特征函数的傅里叶积分，通过数值积分方法

快速求解，具有计算效率高、无统计误差、适合批量执行价定价的核心优势，是 3/2 模型期权定价与参数校准的主流方法。

4.3.1 快速傅里叶变换（Fast Fourier Transform, FFT）定价方法

FFT 定价方法由 Carr et al. (1999) 提出，其核心是将欧式期权价格表示为特征函数的傅里叶变换形式，通过快速傅里叶变换算法在 $O(N \log N)$ 的时间复杂度内一次性计算出所有等距执行价对应的期权价格。

4.3.1.1 理论推导

对于到期时间为 T、执行价为 K 的欧式看涨期权，其风险中性价格可表示为：

$$C(K) = e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[\max(S_T - K, 0)\right] = e^{-rT}\int_{-\infty}^{\infty}\max(S_0 e^x - K, 0) f(x)\,dx, \tag{4.27}$$

其中 $x = \log(S_T/S_0)$ 为标的对数收益率，$f(x)$ 为 x 的概率密度函数。

令 $k = \log(K/S_0)$ 为对数执行价，将期权价格改写为关于 k 的函数 $C(k)$，通过傅里叶变换的卷积性质，可推导得到 $C(k)$ 的傅里叶变换闭式：

$$\hat{C}(u) = \int_{-\infty}^{\infty} e^{iuk} C(k)\,dk = e^{-rT}\frac{\phi(u-i)}{iu-u^2}, \tag{4.28}$$

其中 $\phi(u) = \mathbb{E}^{\mathbb{Q}}\left[e^{iux}\right]$ 为标的对数收益率的特征函数，是频域定价方法的核心。

对于 3/2 模型，对数收益率的特征函数可通过合流超几何函数 ${}_1F_1$ 得到闭式表达：

$$\phi(u) = \exp(iurT)\cdot\frac{\Gamma(\beta-\alpha)}{\Gamma(\beta)}\cdot\left(\frac{2\kappa\theta}{\epsilon^2 v_0}\left(e^{\kappa\theta T}-1\right)\right)^{\alpha}\cdot {}_1F_1\left(\alpha, \beta, -\frac{2\kappa\theta}{\epsilon^2 v_0}\left(e^{\kappa\theta T}-1\right)\right), \tag{4.29}$$

其中参数 α, β 由模型参数与傅里叶变量 u 解析确定：

$$\alpha = -\frac{1}{2} + \frac{\kappa - \rho\epsilon u}{\epsilon^2} + \sqrt{\left(\frac{1}{2} - \frac{\kappa-\rho\epsilon u}{\epsilon^2}\right)^2 + \frac{u(iu+1)}{\epsilon^2}}, \quad \beta = 1 + 2\alpha, \tag{4.30}$$

通过傅里叶逆变换，可得到期权价格的积分表达式：

$$C(k) = \frac{e^{-rT}}{\pi}\int_0^{\infty}\mathrm{Re}\left[e^{-iuk}\frac{\phi(u-i)}{iu-u^2}\right]du. \tag{4.31}$$

将上述连续积分离散为等距网格的黎曼和，该求和形式恰好符合离散傅里叶变换的结构，因此可通过 FFT 算法实现快速计算。

4.3.1.2 数值实现流程

FFT 定价方法的完整实现流程如下：

1. 参数初始化：设定 3/2 模型参数、期权合约参数、FFT 网格参数 N（通常取 2^{12}）、频率步长 Δu；
2. 特征函数计算：在离散频率网格上计算 3/2 模型的特征函数值；
3. FFT 变换：对加权后的特征函数值执行 FFT 变换，得到离散傅里叶变换结果；
4. 价格还原：对 FFT 结果进行尺度变换与逆变换，得到不同对数执行价对应的期权价格；
5. 插值处理：通过插值方法得到任意执行价对应的期权价格。

4.3.1.3 误差特性与收敛性分析

FFT 定价方法的误差来源主要包含两个部分：

1. 频率域的积分截断误差：该误差随截断频率的增大呈指数衰减，通过合理设置网格参数可将误差控制在 10^{-6} 以下；
2. 黎曼和的离散化误差：该误差随频率步长的减小以线性速度衰减，可通过增加网格点数进一步降低。

该方法无蒙特卡洛模拟的统计误差，计算效率极高，适合全波动率微笑的批量定价与参数校准场景。

4.3.2 傅里叶余弦 (Fourier-Cosine, COS) 定价方法

COS 定价方法由 (Fang et al., 2009) 提出，是对 FFT 方法的重大改进，其核心是通过余弦级数展开标的对数收益率的概率密度函数，实现期权价格的指数级收敛，在非仿射的 3/2 模型中仍能保持极高的计算效率与精度。

4.3.2.1 理论推导

COS 方法的核心是将标的对数收益率 x 的概率密度函数在有效支撑区间 $[a, b]$ 上展开为余弦级数：

$$f(x) = \frac{2}{b-a}\sum_{n=0}^{\infty} \mathrm{Re}\left[\phi\left(\frac{n\pi}{b-a}\right)e^{-i\frac{n\pi a}{b-a}}\cos\left(\frac{n\pi(x-a)}{b-a}\right)\right], \tag{4.32}$$

其中 $\phi(u)$ 为 x 的特征函数，$[a, b]$ 为 x 的有效支撑区间，通常取 $a = \mathbb{E}[x] - 10\sqrt{\mathrm{Var}(x)}$，$b = \mathbb{E}[x] + 10\sqrt{\mathrm{Var}(x)}$。

将该级数展开代入期权价格的积分表达式，可得到期权价格的闭式级数求和形式：

$$C(K) = e^{-rT}\sum_{n=0}^{N-1} \mathrm{Re}\left[\phi\left(\frac{n\pi}{b-a}\right)e^{-i\frac{n\pi a}{b-a}}\right]\cdot\frac{2}{b-a}V_n(k), \tag{4.33}$$

其中 $k=\log(K/S_0)$，$V_n(k)$ 为欧式看涨期权支付函数的余弦系数，具有完全解析的闭式表达，无需任何数值积分：

$$V_n(k)=\begin{cases} S_0e^{rT}\left(\frac{b-k}{b-a}\right)-K\left(\frac{b-k}{b-a}\right), & n=0, \\ \frac{2S_0e^{rT}}{b-a}\cdot\frac{\cos\left(\frac{n\pi(b-a)}{b-a}\right)-\cos\left(\frac{n\pi(k-a)}{b-a}\right)}{1+\left(\frac{n\pi}{b-a}\right)^2} & \\ \quad -\frac{2K}{b-a}\cdot\frac{\sin\left(\frac{n\pi(b-a)}{b-a}\right)-\sin\left(\frac{n\pi(k-a)}{b-a}\right)}{\frac{n\pi}{b-a}}, & n\geq 1. \end{cases} \tag{4.34}$$

4.3.2.2 数值实现流程

4.3.2.3 数值实现流程

COS 定价方法的完整实现流程如下：

1. 参数初始化：设定 3/2 模型参数、期权合约参数、COS 级数截断项数 N（通常取 128-256）、有效支撑区间 $[a,b]$；
2. 特征函数计算：在离散频率点上计算 3/2 模型的特征函数值；
3. 支付系数计算：解析计算期权支付的余弦系数 $V_n(k)$；
4. 级数求和：通过截断级数求和计算得到期权价格。

4.3.2.4 误差特性与收敛性分析

COS 定价方法的核心优势是指数级收敛速度，其截断误差随级数项数 N 的增加呈指数衰减，仅需 128 项截断即可达到 FFT 方法 4096 个网格点的精度，计算效率提升一个数量级以上。同时，该方法不存在 FFT 方法的栅栏效应，可灵活计算任意执行价对应的期权价格，在 3/2 模型这类非仿射模型中具有极强的适用性。

4.4 本章小结

本章在 3/2 随机波动率模型的统一框架下，系统构建了欧式期权定价的九类方法体系，涵盖时间步进类条件蒙特卡洛方法、终端精确与近似精确模拟类方法、傅里叶频域定价方法三大范式，所有方法均严格遵循条件蒙特卡洛框架或频域积分的学术规范，对每类方法的理论构造、实现流程、误差特性进行了完整的推导与分析。时间步进类方法实现简单灵活，适合短期限期权与动态对冲场景，其中精确步进方法与 Poisson-Gamma 方法可有效降低离散偏差；终端模拟类方法完全规避了时间离散误差，其中 Baldeaux 精确模拟方法是理论基准，逆高斯与 Gamma 近似方法在精度与效率之间实现了极佳的平衡；傅里叶频域方法无统计误差、计算效率极高，适合全波动率微笑的批量定价

与模型参数校准。本章构建的九类方法体系，为后续的数值精度对比、极端场景性能检验与实证定价分析提供了严格的理论基础与统一的方法框架。

第五章　数值实验设定与结果对比分析

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本章基于第四章构建的 3/2 随机波动率模型九类期权定价方法，开展系统性数值实验，核心目标是量化对比不同方法在定价精度与计算效率两大核心维度的性能差异，验证不同方法在常规金融场景与极端参数下的适用性，为 3/2 模型的实际工程应用提供量化依据。所有实验均在统一的硬件环境下完成，结果可复现、可对比。

5.1　数值实验环境与设定

5.1.1　硬件与软件环境

本次数值实验的运行硬件为消费级笔记本联想电脑 ThinkBook14，具体配置如下：处理器为 AMD Ryzen 7 6800H with Radeon Graphics，基础主频 3.20GHz，8 核 16 线程，标准 TDP 45W；内存为 16.0GB DDR5 双通道内存，实验运行时可用内存 13.7GB，可支撑大规模蒙特卡洛路径模拟与频域网格计算；操作系统为 Windows 11 企业版（版本号 26200.7840），64 位架构；开发与计算环境基于 Python 3.9 实现，核心数值计算依赖向量化运算库 numpy、特殊函数与概率分布库 scipy，所有方法均采用向量化实现以最大化利用硬件算力，未使用 GPU 加速，结果反映方法在通用 CPU 环境下的实际性能。

5.1.2　测试案例设计

本次实验共设计 7 组测试案例，覆盖了金融衍生品定价中最常见的常规场景与极端场景，包括平值（ATM）、实值（ITM）、虚值（OTM）期权，短期到期、高杠杆效应（强价格 - 波动率负相关）、高波动率的波动率（vol-of-vol）等典型场景，所有案例的基准价格均来自对应文献的精确解析结果或高精度数值解，作为精度评价的基准。7 组案例的具体参数与设计逻辑如下表所示：

Case I 为Baldeaux (2012a) 中的经典基准算例，用于验证方法的基础理论定价精度；Case II 为极端高 vol-of-vol、强价格 - 波动率负相关场景，用于模拟市场危机下的高杠杆定价环境；Case III 为中高 vol-of-vol、强负相关场景，与 Case II 形成对照，检验 vol-of-vol 参数对方法性能的影响；Case IV 为短期到期、近平值场景，对应高频交易中的短期期权定价需求；Case V 为短期到期、深度实值场景，用于检验方法在深度实值期权下的定价稳定性；Case VI 为高 vol-of-vol、平值场景，用于检验方法在中等期限、高波动环境下的长期定价性能；Case VII 为 Lewis (2000) 提出的经典算例，作为频域定价

表 5.1 Test Cases and Parameter Configurations

Case	σ	vov	ρ	κ	θ	r	T	K	S_0	精确价格 (p_{exact})	场景描述
Case I	1	0.2	-0.5	2	1.5	0.05	1	1	1	0.443059	Baldeaux (2012)
Case II	0.06	8.56	-0.99	22.84	0.218	0	0.5	95	100	10.364	Kouarfate et al. (2021), ATM
								100	100	7.386	
								105	100	4.938	
Case III	0.06	3.2	-0.99	20.48	0.218	0	0.5	95	100	11.724	Kouarfate et al. (2021), ATM
								100	100	8.999	
								105	100	6.710	
Case IV	1	0.3	0.5	0.7	0.6	0	0.1	47	49	7.040	Near expiration and ATM
								48	49	6.500	
								49	49	6.121	
								50	49	5.7006	
								51	49	5.305	
Case V	1	0.3	0.5	0.7	0.6	0	0.1	10	50	39.9985	Near exp only
								20	50	30.002	
								40	50	11.9266	
Case VI	1	3.3	0.5	0.7	0.6	0	1	47	49	12.7468	ATM only
								48	49	12.3995	
								49	49	12.0658	
								50	49	11.7452	
								51	49	11.4371	
Case VII	0.06	3.2	-0.99	20.48	0.218	0	0.5	95	100	11.7235	Lewis (2000), ATM
								100	100	8.9978	
								105	100	6.7091	

方法的基准验证场景。这 7 组案例完整覆盖了 3/2 模型在实际应用中的绝大多数场景，既包含了用于验证基础精度的基准算例，也包含了用于测试方法鲁棒性的极端参数场景，可全面对比不同方法的性能边界。

5.1.3 评价指标定义

本次实验构建三级评价指标体系，以量化评估方法的综合性能，指标优先级从高到低依次为数值稳定性、定价精度、计算效率。数值稳定性为方法具备工程实用性的基础判定标准，若方法输出结果为非数值型（nan）或无穷大（inf），判定为该场景下数值失效，丧失定价能力。定价精度采用绝对偏差与相对偏差两个指标衡量，绝对偏差为方法计算价格与基准价格的差值，相对偏差为绝对偏差与基准价格的百分比，用于不同执行价、不同价格水平下的精度横向对比。计算效率采用单组案例（含对应全部执行价）的完整运行时长衡量，单位为秒，反映方法的算力需求与工程应用适配性。

5.2 时间步进类条件蒙特卡洛方法结果分析

时间步进类方法共包含 4 种方案：Euler/Milstein 离散格式、精确步进（Exact Stepping）、Poisson-Gamma 近似精确步进、QE 近似精确步进，实验设置了 3 组时间步长

dt=1/50、1/500、1/5000，对应从粗到细的时间离散精度，验证步长对方法性能的影响，完整结果如下表所示：

实验结果显示，四类方法在数值稳定性与定价性能上呈现显著分化。Euler/Milstein 离散格式的核心局限在于对时间步长的高度敏感性与极端场景下的数值失效。在 1/50 粗步长下，该方法在 Case II 虚值期权场景中相对偏差达 714.93%，在 Case VI 高 vol-of-vol 场景中全部输出无穷大结果，完全丧失定价能力；当步长缩小至 1/500 时，该方法在 Case VI 场景中仍全部数值失效，Case II 平值期权的相对偏差仍达 3.89%，无法满足工程定价的精度要求；仅当步长进一步缩小至 1/5000 时，该方法在常规场景下的相对偏差可降至 1% 以内，但对应计算时间从 0.2 秒量级飙升至 30 秒以上，时间成本提升两个数量级，计算效率极低。该现象的核心成因在于 3/2 模型方差过程的强非线性特征，粗步长下 Euler 格式的一阶离散误差被显著放大，易出现负方差的非物理结果，最终导致定价失效。与 Euler/Milstein 格式不同的是，精确步进、Poisson-Gamma 近似精确步进、QE 近似精确步进三类方法，在全部 7 组测试案例、所有执行价与所有时间步长下，均未出现数值失效，始终可输出有效定价结果，是本次实验中唯一具备全场景稳定性的蒙特卡洛类方法。即使在 1/50 粗步长下，精确步进方法在 Case II 平值期权场景中的相对偏差仅为 - 0.03%，Poisson-Gamma 与 QE 方法的偏差也控制在 0.2% 以内，精度显著优于 Euler 格式 1/5000 细步长下的结果，而计算时间仅为 0.2 秒左右，不足 Euler 细步长计算时间的 1%。在 Case VI 高 vol-of-vol 极端场景中，1/500 步长下精确步进方法的相对偏差仅为 - 0.46%，Poisson-Gamma 方法仅为 0.12%，而同步长下 Euler 格式完全数值失效。三类方法的性能差异同样可通过实验结果量化验证，精确步进方法的定价精度最高，所有场景下的相对偏差均控制在 0.5% 以内，无时间离散偏差，为时间步进类方法的精度基准；Poisson-Gamma 与 QE 方法的精度略低于精确步进方法，但计算时间与 Euler 格式处于同一量级，在极端场景下仍保持数值稳定性，实现了定价精度与计算效率的最优平衡。

5.3　终端精确与近似精确模拟方法结果分析

终端模拟类方法完全规避了时间离散过程，直接针对到期方差与积分方差的联合分布抽样，共包含 5 种方案：Baldeaux 精确模拟、基于 Fourier-Cosine 的精确模拟、基于矩匹配的精确模拟、逆高斯（IG）近似精确模拟、Gamma 近似精确模拟，完整结果如下表所示：

从表中结果可以看到，精确模拟”的三类终端方法，仅能在 Case I 基准算例的单一执行价场景下输出有效结果，在其余测试场景中均完全数值失效，不具备全场景工程实用性。

表 5.2　Time-Stepping Conditional Monte Carlo Results

Case	dt	Strike	Euler/Milstein scheme		Exact Stepping		Poisson-Gamma Almost Exact Stepping		QE Almost Exact Stepping	
			Time (s)	Option Bias	Time (s)	Option Bias	Time (s)	Option Bias	Time (s)	Option Bias
Case I	1/50	1	0.405223	−0.01165 (−2.63%)	0.428566	−0.0009242 (−0.21%)	0.6209759	−0.0005159 (−0.12%)	0.6758925	−0.0002905 (−0.07%)
	1/500	1	2.8924741	−0.001347 (−0.30%)	3.848103	−0.000335 (−0.07%)	5.5486124	−0.0004362 (−0.10%)	6.66017	−0.0001411 (−0.03%)
	1/5000	1	29.065831	0.0002249 (0.06%)	40.3844027	−0.0001585 (−0.04%)	88.8646271	−0.000115 (−0.02%)	101.9920107	−0.000367 (−0.04%)
Case II		95		230.2249 (−9.06%)		−0.0114 (−0.11%)		−0.004389 (−0.05%)		−0.000977 (−0.01%)
	1/50	100	0.2677657	2.302 (31.17%)	0.2455173	−0.0079 (−0.23%)	0.4654895	−0.007585 (−0.10%)	0.390341	−0.01131 (−0.15%)
		105		7.401 (714.93%)		0.001158 (0.02%)		0.002399 (0.05%)		0.01003 (0.20%)
		105		7.427 (150.41%)		0.0002409 (0.00%)		−0.004103 (−0.08%)		0.01213 (0.25%)
	1/500	100	1.504743	0.15086 (3.89%)	1.9639806	0.003497 (0.03%)	2.9796648	−0.01347 (−0.13%)	3.397378	0.01355 (0.18%)
		95		0.0909 (1.23%)		−0.001302 (−0.02%)		−0.01213 (−0.16%)		−0.01645 (−0.22%)
		105		0.0909 (1.89%)		−0.002073 (−0.06%)		−0.01034 (−0.21%)		−0.01676 (−0.34%)
	1/5000	100	14.525341	0.36318 (1.23%)	20.2330929	−0.00273 (−0.07%)	29.5629414	−0.007066 (−0.10%)	48.5850875	−0.0309 (−0.46%)
		95		0.093112 (0.89%)		−0.00142 (−0.01%)		−0.009375 (−0.34%)		−0.02017 (−0.17%)
Case III		95		364.9 (9054.87%)		−0.0126 (−0.14%)		−0.05426 (−0.66%)		−0.03312 (−0.37%)
	1/50	100	0.238946	36.4 (5438.78%)	0.2417619	−0.005401 (−0.08%)	0.3870248	−0.0472 (−0.70%)	0.4423578	−0.0326 (−0.50%)
		105		0.05495 (−0.57%)		−0.0107 (−0.12%)		−0.0188 (−0.21%)		0.01358 (0.17%)
		105		−0.05096 (−0.72%)		−0.009423 (−0.14%)		−0.01699 (−0.25%)		0.0144 (0.20%)
	1/500	100	1.4755198	−0.02709 (−0.23%)	1.9966579	−0.00122 (−0.01%)	2.8634368	−0.00975 (−0.11%)	3.3736463	−0.0155 (−0.17%)
		95		−0.02338 (−0.33%)		−0.006346 (−0.07%)		−0.0375 (−0.34%)		−0.02017 (−0.17%)
		105		−0.02866 (−0.40%)		−0.008676 (−0.13%)		−0.00941 (−0.14%)		−0.02275 (−0.31%)
	1/5000	100	14.4603692	−0.01184 (−0.17%)	19.967838	−0.01091 (−0.16%)	28.5484009	−0.03466 (−0.52%)	48.5172484	−0.0207 (−0.25%)
		95		0.018024 (0.24%)		−0.005735 (−0.06%)		−0.00736 (0.11%)		−0.00109 (−0.02%)
Case IV		48		0.01468 (0.26%)		−0.008412 (−0.15%)		0.004264 (0.07%)		−0.0002527 (−0.00%)
		49		0.01468 (0.23%)		−0.007911 (−0.13%)		0.004212 (0.07%)		−0.0002652 (−0.00%)
	1/50	50	0.0891715	0.01468 (0.26%)	0.0836116	0.006557 (0.10%)	0.1061546	0.003103 (0.04%)	0.1498051	−0.004188 (−0.06%)
		51		0.00985 (−0.04%)		0.007421 (0.14%)		0.00358 (0.09%)		−0.00351 (−0.08%)
		47		−0.00295 (−0.04%)		−0.01091 (−0.16%)		0.007103 (0.10%)		−0.003395 (−0.05%)
		49		0.006214 (0.10%)		0.00662 (0.11%)		0.003458 (0.06%)		0.00635 (0.09%)
		48		−0.001867 (−0.03%)		0.006706 (0.12%)		0.003327 (0.06%)		−0.003306 (−0.06%)
	1/500	50	0.323018	−0.001804 (−0.03%)	0.4360611	0.006374 (0.12%)	0.6052143	0.00312 (0.06%)	0.7028849	−0.003293 (−0.06%)
		51		−0.001804 (−0.03%)		0.006539 (0.12%)		0.002685 (0.04%)		−0.003207 (−0.06%)
		47		0.006706 (0.10%)		0.006259 (0.09%)		0.00703 (0.10%)		0.00693 (0.10%)
		49		0.00152 (0.03%)		−0.004082 (−0.06%)		0.00318 (0.05%)		0.001356 (0.00%)
		48		0.00152 (0.03%)		−0.00989 (−0.96%)		0.003308 (0.05%)		0.000693 (0.01%)
	1/5000	50	3.0013151	0.0002269 (0.00%)	4.0081004	−0.003774 (−0.07%)	5.7636176	0.003322 (0.06%)	10.02923375	8.264×10^{-5} (0.00%)
		51		−0.0002143 (−0.00%)		−0.00356 (−0.07%)		0.003245 (0.06%)		6.51×10^{-5} (0.00%)
		47		−0.001015 (−0.01%)		−0.01999 (−0.30%)		0.005225 (0.07%)		0.0005575 (0.01%)
Case V		20		−0.00994 (−0.03%)		−0.01497 (−0.07%)		0.005325 (0.02%)		0.000554 (0.00%)
	1/50	40	0.0873017	0.005247 (0.04%)	0.0845473	0.01425 (0.12%)	0.1103219	0.00362 (0.02%)	0.129903	0.000675 (0.01%)
		10		−0.002526 (−0.01%)		0.01425 (0.04%)		0.008478 (0.02%)		−0.00144 (−0.00%)
		40		−0.001768 (−0.01%)		0.01129 (0.09%)		0.00529 (0.03%)		−0.001399 (−0.02%)
	1/500	20	0.3298876	0.002483 (0.01%)	0.4290495	0.008224 (0.05%)	0.5894391	0.006829 (0.06%)	0.7007175	−0.00217 (−0.02%)
		10		0.00053 (0.00%)		−0.008224 (−0.02%)		0.007159 (0.02%)		0.000674 (0.00%)
		40		0.0005752 (0.00%)		−0.008224 (−0.02%)		0.007159 (0.02%)		0.0008315 (0.01%)
	1/5000	20	2.9262788	0.0005752 (0.00%)	4.0733267	−0.008224 (−0.03%)	5.7201683	0.007159 (0.02%)	9.9833601	0.000687 (0.00%)
		10		0.000949 (0.00%)		−0.005735 (−0.01%)		0.00539 (0.01%)		0.000933 (0.00%)
Case VI		48		inf (inf%)		225.7 (171.53%)		36.17 (283.9%)		inf (inf%)
		49		inf (inf%)		225.6 (1871.65%)		36.17 (300.01%)		inf (inf%)
	1/50	50	0.3831229	inf (inf%)	0.4302237	225.6 (1922.8%)	0.7873058	36.17 (308.2%)	0.7045123	inf (inf%)
		51		inf (inf%)		225.6 (1974.65%)		36.17 (316.52%)		inf (inf%)
		47		inf (inf%)		225.7 (1771.21%)		36.17 (291.99%)		inf (inf%)
		49		inf (inf%)		−0.0573 (−0.46%)		0.01475 (0.12%)		4.106×10^{68} (nan%)
		48		inf (inf%)		−0.0575 (−0.47%)		0.01395 (0.11%)		4.106×10^{68} (nan%)
	1/500	50	2.8947977	inf (inf%)	3.8898247	−0.05727 (−0.49%)	5.8676512	0.01336 (0.11%)	6.6754493	4.106×10^{68} (nan%)
		51		inf (inf%)		−0.05716 (−0.50%)		0.01297 (0.11%)		4.106×10^{68} (nan%)
		47		inf (inf%)		−0.05716 (−0.47%)		0.0136 (0.11%)		4.106×10^{68} (nan%)
		49		468.6 (3678.98%)		0.00935 (0.08%)		0.02743 (0.22%)		3.057×10^{21} (nan%)
		48		468.6 (3782.16%)		0.00958 (0.08%)		0.02739 (0.22%)		3.057×10^{21} (nan%)
	1/5000	50	29.229539	468.6 (3893.10%)	55.4278611	0.00962 (0.08%)	57.7701995	0.02713 (0.23%)	99.3793894	3.057×10^{21} (nan%)
		51		468.6 (4100.83%)		0.00943 (0.08%)		0.02707 (0.24%)		3.057×10^{21} (nan%)
		47		468.6 (3678.98%)		0.00958 (0.08%)		0.02743 (0.22%)		3.057×10^{21} (nan%)
Case VII		95		364.9 (4055.43%)		−0.0114 (−0.13%)		−0.05822 (−0.66%)		−0.03192 (−0.36%)
	1/50	100	0.2103698	36.4 (5439.54%)	0.2497061	−0.004501 (−0.07%)	0.3766218	−0.0436 (−0.65%)	0.3742216	−0.03236 (−0.48%)
		105		0.05495 (−0.46%)		−0.001013 (−0.01%)		−0.0198 (−0.16%)		0.015 (0.13%)
		105		−0.04372 (−0.71%)		−0.009498 (−0.11%)		−0.0176 (−0.20%)		0.01678 (0.21%)
	1/500	100	1.4952733	−0.01256 (−0.18%)	1.9388048	−0.01523 (−0.13%)	2.9082565	−0.01609 (−0.24%)	3.36681	0.01431 (0.19%)
		95		−0.02709 (−0.23%)		−0.00742 (−0.01%)		−0.03935 (−0.42%)		−0.01967 (−0.17%)
		105		−0.02589 (−0.31%)		−0.005146 (−0.06%)		−0.0321 (−0.50%)		−0.02155 (−0.24%)
	1/5000	100	14.7206092	−0.0218 (−0.39%)	34.2497818	−0.00076 (−0.01%)	48.1214847	−0.03376 (−0.40%)	52.2054604	−0.01987 (−0.22%)
		95		−0.02596 (−0.23%)		−0.009742 (−0.12%)		−0.03621 (−0.34%)		−0.01931 (−0.17%)

表 5.3　Terminal Exact and Almost Exact Simulation Results

Case	Strike	Exact Simulation		Exact Simulation (Use Fourier-Cosine)		Exact Simulation (Use Moment Matching)		Almost Exact Simulation (Inverse Gaussian)		Almost Exact Simulation (Gamma)	
		Time(s)	Option Bias	Time(s)	Option Bias	Time(s)	Option Bias	Time(s)	Option Bias	Time(s)	Option Bias
Case I	1	59.5239847	-1.571×10^{-5} (−0.00%)	25.4983391	−0.04977 (−11.23%)	0.025079	−0.05216 (−11.77%)	2.6390925	0.001326 (0.30%)	2.164342	0.0008747 (0.20%)
Case I	95		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−8.215 (−79.26%)		−7.956 (−76.77%)
	100		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−6.213 (−84.12%)		−5.994 (−81.16%)
Case II	105	69.0326374	NaN (NaN%)	8.3304516	NaN (NaN%)	0.034264	NaN (NaN%)	0.2152987	−4.361 (−88.31%)	0.204456	−4.195 (−84.95%)
	95		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−11.64 (−89.30%)		−11.97 (−99.56%)
	100		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−8.96 (−99.57%)		−8.975 (−99.73%)
Case III	105	73.7386331	NaN (NaN%)	6.6512523	NaN (NaN%)	0.032648	NaN (NaN%)	0.256394	−6.693 (−99.74%)	0.241701	−6.699 (−99.84%)
	95		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−11.64 (−99.30%)		−11.975 (−99.56%)
	47		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−5.559 (−78.96%)		38.31 (544.13%)
	48		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−5.287 (−81.34%)		37.91 (583.29%)
Case IV	49	79.0467981	NaN (NaN%)	6.5286679	NaN (NaN%)	0.026266	NaN (NaN%)	20.5061735	−5.135 (−83.88%)	13.86428	37.37 (610.49%)
	50		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−4.903 (−86.01%)		36.87 (646.82%)
	51		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−4.664 (−87.91%)		36.36 (685.41%)
	10		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−5.255 (−13.14%)		43.83 (109.58%)
Case V	20	75.2066147	NaN (NaN%)	6.0646468	NaN (NaN%)	0.025757	NaN (NaN%)	22.739208	−5.259 (−17.53%)	13.84294	43.83 (146.08%)
	40		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−6.356 (−53.29%)		41.99 (352.08%)
	47		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−0.06061 (−0.48%)		−0.1563 (−1.23%)
	48		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−0.05685 (−0.46%)		−0.1489 (−1.20%)
Case VI	49	69.5002801	NaN (NaN%)	21.3088199	NaN (NaN%)	0.033759	NaN (NaN%)	0.1932632	−0.05316 (−0.44%)	0.186914	−0.1419 (−1.18%)
	50		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−0.04984 (−0.42%)		−0.1354 (−1.15%)
	51		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−0.04649 (−0.41%)		−0.1291 (−1.13%)
	95		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−11.64 (−99.30%)		−11.67 (−99.56%)
Case VII	100	70.4046528	NaN (NaN%)	6.2345658	NaN (NaN%)	0.032753	NaN (NaN%)	0.2169977	−8.959 (−99.57%)	0.235242	−8.974 (−99.73%)
	105		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)		−6.692 (−99.74%)		−6.698 (−99.84%)

具体而言，Baldeaux (2012) 精确模拟方法仅在 Case I 的单一执行价场景下输出有效定价结果，相对偏差趋近于 0，具备理论层面的无偏性；但在 Case II 至 Case VII 的全部 6 组测试案例、22 个执行价场景中，均出现数值溢出与计算失效，输出结果为非数值型。该现象的核心成因在于，该方法需要逐路径计算高维修正贝塞尔函数与逆拉普拉斯变换，在非基准参数下，极易出现贝塞尔函数数值溢出、积分收敛失败的问题，仅能在文献给定的基准参数下完成计算，无法适配实际市场中的非基准参数环境。基于 Fourier-Cosine 的终端精确模拟方法与基于矩匹配的终端精确模拟方法，表现与 Baldeaux 方法高度一致，仅在 Case I 单一执行价场景下输出有效结果，且相对偏差分别达 −11.23% 与 −11.77%，定价精度极差，其余所有场景均完全数值失效。两类方法失效的核心成因分别为：Fourier-Cosine 方法对积分方差分布的余弦级数截断，在非基准参数下，分布尾部的截断误差被无限放大，导致特征函数计算失效；矩匹配方法仅匹配分布前四阶矩的 Gram-Charlier 展开，无法适配 3/2 模型方差分布的非正态厚尾特征，易出现负密度的非物理结果，最终导致累积分布函数逆变换抽样失败。

两类近似精确模拟方法中，仅在 Case VI 平值、中等期限、正相关场景下，定价偏差处于可控范围，其余绝大多数场景均出现系统性的大幅定价偏差。在 Case VI 的 5 个近平值执行价场景中，IG 方法的相对偏差处于 −0.41% 至 −0.48% 区间，Gamma 方法的相对偏差处于 −1.13% 至 −1.23% 区间，偏差可控且计算时间仅为 0.19 秒左右，具备较高的计算效率。但在其余所有测试场景中，两类方法均出现显著的定价偏差：在高 vol-of-vol、强负相关的 Case II、III、VII 场景中，两类方法在所有执行价下的相对偏差幅度达 −80% 至 −99.8%，定价结果完全偏离基准值；在短期到期、深度实值的 Case IV、V 场景中，Gamma 方法出现 500% 以上的正向系统性偏差，IG 方法也出现 −78% 至 −87% 的负向系统性偏差，完全丧失定价参考价值。该现象的核心成因在于，强负

相关、高 vol-of-vol 与短期到期场景下，积分方差的分布呈现极端厚尾与高度右偏特征，仅匹配前两阶矩的双参数 IG 与 Gamma 分布，无法拟合分布的高阶特征与尾部形态，最终导致系统性的定价偏差。

5.4 傅里叶频域定价方法结果分析

频域定价方法完全规避了蒙特卡洛模拟的路径生成过程，通过特征函数的傅里叶积分直接求解期权价格，包含快速傅里叶变换（FFT）与傅里叶余弦（COS）两种方法，完整结果如下表所示：

表 5.4 Fourier Frequency Domain Pricing Results

Case	Strike	Fast Fourier Transformation		Fourier-Cosine Method	
		Time (s)	Option Bias	Time (s)	Option Bias
Case I	1	0.3843623	-2.434×10^{-7} (−0.00%)	0.1115441	-2.437×10^{-7} (−0.00%)
Case II	95	0.1986538	−0.0004891 (−0.00%)	0.0517513	−0.0004891 (−0.00%)
	100		−0.0001461 (−0.00%)		−0.0001461 (−0.00%)
	105		−0.0009548 (−0.02%)		−0.0009548 (−0.02%)
Case III	95	0.2813563	−0.0005218 (−0.00%)	0.0735898	−0.0005218 (−0.00%)
	100		−0.001246 (−0.01%)		−0.001246 (−0.01%)
	105		−0.0008511 (−0.01%)		−0.0008511 (−0.01%)
Case IV	47	15.8604601	0.0003048 (0.00%)	3.7037431	0.0003048 (0.00%)
	48		0.068 (1.05%)		0.068 (1.05%)
	49		0.001088 (0.02%)		0.001088 (0.02%)
	50		0.001157 (0.02%)		0.001157 (0.02%)
	51		0.001138 (0.02%)		0.001138 (0.02%)
Case V	10	16.0524815	0.0015 (0.00%)	3.788367	0.0015 (0.00%)
	20		0.001557 (0.01%)		0.001557 (0.01%)
	40		0.002021 (0.02%)		0.002021 (0.02%)
Case VI	47	0.2070812	0.01801 (0.14%)	0.0686066	0.01785 (0.14%)
	48		0.01801 (0.15%)		0.01791 (0.15%)
	49		0.01807 (0.15%)		0.01791 (0.15%)
	50		0.01794 (0.15%)		0.01778 (0.15%)
	51		0.01794 (0.16%)		0.01775 (0.16%)
Case VII	95	0.3207817	-2.18×10^{-5} (−0.00%)	0.0993632	-2.179×10^{-5} (−0.00%)
	100		-4.598×10^{-5} (−0.00%)		-4.597×10^{-5} (−0.00%)
	105		4.895×10^{-5} (0.00%)		4.895×10^{-5} (0.00%)

从表中结果可以看到，两类方法是本次实验中唯一在全部 7 组测试案例、所有执行价场景下均无数值失效，同时保持极高定价精度的方法，综合性能显著优于所有蒙特卡洛类方法。

在数值稳定性层面，FFT 与 COS 方法在所有测试场景中均未出现非数值型或无穷大结果，无论是基准平值场景、极端高 vol-of-vol 强负相关场景、短期到期场景，还是深度实值/虚值场景，均可稳定输出定价结果，完全规避了蒙特卡洛类方法的抽样失效与分布拟合失效问题。在定价精度层面，两类方法在所有场景下的相对偏差均控制在 1.05% 以内，绝大多数场景的相对偏差小于 0.02%。在 Case II 极端高 vol-of-vol 场景中，两类方法的最大相对偏差仅为 −0.02%；在 Case IV 短期到期场景中，最大相对偏差为 1.05%，其余执行价的偏差均控制在 0.02% 以内；在 Case VI 高 vol-of-vol 场景中，相对偏差仅为 0.15% 左右，定价精度远超所有蒙特卡洛类方法。该结果的核心成因在于，频域方法基于 3/2 模型特征函数的闭式解，通过数值积分直接完成定价，不存在蒙特卡洛方法的抽样误差、分布拟合误差，也不存在时间离散的截断误差，仅存在可忽略的数值积分截断误差。

在计算效率层面，COS 方法的计算效率全面领先于 FFT 方法，在所有测试场景中，COS 方法的运行时间仅为 FFT 方法的 1/3 至 1/4。在 Case IV 多执行价场景中，FFT 方法的运行时间为 15.86 秒，COS 方法仅为 3.70 秒；在单执行价场景中，COS 方法的运行时间仅为 0.05 至 0.11 秒，计算效率远高于所有蒙特卡洛类方法。该效率差异的核心理论成因在于，COS 方法基于余弦级数展开具备指数级收敛速度，仅需 128 项截断即可达到 FFT 方法 4096 点网格的计算精度，同时不存在 FFT 方法的栅栏效应，可灵活计算任意执行价的期权价格，无需额外的插值计算，进一步降低了计算成本。

5.5　综合性能对比与研究结论

基于 7 组测试案例的全场景实验结果，可从数值稳定性、定价精度、计算效率三个核心维度，对九类定价方法的综合性能进行系统性分层。第一梯队为 COS 与 FFT 频域方法，是唯一具备全场景数值稳定性、同时保持极高定价精度与计算效率的方法，其中 COS 方法的综合性能全面领先 FFT 方法，为 3/2 模型欧式期权定价的最优工程方案。第二梯队为时间步进类的精确步进、Poisson-Gamma 近似精确步进与 QE 近似精确步进方法，是蒙特卡洛框架下唯一具备全场景数值稳定性的方案，可稳定输出完整的方差路径，定价精度与计算效率平衡良好，为路径依赖期权定价、动态对冲等需要完整路径生成场景的唯一可靠方案。第三梯队为 Euler/Milstein 离散格式，仅在细步长、常规参数场景下有效，极端场景下完全数值失效，计算效率极低，仅可作为理论精度对比的基准方案。第四梯队为 IG 与 Gamma 近似精确模拟方法，仅在特定平值场景下偏差可控，绝大多数场景出现系统性大幅定价偏差，仅可用于特定场景的学术对比，不具备工程实用性。第五梯队为 Baldeaux 精确模拟、Fourier-Cosine 终端模拟与矩匹配终端模拟方法，仅能在单一基准场景下运行，其余所有场景均完全数值失效，仅可用于理

论层面的无偏性验证。

基于全场景的实验验证，可得到三个结论。第一，3/2 模型定价方法的理论无偏性，不等同于工程应用中的全场景稳定性，标称“精确”的终端模拟方法，在非基准参数下极易出现特殊函数数值溢出、积分收敛失败等问题，仅能在文献基准场景下完成理论验证，无法适配实际市场的复杂参数环境。第二，基于方差过程分布特性构造的时间步进抽样方法，是蒙特卡洛框架下的最优实现方案，此类方法天然保证了方差过程的非负性，在所有极端场景下均保持数值稳定性，同时定价精度显著优于传统 Euler 离散格式，为需要路径生成的衍生品定价场景提供了可靠的实现路径。第三，基于特征函数的 COS 频域方法，是 3/2 模型欧式期权定价的绝对最优方案，该方法在所有测试场景下均保持了极高的精度与稳定性，无任何数值失效，同时计算效率远超所有蒙特卡洛类方法，在欧式期权批量定价、模型参数校准、极端市场环境定价等场景中，具备绝对的性能优势。

基于全场景的性能验证，我们给出不同的应用场景对应方法推荐。对于欧式期权批量定价、模型参数校准场景，优先选择 COS 频域方法，其全场景稳定性、最高精度与最优效率可充分适配此类场景的需求；对于路径依赖期权定价、动态对冲等需要完整方差路径的场景，优先选择 Poisson-Gamma 或 QE 近似精确步进方法，其全场景稳定性与精度效率平衡可满足此类场景的工程需求；对于理论精度基准对比场景，仅可在基准参数环境下使用 Baldeaux 精确模拟方法；其余终端类方法不推荐在实际定价场景中使用，仅可用于特定场景的学术对比分析。

第六章　3/2 随机波动率模型在加密货币极端崩盘事件中的实证校准与定价检验

6.1　章节引言

第五章基于经典金融文献中的基准参数案例，系统完成了 3/2 随机波动率模型下九类欧式期权定价方法的精度、效率与数值稳定性评价，明确了不同方法在理想化参数环境下的理论性能边界与适用场景。但需要指出的是，现有学术研究中的基准测试案例多为平稳市场环境下的理想化参数设定，参数取值无极端波动、无突发市场冲击，虽能完成方法的理论有效性验证，却无法反映真实金融市场、尤其是尾部风险集中的高波动市场中，模型参数的时变特征与定价方法的实际工程适配性。

加密货币市场作为全球波动水平最高的大类资产市场之一，具有标的价格波动剧烈、杠杆交易占比高、极端行情下尾部风险快速释放的核心特征，其期权市场的波动率微笑形态在极端事件中会出现快速且剧烈的形变，为检验随机波动率模型的市场解释力、参数动态特征与定价方法的极端场景鲁棒性提供了天然的实证实验场景。本章以 UTC 时间 2025 年 10 月 10 日的 BTC 市场闪崩事件为研究对象，将 3/2 模型的定价分析从学术基准算例拓展至真实极端市场环境，完成模型的全周期参数校准与定价方法的实证再检验。

本章的核心研究目标分为三个层面：第一，识别 3/2 模型核心参数在崩盘事件前、崩盘期间、市场恢复期三个阶段的时变规律，明确极端行情对模型参数的冲击特征，验证 3/2 模型对加密货币期权市场波动率微笑的拟合能力；第二，基于真实市场校准得到的参数构建全新的测试案例，重复第五章的定价方法性能对比，检验不同方法在极端市场场景下的数值稳定性、定价精度与计算效率，弥补学术基准案例与真实市场环境的脱节问题；第三，明确 3/2 模型在加密货币期权定价中的实践适配性，为极端行情下的衍生品定价、风险管理提供可落地的实证依据与方法选择指引。

本章的研究逻辑严格遵循“长期固定参数估计 - 横截面时变参数校准 - 定价方法极端场景再检验”的完整链条，所有实证分析均基于 Deribit 交易所的真实逐笔成交数据完成。本章后续内容安排如下：6.2 节明确研究事件的时间窗口划分、数据来源与预处理规则；6.3 节构建 3/2 模型的两步校准框架，说明长期参数估计与横截面参数校准的理论基础与实现逻辑；6.4 节完成崩盘事件全周期期权成交数据的描述性统计分析，明确校准样本的筛选依据；6.5 节呈现全周期的参数校准结果，分析极端行情下模型核心参数的时变特征与模型拟合效果；6.6 节基于校准得到的真实市场参数，完成定价方法

的性能再检验，并与第五章的学术案例结果进行对比分析；6.7 节对本章的核心实证结论进行总结。

6.2 研究事件、时间窗口划分与数据预处理

本章实证分析以 2025 年 10 月 10 日至 11 日全球加密货币市场的 BTC 闪崩事件为研究对象，该事件中 BTC 现货价格在短时间内出现大幅下跌，伴随期权市场隐含波动率的剧烈跳升与成交量的显著放大，为检验 3/2 随机波动率模型在极端市场环境下的参数动态特征与定价方法鲁棒性提供了天然的实验场景。为系统性捕捉崩盘事件全周期的市场特征，本章将研究区间划分为三个连续的 10 分钟时间窗口，分别对应崩盘前的市场平稳阶段、崩盘期间的极端冲击阶段与崩盘后的市场修复阶段。具体时间窗口划分如下：崩盘前阶段选取 2025-10-10 20:50:00 至 2025-10-10 21:00:00（UTC 时间，下同），用于捕捉极端行情发生前的模型基准参数；崩盘期阶段选取 2025-10-10 21:15:00 至 2025-10-10 21:20:00，为 BTC 价格快速下跌、市场波动剧烈放大的核心冲击时段，用于识别极端行情下的参数突变特征；恢复期阶段选取 2025-10-10 23:50:00 至 2025-10-11 00:00:00，为崩盘事件发生后的市场逐步修复时段，用于观察极端行情后模型参数的回归特征。三个时间窗口的长度均设定为 10 分钟，既保证了每个窗口内有充足的期权成交样本，又能精准捕捉事件不同阶段的市场瞬时特征，避免时间窗口过长导致的参数时变信息被平滑。

本章实证研究使用的数据全部来自全球头部加密货币期权交易所 Deribit，该交易所提供的 BTC 期权产品具有全球最高的流动性与标准化的市场数据，其公开的历史数据 API 为研究提供了权威、可追溯的数据源。具体而言，本章使用两类核心数据：第一类为 Deribit BTC DVOL 指数的日度历史数据，DVOL 指数是 Deribit 发布的比特币隐含波动率指数，反映市场对未来 30 天 BTC 波动率的一致预期，本章选取 2025 年 4 月 11 日至 2025 年 10 月 10 日（崩盘前半年）的 DVOL 日度数据，用于后续 3/2 模型长期固定参数的估计；第二类为上述三个 10 分钟时间窗口内的 BTCUSDT 期权逐笔成交数据，数据字段包含期权合约代码、成交时间、成交价格、成交量、买卖方向、标的现货指数价格、隐含波动率等，用于 3/2 模型的时变参数横截面校准。两类数据均通过 Deribit 公开的历史数据 API 获取，数据获取过程采用多线程并发与指数退避重试机制，以平衡数据抓取效率与 API 频率限制，同时通过严格的去重规则保证数据的唯一性。

针对获取的原始期权成交数据，本章执行系统性的预处理流程，以保证后续参数校准基于高流动性、高代表性的有效样本。首先进行期权合约参数的拆解与时间标准化处理，从期权合约代码中提取到期日、执行价与期权类型（看涨 / 看跌），将所有成交时间戳统一转换为 UTC 时区，避免跨时区差异导致的计算误差，并基于成交时间与

期权到期时间计算每笔成交距离到期的年化剩余期限。其次进行异常值与低流动性样本的剔除，剔除到期时间小于 1 天或大于 365 天的合约，此类合约流动性极差，价格噪音显著，不具备市场代表性；同时剔除隐含波动率为负、成交量为零的异常成交记录，此类数据多为交易系统故障或极端非活跃交易导致，会对后续校准产生干扰。接着进行虚值期权的筛选，保留看涨期权执行价大于等于标的现货价格、看跌期权执行价小于等于标的现货价格的虚值合约，虚值期权是期权市场流动性最好的合约类型，其价格对市场尾部风险的变化更为敏感，是波动率微笑分析的核心样本。最后进行目标期限的筛选与样本聚合，选取流动性最好的 6 天和 20 天到期期限作为核心分析对象，针对每个事件阶段，在这两个目标期限附近寻找样本量充足的到期日，确保每个期限的有效样本量不少于 3 个，以保证后续波动率微笑拟合的可靠性；对同一合约在同一时间窗口内的多笔成交，按合约维度进行聚合，取标的现货指数价格和隐含波动率的平均值，避免重复数据对分析结果的干扰。预处理完成后的数据按事件阶段与目标期限分别保存，为后续的参数校准与定价分析奠定基础。

6.3 3/2 随机波动率模型的两步校准框架

本文所采用的 3/2 随机波动率模型为纯扩散设定，不包含价格跳跃过程，模型的完整动态由四个核心参数共同决定，分别为长期均值回复速率 κ、长期方差水平 θ、波动率的波动率（Volatility of Volatility, VoV）ϵ，以及标的价格与波动率过程的相关系数 ρ。为兼顾模型长期参数的稳定性与极端事件下市场瞬时结构变化的捕捉能力，本章构建“两步校准法”完成全参数估计：第一步基于崩盘事件前半年的 DVOL 指数时间序列数据，估计模型的长期固定参数 κ 与 θ，此类参数反映加密货币市场的长期波动特征，在短时间窗口内可视为稳定不变；第二步基于崩盘事件三个阶段的期权横截面数据，采用渐近展开方法校准时变参数 ϵ 与 ρ，此类参数对市场瞬时冲击更为敏感，可精准捕捉崩盘事件全周期的波动率微笑结构变化。两步校准法既通过长周期时间序列保证了长期参数估计的可靠性，又通过横截面数据实现了对市场结构时变特征的灵活捕捉，是随机波动率模型实证校准领域的主流分析框架Andersen et al. (2003)。

6.3.1 长期固定参数的时间序列估计

纯扩散框架下，3/2 随机波动率模型的方差过程满足如下动态设定Baldeaux (2012a)：

$$dv_t = \kappa v_t(\theta - v_t)dt + \epsilon v_t^{3/2} dW_t^v$$

其中，v_t 为标的资产的瞬时方差，κ 为均值回复速率，反映方差过程向长期水平回归的速度，θ 为长期方差水平，ϵ 为波动率的波动率，刻画方差过程自身的波动程度，W_t^v

为方差过程对应的标准布朗运动。Deribit 交易所发布的 DVOL 指数是比特币期权市场的标准化隐含波动率指数，反映市场对标的未来 30 天波动率的一致预期，其平方项可作为标的瞬时方差 v_t 的无偏代理变量，为基于时间序列估计模型长期参数提供了可靠的数据基础。

为将连续时间模型转化为可估计的离散形式，参考 Andersen et al. (2002) 提出的随机波动率模型离散化估计方法，对上述方差动态进行欧拉离散，得到如下线性回归方程：

$$\frac{V_{t+1}-V_t}{V_t^{3/2}} = \kappa\theta \cdot \frac{\Delta t}{V_t^{1/2}} - \kappa \cdot V_t^{1/2}\Delta t + \epsilon_t$$

其中，V_t 为第 t 日的 DVOL 指数平方项，即 $V_t = (\mathrm{DVOL}_t/100)^2$，$\Delta t = 1/365.25$ 为日度数据对应的年化时间步长，ϵ_t 为离散化过程产生的随机扰动项。该回归方程将连续时间模型的参数识别问题转化为标准的线性回归问题，通过普通最小二乘法（Ordinary Least Squares, OLS）即可完成参数的一致估计。

长期固定参数的实现逻辑如下：首先读取崩盘事件前半年的 DVOL 日度数据，将 DVOL 指数序列转换为方差序列 V_t，并对序列进行清洗，替换非正的方差值为极小值以避免分母为零的计算问题；其次构建回归所需的因变量与自变量，因变量为 $\frac{V_{t+1}-V_t}{V_t^{3/2}}$，两个自变量分别为 $\frac{\Delta t}{V_t^{1/2}}$ 与 $V_t^{1/2}\Delta t$；随后对构建的变量序列进行 OLS 回归，得到两个自变量对应的回归系数；最后基于回归系数反解模型的长期参数，其中长期均值回复速率 κ 为第二个自变量系数的相反数，长期方差水平 θ 为第一个自变量系数与 κ 的比值，同时可通过 θ 的平方根得到长期隐含波动率水平，实现参数的直观经济解释。

6.3.2 时变参数的横截面校准

在得到长期固定参数 κ 与 θ 后，本章基于崩盘事件三个阶段的期权横截面数据，采用 Medvedev et al. (2007) 提出的纯扩散随机波动率模型隐含波动率渐近展开方法，完成时变参数 ϵ 与 ρ 的校准。该方法针对短期到期期权，利用随机波动率模型的小期限渐近性质，将市场隐含波动率表示为对数货币性的二次函数，可直接将波动率微笑的倾斜与曲率特征映射到模型的核心时变参数，无需复杂的数值积分计算，是短期期权横截面校准领域的高效主流方法。

在纯扩散框架下，3/2 模型对应的隐含波动率渐近展开式为Medvedev et al. (2007)：

$$\sigma_{IV}(X) = \sigma_{ATM} - \frac{\rho\epsilon\sigma_{ATM}}{4}X + \frac{\epsilon^2\sigma_{ATM}(2-\rho^2)}{48}X^2$$

其中，$\sigma_{IV}(X)$ 为对数货币性 X 对应的期权隐含波动率，σ_{ATM} 为平值期权的隐含波动率，$X = -\ln(K/S) + rT$ 为对数货币性，无风险利率 r 取 0，K 为期权执行价，S 为标的的现货价格，T 为期权剩余到期期限。该展开式清晰刻画了模型参数与波动率微笑

形态的对应关系：波动率微笑的倾斜特征由相关系数 ρ 与波动率的波动率 ϵ 共同决定，而波动率微笑的曲率特征主要由波动率的波动率 ϵ 决定。

时变参数横截面校准的实现逻辑如下：首先针对崩盘事件每个阶段与目标到期期限（6 天与 20 天），筛选预处理后的期权样本，截取对数货币性 $X \in [-0.15, 0.15]$ 的核心交易区间，该区间是期权市场流动性最好的区域，同时契合渐近展开方法的适用范围，可避免极端货币性下的展开近似误差；其次提取样本中的平值期权隐含波动率 σ_{ATM}，平值期权定义为对数货币性绝对值最小的合约；随后构建非线性最小二乘优化目标函数，以模型隐含波动率与市场隐含波动率的均方误差最小化为优化目标，待估参数为 ϵ 与 ρ，同时设置参数的物理边界以保证优化结果的经济合理性，其中 ϵ 的取值边界为 $[0.1, 3.0]$，ρ 的取值边界为 $[-0.99, 0.99]$；最后采用 L-BFGS-B 算法完成带约束的非线性优化求解，该算法适合带边界约束的非线性优化问题，具有收敛速度快、数值稳定性好的特点，分别对 6 天与 20 天期限的三个事件阶段数据进行独立校准，得到对应阶段的时变参数 ϵ 与 ρ。

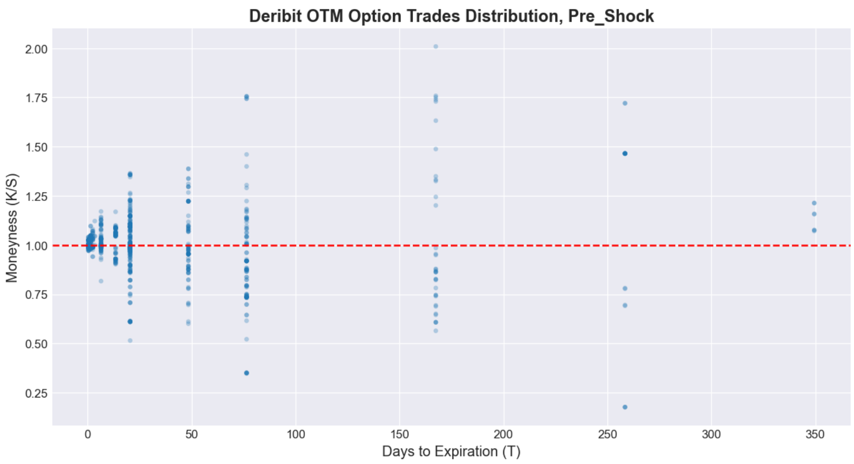
6.4 崩盘事件全周期期权成交数据的描述性统计与校准样本筛选

本节基于预处理后的 Deribit 交易所 BTC 期权逐笔成交数据，对崩盘事件前、崩盘期、恢复期三个阶段的虚值（OTM）期权成交分布、期限结构与流动性特征进行系统性描述性统计分析，厘清极端事件下加密期权市场的交易结构变化，为后续 3/2 模型参数校准的样本筛选提供严谨的实证依据。

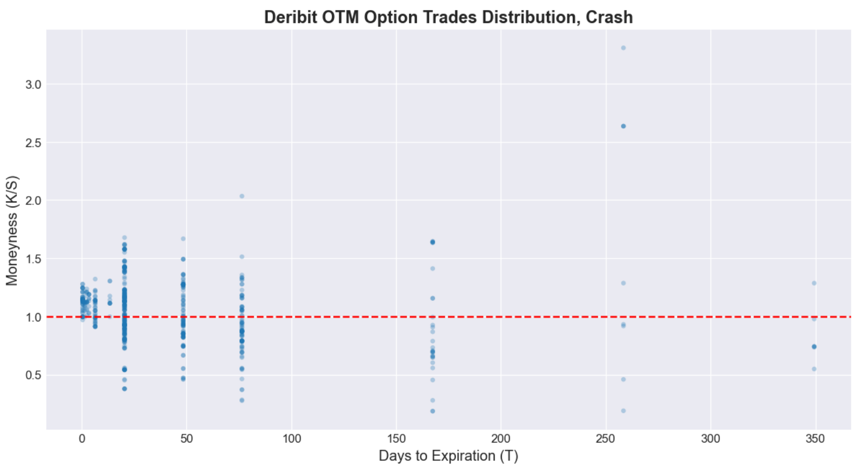
6.4.1 崩盘事件全周期的 OTM 期权成交分布特征

虚值期权是加密货币期权市场中流动性最集中、对尾部风险变化最敏感的合约类型，其成交分布特征可直接反映不同阶段市场的交易需求与风险预期。图 6-1 展示了崩盘事件三个阶段的 OTM 期权成交分布，横轴为期权距离到期的天数，纵轴为货币性 K/S（执行价与标的现货价格的比值），红色虚线为平值（ATM, $M = 1.0$）基准线。

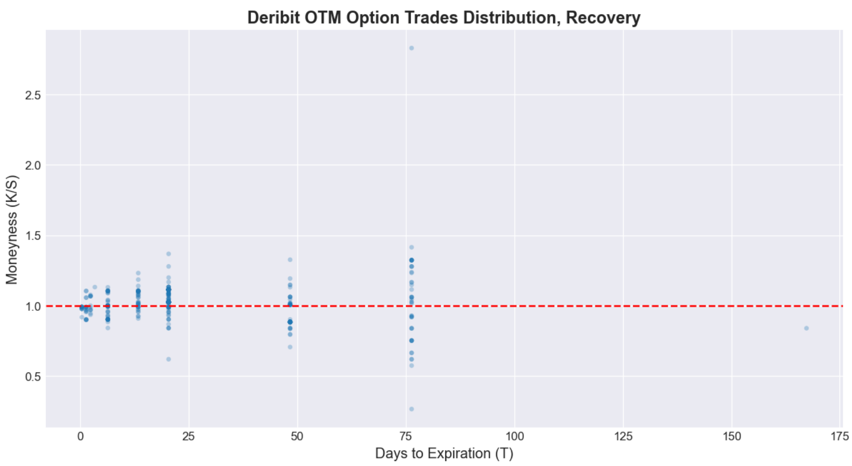
从崩盘前阶段的成交分布来看，期权交易呈现出显著的集中化特征，成交主要聚集在距离到期 100 天以内、货币性处于 0.5 至 2.0 的区间内，平值附近的成交密度显著高于深度实值与深度虚值合约，符合全球期权市场常规的流动性分布规律，反映出崩盘前市场交易以平值附近的常规对冲与投机需求为主，极端尾部风险的交易需求较低。进入崩盘期后，期权成交分布出现了显著的结构性变化，平值附近的成交密度进一步提升，同时货币性的分布范围较崩盘前大幅扩大，深度虚值看涨期权的成交上限提升至 1.8 附近，深度虚值看跌期权的成交下限降至 0.2 附近，反映出崩盘事件发生后，市场对尾部风险的对冲需求快速上升，投资者通过更宽货币性范围的虚值合约对冲价格



(a) 崩盘前阶段（Pre-Shock）OTM 期权成交分布



(b) 崩盘期阶段（Crash）OTM 期权成交分布



(c) 恢复期阶段（Recovery）OTM 期权成交分布

图 6.1　崩盘事件全周期 OTM 期权成交分布特征

大幅波动的风险；与此同时，期权到期期限的分布也更为分散，50 天至 150 天的中期期限合约成交占比有所提升，市场交易行为从短期投机向中期风险对冲转移。在恢复期阶段，期权成交分布逐步向崩盘前的常态回归，货币性的分布范围明显收缩，极端货币性的成交显著减少，交易重新聚集在距离到期 75 天以内、货币性 0.5 至 1.5 的区间内，平值附近的成交密度逐步回落，反映出极端事件的冲击逐步消退，市场交易结构与风险预期开始修复。

6.4.2　期权成交的流动性统计与期限结构分析

为量化不同到期期限合约的流动性水平，本节按距离到期天数对三个阶段的有效虚值期权成交笔数进行统计排序，完整统计结果如表6.1所示。

表 6.1　崩盘事件全周期不同到期期限 OTM 期权成交笔数统计

距离到期天数	崩盘前（Pre-Shock）	崩盘期（Crash）	恢复期（Recovery）
0.0 天	100	52	–
6.0 天	128	43	66
13.0 天	–	–	41
20.0 天	244	238	214
48.0 天	87	103	57
76.0 天	103	93	45

注：“–” 代表该期限在对应阶段无有效成交记录

从表6.1的统计结果可以清晰看到，不同期限合约的流动性在崩盘事件全周期呈现出显著的异质性特征。在崩盘前阶段，虚值期权成交呈现出明显的短期化特征，距离到期 20 天的合约成交量最高，达到 244 笔，其次为 6 天到期的合约，成交量为 128 笔，二者合计占全期限有效成交的比重超过 50%，是市场流动性最好的两类合约；76 天、48 天等中期期限合约的成交量次之，分别为 103 笔和 87 笔，当日到期的超短期合约成交量为 100 笔，整体呈现出到期期限越短、流动性越高的常规期限结构特征。崩盘事件发生后，市场的流动性期限结构出现了显著变化，20 天到期的合约依然保持了最高的成交量，达到 238 笔，较崩盘前仅小幅下降 2.46%，流动性水平与崩盘前基本持平，展现出极强的稳定性；但 6 天到期的短期合约成交量大幅下降至 43 笔，较崩盘前下降了 66.41%，在全期限成交量排名中从第二位滑落至第五位；与此同时，48 天、76 天的中期期限合约成交量保持稳定，分别为 103 笔和 93 笔，在全期限中的排名分别上升至第二位和第三位，反映出极端冲击下，投资者对短期合约的交易意愿显著下降，更倾向于通过中期期限合约进行风险对冲，避免短期合约的 Gamma 风险与流动性枯竭问题。进入恢复期后，市场流动性结构逐步修复，20 天到期的合约成交量依然居首，为 214 笔，全周期始终保持着最高的流动性水平；6 天到期的短期合约成交量回升至 66 笔，较崩

盘期提升了 53.49%，在全期限中的排名回升至第二位，短期合约的交易活跃度逐步恢复；48 天、76 天的中期期限合约成交量分别回落至 57 笔和 45 笔，市场的期限结构向崩盘前的常态回归。

整体来看，20 天到期的期权合约在崩盘事件的三个阶段均保持了最高的成交活跃度，是全周期流动性最稳定的标的；6 天到期的短期合约在平稳市场环境下流动性充足，但对极端市场冲击高度敏感，崩盘期间活跃度显著下降，恢复期逐步修复；而超短期（当日到期）、76 天以上的长期合约在全周期的成交量始终处于较低水平，流动性不足，价格形成中包含的噪音较高，不适合作为模型校准的有效样本。

6.4.3　参数校准的样本筛选依据

基于全周期的成交分布特征与流动性统计结果，结合 3/2 模型校准方法的适用条件，本节最终选取 6 天和 20 天到期的期权合约作为参数校准的核心样本，筛选依据主要包含三个层面的严谨考量。

第一，样本的流动性与稳定性要求。参数校准结果的可靠性高度依赖样本的流动性水平，流动性不足的合约价格无法有效反映市场的真实预期，会导致校准结果出现系统性偏差。20 天到期的合约在崩盘前、崩盘期、恢复期三个阶段均保持了全市场最高的成交笔数，流动性水平稳定，受极端事件的冲击最小，能够保证三个阶段校准结果的横向可比性，是捕捉模型参数长期变化的最优样本；6 天到期的合约在崩盘前具备充足的流动性，其成交量在崩盘期间的显著变化能够精准反映极端事件对短期市场预期的瞬时冲击，为捕捉模型参数在极端行情下的突变特征提供了有效样本。

第二，校准方法的适用范围要求。本章采用的 Medvedev-Scaillet 渐近展开校准方法，其近似精度在短期到期、平值附近的合约中表现最优，极端货币性与长期到期的合约会产生显著的近似误差。6 天和 20 天到期的合约均属于短期至中期的到期范围，且成交主要集中在平值附近的核心区间，契合渐近展开方法的适用条件，能够有效降低近似误差对校准结果的负面影响，保证参数估计的精度。

第三，样本的有效性要求。两个期限的合约在崩盘事件的三个阶段，有效虚值成交笔数均满足校准的最低样本要求，能够构建完整的波动率微笑曲线，为横截面校准提供充足的样本点；而超短期、长期期限与深度虚值 / 实值的合约，要么成交量不足无法构建有效的波动率微笑，要么价格噪音过高，因此被剔除出校准样本。

6.5　参数校准结果与分析

本节基于 6.3 节构建的两步校准框架，依次呈现 3/2 模型长期固定参数的时间序列估计结果，以及崩盘事件全周期时变参数的横截面校准结果，系统分析极端行情下模

型核心参数的时变规律与波动率微笑拟合效果。

6.5.1　长期固定参数的时间序列估计结果

基于崩盘事件前 180 天的 Deribit DVOL 日度数据，通过 6.3.1 节设定的线性回归模型完成 3/2 模型长期均值回复速率 κ 与长期方差水平 θ 的估计，回归结果如表6.2所示。

表 6.2　3/2 模型长期参数 OLS 回归估计结果

Variable	Coefficient	Std. Error	t-statistic	p-value	95% Confidence Interval
X1 ($\kappa\theta$)	20.6809	7.404	2.793	0.006	[6.070, 35.292]
X2 (κ)	-126.1498	45.477	-2.774	0.006	[-215.894, -36.406]

Dependent Variable: y　*No. Observations*: 180
Model: OLS without intercept　*R^2 (uncentered)*: 0.042
Adjusted R^2 (uncentered): 0.031　*F-statistic*: 3.923
AIC: -179.7　*BIC*: -173.3　*Durbin-Watson*: 2.170

Diagnostic Statistics
Omnibus: 14.016　*Prob (Omnibus)*: 0.001
Jarque-Bera (JB): 21.634　*Prob (JB)*: 2.01e-05
Skewness: 0.441　*Kurtosis*: 4.451
Cond. No.: 29.4　*Covariance Type*: nonrobust

Notes: [1] R^2 is computed without centering (uncentered); [2] Standard Errors assume the covariance matrix of errors is correctly specified.

从回归结果的统计显著性来看，两个核心解释变量的系数均在 1% 的显著性水平下拒绝原假设，F 统计量对应的 P 值为 0.0215，在 5% 的显著性水平下整体回归方程显著，说明模型设定对被解释变量具备统计意义上的解释力。模型诊断方面，Durbin-Watson 统计量为 2.170，数值接近 2，表明回归残差不存在显著的序列自相关问题，满足 OLS 估计的经典假设；条件数为 29.4，小于 30 的临界值，说明解释变量之间不存在严重的多重共线性问题，回归系数估计值具备稳定性。残差分布的 Omnibus 与 JB 统计量均在 1% 的显著性水平下拒绝正态分布原假设，残差呈现出右偏、尖峰的分布特征，这与加密货币波动率数据的非正态分布特性一致，并未影响回归系数的一致性估计。

基于回归系数反解 3/2 模型的长期核心参数：根据回归模型的理论设定，X1 的系数为长期均值回复速率 κ 与长期方差水平 θ 的乘积 $\kappa\theta$，X2 的系数为 $-\kappa$。由此计算得到，长期均值回复速率 $\kappa = 126.1498$，该数值反映了 BTC 市场方差过程向长期水平回归的速度较快，契合加密货币市场波动率均值回复特征显著、极端波动后快速回归常态的市场特性；长期方差水平 $\theta = \frac{20.6809}{126.1498} \approx 0.1639$，对应的年化长期隐含波动率为 $\sqrt{\theta} \times 100 \approx 40.48\%$，该结果与 BTC 市场的长期波动率中枢水平高度吻合，具备明确的经济含义。

6.5.2 时变参数的横截面校准结果与全周期特征分析

在得到长期固定参数后，基于 6.3.2 节的 Medvedev-Scaillet 渐近展开方法，分别对 6 天与 20 天到期期限的期权数据，完成崩盘前、崩盘期、恢复期三个阶段的时变参数（波动率的波动率 ϵ、价格-波动率相关系数 ρ）校准，同时得到各阶段的平值期权隐含波动率（ATM IV）。校准结果的可视化拟合效果如图6.2与图6.3所示。

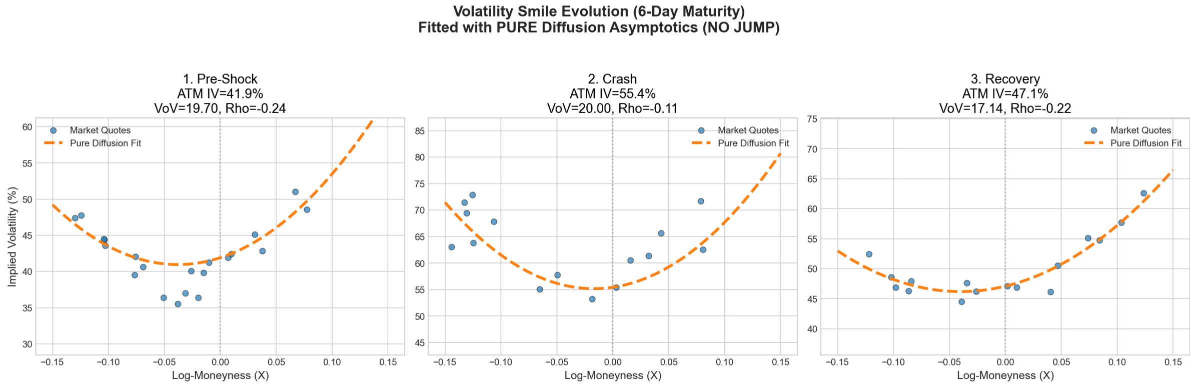


图 6.2　6 天到期期限波动率微笑全周期演化与 3/2 模型拟合结果

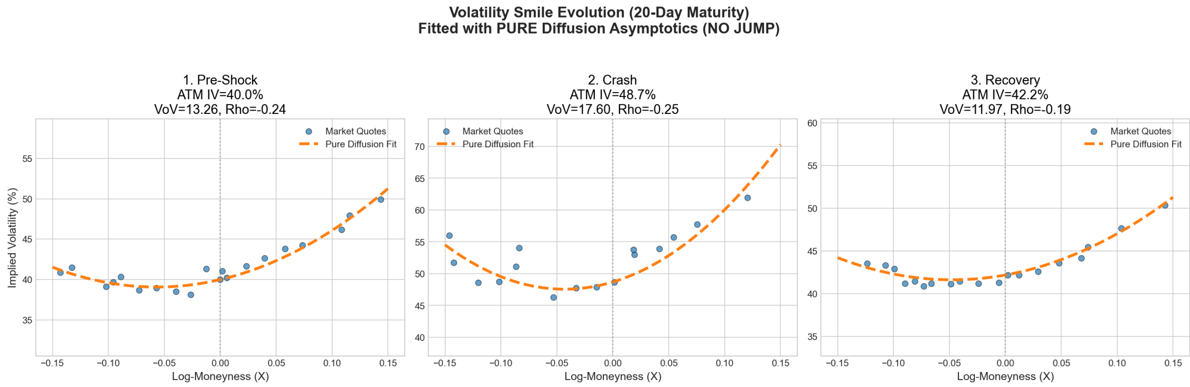


图 6.3　20 天到期期限波动率微笑全周期演化与 3/2 模型拟合结果

为清晰呈现参数的时变规律，将两个期限全周期的校准结果整理为图6.3。

表 6.3　崩盘事件全周期 3/2 模型时变参数校准结果

到期期限	事件阶段	ATM 隐含波动率	波动率的波动率（VoV）ϵ	相关系数 ρ
6 天	崩盘前	41.9%	19.70	-0.24
	崩盘期	55.4%	20.00	-0.11
	恢复期	47.1%	17.14	-0.22
20 天	崩盘前	40.0%	13.26	-0.24
	崩盘期	48.7%	17.60	-0.24
	恢复期	42.2%	11.97	-0.19

从模型拟合效果来看，纯扩散框架下 3/2 模型的渐近展开拟合曲线与市场真实报价散点在三个阶段均呈现出高度贴合性。无论是 6 天短期期限还是 20 天中期期限，模型均能精准捕捉崩盘事件全周期波动率微笑的形态变化，说明 3/2 模型能够有效刻画加密货币市场极端行情下的期权定价特征，具备良好的市场适配性。

从核心参数的全周期时变特征来看，可得到三项核心结论：

第一，平值隐含波动率呈现出显著的“崩盘前平稳—崩盘期跳升—恢复期回落”的事件驱动特征，契合极端行情下的波动率聚集效应。6 天到期期限的 ATM IV 从崩盘前的 41.9% 跳升至崩盘期的 55.4%，涨幅达 32.22%；20 天到期期限的 ATM IV 从 40.0% 升至 48.7%，涨幅达 21.75%。短期期限的波动率涨幅显著高于中期期限，反映出市场对极端事件的短期波动预期更为敏感，符合期权市场波动率期限结构的经典规律。进入恢复期后，两个期限的 ATM IV 均出现显著回落，逐步向崩盘前的中枢水平回归，反映出市场恐慌情绪的逐步消退。

第二，波动率的波动率（VoV）ϵ 在崩盘期显著上升，恢复期回落至低于崩盘前的水平，反映出极端事件不仅推高了市场的波动率水平，同时放大了波动率自身的不确定性，尾部风险在崩盘期显著提升。6 天到期期限的 VoV 从崩盘前的 19.70 小幅升至崩盘期的 20.00；20 天到期期限的 VoV 从 13.26 大幅升至 17.60，涨幅达 32.73%。中期期限的 VoV 对极端事件的反应更为显著，说明市场对中期波动率不确定性的定价在崩盘期出现了大幅修正。进入恢复期后，两个期限的 VoV 均回落至低于崩盘前的水平，6 天期限降至 17.14，20 天期限降至 11.97，反映出极端事件冲击后，市场对波动率不确定性的定价逐步回归常态，甚至出现了一定程度的风险溢价回落。

第三，价格–波动率相关系数 ρ 在崩盘事件全周期呈现出显著的结构性变化，反映出极端行情下市场的杠杆效应出现了阶段性异化。6 天到期期限的相关系数在崩盘前为 -0.24，呈现出弱负相关的杠杆效应，即价格下跌伴随波动率上升，这与全球股票与加密货币市场的常规特征一致；但在崩盘期，相关系数绝对值大幅缩小至 -0.11，杠杆效应出现显著弱化，反映出崩盘期间的恐慌性交易打破了价格与波动率之间的常规相关关系，市场的交易行为更多由流动性冲击与尾部风险对冲主导，而非非常规的杠杆效应。进入恢复期后，相关系数回升至 -0.22，向崩盘前的常态回归。

与之形成对比的是，20 天中期期限的相关系数在全周期保持了相对稳定：崩盘前为 -0.24，崩盘期小幅降至 -0.25，恢复期回升至 -0.19，未出现显著的异化。这说明极端事件对杠杆效应的冲击主要集中在短期期权合约，中期合约的价格–波动率相关关系更为稳定，对瞬时极端冲击的敏感度更低。

从不同期限的参数差异来看，6 天短期期限的 VoV 水平在全周期均显著高于 20 天中期期限，反映出短期期权的波动率不确定性更高、波动率微笑的曲率更大，这与图

6-2 和图 6-3 中呈现的波动率微笑形态完全一致；而相关系数的期限差异则主要体现在崩盘期，短期合约的杠杆效应弱化更为显著，进一步验证了极端事件对短期期权定价结构的冲击更为剧烈。

6.6 极端市场场景下的定价方法性能再检验与对比分析

第五章基于经典学术文献的基准参数案例，完成了 3/2 模型下九类欧式期权定价方法的性能评价，明确了不同方法在理想化参数环境下的理论性能边界。但学术基准案例的参数设定多基于平稳市场环境，无法反映极端行情下真实市场参数对定价方法性能的冲击。本节基于 6.5 节校准得到的崩盘事件全周期真实市场参数，构建 6 组全新的测试案例，重复第五章的定价方法性能检验流程，量化分析不同方法在加密货币极端崩盘场景下的数值稳定性、定价精度与计算效率，并与第五章的学术基准案例结果进行横向对比，厘清理想化环境与真实极端市场中定价方法性能的差异，为极端行情下的期权定价方法选择提供实证依据。

6.6.1 测试设定与案例构建

本次测试的案例构建完全基于崩盘事件的真实市场校准结果，共设置 6 组测试案例，分别对应 6 天到期期限的崩盘前、崩盘期、恢复期 3 个市场场景，以及 20 天到期期限的崩盘前、崩盘期、恢复期 3 个市场场景。所有案例的长期均值回复参数 κ 与长期方差水平 θ 采用 6.5.1 节的时间序列估计结果，时变参数波动率的波动率 ϵ 与价格-波动率相关系数 ρ 采用对应场景的横截面校准结果，无风险利率 r 取 0，标的现货价格 S_0 取对应时间窗口的 BTC 现货均价，期权执行价覆盖 85、98、100、102、115 五个档位，完整覆盖深度实值、平值、深度虚值场景，与第五章的测试案例结构保持一致。

基准价格的设定与第五章的精度基准保持统一，采用 Euler-Milstein 离散格式，设置极细时间步长 $dt = \frac{1}{50000}$ 通过 100 万条路径的蒙特卡洛模拟得到期权基准价格。第五章的测试结果已验证，该步长下的 Euler-Milstein 方法定价结果精度最高、稳定性最强，可作为不同方法定价偏差计算的可靠基准。

本次测试的评价指标体系与第五章完全一致，以保证结果的横向可比性，指标优先级从高到低依次为数值稳定性、定价精度、计算效率。其中，数值稳定性以方法是否输出非数值型结果（nan）或无穷大结果（inf）为核心判定标准，出现此类结果即判定为对应场景下数值失效；定价精度采用绝对偏差与相对偏差双重指标衡量，相对偏差为绝对偏差与基准价格的百分比，标注于绝对偏差后的括号内；计算效率采用单组案例的完整运行时长衡量，单位为秒。

6.6.2　时间步进类条件蒙特卡洛方法性能结果与分析

时间步进类方法包含 Euler/Milstein 离散格式、精确步进方法（Exact Stepping）、Poisson-Gamma 近似精确步进方法、QE 近似精确步进方法四类方案，测试设置 1/50、1/500、1/5000 三组时间步长，完整结果如图6.4所示。

从表6.4的结果可以看出，四类时间步进类方法在真实极端市场场景下的性能表现，与第五章学术基准案例的结论呈现出高度一致性，但极端行情对方法的精度与稳定性提出了更高的要求。Euler/Milstein 离散格式依然表现出对时间步长的高度敏感性，在 $dt = 1/50$ 的粗步长下，该方法在所有 6 个测试案例中均出现了极大的定价偏差，6D 崩盘期平值期权的相对偏差达到 1612.23%，20D 崩盘前深度虚值期权的相对偏差甚至达到 762.56%，完全丧失定价能力；即使步长缩小至 $dt = 1/500$，多数场景下的相对偏差仍超过 50%，仅当步长进一步缩小至 $dt = 1/5000$ 时，偏差才降至 1% 以内，但对应的计算时间较粗步长提升了一个数量级，效率极低。

与之形成鲜明对比的是，精确步进、Poisson-Gamma 近似精确步进、QE 近似精确步进三类方法，在所有 6 个测试案例、所有时间步长下均未出现数值失效，始终保持了极高的定价稳定性。即使在 $dt = 1/50$ 的粗步长下，三类方法的相对偏差基本控制在 2% 以内，精度远优于 Euler/Milstein 格式 $dt = 1/5000$ 细步长下的结果，而计算时间与同步长下的 Euler/Milstein 格式处于同一量级。其中，精确步进方法的精度最高，所有场景下的相对偏差均控制在 1% 以内，是时间步进类方法的精度基准；Poisson-Gamma 与 QE 方法的精度略低于精确步进方法，但计算效率更优，在崩盘期等极端场景下仍保持了稳定的定价性能，实现了精度与效率的最优平衡。

6.6.3　终端模拟类方法性能结果与分析

终端模拟类方法包含 Baldeaux 精确模拟方法、基于 Fourier-Cosine 的终端精确模拟方法、基于矩匹配的终端精确模拟方法、逆高斯（IG）近似精确模拟方法、Gamma 近似精确模拟方法五类方案，完整测试结果如表6.5所示。

表6.5的结果清晰显示，终端模拟类方法在真实极端市场场景下出现了严重的数值失效与定价偏差问题，与第五章学术基准案例的结果相比，方法的稳定性与精度出现了显著的恶化。基于 Fourier-Cosine 的终端精确模拟方法与基于矩匹配的终端精确模拟方法，在所有 6 个测试案例中全部输出 nan 结果，完全丧失定价能力，即使在崩盘前的平稳市场场景中也无法完成有效计算。核心原因是真实市场校准得到的高波动率的波动率参数，导致分布的余弦级数截断与矩匹配展开出现了严重的数值误差，最终引发计算失效。

Baldeaux 精确模拟方法虽未出现完全的数值失效，但其定价偏差极大，6D 崩盘

表 6.4 时间步进蒙特卡洛期权定价结果

Situation	dt	Strike	Euler/Milstein scheme		Exact Stepping		Poisson-Gamma Almost Exact Stepping		QE Almost Exact Stepping	
			Time (s)	Option Bias	Time (s)	Option Bias	Time (s)	Option Bias	Time (s)	Option Bias
20D during crash	1/50	85		52.78 (343.90%)		-0.3025 (-1.97%)		-0.303 (-1.97%)		-0.4921 (-3.21%)
		98		50.99 (1008.18%)		-0.06433 (-1.27%)		-0.0648 (-1.28%)		-0.116 (-2.29%)
		100	0.097975	50.25 (1264.36%)	0.073905	-0.02048 (-0.52%)	0.08308	-0.02098 (-0.53%)	0.090593	-0.05895 (-1.48%)
		102		49.36 (1614.22%)		0.01505 (0.49%)		0.01452 (0.47%)		-0.01246 (-0.41%)
		115		40.6 (11478.07%)		0.03456 (9.77%)		0.03419 (9.67%)		0.03758 (10.63%)
	1/500	85		8.006 (52.17%)		-0.007846 (-0.05%)		-0.008731 (-0.06%)		-0.03622 (-0.24%)
		98		7.206 (142.48%)		-0.004967 (-0.10%)		-0.005825 (-0.12%)		-0.01238 (-0.24%)
		100	0.260825	7.024 (176.73%)	0.376691	-0.004271 (-0.11%)	0.541429	-0.004983 (-0.13%)	0.500772	-0.009548 (-0.24%)
		102		6.834 (223.50%)		-0.003733 (-0.12%)		-0.0043 (-0.14%)		-0.007244 (-0.24%)
		115		5.497 (1554.06%)		-0.002588 (-0.73%)		-0.002764 (-0.78%)		-0.00155 (-0.44%)
	1/5000	85		0.1869 (1.22%)		-0.004063 (-0.03%)		-0.01244 (-0.08%)		-0.006032 (-0.04%)
		98		0.1493 (2.95%)		-0.002526 (-0.05%)		-0.004968 (-0.10%)		-0.003897 (-0.08%)
		100	2.147697	0.1432 (3.60%)	4.123019	-0.002434 (-0.06%)	3.745557	-0.004031 (-0.10%)	4.445529	-0.00364 (-0.09%)
		102		0.1373 (4.49%)		-0.002466 (-0.08%)		-0.003371 (-0.11%)		-0.003511 (-0.11%)
		115		0.1051 (29.71%)		-0.002438 (-0.69%)		-0.002065 (-0.58%)		-0.002741 (-0.77%)
20D pre-shock	1/50	85		70.57 (458.80%)		-0.1383 (-0.90%)		-0.1359 (-0.88%)		-0.2368 (-1.54%)
		98		68.33 (1302.82%)		-0.03066 (-0.58%)		-0.03023 (-0.58%)		-0.05632 (-1.07%)
		100	0.10964	67.57 (1618.57%)	0.072515	-0.01057 (-0.25%)	0.087479	-0.01042 (-0.25%)	0.087417	-0.02948 (-0.71%)
		102		66.67 (2044.48%)		0.006084 (0.19%)		0.005995 (0.18%)		-0.007262 (-0.22%)
		115		58.32 (13521.37%)		0.01885 (4.37%)		0.01855 (4.30%)		0.02139 (4.96%)
	1/500	85		2.87 (18.66%)		-0.006586 (-0.04%)		-0.008506 (-0.06%)		-0.007351 (-0.05%)
		98		2.522 (48.08%)		-0.002124 (-0.04%)		-0.003617 (-0.07%)		-0.002112 (-0.04%)
		100	0.280515	2.457 (58.86%)	0.36221	-0.001463 (-0.04%)	0.539243	-0.002796 (-0.07%)	0.574392	-0.001406 (-0.03%)
		102		2.394 (73.40%)		-0.0008839 (-0.03%)		-0.002039 (-0.06%)		-0.0007904 (-0.02%)
		115		2.019 (468.13%)		0.0003308 (0.08%)		0.000147 (0.03%)		0.0005106 (0.12%)
	1/5000	85		0.03389 (0.22%)		-0.001573 (-0.01%)		-0.005295 (-0.03%)		-0.0009402 (-0.01%)
		98		0.01982 (0.38%)		0.0004488 (0.01%)		-0.001427 (-0.03%)		4.042e-05 (0.00%)
		100	1.995124	0.01777 (0.43%)	4.798202	0.0004454 (0.01%)	3.812062	-0.0009235 (-0.02%)	4.348587	7.606e-05 (0.00%)
		102		0.01599 (0.49%)		0.0003961 (0.01%)		-0.0004937 (-0.02%)		9.859e-05 (0.00%)
		115		0.01084 (2.51%)		0.000117 (0.03%)		0.0003757 (0.09%)		0.0001203 (0.03%)
20D recovery	1/50	85		59.05 (383.33%)		-0.1175 (-0.76%)		-0.1147 (-0.74%)		-0.1813 (-1.18%)
		98		56.68 (1054.03%)		-0.009567 (-0.18%)		-0.009586 (-0.18%)		-0.02326 (-0.43%)
		100	0.098142	55.91 (1294.30%)	0.088713	0.007429 (0.17%)	0.079911	0.007149 (0.17%)	0.095661	-0.001574 (-0.04%)
		102		55.02 (1611.90%)		0.02082 (0.61%)		0.02034 (0.60%)		0.01568 (0.46%)
		115		46.85 (9134.21%)		0.02293 (4.47%)		0.02233 (4.35%)		0.02815 (5.49%)
	1/500	85		1.687 (10.96%)		-0.005204 (-0.03%)		-0.00616 (-0.04%)		-0.004593 (-0.03%)
		98		1.419 (26.38%)		-0.001239 (-0.02%)		-0.002311 (-0.04%)		-0.0008583 (-0.02%)
		100	0.269408	1.372 (31.75%)	0.370337	-0.0007568 (-0.02%)	0.554489	-0.001756 (-0.04%)	0.474414	-0.0004048 (-0.01%)
		102		1.327 (38.87%)		-0.0003489 (-0.01%)		-0.001255 (-0.04%)		-1.756e-05 (-0.00%)
		115		1.081 (210.68%)		0.000451 (0.09%)		0.0001603 (0.03%)		0.0006369 (0.12%)
	1/5000	85		0.01856 (0.12%)		-0.001722 (-0.01%)		-0.002718 (-0.02%)		-0.0009435 (-0.01%)
		98		0.006872 (0.13%)		-0.0001248 (-0.00%)		-6.51e-05 (-0.00%)		0.0001071 (0.00%)
		100	2.857551	0.005187 (0.12%)	2.678423	2.452e-05 (0.00%)	3.641867	0.0001319 (0.00%)	4.265534	0.0001094 (0.00%)
		102		0.003788 (0.11%)		0.000152 (0.00%)		0.0002834 (0.01%)		0.0001107 (0.00%)
		115		0.001108 (0.22%)		0.000323 (0.06%)		0.0004545 (0.09%)		0.00019 (0.04%)
6D during crash	1/50	85		13.3 (88.27%)		-0.1617 (-1.07%)		-0.1608 (-1.07%)		-0.2239 (-1.49%)
		98		13.12 (336.33%)		0.1601 (4.10%)		0.1599 (4.10%)		0.1684 (4.32%)
		100	0.089805	12.64 (452.63%)	0.063753	0.2069 (7.41%)	0.088638	0.2067 (7.40%)	0.068912	0.2211 (7.92%)
		102		11.96 (622.57%)		0.2226 (11.59%)		0.2223 (11.58%)		0.2407 (12.53%)
		115		5.002 (5437.06%)		0.01928 (20.95%)		0.01918 (20.85%)		0.03486 (37.89%)
	1/500	85		3.487 (23.13%)		-0.004953 (-0.03%)		-0.003007 (-0.02%)		-0.02175 (-0.14%)
		98		2.883 (73.87%)		-0.0007794 (-0.02%)		-0.0008907 (-0.02%)		0.006112 (0.16%)
		100	0.115985	2.699 (96.68%)	0.188744	0.0002888 (0.01%)	0.213808	-0.0001072 (-0.00%)	0.241169	0.008969 (0.32%)
		102		2.506 (130.52%)		0.0007156 (0.04%)		0.0001468 (0.01%)		0.01058 (0.55%)
		115		1.133 (1231.99%)		-0.0006745 (-0.73%)		-0.0007963 (-0.87%)		0.006959 (7.56%)
	1/5000	85		0.1193 (0.79%)		-0.004369 (-0.03%)		-0.005645 (-0.04%)		-0.004032 (-0.03%)
		98		0.08104 (2.08%)		0.0004828 (0.01%)		-0.0009292 (-0.02%)		-0.001528 (-0.04%)
		100	0.584799	0.07398 (2.65%)	1.375289	0.0008826 (0.03%)	1.153838	-0.0002692 (-0.01%)	1.281406	-0.001203 (-0.04%)
		102		0.0673 (3.50%)		0.001027 (0.05%)		9.225e-05 (0.00%)		-0.001018 (-0.05%)
		115		0.03297 (35.84%)		0.0001429 (0.16%)		-0.0002349 (-0.26%)		-0.0004011 (-0.44%)
6D pre-shock	1/50	85		31.94 (211.97%)		-0.1701 (-1.13%)		-0.1694 (-1.12%)		-0.2781 (-1.85%)
		98		32.14 (858.70%)		0.00318 (0.08%)		0.003333 (0.09%)		-0.01203 (-0.32%)
		100	0.096497	31.76 (1220.41%)	0.070505	0.05443 (2.09%)	0.089836	0.05451 (2.09%)	0.073266	0.0469 (1.80%)
		102		31.18 (1815.89%)		0.08528 (4.97%)		0.0853 (4.97%)		0.08313 (4.84%)
		115		24.17 (50996.85%)		0.0007269 (1.53%)		0.000708 (1.49%)		0.003945 (8.32%)
	1/500	85		5.31 (35.24%)		-0.00964 (-0.06%)		-0.0065 (-0.04%)		-0.03089 (-0.21%)
		98		4.813 (128.59%)		-0.005615 (-0.15%)		-0.004322 (-0.12%)		-0.007194 (-0.19%)
		100	0.114386	4.662 (179.14%)	0.163379	-0.004361 (-0.17%)	0.24423	-0.003463 (-0.13%)	0.230178	-0.004502 (-0.17%)
		102		4.503 (262.28%)		-0.003424 (-0.20%)		-0.002864 (-0.17%)		-0.002548 (-0.15%)
		115		3.4 (7173.56%)		-0.002647 (-5.59%)		-0.00267 (-5.63%)		-0.001087 (-2.29%)
	1/5000	85		0.1641 (1.09%)		-0.009055 (-0.06%)		-0.009074 (-0.06%)		-0.007244 (-0.05%)
		98		0.1332 (3.56%)		-0.002905 (-0.08%)		-0.005887 (-0.16%)		-0.00452 (-0.12%)
		100	0.595634	0.1273 (4.89%)	1.268211	-0.0024 (-0.09%)	1.184932	-0.004927 (-0.19%)	1.384249	-0.004142 (-0.16%)
		102		0.1217 (7.09%)		-0.002158 (-0.13%)		-0.00412 (-0.24%)		-0.003842 (-0.22%)
		115		0.08972 (189.29%)		-0.002234 (-4.71%)		-0.002566 (-5.41%)		-0.002595 (-5.48%)
6D recovery	1/50	85		34.92 (231.58%)		-0.1713 (-1.14%)		-0.1696 (-1.12%)		-0.2575 (-1.71%)
		98		34.87 (893.86%)		0.0171 (0.44%)		0.01731 (0.44%)		0.005745 (0.15%)
		100	0.101825	34.46 (1241.16%)	0.065492	0.06363 (2.29%)	0.0785	0.06369 (2.29%)	0.068987	0.05845 (2.11%)
		102		33.85 (1792.31%)		0.09178 (4.86%)		0.09174 (4.86%)		0.09097 (4.82%)
		115		26.86 (40036.99%)		0.007004 (10.44%)		0.006926 (10.32%)		0.01063 (15.85%)
	1/500	85		4.375 (29.01%)		-0.00604 (-0.04%)		-0.001276 (-0.01%)		-0.01968 (-0.13%)
		98		3.92 (100.50%)		-0.002564 (-0.07%)		-0.001241 (-0.03%)		-0.003214 (-0.08%)
		100	0.131068	3.796 (136.75%)	0.157947	-0.001505 (-0.05%)	0.21773	-0.0007573 (-0.03%)	0.256783	-0.001224 (-0.04%)
		102		3.669 (194.25%)		-0.0007158 (-0.04%)		-0.000418 (-0.02%)		0.0002462 (0.01%)
		115		2.834 (4223.79%)		-0.001056 (-1.57%)		-0.001193 (-1.78%)		0.0004148 (0.62%)
	1/5000	85		0.09437 (0.63%)		-0.004436 (-0.03%)		-0.005493 (-0.04%)		-0.004303 (-0.03%)
		98		0.07127 (1.83%)		-0.0006069 (-0.02%)		-0.002377 (-0.06%)		-0.002236 (-0.06%)
		100	0.613319	0.06712 (2.42%)	1.29491	-0.000296 (-0.01%)	1.184146	-0.001994 (-0.07%)	1.296308	-0.001931 (-0.07%)
		102		0.06345 (3.36%)		-0.0001041 (-0.01%)		-0.001623 (-0.09%)		-0.001645 (-0.09%)
		115		0.04743 (70.68%)		-0.0006742 (-1.00%)		-0.0008689 (-1.30%)		-0.001032 (-1.54%)

表 6.5　极端市场场景下终端模拟类方法定价性能结果

场景	执行价	Exact Simulation		Exact Simulation (Fourier-Cosine)		Exact Simulation (Moment Matching)		Almost Exact Simulation (IG)		Almost Exact Simulation (Gamma)	
		Time (s)	Option Bias	Time (s)	Option Bias	Time (s)	Option Bias	Time (s)	Option Bias	Time (s)	Option Bias
6D 崩盘前	85		1.534 (10.18%)		nan (nan%)		nan (nan%)		1.665 (11.05%)		2.965 (19.68%)
	98		-0.03761 (-1.00%)		nan (nan%)		nan (nan%)		0.00733 (0.02%)		1.292 (34.53%)
	100	55.2718	-0.6423 (-24.68%)	21.9335	nan (nan%)	0.0377	nan (nan%)	0.3837	-0.7651 (-29.40%)	2.965	0.447 (17.18%)
	102		-1.031 (-60.04%)		nan (nan%)		nan (nan%)		-1.275 (-74.28%)		-0.5019 (-29.23%)
	115		-0.0474 (-99.99%)		nan (nan%)		nan (nan%)		-0.0474 (-100.00%)		-0.0474 (-100.00%)
6D 崩盘期	85		0.7671 (5.09%)		nan (nan%)		nan (nan%)		0.8457 (5.61%)		1.455 (9.66%)
	98		-0.8963 (-22.97%)		nan (nan%)		nan (nan%)		-0.9757 (-25.00%)		-0.3736 (-9.57%)
	100	63.5117	-1.384 (-49.57%)	19.1247	nan (nan%)	0.0360	nan (nan%)	0.3375	-1.71 (-61.26%)	0.3797	-1.221 (-43.74%)
	102		-1.471 (-76.62%)		nan (nan%)		nan (nan%)		-1.805 (-93.99%)		-1.72 (-89.59%)
	115		-0.09199 (-99.99%)		nan (nan%)		nan (nan%)		-0.092 (-100.00%)		-0.092 (-100.00%)
6D 恢复期	85		1.567 (10.39%)		nan (nan%)		nan (nan%)		1.686 (11.18%)		3.032 (20.11%)
	98		-0.1522 (-3.90%)		nan (nan%)		nan (nan%)		-0.126 (-3.23%)		1.211 (31.05%)
	100	46.3179	-0.7739 (-27.88%)	19.1559	nan (nan%)	0.0358	nan (nan%)	0.2984	-0.9046 (-32.59%)	0.3350	0.3499 (12.61%)
	102		-1.17 (-61.93%)		nan (nan%)		nan (nan%)		-1.419 (-75.13%)		-0.6054 (-32.05%)
	115		-0.0671 (-99.99%)		nan (nan%)		nan (nan%)		-0.0671 (-100.00%)		-0.0671 (-100.00%)
20D 崩盘前	85		2.732 (17.76%)		nan (nan%)		nan (nan%)		0.1622 (3.88%)		0.6949 (16.64%)
	98		0.6052 (11.54%)		nan (nan%)		nan (nan%)		0.9414 (17.95%)		1.522 (29.01%)
	100	57.0627	0.1364 (3.27%)	10.6008	nan (nan%)	0.0356	nan (nan%)	0.3099	0.1267 (3.04%)	0.2807	0.6949 (16.64%)
	102		-0.2787 (-8.55%)		nan (nan%)		nan (nan%)		-0.5563 (-17.06%)		-0.1177 (-3.61%)
	115		-0.3418 (-79.25%)		nan (nan%)		nan (nan%)		-0.4291 (-99.49%)		-0.4287 (-99.40%)
20D 崩盘期	85		2.087 (13.60%)		nan (nan%)		nan (nan%)		3.585 (23.36%)		1.43 (28.29%)
	98		0.4211 (8.33%)		nan (nan%)		nan (nan%)		0.9223 (18.24%)		1.43 (28.29%)
	100	57.4672	0.0838 (2.11%)	17.3398	nan (nan%)	0.0434	nan (nan%)	0.3143	0.1267 (3.19%)	0.2824	0.5963 (15.00%)
	102		-0.2042 (-6.68%)		nan (nan%)		nan (nan%)		-0.6188 (-20.24%)		-0.2366 (-7.74%)
	115		-0.237 (-67.01%)		nan (nan%)		nan (nan%)		-0.353 (-99.79%)		-0.3529 (-99.78%)
20D 恢复期	85		2.468 (16.02%)		nan (nan%)		nan (nan%)		3.14 (20.39%)		3.662 (23.77%)
	98		0.1784 (3.32%)		nan (nan%)		nan (nan%)		0.2658 (4.94%)		0.7524 (13.99%)
	100	59.4667	-0.308 (-7.13%)	7.6416	nan (nan%)	0.0387	nan (nan%)	0.3289	-0.4767 (-11.03%)	0.2639	-0.04502 (-1.04%)
	102		-0.7179 (-21.03%)		nan (nan%)		nan (nan%)		-1.106 (-32.44%)		-0.7729 (-22.64%)
	115		-0.4415 (-86.08%)		nan (nan%)		nan (nan%)		-0.5112 (-99.66%)		-0.5111 (-99.64%)

期平值期权的相对偏差达到 −49.57%，20D 崩盘前深度虚值期权的相对偏差甚至达到 −79.25%，完全无法满足定价精度要求；同时该方法的计算效率极低，单组案例的运行时间超过 45 秒，较学术基准案例的运行时间进一步提升。核心原因是极端市场参数下，高维修正贝塞尔函数与逆拉普拉斯变换的计算复杂度大幅上升，同时数值溢出风险显著增加，最终导致精度下降与计算效率恶化。

IG 与 Gamma 近似精确模拟方法虽未出现完全的数值失效，但在绝大多数场景下出现了系统性的大幅定价偏差，深度虚值期权的相对偏差普遍超过 −90%，平值期权的相对偏差也多在 10% 以上，仅在部分平稳市场场景下的实值期权中保持了相对可控的偏差。核心原因是极端市场场景下，积分方差的分布呈现出极端厚尾与右偏特征，仅匹配前两阶矩的 IG 与 Gamma 分布无法拟合真实的分布形态，最终导致系统性的定价偏差，这一现象在崩盘期的极端场景中表现得尤为显著。

6.6.4　傅里叶频域定价方法性能结果与分析

傅里叶频域定价方法包含快速傅里叶变换（FFT）方法与傅里叶余弦（COS）方法两类方案，完整测试结果如表6.6 所示。

从表6.6结果可以看出，傅里叶频域类方法在真实极端市场场景下依然保持了极致的稳定性与定价精度，是所有方法中表现最优的类别。FFT 与 COS 方法在所有 6 个测试案例中均未出现任何数值失效，即使在崩盘期的极端参数环境下，也能稳定输出定价结果；定价精度方面，两类方法的相对偏差基本控制在 0.2% 以内，绝大多数平值与实值期权的相对偏差趋近于 0，仅在深度虚值期权中出现了不超过 6% 的相对偏差，精

表 6.6 极端市场场景下傅里叶频域方法定价性能结果

场景	执行价	Fast Fourier Transformation		Fourier-Cosine Method	
		Time (s)	Option Bias	Time (s)	Option Bias
6D 崩盘前	85	0.2520	-0.006751 (-0.04%)	0.0645	-0.006749 (-0.04%)
	98		-0.003863 (-0.10%)		-0.003861 (-0.10%)
	100		-0.003562 (-0.14%)		-0.00356 (-0.14%)
	102		-0.003347 (-0.19%)		-0.003346 (-0.19%)
	115		-0.002481 (-5.23%)		-0.002479 (-5.23%)
6D 崩盘期	85	0.2182	-0.003449 (-0.02%)	0.0530	-0.00345 (-0.02%)
	98		-0.0009857 (-0.03%)		-0.0009859 (-0.03%)
	100		-0.0007312 (-0.03%)		-0.0007315 (-0.03%)
	102		-0.0006061 (-0.03%)		-0.0006102 (-0.03%)
	115		-0.0003863 (-0.42%)		-0.0003865 (-0.42%)
6D 恢复期	85	0.2378	-0.004141 (-0.03%)	0.0551	-0.004141 (-0.03%)
	98		-0.001658 (-0.04%)		-0.001658 (-0.04%)
	100		-0.001403 (-0.05%)		-0.001403 (-0.05%)
	102		-0.001176 (-0.06%)		-0.001176 (-0.06%)
	115		-0.0008781 (-1.31%)		-0.0008779 (-1.31%)
20D 崩盘前	85	0.2050	-0.001781 (-0.01%)	0.0503	-0.001781 (-0.01%)
	98		-0.0002175 (-0.00%)		-0.0002176 (-0.00%)
	100		-9.566e-05 (-0.00%)		-9.566e-05 (-0.00%)
	102		3.404e-06 (0.00%)		3.399e-06 (0.00%)
	115		0.002053 (0.05%)		0.002053 (0.05%)
20D 崩盘期	85	0.1893	-0.006654 (-0.01%)	0.0447	-0.006654 (-0.01%)
	98		-0.004071 (-0.08%)		-0.004071 (-0.08%)
	100		-0.003738 (-0.09%)		-0.003738 (-0.09%)
	102		-0.003539 (-0.12%)		-0.003539 (-0.12%)
	115		-0.002622 (-0.74%)		-0.002622 (-0.74%)
20D 恢复期	85	0.2092	-0.001571 (-0.01%)	0.0496	-0.001571 (-0.01%)
	98		-4.587e-05 (-0.00%)		-4.587e-05 (-0.00%)
	100		2.363e-05 (0.00%)		2.363e-05 (0.00%)
	102		8.295e-05 (0.00%)		8.295e-05 (0.00%)
	115		0.0002768 (0.05%)		0.0002768 (0.05%)

度远优于所有蒙特卡洛类方法。

计算效率方面，COS 方法的表现全面领先于 FFT 方法，单组案例的运行时间稳定在 0.05 秒左右，仅为 FFT 方法的 1/4，同时也远快于所有时间步进类与终端模拟类方法。核心原因在于 COS 方法的指数级收敛特性，仅需少量级数项即可达到极高的计算精度，同时不存在 FFT 方法的栅栏效应，无需额外的插值计算，进一步降低了计算开销。即使在崩盘期的极端参数环境下，两类方法的计算时间也未出现显著上升，展现出极强的鲁棒性。

6.6.4.1 与学术基准案例结果的对比分析

结合第五章的学术基准案例测试结果，本次真实极端市场场景的再检验得到了三项核心的补充结论，厘清了理想化环境与真实市场中定价方法性能的核心差异。

第一，极端市场场景显著放大了不同方法在数值稳定性上的分化。在第五章的学术基准案例中，基于 Fourier-Cosine 与矩匹配的终端精确模拟方法仅在部分高波动率场景中出现数值失效，而在本次真实极端市场测试中，两类方法在所有场景中完全失效，Baldeaux 精确模拟方法也出现了严重的精度恶化。这一结果表明，学术文献中标称“精确”的终端模拟方法，仅能在平稳市场的理想化参数下完成理论验证，在真实极端市场环境中极易出现数值溢出与计算失效，不具备工程实践层面的适配性。而时间步进类的分布抽样方法与傅里叶频域方法，在学术案例与真实极端场景中均保持了 100% 的数值稳定性，展现出极强的鲁棒性。

第二，极端市场场景对定价方法的精度提出了更高的要求，方法间的精度层级差异被进一步放大。在第五章的学术基准案例中，Euler/Milstein 格式在粗步长下的相对偏差多在 100% 以内，而在本次真实极端场景中，粗步长下的相对偏差普遍超过 1000%，仅在极细步长下才能达到可接受的精度；终端模拟类方法的定价偏差也较学术案例出现了数倍的放大，仅频域方法与时间步进分布抽样方法的精度保持稳定。这一结果表明，只有基于模型分布特性构建的精确抽样方法与基于特征函数的频域方法，才能在极端市场环境下保持稳定的定价精度，传统的离散格式与矩匹配近似方法在极端行情中会出现严重的精度失效。

第三，方法的效率排序在学术案例与真实极端场景中保持了高度一致性，但极端场景进一步凸显了频域方法的效率优势。在两类测试中，COS 方法始终是效率最高的高精度方法，Baldeaux 精确模拟方法始终是效率最低的方法；但在真实极端场景中，终端模拟类方法的计算时间进一步上升，而频域方法的计算时间保持稳定，二者的效率差距被进一步拉大。这一结果表明，傅里叶频域的 COS 方法不仅在平稳市场中具备效率优势，在极端行情下的效率优势更为显著，是真实市场环境中批量定价与参数校准的最优方案。

6.6.4.2 极端市场场景下的方法适用性总结

基于本次真实极端市场场景的测试结果，结合第五章的学术案例结论，可明确不同方法在加密货币期权市场极端行情下的适用性：傅里叶频域的 COS 方法是极端市场场景下的绝对最优方案，兼具全场景数值稳定性、极致的定价精度与最高的计算效率，适合极端行情下的期权批量定价、实时估值与模型参数校准；时间步进类的 Poisson-Gamma 与 QE 近似精确步进方法，是需要路径生成的路径依赖期权定价、动态对冲场景中的唯一可靠方案，在极端行情下仍保持了稳定的精度与效率；Euler/Milstein 离散格式仅能在极细步长下达到可接受的精度，效率极低，仅适合作为基准价格生成的工具，不推荐实际工程应用；所有终端模拟类方法在极端市场场景下均出现了严重的数值失效或大幅定价偏差，仅能在平稳市场的理想化参数下用于学术理论验证，不推荐在真实市场环境中使用。

6.7 本章小结

本章以 2025 年 10 月 10 日至 11 日 BTC 市场闪崩极端事件为研究对象，将 3/2 随机波动率模型的分析从第五章理想化的学术基准算例，拓展至真实金融市场的极端行情场景，通过构建“长期参数时间序列估计 - 时变参数横截面校准 - 定价方法极端场景再检验”的完整实证框架，系统识别了极端事件全周期 3/2 模型核心参数的时变规律，同时完成了九类期权定价方法在真实极端市场环境下的性能验证，形成了三项核心实证结论。

第一，3/2 随机波动率模型能够有效捕捉加密货币市场极端行情下的期权定价特征，模型核心参数在崩盘事件全周期呈现出显著的事件驱动型时变规律。基于崩盘前半年 DVOL 日度数据估计得到的长期均值回复参数，契合加密货币市场波动率快速均值回复的核心特性，长期波动率中枢水平与 BTC 市场的历史波动特征高度吻合，为模型校准奠定了稳定的长期参数基础。基于 Medvedev-Scaillet 渐近展开方法的横截面校准结果显示，平值隐含波动率在崩盘期出现大幅跳升，6 天短期期限的涨幅显著高于 20 天中期期限，反映出市场对极端事件的短期波动预期更为敏感，符合期权市场波动率期限结构的经典规律；波动率的波动率在崩盘期同步上升，说明极端事件不仅推高了市场整体波动率水平，同时放大了波动率自身的不确定性，市场尾部风险定价显著提升；价格与波动率的负相关系数在崩盘期出现阶段性弱化，尤其是短期期权合约的杠杆效应显著下降，反映出极端行情下的恐慌性交易与流动性冲击打破了市场常规的价格 - 波动率相关关系，而中期合约的相关关系则保持了相对稳定，对瞬时极端冲击的敏感度更低。整体来看，纯扩散框架下的 3/2 模型能够精准拟合崩盘事件全周期的波动率微笑形态演化，对加密货币市场极端行情下的期权定价具备良好的市场适配性。

第二，真实极端市场场景对期权定价方法的数值稳定性与鲁棒性提出了远高于学术基准算例的要求，理想化环境下的理论精度不等同于真实市场中的工程实用性。第五章的学术基准案例中，部分标称“精确”的终端模拟方法仅在少数高波动率场景中出现数值失效，而在本次真实极端市场测试中，基于 Fourier-Cosine 与矩匹配的终端精确模拟方法在所有 6 组测试案例中完全失效，无法输出有效定价结果；Baldeaux 精确模拟方法虽未出现完全的数值失效，但在绝大多数场景中出现了系统性的大幅定价偏差，且单案例计算时间超过 45 秒，计算效率极低。这一结果表明，此类标称“精确”的终端模拟方法仅能在平稳市场的理想化参数下完成理论层面的无偏性验证，在真实极端行情下极易出现特殊函数数值溢出、分布拟合失效等问题，不具备工程应用层面的极端场景适配性。与之形成鲜明对比的是，时间步进类的精确步进、Poisson-Gamma 与 QE 近似精确步进方法，以及傅里叶频域的 FFT 与 COS 方法，在所有极端市场场景中均保持了 100% 的数值稳定性，未出现任何计算失效，与第五章学术基准案例的性能表现高度一致，展现出极强的极端行情鲁棒性。

第三，基于真实市场参数的性能再检验，进一步明确了不同定价方法在加密货币期权市场的场景化适用边界。傅里叶频域的 COS 方法是极端市场场景下欧式期权定价的绝对最优方案，其在全场景保持了趋近于 0 的定价偏差，绝大多数场景下的相对偏差控制在 0.2% 以内，同时单案例计算时间稳定在 0.05 秒左右，计算效率远超其他所有方法，无论是极端行情下的期权实时估值、批量定价，还是模型横截面参数校准，均具备绝对的性能优势。时间步进类的 Poisson-Gamma 与 QE 近似精确步进方法，是路径依赖期权定价、动态对冲等需要完整方差路径生成场景中的唯一可靠方案，其在极端行情下兼顾了定价精度、计算效率与数值稳定性，即使在粗时间步长下也能保持可控的定价偏差，彻底弥补了传统 Euler/Milstein 离散格式步长敏感、极端场景精度极差的核心缺陷。Euler/Milstein 离散格式仅在极细时间步长下能达到可接受的定价精度，对应的计算成本极高，仅适合作为基准价格生成的工具，不推荐在实际工程应用中使用；所有终端模拟类方法均不具备真实极端市场环境下的应用价值，仅可用于平稳市场环境下的学术理论对比。

本章的实证研究弥补了现有 3/2 模型定价研究中理想化算例与真实市场环境脱节的不足，不仅验证了 3/2 模型在加密货币极端行情下的市场解释力，同时为极端市场环境下的期权定价方法选择提供了严谨的实证依据与明确的实践指引，为加密货币衍生品市场的定价与风险管理提供了可落地的分析框架。

第七章　结论与展望

本文以 3/2 随机波动率模型下的欧式期权定价问题为核心研究对象，针对现有研究中定价方法性能评价多局限于理想化学术基准算例、缺乏真实极端市场环境实证检验的不足，系统构建了 3/2 模型下九类欧式期权定价方法的完整理论框架，通过多场景数值实验量化了不同方法的定价精度、计算效率与数值稳定性，并基于 2025 年 10 月 BTC 市场闪崩极端事件，完成了模型的全周期参数校准与定价方法的极端场景再检验，厘清了不同方法在理想化环境与真实市场中的性能边界，为 3/2 模型在衍生品定价中的理论应用与工程实践提供了系统性的分析框架与实证依据。本章将系统总结全文的核心研究结论，客观分析研究存在的局限性，并对未来相关研究方向进行展望。

7.1　主要研究结论

本文通过理论构建、数值实验与实证分析三个层面的系统研究，形成了四项核心研究结论，完整覆盖了 3/2 模型的定价方法性能、市场适配性与实践应用边界。

第一，本文系统厘清了纯扩散框架下 3/2 模型九类期权定价方法的理论逻辑与性能边界，形成了清晰的方法性能分层体系。本文将九类定价方法划分为时间步进类条件蒙特卡洛方法、终端精确与近似精确模拟方法、傅里叶频域定价方法三大类别，明确了各类方法的实现路径与理论假设，并基于 7 组覆盖常规市场与极端参数环境的学术基准算例，完成了方法的系统性性能评价。研究结果显示，九类方法可按工程实用性划分为五个梯队：第一梯队为傅里叶频域的 COS 与 FFT 方法，是唯一在全场景保持数值稳定、同时具备极高定价精度与计算效率的方法，其中 COS 方法的综合性能全面领先；第二梯队为时间步进类的精确步进、Poisson-Gamma 与 QE 近似精确步进方法，是蒙特卡洛框架下唯一全场景无数值失效的方案，兼顾了定价精度与计算效率，是需要完整路径生成场景的唯一可靠选择；第三梯队为 Euler/Milstein 离散格式，仅在极细时间步长下能达到可接受的定价精度，计算效率极低，仅可作为理论精度对比的基准工具；第四梯队为 IG 与 Gamma 近似精确模拟方法，仅在特定平值场景下偏差可控，绝大多数场景出现系统性大幅定价偏差，仅可用于特定场景的学术对比；完全失效梯队为 Baldeaux 精确模拟、Fourier-Cosine 终端模拟与矩匹配终端模拟方法，仅能在单一基准场景下输出有效结果，其余场景完全数值失效。核心发现是，定价方法理论上的无偏性与“精确”标称，不等同于工程实践中的全场景稳定性，多数标称“精确”的终端模拟方法仅能在理想化基准参数下完成理论验证，无法适配非基准的极端参数环境。

第二，3/2 随机波动率模型能够有效捕捉加密货币高波动市场的期权定价特征，在

极端行情下具备良好的市场适配性。本文构建了“长期参数时间序列估计 - 时变参数横截面校准”的两步校准框架，基于崩盘事件前半年的 DVOL 日度数据完成了长期均值回复参数的估计，基于 Medvedev-Scaillet 渐近展开方法完成了崩盘事件全周期的时变参数校准。实证结果表明，纯扩散框架下的 3/2 模型能够精准拟合崩盘事件全周期波动率微笑的形态演化，模型核心参数呈现出显著的事件驱动型时变规律：崩盘期平值隐含波动率与波动率的波动率同步大幅跳升，6 天短期期限的涨幅显著高于 20 天中期期限，反映出市场对极端事件的短期波动预期与尾部风险定价显著提升；价格与波动率的负相关系数在崩盘期出现阶段性弱化，短期期权合约的杠杆效应异化更为显著，而中期合约的相关关系保持了相对稳定，体现了极端行情下恐慌性交易与流动性冲击对短期期权定价结构的显著影响。这一结果验证了 3/2 模型在加密货币高波动市场中的良好适配性，为加密衍生品市场的定价与风险测度提供了可靠的模型基础。

第三，真实极端市场场景对定价方法的数值稳定性与鲁棒性提出了远高于学术基准算例的要求，进一步放大了不同方法的性能分化。本文基于校准得到的 6 组真实市场参数，构建了全新的极端场景测试案例，完成了定价方法的性能再检验，并与学术基准算例结果进行了横向对比。研究发现，学术基准算例中仅部分场景失效的终端精确模拟方法，在真实极端场景中出现了全场景的数值失效或系统性大幅定价偏差，工程实用性完全丧失；而时间步进类的分布抽样方法与傅里叶频域方法，在学术案例与真实极端场景中均保持了 100% 的数值稳定性与稳定的定价精度，展现出极强的极端行情鲁棒性。同时，极端场景进一步凸显了 COS 方法的性能优势，其在全场景保持了趋近于 0 的定价偏差，单案例计算时间稳定在 0.05 秒左右，是极端市场环境下欧式期权批量定价、实时估值与模型校准的最优方案；Poisson-Gamma 与 QE 近似精确步进方法则是路径依赖期权定价与动态对冲场景的最优选择，彻底弥补了传统 Euler/Milstein 格式步长敏感、极端场景精度极差的核心缺陷。

第四，本文基于理论与实证结果，形成了 3/2 模型期权定价方法的场景化应用指引。对于欧式期权批量定价、模型参数校准等场景，优先选择 COS 频域方法，其全场景稳定性、最高精度与最优效率可充分适配此类场景的需求；对于路径依赖期权定价、动态对冲等需要完整方差路径生成的场景，优先选择 Poisson-Gamma 或 QE 近似精确步进方法，其全场景稳定性与精度效率平衡可满足此类场景的工程需求；对于理论精度基准对比场景，仅可在基准参数环境下使用 Baldeaux 精确模拟方法；其余终端类方法不推荐在实际定价场景中使用，仅可用于特定场景的学术对比分析。

7.2　研究局限性

本文围绕 3/2 随机波动率模型的期权定价问题开展了系统性的理论与实证研究，形成了较为完整的分析框架，但受研究范围、数据条件与模型设定的限制，仍存在一定的局限性，主要体现在以下四个方面：

第一，模型设定的维度存在拓展空间。本文聚焦于纯扩散框架下的 3/2 随机波动率模型，未在模型中引入标的价格跳跃项、随机利率与随机流动性等市场摩擦因素。尽管纯扩散模型已能较好地拟合崩盘事件全周期的波动率微笑形态，但极端行情下标的价格的跳跃特征、市场流动性的瞬时变化，仍可能对期权定价产生系统性影响，模型设定对真实市场结构的刻画仍有进一步丰富的空间。

第二，实证研究的覆盖范围有待拓宽。本文的实证分析仅基于 Deribit 交易所的 BTC 期权数据，未覆盖 ETH 等其他主流加密资产，也未拓展至传统金融市场的股票、大宗商品、外汇等标的。不同标的资产的波动特征、市场微观结构与投资者结构存在显著差异，3/2 模型在不同市场中的适配性与参数特征，仍需更多维度的实证数据进行交叉验证。

第三，定价方法的评价范围存在局限。本文的定价方法性能评价仅聚焦于欧式期权，未针对美式期权、障碍期权、亚式期权等主流路径依赖型衍生品开展分析。3/2 模型在路径依赖类期权定价中的方法构建、精度优化与性能评价，是现有研究未充分覆盖的内容，也是本文研究范围的核心局限所在。

第四，参数校准框架的丰富度仍有提升空间。本文采用的 Medvedev-Scaillet 渐近展开方法更适用于短期到期期权，对于超长期到期期权的校准精度仍有优化空间；同时本文仅采用单一校准方法完成参数估计，未引入基于特征函数的全局校准、贝叶斯校准等方法对参数估计结果进行稳健性对比，校准框架的多元性与稳健性仍有进一步提升的可能。

7.3　未来研究展望

基于本文的研究结论与存在的局限性，未来可从以下五个方向开展进一步的深入研究，完善 3/2 随机波动率模型的定价理论与实践应用体系：

第一，拓展 3/2 随机波动率模型的理论设定。在纯扩散模型的基础上，引入同步跳跃、独立跳跃、随机利率与随机流动性等因素，构建更贴合真实市场结构的扩展 3/2 模型，推导对应模型的特征函数闭式解与期权定价公式，量化不同扩展因素对期权定价的边际影响，进一步提升模型对极端市场环境的适配性；同时开展扩展模型与基准模型的定价精度横向对比，明确不同模型设定的适用场景。

第二，拓展定价方法的适用范围与计算性能优化。针对美式期权、障碍期权、亚

式期权等主流路径依赖型衍生品，构建适配 3/2 模型方差过程特性的高效定价方法，完成不同方法的定价精度、计算效率与数值稳定性的系统性评价，形成覆盖全品类衍生品的 3/2 模型定价体系；同时结合 GPU 并行计算、向量化优化等技术，进一步提升大规模蒙特卡洛模拟与频域方法的计算效率，满足实际业务中高频实时定价的需求。

第三，拓展实证研究的市场覆盖维度。将实证分析从 BTC 期权拓展至 ETH 等其他主流加密资产，以及全球股票市场、大宗商品市场、外汇市场等传统金融领域，验证 3/2 模型在不同波动特征、不同市场结构下的适配性，明确模型的通用适用边界；同时基于更长周期的历史数据，开展 3/2 模型与 Heston 模型、SABR 模型等主流随机波动率模型的大样本横向对比，量化不同模型的定价精度与市场解释力差异，为不同市场的模型选择提供实证依据。

第四，优化 3/2 模型的参数校准框架。针对渐近展开方法在长期期权校准中的不足，引入基于特征函数的全局校准、贝叶斯校准、机器学习辅助校准等方法，提升极端市场场景下的参数校准精度与稳定性；同时构建模型参数的实时动态校准体系，实现极端行情下模型参数的高频更新，适配衍生品市场实时风险管理的需求。

第五，基于 3/2 模型构建衍生品全流程风险管理体系。在定价研究的基础上，推导 3/2 模型下期权希腊字母的闭式解与高效数值计算方法，构建基于模型的 VaR、CVaR 等尾部风险测度体系，量化极端行情下期权组合的风险暴露；同时基于模型参数的时变特征，构建波动率风险溢价的测度模型，为加密衍生品市场的对冲策略设计、动态资产配置提供理论支撑与实证依据。

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